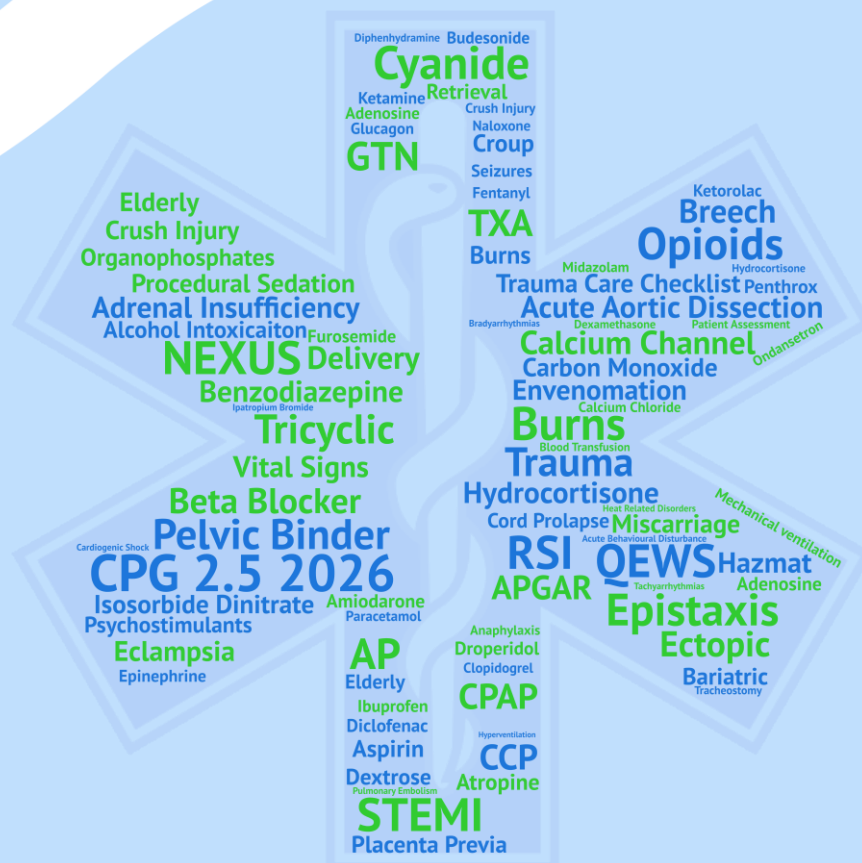


Ambulance Service Clinical Practice Guidelines

CPG 2026
V2.5

- ✓ INTRODUCTION
- ✓ CLINICAL PRACTICE GUIDELINE
- ✓ FORMULARY
- ✓ CLINICAL SCOPE OF PRACTICE
- ✓ REFERENCES



VERSION CONTROL

VERSION HISTORY			
Version number	Description of work completed	Name of author/s or reviser/s	Completion date
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Message from the CEO

Dear Colleagues

The delivery of high-quality evidence-based healthcare is essential for improving patient outcomes and ensuring the best possible clinical practice as part of our drive to be a world leading high performing ambulance service, including in both access and effectiveness.

These Clinical Practice Guidelines (CPG's) have been developed to provide you with clear, up to date, and practical recommendations for diagnosis, treatment, and management of a range of conditions faced in the prehospital setting.

The purpose of these HMCAS Clinical Practice Guidelines is to:

- Aim to support clinical staff in making informed decisions while also promoting consistency in care across diverse health care settings
- Provide a common clinical practice framework for clinical staff from different healthcare systems
- Ensure consistent application of evidence, latest scientific research, best practice, clinical trials and expert opinion-based clinical management.
- Provide guidance to facilitate individual patient needs when it comes to decision making for infrequently encountered or complex clinical conditions.
- Facilitate a defence against alleged malpractice.

The Guidelines reflect a patient centered and practical approach and where appropriate, link in with receiving hospital treatment pathways. The CPGs cannot cover all eventualities encountered within the complex and dynamic environment of prehospital care. Safety – Benefit – Effectiveness trade-offs have to be made on a daily basis. At times, clinicians will have to undertake care for conditions not covered or use available tools and techniques for situations beyond the published guidelines.

The guiding principle for those situations is to provide safe and effective patient care by concentrating on core patient outcomes:

- Saving and preserving of life
- Relieving suffering
- Facilitate safe transfer to a specialist center or hospital
- Maximizing recovery from critical injury or illness

HMCAS guidelines will be reviewed regularly taking into account changing national and international guidelines, feedback from clinical staff of all grades as well as hospital-based colleagues. Best Practice Guidelines are an essential aspect of effective clinical governance, and these Clinical Practice Guidelines form the corner stone of evidence-based practice, clinical audit, pathway review and pathway development. For the Ambulance Service to learn, feedback from clinical practice is required. Clinicians have a professional responsibility to highlight deviations and situations not covered by the CPGs to the medical directorate for the organisation to review its practice and adapt.

Ambulance Paramedics remain the backbone of the ambulance service, providing care in all patient contacts. The Clinical Guidelines address the vast majority of patient contacts, where prehospital intervention is required to be performed by an ambulance paramedic. Should a deviation from the clinical Guidelines be required, wherever possible, the attending Ambulance Paramedic should discuss the patient's needs with the Clinical Coordinator based within the National Command Centre.

Critical Care Paramedics target a smaller number of patients who may require more advanced interventions. They will face clinical situations that cannot be predicted and may have to, on occasions, act outside of these published guidelines to manage a patient according to the principles mentioned above. The wealth of experience, within our staff, remains essential to optimising patient outcomes and therefore our clinical guidelines outline variations in clinical practice between Ambulance Paramedics and Critical Care Paramedics.

I encourage you to integrate these CPG's into your day to day clinical practice and thank you and those who develop and review these guidelines; for all you do!

Mr. Daren Mochrie
Chief Executive Officer
HMC Ambulance Service



Message from the Medical Director

Over the past years, and since publishing the last version of the HMC Ambulance Service (HMC AS) clinical practice guidelines, much has happened to the service. In order to respond to rising clinical need within the State of Qatar, the service has grown in size and has diversified into specialized spheres of operations. In order to maintain the appropriate response to calls, by either the public or the various medical facilities, the service now has more hubs, spokes and clinical units to accommodate the widest spectrum of clinical need.

HMC AS' clinical staff is the greatest asset the service has, therefore maintaining HMC AS clinicians' professional standing through appropriate guidance, training and education is of the highest importance. To that end, this latest version of HMC AS' clinical practice guidelines has been developed to support that theme. These guidelines are based on the latest evidence and best practice in the pre-hospital field. They have been updated through review of the latest published evidence and the learning from the HMC AS' clinical frontline.

Finally, HMC AS clinicians should view these guidelines as clinical intervention recommendations to help with their clinical decision-making.

Dr. Loua Asad Hanna Al Shaikh

Group Medical Director

HMC Ambulance Service

Specific Clinical Privileges

STAFF CIRCULAR HMCAS REF. NO. AS/SC 17-27

In addition to established procedure as per HMCAS Clinical Guidelines, Critical Care and Ambulance Paramedics are privileged to autonomously carry out the clinical activities below subject to having undertaken in-service training and being deemed competent (signed-off by the Head of Professions) or following an individual command decision by the on-call Silver Clinical.

PROCEDURE	AP	CCP	COMMENTS
Care, maintenance and management of chest tubes	Carry out	Carry out	
Use of ventilators	Assist	Carry out	Any ventilator in service use
Use of invasive and non-invasive ventilation modes	Assist	Carry out	
Adjustment of ventilator settings to implement lung protective ventilation or action on blood gases to acceptable parameters	Assist	Carry out	Silver Clinical discussion required
Care, use, maintenance and management of central lines	Assist	Carry out	
Care, use, maintenance and management of arterial lines	Assist	Carry out	
Management of ECMO patients as directed by medical team	Assist	Carry out	
Care, maintenance, and management of indwelling urinary catheters	Assist	Carry out	
Insertion of indwelling urinary catheters	Assist	Carry out	
Use of point of care diagnostic devices for blood gas, electrolytes and haemoglobin, basic interpretation and actioning on results	Assist	Carry out	
Basic and advanced nursing care (e.g. Trach care, eye and mouth care, pressure care)	Assist	Carry out	
Transport of Level 2 and 3 patients	Carry out	Carry out	Level 3 patients under indirect medical supervision at the discretion of Silver Clinical or on-call Transfer and Retrieval ICU Consultant
Continuation of drug therapy, outside service specific formulary	Assist	Carry out	
Continuation of drug therapy outside service specific formulary where prescribed by the referral hospital	Assist	Carry out	
Use of syringe and IV drivers	Assist	Carry out	
Initiation, on independent assessment of patient, of drug therapy within service specific formulary	Assist	Carry out	

Assessment of in-hospital level 2 and 3 patients for transfer and basic physiological manipulation to improve clinical stability before and during transport	Assist	Carry out	
Assessment using ultrasound	Assist	Carry out	
Basic interpretation of radiological interventions for the purpose of verifying the correct position of indwelling medical devices	Assist	Carry out	

SYMBOL	PRIVILEGING LEVEL
	LEVEL 1 PRIVILEGES – First Responder (Basic Life Support)
	LEVEL 2 PRIVILEGES – reduced scope Ambulance Paramedic
	LEVEL 3 PRIVILEGES – full scope Ambulance Paramedic
	LEVEL 4 PRIVILEGES – full scope Critical Care Paramedic

All HMCAS staff are assigned a privileging level based on their area of operations and their clinical exposure. The symbols above depict each of the four privileging levels and are utilised throughout this document to define the capabilities of staff in terms of skills and medication administration for emergency and non-emergency clinical situations. Staff are restricted to practice within their privileged scope only. It is the responsibility of the individual staff member to ensure they maintain licensure with DHP (where required) and to remain current through CPD and self-directed learning.

Treatment in the algorithms is divided into the two colours below, indicating scope of practice limited to CCPs only and what can be carried out by both CCPs and APs. Although the treatment algorithms do not differentiate treatment for Level 1 and Level 2 practitioners, these practitioners are limited to their scope of practice as per their privileging letter.

 - **AP and CCP**

 - **CCP only**

ABBREVIATIONS

Of the accepted abbreviations, the use of upper and/or lower case is acceptable.

>	Greater than	RHF	Right Heart Failure
≥	Greater than or Equal to	gm	Gram
<	Less than	h/o	History of
≤	Less than or Equal to	Hb, Hgb	Haemoglobin
AAA	Abdominal Aortic Aneurysm	HHS	Hyperosmolar Hyperglycaemic Syndrome
ABD	Acute Behavioural Disturbance	HIV	Human Immunodeficiency Virus
ABC	Airway Breathing Circulation	HR	Heart Rate
ACF	Antecubital fossa	HTN	Hypertension
ACS	Acute Coronary Syndrome	hx	History
AF	Atrial Fibrillation	I:E Ration	Inspiratory Expiratory Ratio
AIDS	Acquired Immune Deficiency Syndrome	ICP	Intracranial Pressure
ant.	Anterior	IDDM	Insulin Dependent Diabetes Mellitus
AP	Ambulance Paramedic	IHD	Ischemic Heart Disease
A-P	Anteroposterior	IM	Intramuscular
APH	Antepartum Haemorrhage	IMV	Intermittent Mandatory Ventilation
ARDS	Adult Respiratory Distress Syndrome	INF	Infusion
ARF	Acute Renal Failure	INH	Inhaled
AVPU	Alert, Verbal, Painful, Unresponsive	Inj	Injection
Bilat or B/L	Bilateral	IO	intraosseous
BIPAP	Bi-Level Positive Airway Pressure	IPPV	Intermittent Positive Pressure Ventilation
BP	Blood Pressure	IPC	Infection Prevention and control
bpm	Beats per minute	IV	Intravenous
BSA	Body Surface Area	JVP	Jugular Venous Pressure
BVM	Bag Valve Mask	kg.	Kilogram
BW	Birth weight	KVO	Keep Vein Open
c/o	Complains of	L/min	Litres per minute
Ca	Cancer	LBBB	Left Bundle Branch Block
CAD	Coronary Artery Disease	LBW	Low Birth Weight
CCP	Critical Care Paramedic	LLQ	Left Lower Quadrant
CHF	Congestive Heart Failure	LMA	Laryngeal Mask Airway
CNS	Central nervous system	LMP	Last Menstrual Period
COPD	Chronic Obstructive Pulmonary Disease	LOC	Level of Consciousness
CPAP	Continuous Positive Airway Pressure	LUQ	Left Upper Quadrant
CPR	Cardiopulmonary Resuscitation	LVF	Left Ventricular Failure
CRF	Chronic Renal Failure	MAP	Mean Arterial Pressure
CSF	Cerebrospinal Fluid	MAS	Meconium Aspiration Syndrome
C-Spine	Cervical Spine	Max	Maximum
CT Scan	Computerised Axial Tomography Scan	mcg	microgram
CVA	Cerebrovascular Accident	MDI	Metred-dose inhaler
CVS	Cardiovascular System	mg	Milligram
D5W	5% Dextrose in Water	MI	Myocardial Infarction
DDx	Differential Diagnosis	min	Minute
DKA	Diabetic Ketoacidosis	MIR	Major Incident Response
DNAR	Do Not Attempt Resuscitation	ml	Millilitre
Dr	Doctor	mmol	Millimoles
DSED	Dual Sequential External Defibrillation	mV	Millivolt
DVT	Deep Venous Thrombosis	N & V	Nausea & Vomiting
Dx	Diagnosis	NA	Not Applicable
ECG	Electrocardiogram	NaCl	Sodium Chloride
ECHO	Echocardiogram	NAD	No Abnormality Detected
EtCO ₂	End Tidal Carbon Dioxide	NEB	Nebuliser

ETT	Endotracheal Tube	NGT	Nasogastric Tube
FiO ₂	Fraction of Inspired Oxygen	NIDDM	Non-insulin dependent diabetes mellitus
GCS	Glasgow Coma Scale	NIV	Non-invasive ventilation
GI	Gastrointestinal	RLQ	Right Lower Quadrant
GTN	Glycerine Trinitrate	RN	Registered Nurse
NRBM	Non-rebreather mask	RR	Respiratory Rate
NSAID	Non-steroid anti-inflammatory drug	RSI	Rapid Sequence Induction
NSR	Normal Sinus Rhythm	RTA	Road Traffic Accident
NSTEMI	Non-ST Elevation Myocardial Infarction	RUQ	Right Upper Quadrant
O/E	On Examination	SAT	Sedation Assessment Tool
O ₂	Oxygen	sec	Second
O ₂ Sat	Oxygen Saturation	SGA	Supraglottic Airway
OD	Overdose	SIDS	Sudden Infant Death Syndrome
PAF	Paroxysmal Atrial Fibrillation	SIMV	Synchronised intermittent mandatory ventilation
PCI	Percutaneous Coronary Intervention	SL	Sublingual (under the tongue)
PE	Pulmonary Embolism	SOB	Shortness of breath
PEA	Pulseless electrical activity	SpO ₂	Oxygen Saturation via pulse oximetry
PEEP	Positive End Expiratory Pressure	STEMI	ST-elevation Myocardial Infarction
PERRLA	Pupils Equal, Round, Reactive to Light and Accommodation	SVD	Spontaneous Vaginal Delivery
pH	Hydrogen Ion Concentration	SVT	Supraventricular Tachycardia
PID	Pelvic Inflammatory Disease	T	Temperature
PIH	Pregnancy Induced Hypertension	TBI	Traumatic Brain Injury
PIP	Peak Inspiratory Pressure	TV	Tidal Volume
PMS	Pulse motor sensory	Tx	Treatment
PND	Paroxysmal Nocturnal Dyspnoea	URTI	Upper Respiratory Tract Infection
po	Per oral, by mouth	UTI	Urinary Tract Infection
PPH	Post-Partum Haemorrhage	VF	Ventricular Fibrillation
prn	As often as necessary	V/Q	Ventilation/perfusion
PROM	Premature Rupture of Membranes	Vol	Volume
PSV	Pressure Support Ventilation	Vtach	Ventricular Tachycardia
pt	Patient	wk	Week
PTB	Pulmonary Tuberculosis	wt	Weight
PTL	Pre-term labour	x/7	Number of days
PV	Per Vaginum	x/12	Number of months
PVC	Premature Ventricular Contraction	x/52	Number of weeks
RBBB	Right Bundle Branch Block	yrs	Years
RBS	Random Blood Sugar		

DO NOT USE ABBREVIATIONS

Abbreviations and Symbols	Potential Problem	Use instead
dc	Could mean "discontinue" or "discharge"	Discontinue or discharge
HS, hs	Hs could mean "half strength" or "bedtime"	Half strength or bedtime
iu (international unit)	Mistaken for iv (intravenous) or the number 10 (ten)	Write "international unit"
od	Could mean "right eye" or "once daily"	Right eye or once daily
trailing zero "x.0 mg" lack of leading zero ".x mg"	Decimal point is missed	Write "x mg" or "0.x mg"
u (unit)	Mistaken for "0" (zero), the number "4" (four) or "cc"	Write "unit"

Acknowledgements

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GET IN TOUCH

If you have any comments, suggestions or feedback relating to the CPGs or clinical practice in general please email HMCAS-Clinicalguidelines@hamad.qa

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CLINICAL APPROACH AND PATIENT ASSESSMENT



CPG 1.1 CLINICAL APPROACH AND ASSESSMENT OF THE ADULT PATIENT

The approach and assessment of adult patients using the primary and secondary survey can be used for both medical and trauma cases, with certain components adjusted based on the presentation of the patient. The aim is to obtain a comprehensive clinical picture of the patient.

RESPONDER SAFETY AND SCENE SIZE-UP

- Standard and environmental precautions
- Observe for any dangers (crew, patient, bystanders)
- Identify number of patients and need for additional support (Ambulances, Police, Civil Defence)
- Consider infectious disease risk

PRIMARY SURVEY

MEDICAL:	TRAUMA:
Check Response (AVPU)	Catastrophic haemorrhage/C-spine control
Assess Airway	Check Response (AVPU)
Assess Breathing	Assess Airway
Assess Circulation	Assess Breathing
	Assess Circulation



Manage Life-Threats as they are identified. Do not delay life-saving interventions. Begin resuscitation if the patient present with no signs of life (unless undeniable death).

SECONDARY SURVEY

HISTORY	Obtain a comprehensive history:	
	O Onset P Provokes Q Quality R Radiates S Severity T Timing	S Signs/Symptoms A Allergies M Medications P Past Medical History L Last oral intake E Events prior
VITAL SIGNS ASSESSMENT	Perform a detailed patient assessment (as appropriate): • Heart rate • Respiratory rate • Blood pressure • Temperature • SpO₂ • AVPU/GCS • RBS (where applicable) • Capillary refill time • Pain assessment • EtCO₂ (when appropriate) • 12-lead ECG (when appropriate) • B.E.F.A.S.T (when appropriate) • Sedation Assessment Tool (when appropriate)	

PHYSICAL EXAMINATION (head-to-toe)	Complete a physical exam of the patient as required based on the chief complaint. A thorough head-to-toe examination is particularly applicable in the trauma patient.
---	--

VITAL SIGN	NORMAL RANGE
Blood pressure	SBP 110 – 140 mmHg/ DBP 60 – 90 mmHg
Heart rate	51 – 100 beats/min
Respiratory rate	12 – 20 breaths/min
Temperature	36 – 37.6°C
RBS	4 – 6.7 mmol/L

HEAD-TO-TOE EXAM

HEAD	
General	<i>Inspect for:</i> lacerations, deformity, facial muscle tone, asymmetry. <i>Palpate for:</i> crepitus, bony tenderness, subcutaneous emphysema.
Eyes	<i>Inspect:</i> pupils, Raccoon eyes
Ears	<i>Inspect:</i> Blood or CSF, Battle's sign
Nose	<i>Inspect:</i> deformity, epistaxis
Mouth	<i>Inspect:</i> loose teeth, bite malocclusion, airway/tongue swelling
Voice	<i>Inspect:</i> hoarseness
NECK	
Inspect	Deformity, lacerations, raised JVP
Palpate	Tracheal position, bony tenderness, carotid pulse, subcutaneous emphysema, lymphadenopathy
CHEST	
Inspect	Expansion, paradoxical movement, accessory muscle use, lacerations, deformity
Palpate	Tenderness, subcutaneous emphysema, bony crepitus
Auscultate	Air entry, breath sounds
ABDOMEN	
Inspect	Laceration, bruising, distention
Palpate	Tenderness, guarding, rigidity, rebound tenderness
PELVIS	
Inspect	Lacerations, bruising, deformity, priapism (spinal trauma)
BACK	
Inspect	Laceration, bruising, deformity
Palpate	Bony tenderness, evidence of a bony step, subcutaneous emphysema
UPPER & LOWER LIMBS	
Inspect	Laceration, bruising, deformity, shortening or rotation
Palpate	Bony tenderness, crepitus, PMS



MONITORING REQUIREMENTS

- Transported patients require basic vital signs to be recorded at a minimum of every 15 minutes or a minimum of 2 sets if patient contact time is less than 15 minutes. Basic vital signs consisting of:
 - HR (4-lead ECG/Pads)
 - RR
 - BP
 - SpO₂
 - LOC (AVPU/GCS)
 - Pain score
 - Temperature (when applicable. E.g. Fever, heat illness)
- Continuous observational and vital signs monitoring should be continued until handover at the receiving facility. Aim to record vital signs every 15 minutes for stable patients and every 5 minutes for unstable/critically ill or injured patients.
- Patients that refuse vital signs assessment or when vital sign assessment is not possible should be clearly documented in the EPCR.
- Continuous EtCO₂ together with basic monitoring is required for the following:
 - Advanced airway placement
 - Administration of Fentanyl, Ketamine or Midazolam
 - The recent administration (within last 1 hour) of any sedative or narcotic medication administered in a healthcare facility or by a carer.
 - Critically ill or injured patient/priority patients
- Defib pads must be applied in the presence of a confirmed AMI and the critically ill or injured patient. Consider placing defib pads in the A/P position.
- Good clinical judgement must be used when assessing and monitoring patients in order to determine the most appropriate clinical approach and treatment.



TRANSPORT

- Ensure scene time is appropriate based on the patient's condition.
- Transport to nearest, most appropriate facility.
- Transport pathway patients to their predefined destination.
- Update the CCD for hospital prenotification for any pathway, critical or priority 1 patients.
- Should a CCP be required and is not on scene, arrange a rendezvous when appropriate. Do not delay transport waiting for a CCP at the rendezvous point if the transport time to hospital is shorter than the rendezvous time.

CPG 1.2

CLINICAL APPROACH AND ASSESSMENT OF THE PAEDIATRIC PATIENT

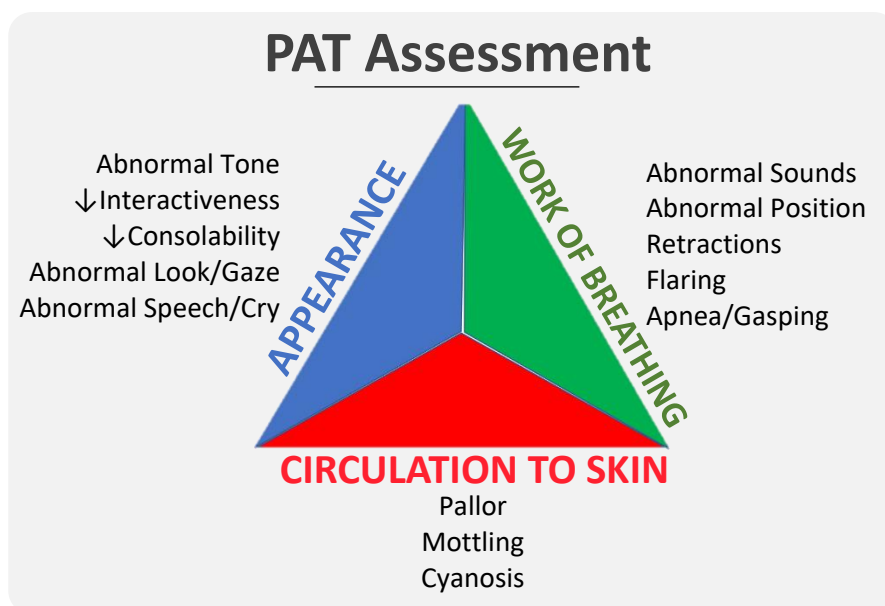
The approach and assessment of paediatric patients using the primary and secondary survey can be used for both medical and trauma cases, with certain components adjusted based on the presentation of the patient. The aim is to obtain a comprehensive clinical picture of the patient in order to guide treatment.

RESPONDER SAFETY AND SCENE SIZE-UP

- Standard and environmental precautions
- Observe for any dangers (crew, patient, bystanders)
- Identify number of patients and need for additional support (Ambulances, Police, Civil Defence)
- Consider infectious disease risk

PRIMARY SURVEY

The paediatric assessment triangle is a standardised tool for the emergency assessment of critically ill and injured paediatrics of all age groups. It is designed to be a quick initial assessment conducted in 60 seconds that does not require any equipment.



Tone	Is the child active, listless or lethargic? Is the child moving spontaneously, sitting or standing as appropriate for age or supporting the weight of their head?
Interactivity	Does the child interact with you or their parents or caregivers? Are you able to gain the child's interest or engage them in play activity? Does the child react to sounds?
Consolability	can the child be comforted by their parents or caregiver? Is the child behaving to external stimuli as you would predict?
Look/Gaze	Is the child able to fix their gaze on their parents, caregiver, or the clinician?
Speech/Cry	Is the child using speech appropriate for age? Does the child have a strong, vigorous cry or a weak cry?
MEDICAL:	TRAUMA:
Check Response (AVPU)	Catastrophic haemorrhage/C-spine control
Assess Airway	Check Response (AVPU)

Assess Breathing	Assess Airway
Assess Circulation	Assess Breathing
	Assess Circulation

! Manage Life-Threats as they are identified. Do not delay life-saving interventions. Begin resuscitation if the patient present with no signs of life (unless undeniable death).

SECONDARY SURVEY	
HISTORY	Obtain a comprehensive history: O Onset P Provokes Q Quality R Radiates S Severity T Timing S Signs/Symptoms A Allergies M Medications P Past Medical History L Last oral intake E Events prior
VITAL SIGNS ASSESSMENT	Perform a detailed patient assessment (as appropriate): - Heart rate - Respiratory rate - Blood pressure - Temperature - SpO₂ - AVPU/GCS - Capillary refill time - Pain assessment - EtCO₂ (when appropriate) - Sedation Assessment Tool (when appropriate) - RBS (where applicable)
PHYSICAL EXAMINATION (head-to-toe)	Complete a physical exam of the patient as required based on the chief complaint. A thorough head-to-toe examination is particularly applicable in the trauma patient. Certain components of the head-to-toe examination may be applicable in the medical patient.

NORMAL VITAL SIGNS RANGE

Age Group	Weight	RR (breaths/min)	HR (beats/min)	SBP (mmHg)	DBP (mmHg)	Temp (°C)	RBS (mmol/L)
Preterm	2kg	40-70	110-170	55-75	35-45	36.5	2.6-6
Term – 2 months	3 – 4kg	35-55	110-160	65-85	45-55	36.5	2.6-6
3 – 7 months	5 – 7kg	35-45	110-160	70-90	50-65	36.5	4-6
8 – 12 months	8 – 11kg	22-38	90-160	80-100	55-65	36.5	4-6
1 – 3 years	12 – 15kg	22-30	80-150	90-105	55-70	36.5	4-6
4 – 5 years	16 – 21kg	20-24	70-120	95-110	60-75	36.5	4-6
6 – 7 years	22 – 28kg	20-24	70-120	95-110	60-75	36.5	4-6
8 – 9 years	29 – 34kg	16-22	60-110	100-120	60-75	36.5	4-6
10 – 11 years	35 – 40kg	16-22	60-110	100-120	60-75	36.5	4-6
12 – 13 years	41 – 45kg	12-20	60-100	110-135	65-85	36.5	4-6
≥ 14 years	46 – 50kg	12-20	60-100	110-135	65-85	36.5	4-6



ESTIMATED WEIGHT CALCULATIONS

0 – 12 months	$(\text{age} \times 0.5) + 4$
1 – 5 years	$(\text{age} \times 2) + 8$
6 – 14 years	$(\text{age} \times 3) + 7$
Circulating blood volume	80 – 85ml/kg

HEAD-TO-TOE EXAM

HEAD	
General	<i>Inspect for:</i> lacerations, deformity, facial muscle tone, asymmetry. <i>Palpate for:</i> crepitus, bony tenderness, subcutaneous emphysema.
Eyes	<i>Inspect:</i> pupils, Raccoon eyes
Ears	<i>Inspect:</i> Blood or CSF, Battle's signs
Nose	<i>Inspect:</i> deformity, epistaxis
Mouth	<i>Inspect:</i> loose teeth, bite malocclusion, airway/tongue swelling
Voice	<i>Inspect:</i> hoarseness
NECK	
Inspect	Deformity, lacerations, raised JVP
Palpate	Tracheal position, bony tenderness, carotid pulse, subcutaneous emphysema, lymphadenopathy
CHEST	
Inspect	Expansion, paradoxical movement, accessory muscle use, lacerations, deformity
Palpate	Tenderness, subcutaneous emphysema, bony crepitus
Auscultate	Air entry, breath sounds
ABDOMEN	
Inspect	Laceration, bruising, distention, priapism (spinal trauma)
Palpate	Tenderness, guarding, rigidity, rebound tenderness
PELVIS	
Inspect	Lacerations, bruising, deformity
BACK	
Inspect	Laceration, bruising, deformity
Palpate	Bony tenderness, evidence of a bony step, subcutaneous emphysema
UPPER & LOWER LIMBS	
Inspect	Laceration, bruising, deformity, shortening or rotation
Palpate	Bony tenderness, crepitus, PMS



MONITORING REQUIREMENTS

- Transported patients require basic vital signs to be recorded at a minimum of every 15 minutes or a minimum of 2 sets if patient contact time is less than 15 minutes. Basic vital signs consisting of:
 - HR (4-lead ECG when applicable)
 - RR
 - BP
 - SpO₂
 - LOC (AVPU/GCS)
 - Pain score
- Continuous observational and vital signs monitoring should be continued until handover at the receiving facility.
- Patients/family members/guardians that refuse vital signs assessment or when vital sign assessment is not possible should be clearly documented in the ePCR.
- Continuous EtCO₂ together with basic monitoring is required for the following patients:
 - Advanced airway placement
 - Administration of Fentanyl, Ketamine or Midazolam
 - Critically ill or injured patient/priority patients
- Defib pads must be applied to the critically ill or injured patient.
- AP's may monitor an IV line in the paediatric patient ≤1 year old but may not establish an IV in the ≤1-year olds unless under CCP supervision.



TRANSPORT

- Ensure scene time is appropriate based on the patient's condition
- Transport to nearest most appropriate facility
- Transport pathway patients to their predefined destination.
- Update the CCD for hospital prenotification for any pathway, critical or priority 1 patients
- Should a CCP be required and is not on scene, arrange a rendezvous when appropriate. Do not delay transport waiting for a CCP at the rendezvous point if the transport time to hospital is shorter than the rendezvous time.

CPG 1.3

CLINICAL APPROACH AND ASSESSMENT OF THE OBSTETRIC PATIENT

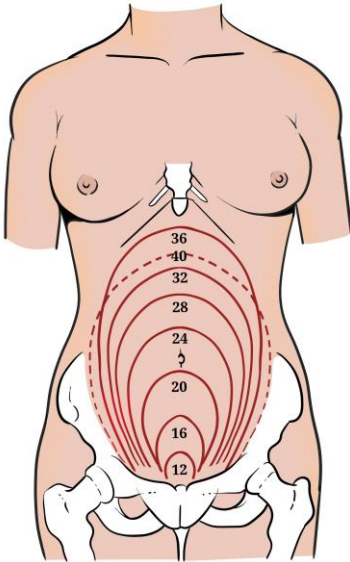
Approach and assess the obstetric patient according to CPG 1.1.

Various physiological changes occur during pregnancy. These changes should be taken into consideration when assessing and managing the obstetric patient. Gathering an obstetric focus history in addition to a standard history (SAMPLE) is of particular importance to identify potential issues that may be encountered.

OBSTETRIC FOCUSED HISTORY

- Gravidity
- Parity
- Gestation
- Previous pregnancy related complications
- Any complications identified with the current pregnancy
- Onset of labour
 - Frequency and duration of contractions
- Membrane rupture
 - Time of membrane rupture
 - Colour of fluid (large volume of blood or meconium present?)
- Past medical history (not obstetric related)

FUNDUS HEIGHT LANDMARKS

12 weeks gestation	Height of symphysis pubis	
20 weeks gestation	Height of the umbilicus	
36 weeks gestation	Height of xiphoid process	
37-40 weeks gestation	Fundus regresses to between 32-36cm	

PHYSIOLOGICAL CHANGES DURING PREGNANCY

HAEMATOLOGY

There is an overall increase in blood volume (approximately 1500ml). During the first trimester, the blood plasma increases by 15% and peaks at 50% above the normal (32 weeks).

CARDIOVASCULAR SYSTEM

The heart enlarges due to an increase in blood volume and increase in cardiac output. The heart rate increases up to 25% by the end of the second trimester. Pressure on the diaphragm displaces the heart upwards, rotating it forwards and displaced laterally. Due to the placenta acting as arteriovenous shunt and peripheral vasodilation, systolic blood pressure may drop slightly (6-8%) with a more marked drop in diastolic blood pressure during the first two trimesters. The following ECG changes occur during pregnancy (no clinical significance):

- Sinus tachycardia with atrial and ventricular ectopics
- ST segment depression and t-wave inversion in the inferior and lateral leads

RESPIRATORY SYSTEM

The upper respiratory tract becomes more vascular with oedema and hyperaemia developing in the nose, pharynx, larynx, trachea and bronchi. Maternal oxygen requirements increase. The thoracic cage increases by 5 – 7 cm in diameter. The diaphragm is displaced upwards by the enlarging uterus. Due to the diaphragm's inability to descend during inspiration, thoracic breathing replaces abdominal breathing in late pregnancy. As oxygen consumption increases minute ventilation increases by 40% with increased respiratory rate and tidal volume from the first trimester to term by a total 45%. Functional residual capacity decreases by 20-30%. Anatomical and physiological changes in both the cardiovascular and respiratory system causes a general feeling of breathlessness particular on slight exertion but also consider underlying pulmonary effects of cardiovascular disease.

UTERUS

The uterus changes from an initial weight of roughly 70g (10ml) to a term weight of 1.1kg (5L). The perfusion of the placenta is dependent on maternal circulation to the uterus and as the uterus increases in size blood flow increases rapidly. The total increase in blood flow to the uterus is up to twenty times normal. By the end of pregnancy, about a sixth of the maternal blood volume is within the uterine circulation.

CPG 1.4 CONSCIOUS STATUS ASSESSMENT

A patient's level of consciousness should be assessed using either the AVPU and/or the Glasgow Coma Scale (GCS), depending on aetiology (medical or trauma). This assessment can be used for both adults and paediatrics.

AVPU SCALE

A	Alert	Person is aware of who they are, where they are and time.
V	Verbal	Person appears drowsy, confused or unresponsive, but responds when spoken to or to a loud shout
P	Pain	Appears unresponsive, but responds with movement or groaning when painful stimulus is applied
U	Unresponsive	No response to stimulus (verbal or painful)

GLASGOW COMA SCALE (GCS)

Adult/Child		Infant	
Eye Opening			
Spontaneous	4	Spontaneous	4
Reacts to speech	3	Reacts to speech	3
Reacts to pain	2	Reacts to pain	2
No response	1	No response	1
Verbal response			
Orientated	5	Babbles, follows objects	5
Confused	4	Irritable, cries	4
Inappropriate words	3	Cries to pain	3
Incomprehensible	2	Moans and grunts	2
No response	1	No response	1
Motor response			
Obeys commands	6	Spontaneous	6
Localises to pain	5	Localises to pain	5
Withdraws from pain	4	Withdraws from pain	4
Flexion response	3	Flexion response	3
Extension response	2	Extension response	2
No response	1	No response	1

- When assessing the level of consciousness, take into account the effects of alcohol or other intoxicating substances that the patient may have consumed.
- Regularly assess the patient's LOC to identify any deterioration or improvement in their LOC status.

CPG 1.5 AIRWAY & RESPIRATORY STATUS ASSESSMENT

The airway and respiratory status assessments form a vital part of patient assessment to identify any potential life-threats. A detailed respiratory status assessment should be conducted for all patients that present with respiratory distress of any severity or that have a respiratory related complaint. A complete assessment will guide patient care.

AIRWAY STATUS ASSESSMENT

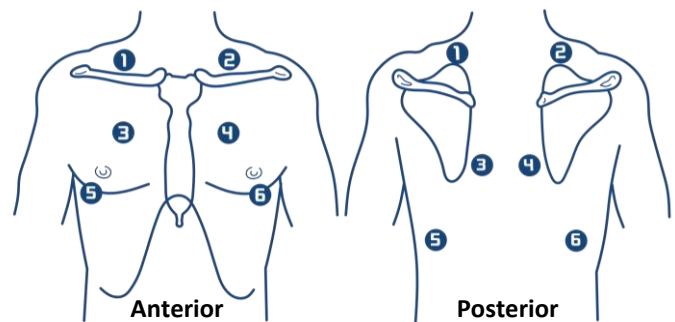
- Conscious patients that can talk have a patent airway.
- A patient's airway should be assessed when:
 - They are unable to speak properly
 - Have a reduced LOC
 - Abnormal sounds coming from the airway
- Any occlusion to the airway should be dealt with immediately.
 - Airway suctioning
 - FBAO removal
 - Placing patient in lateral position
 - Basic airway manoeuvres/devices
 - Advanced airway placement

RESPIRATORY STATUS ASSESSMENT

A comprehensive respiratory status assessment involves multiple components. Paramedic must identify what is abnormal and the severity of the abnormality when assessing a patients' respiratory status and manage appropriately.

CHEST AUSCULTATION

- Position the patient upright if possible (dependent on patients' condition)
- Ask patient to breath normally through their mouth
- Listen to both sides of the chest and all lung fields in a methodical manner and compare.



RESPIRATORY STATUS ASSESSMENT		
COMPONENTS	NORMAL	RESPIRATORY DISTRESS
Conscious state	Alert	Altered
General appearance	Calm and quiet	Distressed, anxious, struggling to breath, exhausted
Speech	Clear, fluent and steady	Difficult, speaks in short sentences or phrases, unable to talk
Respiratory rate	Adult: 12-20 Paediatric: age dependent (see CPG 1.2 for normal values)	Adult: > 20 Paediatric: age dependent (see CPG 1.2 for normal values)
Respiratory rhythm	Regular even cycles	No respiratory pause, prolonged expiratory phase
Respiratory effort	Minimal with little chest or abdominal movement	Marked chest movement, accessory muscle use
Skin	Pink	Pale, sweaty, cyanosis (late and serious sign)
Pulse rate	Adult: 50-100 Paediatric: age dependent (see CPG 1.2 for normal values)	Tachycardia, bradycardia (late and serious sign)
Breath sounds	Clear (no abnormal sounds)	Stridor, wheeze, crackles, abnormal breath sounds, decreased or absent

CPG 1.6 PERFUSION STATUS ASSESSMENT

Early assessment of a patients' perfusion status is a crucial part of patient care. Inadequate perfusion will negatively impact on a patients' blood pressure, pulse rate and level of consciousness. Early identification and appropriate management to correct perfusion abnormalities are essential.

ADULT PERFUSION STATUS ASSESSMENT				
	SKIN	PULSE	SBP	LOC
Adequate perfusion	Warm, pink, dry	51 – 100 bpm	>100 mmHg	Alert and orientated
Borderline perfusion	Cool, pale, clammy	51 – 100 bpm	80 – 100 mmHg	Alert and orientated
Inadequate perfusion	Cool, pale, clammy	<50 bpm or >120 bpm	60 – 80 mmHg	Either alert or altered LOC
Grossly inadequate perfusion	Cool, pale, clammy	<50 bpm or >120 bpm	<60 mmHg or unrecordable	Altered LOC or unconscious
No perfusion	Cool, pale, clammy	No palpable pulses	Unrecordable	Unconscious

PAEDIATRIC PERFUSION STATUS ASSESSMENT				
	SKIN	PULSE	SBP	LOC
Adequate perfusion	Warm, pink, dry	80 – 160 bpm	>70 mmHg	Alert and interacting normally for age
Borderline perfusion	Cool, pale, clammy	60 – 180 bpm	50 – 70 mmHg	Alert and interacting normally for age
Inadequate perfusion	Cool, pale, clammy	<60 bpm	40 – 50 mmHg	Alert and interacting normally for age or confused
Grossly inadequate perfusion	Cool, pale, clammy	<50 bpm	<40 mmHg or unrecordable	Altered LOC or unconscious
No perfusion	Cool, pale, clammy	No palpable pulses	Unrecordable	Unconscious

CPG 1.7 MENTAL STATUS ASSESSMENT

A mental status assessment is designed to systematically evaluate a patient's thought processes at a particular point in time in order to guide care. Patients presenting with behavioural abnormalities should have a mental status assessment conducted if possible. It is important to note that a mental status assessment is not used to diagnose a specific condition.

MENTAL STATUS ASSESSMENT

- Patients presenting with a behavioural abnormality should be assessed in an attempt to identify the cause of the presenting signs and symptoms. Approach these patients in a compassionate, non-threatening manner.
- Exclude and/or manage causes of abnormal behaviour where possible (e.g. alcohol/drug intoxication, CVA).
- Only attempt to treat the patient if it is safe to do so.
- Use the mental status assessment guide to conduct a patient assessment. The assessment should be conducted in a highly respectful and empathetic manner. Do not be disrespectful, judgemental or interrogatory as this will most likely worsen the circumstances.
- Be cognisant that different cultures hold different beliefs and view mental illness differently. Some may be accepting whereas others may see it as taboo.

MENTAL STATUS ASSESSMENT GUIDE

Appearance	• Grooming • Posture • Build • Clothing • Cleanliness	Thought form	• Amount • Rate • Derailment • Flight of ideas
Behaviour	• Eye contact • Mannerisms • Gait • Activity level	Thought content	• Disturbances • Delusions • Suicidal • Obsessions
Speech	• Rate • Volume • Pitch • Tone • Flow • Pressure	Perception	• Illusions • Thought insertion • Broadcasting • Hallucinations: - Auditory, olfactory, tactile, visual, gustatory
Mood	Emotion described as: • Anxious • Depressed • Cheerful	Insight & judgement	• Cognition • Illness • Understanding • Cause & effect
Affect	Emotion observed as: • Restrictive • Blunted • Labile		

CPG 1.8 QATAR EARLY WARNING SCORE (QEWS)

The QEWS is used to identify clinically at-risk patients. A QEWS should be calculated once the first set of basic vital signs has been obtained. A score of ≥ 4 is a marker of increased clinical risk to the patient. Request for CCP assistance if required based on the clinical presentation of the patient. Manage the patient according to the relevant CPG.

HMCAS QEWS TABLE							
QEWS	3	2	1	0	1	2	3
RESPIRATORY RATE	< 8			9–20	21–25	26–35	> 36
HEART RATE	< 31	31– 40	41– 50	51–100	101–110	111–129	> 129
SYSTOLIC BP	< 70	71– 89	90 –109	110–179	180–199	> 200	
TEMPERATURE		C(<36°C)		N(36°–38°C)		H(> 38°C)	
SpO ₂ on Air	< 90%	90 – 92%	93 – 94%	$\geq 95\%$			
AVPU	U	P	V	A			
A score of 4 or more is a marker of an increased risk to the patient.							TOTAL =



CPG 2.1-2.8

CARDIAC ARREST RESUSCITATION



CPG 2.1 ADULT MEDICAL CARDIAC ARREST

Cardiac arrest management is a time critical event. A potentially successful outcome following a cardiac arrest is determined by key interventions, represented by the “Chain of survival”:

1. Early recognition of cardiac arrest and call for backup
2. Early CPR with emphasis on effective chest compressions
3. Early defibrillation if indicated
4. Effective and timely advanced life support
5. Integrated post-cardiac arrest care



CLINICAL PRESENTATION

No signs of life:

- Unresponsive
- No breathing or no normal breathing (agonal breathing)



MANAGEMENT

The management principles during medical cardiac arrest are:

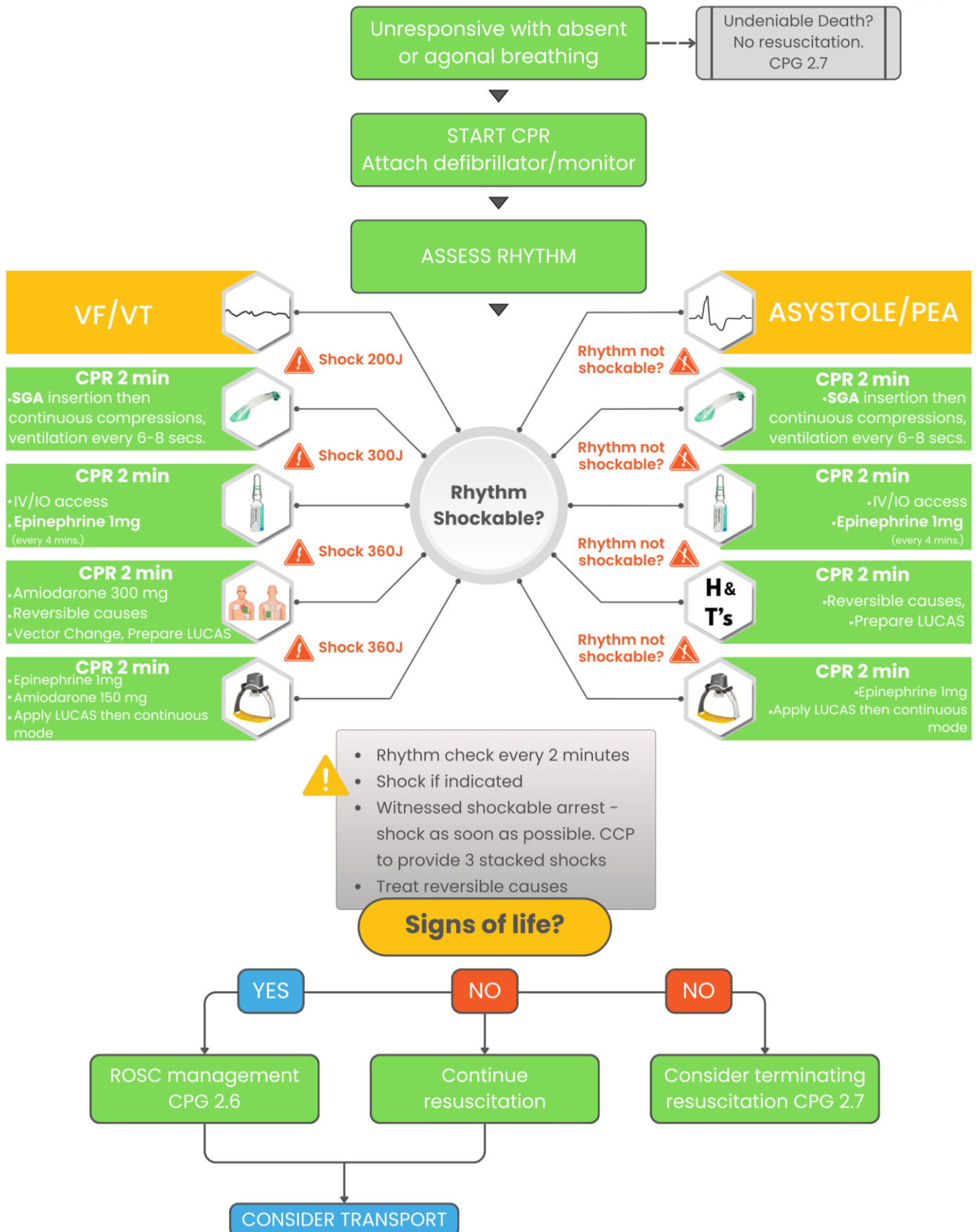
- Ensuring high quality compressions (rate, depth, recoil, position, compress at least one-third of the chest diameter and minimal interruptions)
- Early defibrillation of VF/VT
- Effective oxygenation and ventilation
- Effective teamwork with clearly identified roles
- Correct reversible causes



SPECIAL NOTES

- Ensure defibrillation pads are placed in the correct position with good contact. Poor pad placement will lead to ineffective defibrillation.
- Vector change defibrillation – APs are to perform a vector change following 3 consecutive defibrillations and the rhythm remains shockable, perform a vector change by changing pad placement to the anterior-posterior position.
- Pad position for anterior-posterior pad placement:
 - Pad 1 – left posterior back, left of spine, just below the scapula.
 - Pad 2 – left anterior chest between midline of chest and left nipple.
- CCPs may perform DSED for refractory VF/VT following 3 consecutive defibrillations without termination of VF/VT or if the maximum joule escalation has been reached (i.e. on shock 4 or more). Once DSED is initiated, all subsequent shocks should be DSED.
- CCPs do not need to provide a vector change defibrillation before proceeding to DSED. CCPs may proceed directly to DSED when indicated.
- CCPs may perform 3 stacked shocks for witnessed cardiac arrest presenting with a shockable rhythm.

- The SGA is the primary airway adjunct to be used during cardiac arrest. CCPs should consider replacing the SGA with an ETT once all necessary interventions have been performed, particularly if there is a high likelihood of ROSC or if transporting the patient.
- CCPs should consider inserting a gastric tube via the SGA or an orogastric tube if an ETT is in place to prevent aspiration.
- Attach an HME/filter to the SGA/ETT.
- EtCO₂ measurement with an advanced airway in place is an effective tool to measure cardiac output and indication of ROSC.
- CPR induced consciousness should be managed with a bolus dose of Ketamine 1mg/kg IV/IO.
- Consider terminating resuscitation after 20 minutes of CPR if no response to treatment and no signs of life present. See CPG 2.7.
- Use of the mechanical ventilator during cardiac arrest is not recommended.



Defibrillation Energy • LP15 (Biphasic): 200J, 300J, 360J • Zoll (Biphasic): 120J, 150J, 200J	Reversible Causes • Hypovolaemia • Hypoxia • Hydrogen ions (acidosis) • Hypoglycaemia • Hypo/hyperkalaemia • Hypothermia • Tension pneumothorax • Tamponade, cardiac • Toxins • Thrombosis, cardiac • Thrombosis, pulmonary
Advanced Airway • SGA as primary airway. • CCPs may consider ETI after approx. 10 min of resuscitation • Waveform capnography must be used with all advanced airways	
CPR • Push hard (~5cm) and fast (100-120/min) and allow complete chest recoil • Hand position on lower half of sternum • Minimise interruptions • Avoid excessive ventilation – ventilate according to LUCAS when applied • Change compressor every 2 minutes • LUCAS application after 6 minutes regardless of presenting rhythm • SGA/ETT airway in place – continuous compressions. If an SGA air leak occurs as a result of continuous compressions, revert to 30:2	  

CPG 2.2 ADULT TRAUMA CARDIAC ARREST

Cardiac arrest management is a time critical event. A potentially successful outcome following a cardiac arrest is determined by key interventions, represented by the “Chain of survival”:

1. Early recognition of cardiac arrest and call for backup
2. Correction of trauma specific reversible causes
3. Early CPR with emphasis on effective chest compressions
4. Early defibrillation if indicated
5. Effective and timely advanced life support
6. Integrated post-cardiac arrest care

CLINICAL PRESENTATION

No signs of life:

- Unresponsive
- No breathing or no normal breathing (agonal breathing)
- No palpable carotid pulse

MANAGEMENT

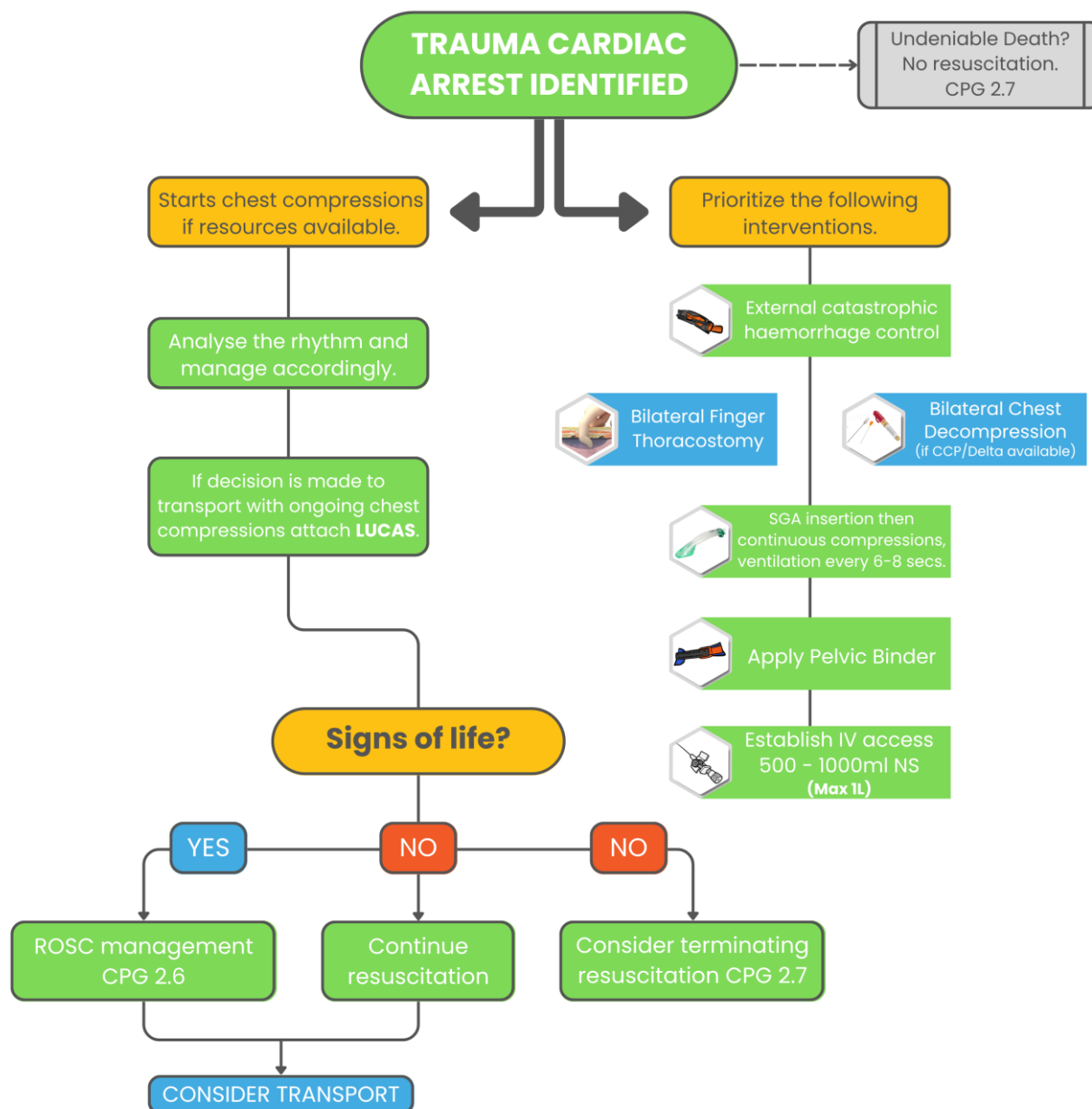
The management principles during trauma cardiac arrest include those of medical cardiac arrest and:

- Controlling catastrophic external haemorrhage
- Correcting hypovolaemia
- Controlling the airway and maximising oxygenation
- Bilateral chest decompression by Delta and bilateral finger thoracostomy by CCP

SPECIAL NOTES

- Ensure defibrillation pads are placed in the correct position with good contact. Poor pad placement will lead to ineffective defibrillation.
- Vector change defibrillation – APs are to perform a vector change following 3 consecutive defibrillations and the rhythm remains shockable, perform a vector change by changing pad placement to the anterior-posterior position.
- Pad position for anterior-posterior pad placement:
 - Pad 1 – left posterior back, left of spine, just below the scapula.
 - Pad 2 – left anterior chest between midline of chest and left nipple.
- CCPs may perform DSED for refractory VF/VT following 3 consecutive defibrillations without termination of VF/VT or if the maximum joule escalation has been reached (i.e. on shock 4 or more). Once DSED is initiated, all subsequent shocks should be DSED.
- CCPs do not need to provide a vector change defibrillation before proceeding to DSED. CCPs may proceed directly to DSED when indicated.
- CCPs may perform 3 stacked shocks for witnessed cardiac arrest presenting with a shockable rhythm.
- The SGA airway is the primary airway adjunct to be used during cardiac arrest. CCPs should consider replacing the SGA with an ETT once all necessary interventions have been performed, particularly if there is a high likelihood of ROSC or if transporting the patient.
- CCPs should consider inserting a gastric tube via the SGA or an orogastric tube if an ETT is in place to prevent aspiration.

- Attach an HME/filter to the SGA/ETT.
- EtCO₂ measurement with an advanced airway in place is an effective tool to measure cardiac output and indication of ROSC.
- CPR induced consciousness should be managed with a bolus dose of Ketamine 1mg/kg IV/IO.
- Perform compressions and correct reversible causes (haemorrhage control etc) simultaneously. If insufficient resources are available, correct reversible causes first before starting compressions.
- Do not administer adrenaline until reversible causes and appropriate volume replacement (at least 500ml normal saline) has occurred.
- Consider terminating resuscitation after 20 minutes of CPR if there is no response to treatment and no signs of life present. See CPG 2.7.
- Use of the mechanical ventilator during cardiac arrest is not recommended.



Defibrillation Energy

- LP15 (Biphasic): 200J, 300J, 360J
- Zoll (Biphasic): 120J, 150J, 200J

Advanced Airway

SGA as primary airway.

- CCPs may consider ETI after approx. 10 min of resuscitation
- Waveform capnography must be used with all advanced airways

CPR

- Push hard (~5cm) and fast (100-120/min) and allow complete chest recoil
- Hand position on lower half of sternum
- Minimise interruptions
- Avoid excessive ventilation – ventilate according to LUCAS when applied
- Change compressor every 2 minutes
- LUCAS should be applied prior to transport if transporting patient to hospital.
- SGA/ETT airway in place – continuous compressions. If an SGA air leak occurs as a result of continuous compressions, revert to 30:2
- *AP may perform needle decompression under supervision in trauma cardiac arrest

Reversible Causes

- Hypovolaemia
- Hypoxia
- Hydrogen ions (acidosis)
- Hypoglycaemia
- Hypo/hyperkalaemia
- Hypothermia
- Tension pneumothorax
- Tamponade, cardiac
- Toxins
- Thrombosis, cardiac
- Thrombosis, pulmonary



CPG 2.3 PAEDIATRIC MEDICAL CARDIAC ARREST

Cardiac arrest management is a time critical event. A potentially successful outcome following a cardiac arrest is determined by key interventions, represented by the “Chain of survival”:

1. Early recognition of cardiac arrest and call for backup
2. Early CPR with emphasis on effective chest compressions and ventilations
3. Early defibrillation if indicated
4. Effective and timely advanced life support
5. Integrated post-cardiac arrest care

CLINICAL PRESENTATION

No signs of life:

- Unresponsive
- No breathing or no normal breathing (agonal breathing)
- No palpable carotid pulse

MANAGEMENT

The management principles during medical cardiac arrest are:

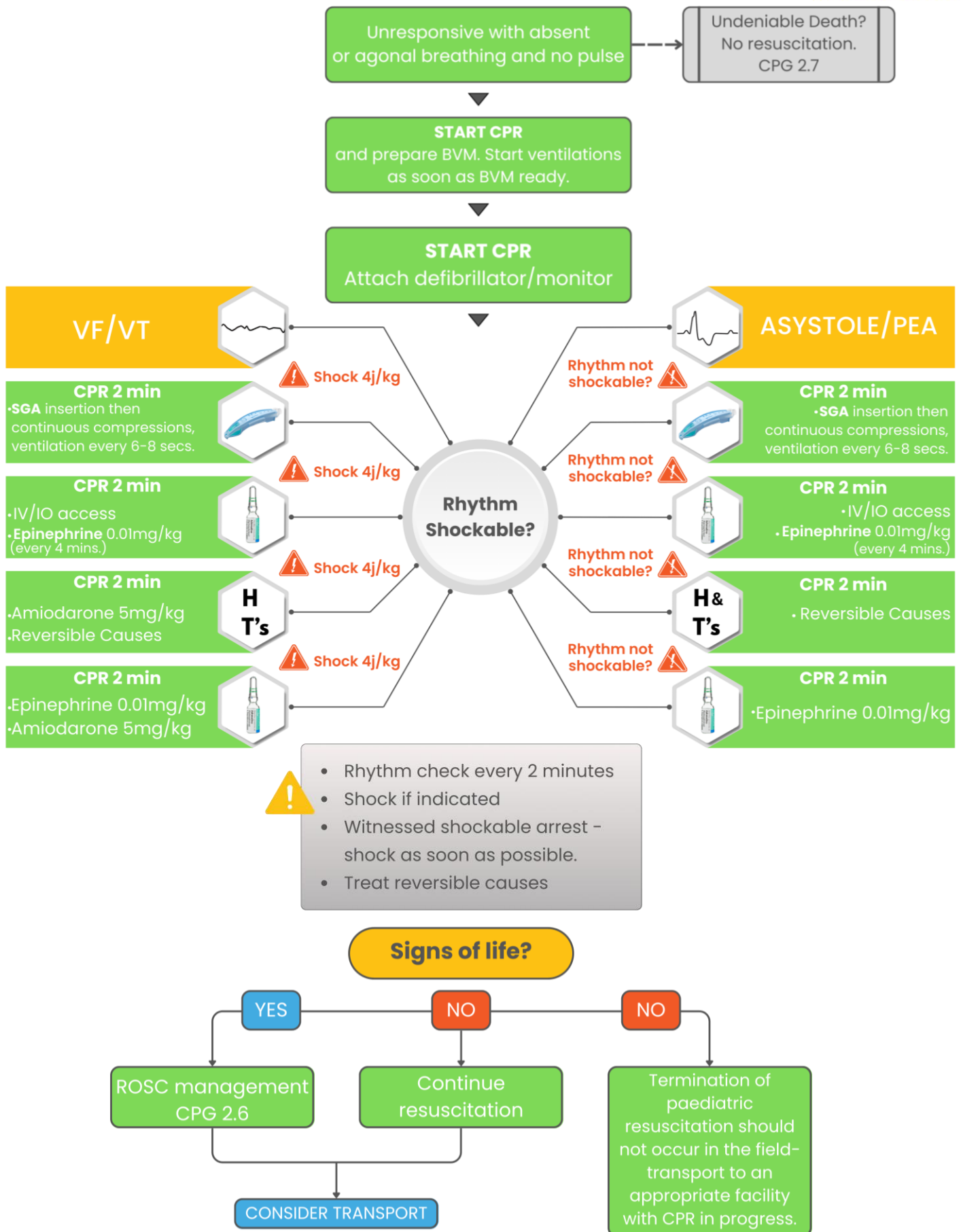
- Ensuring high quality compressions (rate, depth, recoil and minimal interruptions)
- Early defibrillation of VF/VT
- Effective oxygenation and ventilation
- Effective teamwork with clearly identified roles
- Correct reversible causes

SPECIAL NOTES

- Ensure defibrillation pads are placed in the correct position with good contact. Poor pad placement will lead to ineffective defibrillation.
- Vector change defibrillation – following 3 consecutive defibrillations and the rhythm remains shockable, consider a vector change by changing pad placement to the anterior-posterior position.
- Pad position for anterior-posterior pad placement:
 - Pad 1 – left posterior back, left of spine, just below the scapula.
 - Pad 2 – left anterior chest between midline of chest and left nipple.
- The SGA airway is the primary airway adjunct to be used during cardiac arrest.
- CCPs should consider inserting a gastric tube via the SGA to prevent aspiration.
- Attach an HME/filter to the SGA/ETT.
- EtCO₂ measurement with an advanced airway in place is an effective tool to measure cardiac output and indication of ROSC.
- CPR induced consciousness should be managed with a bolus dose of Ketamine 1mg/kg.
- Termination of resuscitation for paediatric patients is not a routine practice and may only occur in special circumstances. See CPG 2.7.
- Use of the mechanical ventilator during cardiac arrest is not recommended.
- CCPs may escalate the joules if required after 3 shocks in increments of 2J for every subsequent shock to a maximum of 10J/kg or adult dosing.

The WAAFELSS mnemonic may be used to help guide care while managing a paediatric cardiac arrest:

W	Weight: 0-12 months = (age x 0.5) + 4, 1-5 years = (age x 2) + 8, 6-14 years = (age x 3) + 7
A	Adrenaline: 0.01mg/kg (0.1ml/kg 1:10,000)
A	Amiodarone: 5mg/kg
F	Fluid: 10-20 ml/kg x 3
E	Energy: 4J/kg & CCPs can escalate to max 10J/kg
L	SGA-size: size 1 = 2-5 kg, size 1.5 = 5-12 kg, size 2 = 10-25 kg, size 2.5 = 25-35 kg. ET-size: uncuffed (age÷4) + 4 and cuffed (age÷4) + 3.5. Depth: (age ÷ 2) + 12
S	Sugar: 2.5ml/kg of 10% Dextrose
S	Systolic BP: (age x 2) + 70



Defibrillation Energy First and subsequent shocks at 4J/kg. CCPs may escalate to a max of 10J/kg or adult dose	Reversible Causes - Hypovolaemia - Hypoxia - Hydrogen ions (acidosis) - Hypoglycaemia - Hypo/hyperkalaemia - Hypothermia - Tension pneumothorax - Tamponade, cardiac - Toxins - Thrombosis, cardiac - Thrombosis, pulmonary
Advanced Airway - SGA as primary airway. - CCPs must consider ETI after approx. 10 min of resuscitation (if suitable ETT size available) - Waveform capnography must be used with all advanced airways	
CPR - Push hard (>1/3 of anteroposterior diameter of chest)) and fast (100-120/min) and allow complete chest recoil - Hand position on lower half of sternum - Minimise interruptions. - Change compressor every 2 minutes - Avoid excessive ventilation - SGA/ETT airway in place – continuous compressions. If an SGA air leak occurs as a result of continuous compressions, revert to 15:2/30:2	

CPG 2.4 PAEDIATRIC TRAUMA CARDIAC ARREST

Cardiac arrest management is a time critical event. A potentially successful outcome following a cardiac arrest is determined by key interventions, represented by the “Chain of survival”:

1. Early recognition of cardiac arrest and call for backup
2. Correction of trauma specific reversible causes
3. Early CPR with emphasis on effective chest compressions
4. Early defibrillation if indicated
5. Effective and timely advanced life support
6. Integrated post-cardiac arrest care



CLINICAL PRESENTATION

No signs of life:

- Unresponsive
- No breathing or no normal breathing (agonal breathing)
- No palpable carotid pulse

Signs of inadequate perfusion

- Pallor and/or central cyanosis
- Pulse < 60 bpm



MANAGEMENT

The management principles during trauma cardiac arrest include those of medical cardiac arrest and:

- Controlling catastrophic external haemorrhage
- Correcting hypovolaemia
- Controlling the airway and maximising oxygenation
- Bilateral chest decompression

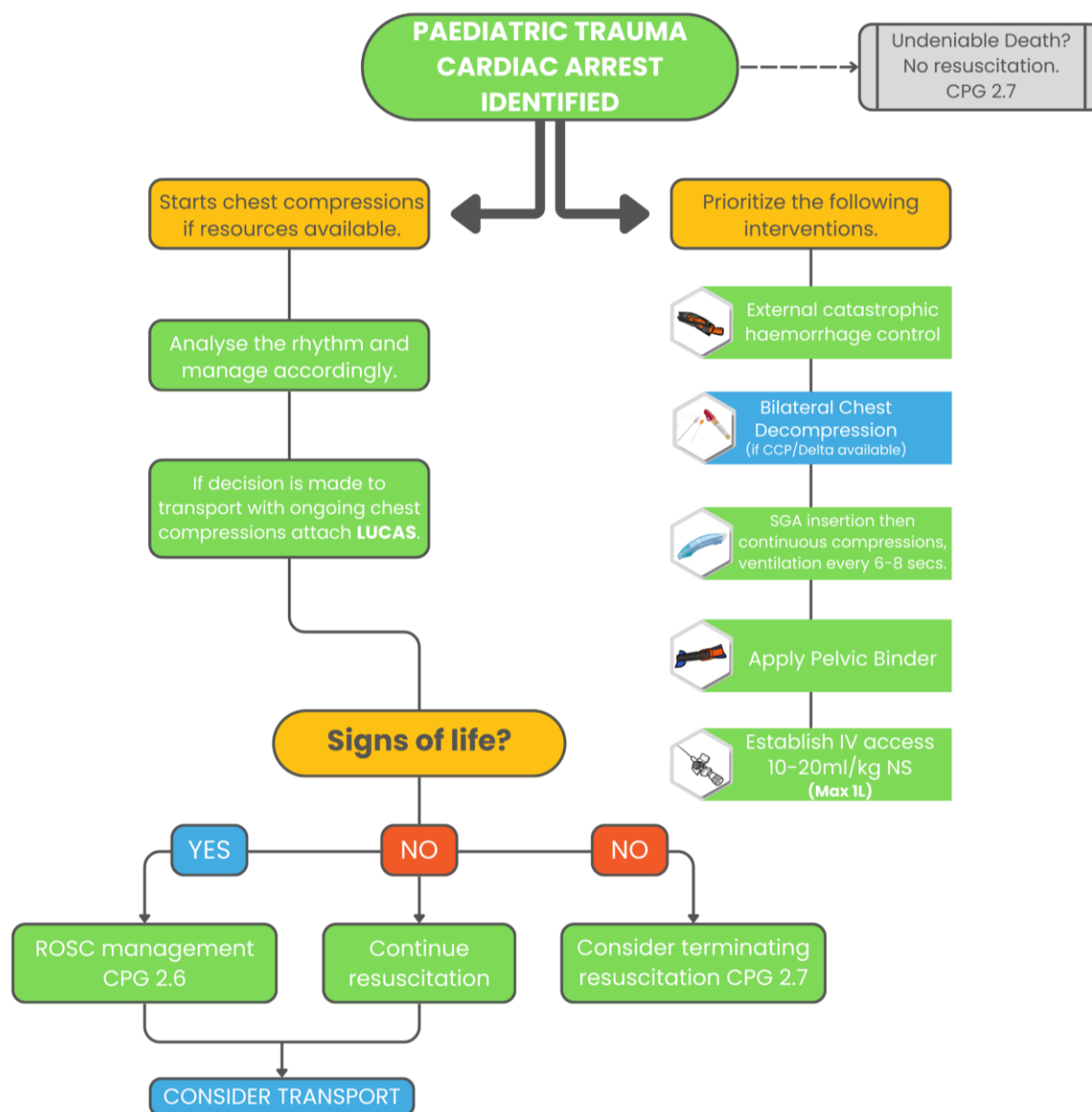


SPECIAL NOTES

- Perform compressions and correct reversible causes (haemorrhage control etc) simultaneously. If insufficient resources are available, correct reversible causes first before starting compressions.
- Ensure defibrillation pads are placed in the correct position with good contact. Poor pad placement will lead to ineffective defibrillation.
- Vector change defibrillation – following 3 consecutive defibrillations and the rhythm remains shockable, consider a vector change by changing pad placement to the anterior-posterior position.
- Pad position for anterior-posterior pad placement:
 - Pad 1 – left posterior back, left of spine, just below the scapula.
 - Pad 2 – left anterior chest between midline of chest and left nipple.
- The SGA airway is the primary airway adjunct to be used during cardiac arrest.
- CCPs should consider inserting a gastric tube via the SGA to prevent aspiration.
- EtCO₂ measurement with an advanced airway in place is an effective tool to measure cardiac output and indication of ROSC.
- Attach an HME/filter to the SGA/ETT.

- CPR induced consciousness should be managed with a bolus dose of Ketamine 1mg/kg.
- Do not administer adrenaline until reversible causes and appropriate volume replacement (10ml/kg bolus of NaCl) has occurred. A fluid bolus of 10ml/kg may be repeated once.
- Termination of resuscitation for paediatric patients is not a routine practice and may only occur in special circumstances. See CPG 2.7.
- Use of the mechanical ventilator during cardiac arrest is not recommended.
- CCPs may escalate the joules if required after 3 shocks in increments of 2J for every subsequent shock to a maximum of 10J/kg or adult dosing.

The WAAFELSS mnemonic may be used to help guide care while managing a paediatric cardiac arrest:	
W	Weight: 0-12 months = (age x 0.5) + 4, 1-5 years = (age x 2) + 8, 6-14 years = (age x 3) + 7
A	Adrenaline: 0.01mg/kg (0.1ml/kg 1:10,000)
A	Amiodarone: 5mg/kg
F	Fluid: 10 ml/kg x 2
E	Energy: 4J/kg & CCPs can escalate to max 10J/kg
L	SGA-size: size 1 = 2-5 kg, size 1.5 = 5-12 kg, size 2 = 10-25 kg, size 2.5 = 25-35 kg. ET-size: uncuffed (age÷4) + 4 and cuffed (age÷4) + 3.5. Depth: (age ÷ 2) + 12
S	Sugar: 2.5ml/kg of 10% Dextrose
S	Systolic BP: (age x 2) + 70



Defibrillation Energy	Reversible Causes
First and subsequent shocks at 4J/kg. CCPs may escalate to a max of 10J/kg or adult dose	- Hypovolaemia - Hypoxia - Hydrogen ions (acidosis) - Hypoglycaemia - Hypo/hyperkalaemia - Hypothermia - Tension pneumothorax - Tamponade, cardiac - Toxins - Thrombosis, cardiac - Thrombosis, pulmonary
Advanced Airway	
- SGA as primary airway. - CCPs must consider ETI after approx. 10 min of resuscitation (if suitable ETT size available) - Waveform capnography must be used with all advanced airways	
CPR	
- Push hard (>1/3 of anteroposterior diameter of chest)) and fast (100-120/min) and allow complete chest recoil - Hand position on lower half of sternum - Minimise interruptions. - Change compressor every 2 minutes - Avoid excessive ventilation - SGA/ETT airway in place – continuous compressions. If an SGA air leak occurs as a result of continuous compressions, revert to 15:2/30:2	



CPG 2.5 RESUSCITATION OF THE NEWBORN & NEONATES

Resuscitation of the new-born requires prompt recognition of when to actively intervene to ensure adequate ventilation, perfusion and to prevent hypothermia.

CLINICAL PRESENTATION

No signs of life:

- Absent or poor tone
- No normal breathing / absent breathing

Signs of inadequate perfusion

- Absent or poor tone
- Unable to establish normal breathing
- Heart rate < 100 beats/minute

ASSESSMENT CONSIDERATIONS

Assessment and the recognition of the need for resuscitation should occur immediately after birth.

Assessment of the new-born includes:

- **Tone** – ability to move limbs appropriately. Poor/absent tone or a floppy new-born will most likely require resuscitation.
- **Breathing** – new-borns may take a few seconds to establish regular breathing. Signs of breathing difficulty include chest recession/retraction, expiratory grunting, and nasal flaring.
- **Heart rate** – new-borns with a heart rate < 100 beats/minute indicate poor perfusion and the need for resuscitation.

A combination of poor tone and floppiness, persistent apnoea and a heart rate < 100 beats/minute indicates the need for urgent resuscitation.

MANAGEMENT

The management aim when resuscitating the new-born is to ensure adequate ventilation, perfusion, and temperature. Important considerations when managing the ABCs include:

Airway

- Keep the head in the neutral position. Place pad under the shoulders.
- Suction is only indicated if there is airway obstruction or if meconium-stained fluid is present
- If suction is required, suction the mouth first then the nose. Gentle wiping of the nose and mouth is often adequate to remove secretions

Breathing

- Oxygen and ventilation are only required in the compromised new-born
- New-borns may appear cyanosed for the first 10 minutes post birth. Should cyanosis persist after 10 minutes, 2L of oxygen may be administered until centrally pink.
- SpO₂ may range from 65% to 95% within the first 10 minutes post birth, gradually increasing to normal within those first 10 minutes. The normal SpO₂ range within the first 10 minutes after birth:
 - 1 minute – 60-65%
 - 2 minutes – 65-70%

- 3 minutes – 70-75%
- 4 minutes – 75-80%
- 5 minutes – 80-85%
- 10 minutes – 85-95%

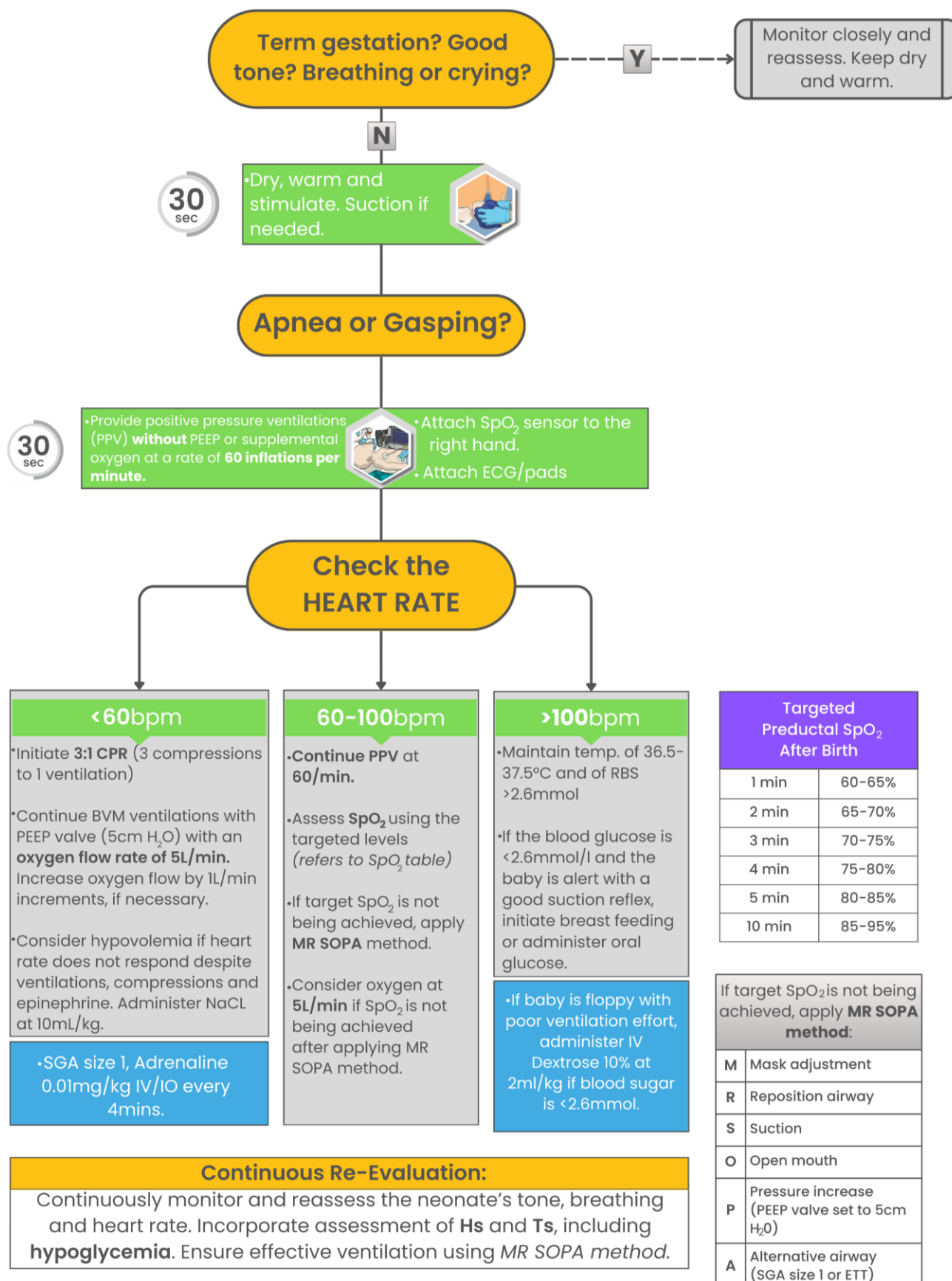
Circulation

- Bradycardia (heart rate <100 bpm) in the new-born is generally due to hypoxia.
- Compressions should only be initiated if the heart rate remains < 60 bpm following 30 seconds of BVM ventilations.
- Rapid improvement following effective BVM ventilations will generally occur and is a sign of adequate ventilation



SPECIAL NOTES

- Resuscitation is futile in pre-term infants < 20 weeks gestation. Resuscitation does not need to be started. Note that these preterm infants may show false signs of life (movement/gasp).
- If uncertain, begin resuscitation.
- If immediate resuscitation is required, clamp and cut the cord and move to a dry area to begin resuscitation.
- Active temperature control is an important aspect of managing the new-born and should be managed appropriately to prevent heat loss.
- Should the new-borns responsiveness remain poor despite adequate ventilation, perfusion and temperature control, assess RBS. Prick the lateral aspect of the heel to obtain blood for RBS assessment.
- A shockable rhythm in the new-born is very rare. Should defibrillation be required, use appropriate weight-based energy of 4 joules/kg.
- Termination of resuscitation for new-borns is not a routine practice and may only occur in special circumstances. See CPG 2.7.



CPG 2.6 POST CARDIAC ARREST (ROSC) CARE

Post ROSC care plays a vital role in improving the potential for a favourable outcome following successful resuscitation. Post ROSC care in the prehospital setting focuses on a few key points below.



MANAGEMENT

The primary management goals following ROSC include:

- Supporting airway, breathing and circulation
- Maintaining cerebral perfusion
- Managing arrhythmias

Continue to manage identified reversible causes:

- Hypoxia
- Hypo/hyperthermia
- Hypovolaemia
- Hypo/hyperkalaemia
- Acidosis
- Tension pneumothorax
- Cardiac tamponade
- Toxins

Airway and Breathing

- Consider inserting an advanced airway if not already done so during the cardiac arrest if the patient's airway is compromised or to maximise oxygenation and ventilation.
- Consider replacing the SGA with an ETT (age dependent) if not done so during the cardiac arrest.
- Ensure adequate tidal volumes based on the patient's ideal body weight. Do not hyperventilate the patient.
- An elevated EtCO₂ is to be expected following ROSC. Do not try to normalise EtCO₂ prehospitally as this requires correction over time in-hospital. Adopt an appropriate ventilation strategy based on the patient's lung pathology and the cause of the cardiac arrest.
- Aim to maintain SpO₂ > 90%.

Circulation

- Consider performing a 12-lead ECG on all ROSC patients following a medical cardiac arrest, particularly if the cause of arrest is suspected to be cardiac related. Patients with a pre-arrest 12-lead ECG indicating a STEMI should be transported to an appropriate facility for PCI.
- BP and MAP should be monitored regularly (minimum of every 5 minutes)
- Maintain SBP > 90 mmHg/MAP > 65 mmHg or MAP of 70-80 for isolated TBI in adults and according to age appropriate SBP range in paediatrics using (age x 2) + 70. Avoid spikes in BP, particularly in trauma. The target MAP is to achieve adequate cerebral perfusion.
- For adults administer fluids as required using 250ml boluses to a maximum of 1-2L depending on the cause of arrest. Limit fluids in the presence of major haemorrhage to a maximum of 1L. For paediatrics administer a fluid bolus of 10-20ml/kg repeated once if required.
- Consider vasopressor/inotrope to maintain BP and MAP. Phenylephrine and Noradrenaline may be considered for adults only. Adrenaline infusions for adults are not for routine use and are only

to be considered for specific indications (see formulary). Adrenaline infusion may be considered for paediatrics requiring vasopressor/inotrope support.

- Provide sedation/analgesia and paralysis as required.
- Correct hypoglycaemia (RBS < 4.0 mmol/L) if present. For adults administer 50ml bolus of Dextrose 10% and reassess after 10 minutes. Repeat if required. Avoid increasing RBS to > 6.7 mmol/L. For newborns, administer 2ml/kg (max 50ml) and for infants/children, administer 2.5ml/kg (max 50ml) of Dextrose 10% and reassess after 10 minutes. Repeat as required to normalise RBS.
- Consider TXA for the trauma patient if indicated and if not already given during the cardiac arrest management.



SPECIAL NOTES

- Avoid rapid movement of the patient for up to 10 minutes following ROSC. Rapid movement may result in re-arrest. Transport should then be initiated if not already loaded in the ambulance.
- ROSC patients should be considered “unstable” when performing an RSI.
- Monitor the patient’s temperature and avoid hypo/hyperthermia.
- Provide hospital prenotification when transporting. Transport to the most appropriate facility according to the cause of arrest.



CPG 2.7 TERMINATION OF RESUSCITATION/DECLARATION OF DEATH

The following criteria provide guidance to APs and CCPs for when not to start CPR and when resuscitation can be discontinued for both adults and paediatrics.

CRITERIA TO NOT START CPR

It is appropriate to not commence CPR for both adults and paediatrics where the patient shows signs of obvious death or injuries incompatible with life such as:

- Decomposition
- Putrefaction
- Rigor mortis
- Dependent lividity
- Decapitation
- Cranial/cerebral destruction
- Hemitorporectomy or similar significant injury
- Incineration
- Evisceration of major organs
- Foetal maceration
- Pre-term infant <20 weeks gestation
- A valid "Do Not Resuscitate" (DNR) Order or at the family's request

APs and CCPs are not required to start CPR if any of the above criteria are met. APs and CCPs do not need to consult with a senior HMCAS clinician to obtain permission to withhold CPR if the above criteria are met.

CRITERIA TO TERMINATE RESUSCITATION

Following the commencement of CPR, resuscitation efforts on adult patients may be terminated if the criteria below is met:

- No breathing and no pulses
- Asystole or broad PEA (agonal rhythm) < 40 bpm for ≥20 minutes
- Reversible causes have been managed
- The patient's family or significant other gives verbal consent to stop CPR (if present on scene)

Where no CCP is on scene, APs must contact the CC: Emergency to confirm termination of resuscitation.

SPECIAL NOTES

- Resuscitation commenced on a paediatric patient should not be terminated prehospitally unless there are extenuating circumstances and Sliver Clinical has been consulted.
- Resuscitation commenced on a pregnant patient should not be terminated prehospitally unless there are extenuating circumstances and Sliver Clinical has been consulted.
- CCPs should assess the futility of resuscitation prior to making the decision to transport a patient with on-going CPR. The routine transport of patients with on-going CPR is not recommended.

CPG 2.8 CARDIAC ARREST - SPECIAL CIRCUMSTANCES

The following considerations and adjustments to treatment should be taken into account when managing a cardiac arrest with special circumstances.

SPECIAL NOTES

HYPERKALAEMIA

- Consider hyperkalaemia for all renal dialysis/impairment patients that are in cardiac arrest where the aetiology of the cardiac arrest is not obvious (e.g. trauma, respiratory).
- Administer 10ml Calcium Chloride 10% as a bolus during resuscitation if hyperkalaemia is suspected.
- Consider repeating dose if cardiac arrest is refractory or prolonged.

HYPOTHERMIA

- If VF/VT persists after 3 shocks, delay any further shocks until the core temperature is $>30^{\circ}\text{C}$.
- Withhold adrenaline if core temperature is $<30^{\circ}\text{C}$.
- Increase adrenaline administration intervals to 6-10 minutes if core temperature is $>30^{\circ}\text{C}$.
- Normal adrenaline administration intervals may be followed if the patient is warmed to normal core temperatures, however this will most likely only occur in the in-hospital setting.
- If PEA is the presenting rhythm and the patients core temperature is $<30^{\circ}\text{C}$, feel for a pulse for at least 60 seconds and assess EtCO₂ to determine if there is any cardiac output. Chest compressions in hypothermic patients with cardiac output may precipitate VF.
- Consider transporting all patients to hospital who are in cardiac arrest as a result of hypothermia.

CPR INDUCED CONSCIOUSNESS

- CPR induced consciousness occurs when chest compressions generate sufficient cerebral perfusion to induce varying levels of consciousness. Variations of CPR induced consciousness may include body movements, eye opening and eye rolling, combativeness and possibly even speech.
- These patients will lose consciousness when compressions are stopped.
- If CPR induced consciousness occurs, stop compressions, and assess for ROSC. If ROSC has not occurred, resume compressions.
- Should CPR induced consciousness continue to occur, CCPs should consider administering Ketamine 1mg/kg IV/IO.

CPG 2.9 ECMO CPR PATHWAY INCLUSION CRITERIA

The below ECMO CPR (e-CPR) checklist should be applied to all cardiac arrest cases to determine whether the patient meets inclusion criteria. Any patient meeting ALL the criteria below should be considered a candidate for e-CPR.

e-CPR INCLUSION CRITERIA CHECKLIST

Inclusion Criteria:

- Medical cardiac arrest
- Age < 60 years
- Witnessed cardiac arrest
- Immediate bystander/paramedic CPR was initiated
- Initial presenting rhythm was either VF / VT / PEA
- Time from onset of cardiac arrest to estimated time of arrival at ED $\leq$ 45 minutes
- EtCO₂ is > 10mmHg

Comorbidity Assessment:

- NO active cancer or haematological disease
- NO cardiac disease (e.g., CHF)
- NO pulmonary disease (e.g., COPD)
- NO renal failure and not on haemodialysis
- NOT a bed bound patient
- NO dementia or advanced neurological disease
- NO prolonged down-time without CPR

OR

- Medical history cannot be determined and the CCP reasonably assumes no significant medical history exists – the patient should be treated as comorbidity free until proven otherwise



CPG 3.1-3.2

NEUROLOGICAL



CPG 3.1 STROKE

Stroke is a time sensitive emergency requiring prompt recognition. Rapid triage, treatment and transport to an appropriate facility are the priority.

CLINICAL PRESENTATION

The common clinical features of stroke include:

- Loss of movement or weakness in a part of the body, especially on one side (hemiparesis/hemiplegia)
- Difficulty speaking (dysphasia)
- Difficulty swallowing (dysphagia)
- Visual disturbances
- Unequal pupils
- Sudden onset headache with associated neurological symptoms
- Altered level of consciousness

BEFAST ASSESSMENT

BALANCE	Does the patient have a sudden loss of balance or are they dizzy?
EYES	Has the patient lost vision in one or both eyes? Or is their vision blurry?
FACE	Ask the patient to smile. Is one side of the face drooping?
ARMS	Ask the patient to raise their arms or squeeze your fingers. Is one arm weaker than the other?
SPEECH	Ask the patient to speak. Is their speech slurred?
TIME	Time last seen normal or of symptom onset

ASSESSMENT CONSIDERATIONS

Rule out other conditions that may mimic stroke:

- Intoxication (alcohol/drugs)
- Hypoxia
- Hyper/hypoglycaemia
- Seizures/post-ictal state
- Cerebral infections (meningitis, encephalitis)
- Middle ear disorders
- Migraine (hemiplegic)
- Electrolyte abnormalities (e.g., hyponatraemia)
- Intracerebral lesions (e.g., tumour, abscess)

Stroke is a time critical emergency. Manage symptomatically and transport rapidly to hospital:

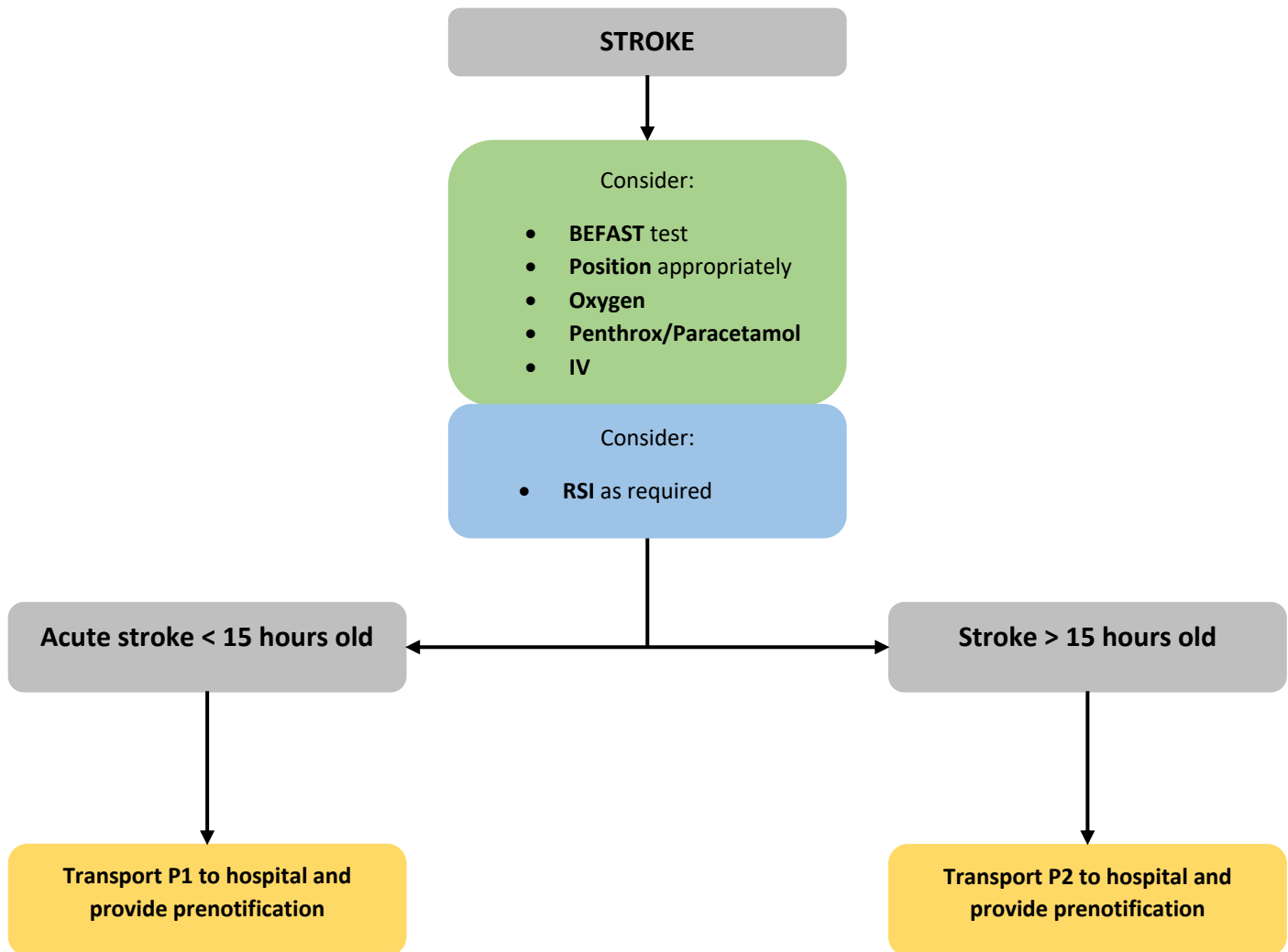
- Keep patient in the lateral position or semi-fowlers position if the patient has a reduced level of consciousness
- Consider RSI if airway/respiratory compromise
- Rule out reversible/mimicking causes
- Determine time of stroke onset:
 - Positive BEFAST test with symptoms < 15 hours old or patients who wake up from sleeping with signs of stroke – transport priority 1 to appropriate facility. Update CC: Emergency to provide prenotification

Positive BEFAST test with symptoms > 15 hours old – transport priority 2 to appropriate facility. Update CC: Emergency to provide prenotification.

All suspected stroke patients with new onset of symptoms, regardless of time of onset of symptoms, should be transported to HGH Emergency Department with prenotification.

SPECIAL NOTES

- APs do not require CCP assistance unless haemodynamic or airway compromise is present.
- Do not delay transport to hospital waiting on scene for a CCP. If a CCP is required, meet the CCP en route to hospital.
- Accurately identifying the time of onset of symptoms or when the patient was last seen normal must be clearly defined in order to determine eligibility for potential reperfusion therapy.
- DO NOT attempt to reduce BP using nitrates if the patient is hypertensive.



CPG 3.2 SEIZURES

Seizures can be classified into focal or generalised seizures. Seizures may be as a result of epilepsy or a recognisable non-epileptic cause. It is important to differentiate between the two and manage the potential cause of the seizures where appropriate.

CLINICAL PRESENTATION

The common clinical presentation in seizures include:

- Visual hallucinations
- Localised twitching of muscles without impaired consciousness
- Nonsensical speech
- Disorientation
- Sudden pause in activity or a fixed gaze
- Nystagmus
- Automatism
- Increased or loss of tone
- Tonic/clonic posturing
- Incontinence

Patients that experience prolonged seizures or status epilepticus are associated with:

- Hypoxia
- Hypercarbia
- Acidosis
- Hyperthermia, hypertension, tachycardia
- Hypo/hyperglycaemia
- Hyperkalaemia

The post-ictal phase may include:

- Confusion
- Fatigue
- Headache
- Nausea

Paediatric seizure activity may manifest differently and include:

- Vacant stare
- Lack of gross muscle tonicity
- Nystagmus, lateral fixed gaze and/or facial muscle twitching

ASSESSMENT CONSIDERATIONS

Non-epileptic seizures are generally provoked by a recognisable cause. If seizures are not thought to be epileptic, rule out and manage potential reversible causes of seizures:

- Hypoxia and hypercarbia
- Hypotension
- Metabolic causes - hypoglycaemia, hyponatraemia, hypocalcaemia, hyperthyroidism, uraemia
- Eclampsia
- Meningitis/encephalitis

- Hyperthermia (febrile seizures)
- Drugs/toxins
- Cerebral pathology (stroke, TBI, tumour)

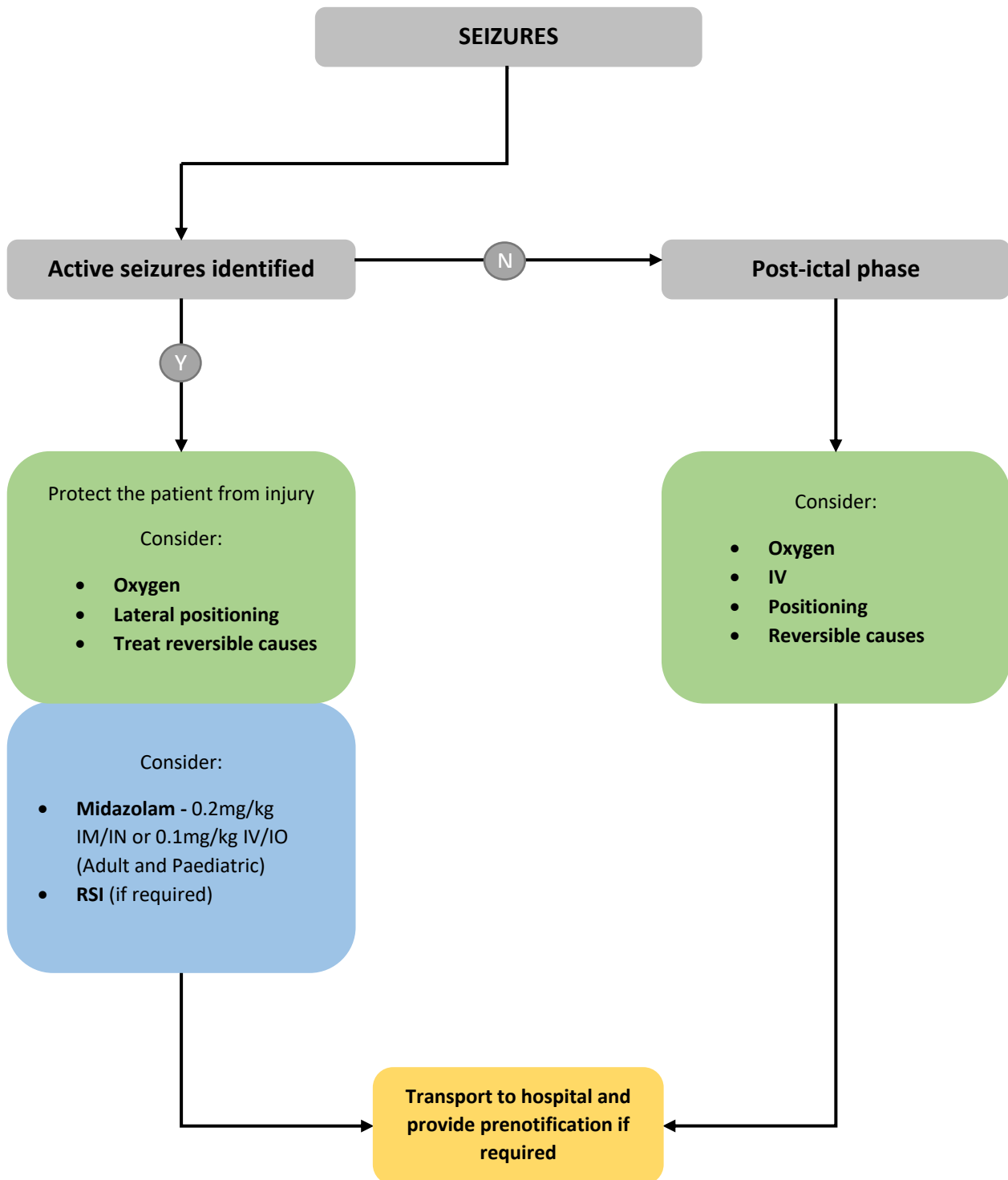
Seizures will often terminate without the need for anticonvulsant medication. Patients with prolonged seizures or status epilepticus is a medical emergency requiring urgent intervention. Management of seizures includes:

- Basic airway manoeuvres
- Keep the patient in a position of safety while actively seizing and prevent further injury to the patient. Do not move the actively seizing patient.
- Consider Midazolam for prolonged seizures/status epilepticus
- Consider RSI for patients with status epilepticus (age dependent)
- Manage reversible causes of non-epileptic seizures
- A patient in the post-ictal phase should be kept in the lateral position if they have a reduced LOC.
- Calm and reassure as post-ictal patients may be confused and non-compliant with monitoring.



SPECIAL NOTES

- If no IV access is available, the IN route is preferred over IM.
- Non-epileptic seizures may require management with Midazolam initially while the potential reversible cause is identified and treated.
- Eclampsia should be considered in the seizing pregnant patient and managed accordingly.





CPG 4.1-4.6

CARDIAC



CPG 4.1 ACUTE CORONARY SYNDROME

Acute Coronary Syndrome (ACS) refers to multiple cardiac conditions resulting from myocardial ischemia. These conditions include ST-elevation myocardial infarction (STEMI), non-ST elevation myocardial infarction (NSTEMI) and unstable angina (UA). It is important to differentiate between ACS and stable angina in order to implement the appropriate treatment.

CLINICAL PRESENTATION

ACS may present with the following:

- Chest pain/discomfort (burning, pressure, tightness)
- Referred pain (to shoulder, arms, jaw)
- Upper epigastric pain
- Dyspnoea
- Diaphoresis
- Pallor
- Nausea and/or vomiting
- Feeling of impending doom

Additionally, complications may arise, such as cardiac failure, cardiogenic shock, and cardiac arrest. Manage these complications according to the relevant CPG as required.

Right ventricular myocardial infarction – it is important to identify and appropriately manage RV myocardial infarction. In addition to the above, patients may present with the following symptoms:

- Hypotension
- Jugular vein distension
- ST-elevation in lead V₄R – which is indicative of RV myocardial infarction.

ASSESSMENT CONSIDERATIONS

When assessing a potential ACS patient, take into account possible high-risk features and risk factors.

High risk features include:

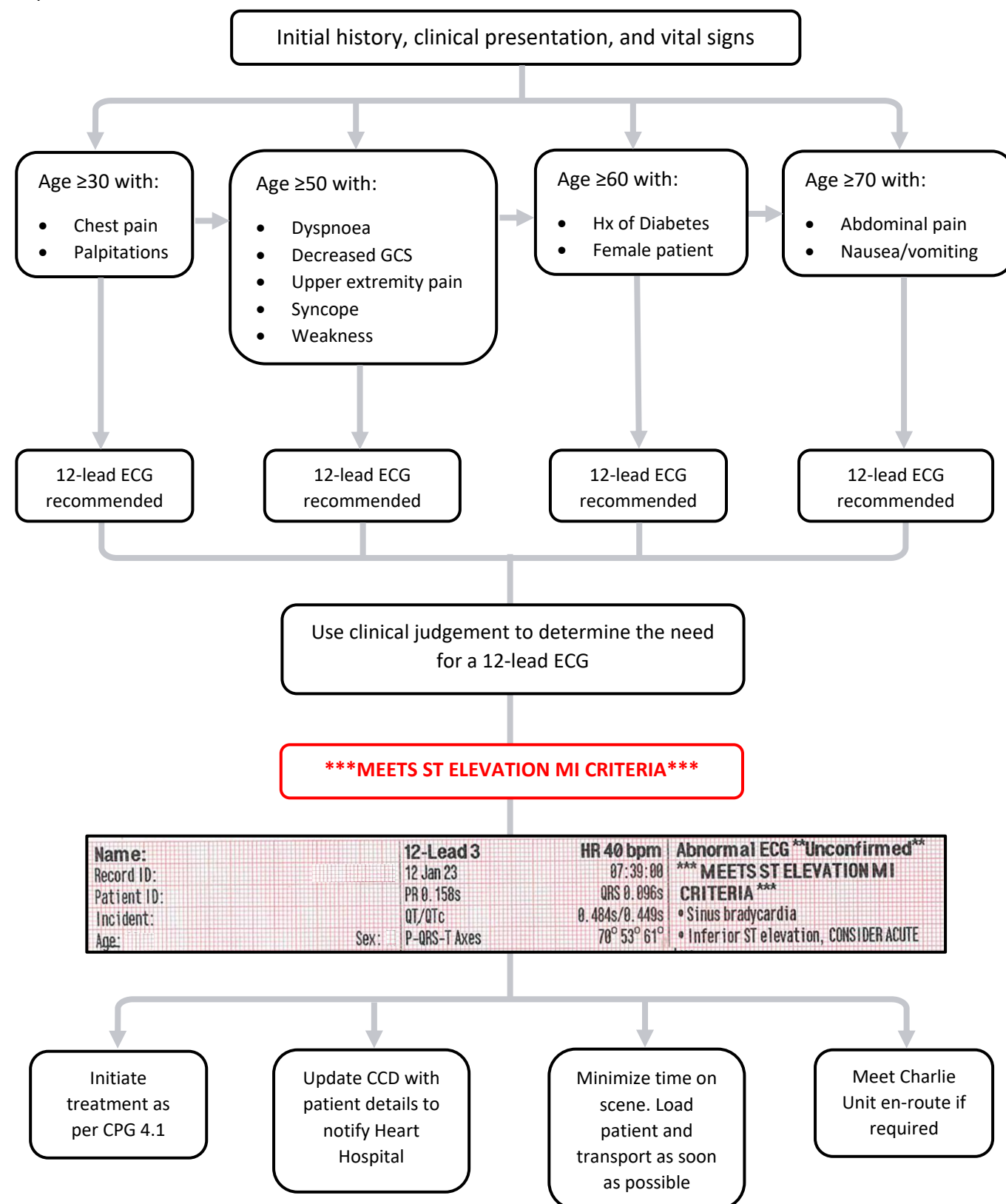
- Repetitive or prolonged chest pain/discomfort
- Persistent or dynamic ST-depression (≥ 0.5 mm) or new T-wave inversion (≥ 2 mm)
- Transient ST-segment elevation (≥ 0.5 mm) in 2 or more contiguous leads
- Hypotension
- Sustained VT
- Syncope
- Left ventricular dysfunction
- Previous PCI or coronary artery bypass graft
- Diabetes and/or renal impairment

ACS risk factors include:

- Elderly
- Smoking
- Hypertension
- Hyperlipidaemia
- History of heart disease
- Family history of ACS

12-LEAD ECG DECISION TREE

This algorithm provides a guideline to assist in identifying patients at greatest risk for ACS and who require a 12-lead ECG.





HEART HOSPITAL DIRECT TRANSFER CRITERIA

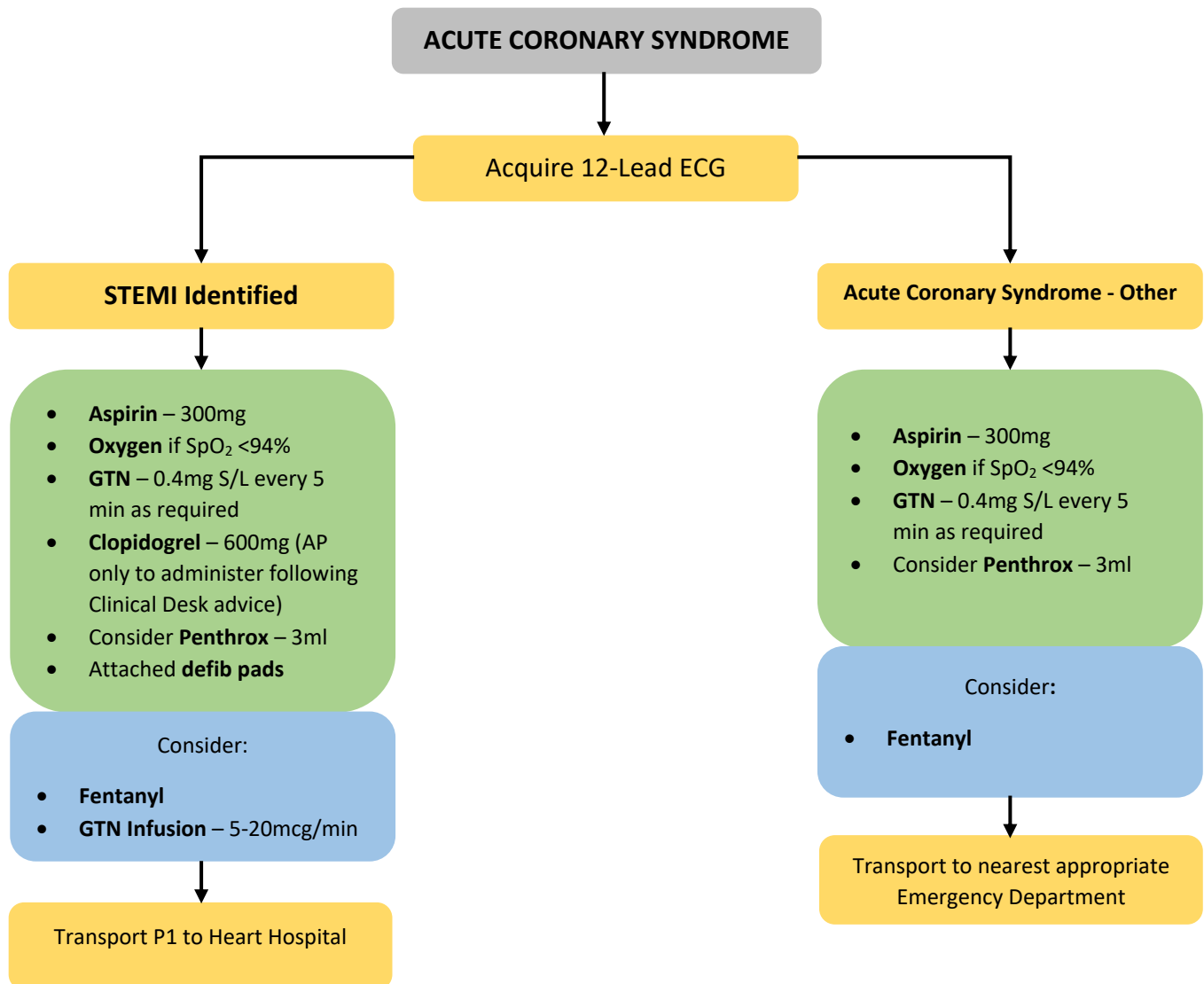
Patients meeting the criteria below are eligible for direct transfer to the Heart Hospital Emergency Department. Liaise with the CC: Emergency to ensure acceptance to heart hospital.

- Adult patients with STEMI identified on 12-lead ECG with supporting history, with or without chest pain
- Adult patients with ST depression in leads V₁ – V₃ identified on 12-lead ECG indicative of a posterior STEMI
- Cardiac arrest WITNESSED by HMCAS staff with ROSC where the pre-arrest 12-lead ECG is indicative of a STEMI
- For other non-clinical criteria refer to SOPs



SPECIAL NOTES

- Do not allow the patient to walk.
- Certain patient groups such as the elderly, women, diabetics, congestive cardiac disease and renal failure patients may present with atypical symptoms.
- Isolated LBBB without symptoms suggestive of a STEMI should be transported to the nearest emergency department, not heart hospital.
- All patients with a STEMI should be transported P1 to heart hospital. Do not delay on scene waiting for CCP for patients with STEMI. Rapidly transport and meet CCP on route.
- A GTN infusion may be considered for the STEMI patient where repeated doses of sublingual GTN and Fentanyl have not provided sufficient relief.
- All patients with a STEMI must have defib pads applied. Consider the A/P position as the primary placement.
- Patients that present as a **SUSPECTED STEMI**, but the initial ECG does not meet STEMI criteria:
 - Manage for ACS, withholding Clopidogrel.
 - CCPs should repeat a 12-lead ECG on arrival at the Emergency department prior to unloading the patient. If the ECG is now diagnostic of STEMI, update the CC: Emergency and transport to Heart Hospital.
 - APs should repeat a 12-lead ECG on arrival at the Emergency department prior to unloading the patient. Contact the CC: Emergency for ECG interpretation. If the ECG is now diagnostic of STEMI, transport to Heart Hospital.
 - If the ECG does not meet criteria, unload the patient and proceed into the emergency department.



CPG 4.2 BRADYARRHYTHMIAS

Bradycardia in adults is defined as a heart rate < 60 bpm and is age dependent in paediatrics. It is important to note that bradycardia may be normal for some patients (e.g. athletes etc). Asymptomatic bradycardia generally does not require treatment.

Bradycardia is classified as either cardiac (SA node, AV node or His-Purkinje system disease) or non-cardiac (environmental, metabolic, or endocrine disorders and toxicology).



CLINICAL PRESENTATION

Symptomatic bradyarrhythmias that require treatment may present with the following:

- Hypotension
- Syncope
- Altered level of consciousness
- Chest pain/discomfort
- Congestive heart failure
- Dyspnoea
- Diaphoresis
- Nausea and/or vomiting
- Dizziness



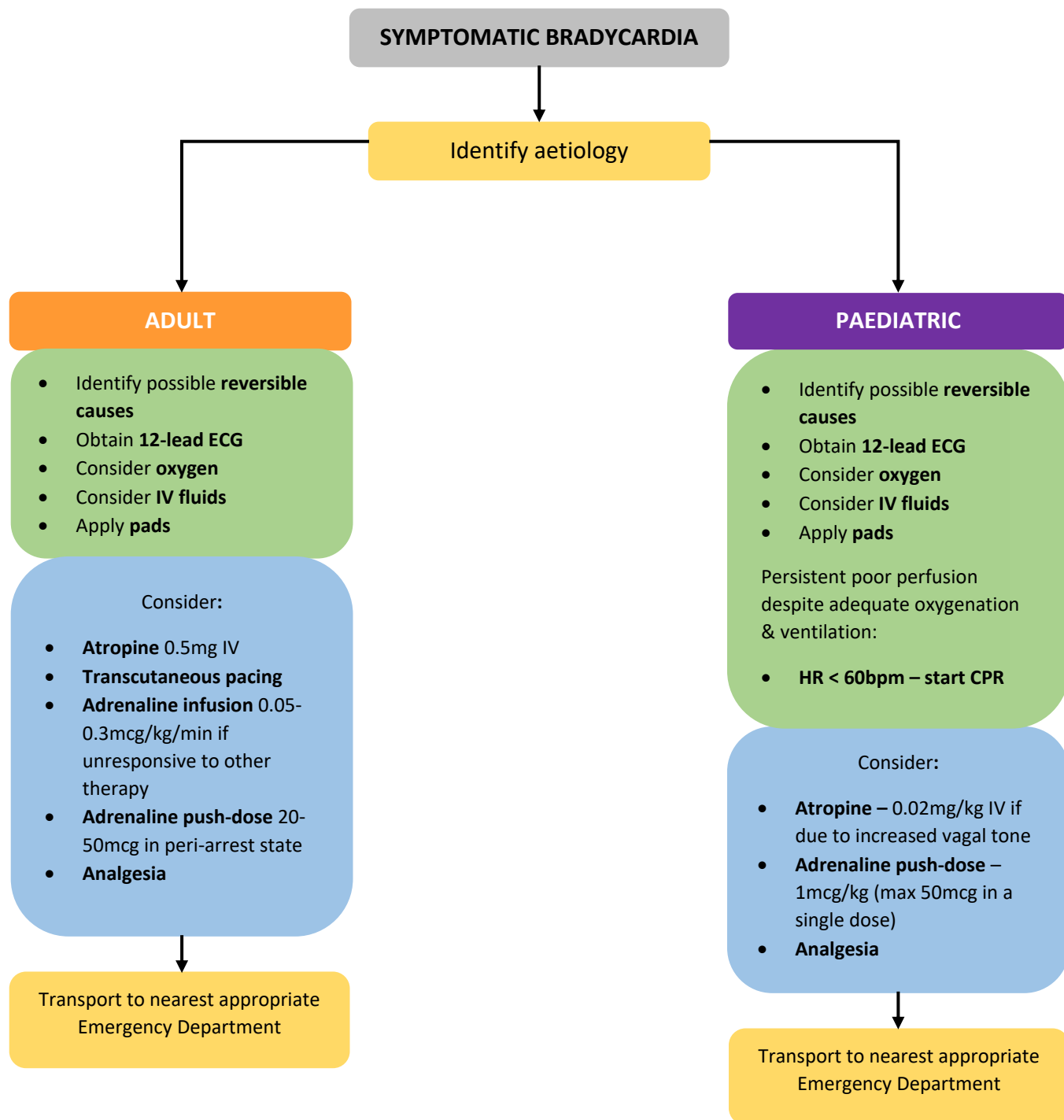
ASSESSMENT CONSIDERATIONS

- Hypoxia is one of the more common causes of bradycardia – ensure that oxygenation and ventilation is maximised to manage this reversible cause.
- Management of non-cardiac symptomatic bradycardia should focus on identifying and treating reversible causes.



SPECIAL NOTES

- When performing transcutaneous pacing, start at a rate of 70bpm and 30mA. Increase energy (to maximum energy allowed by device as required) until electrical and mechanical capture obtained. If BP does not improve following capture, increase rate up to a maximum of 90bpm.
- Anterior-Posterior pad placement is preferred for transcutaneous pacing whenever possible.
- Provide light sedation as required when pacing a patient (see CPG 11.3). Adjust sedation dosing based on haemodynamic status.
- Do not prioritise procedural sedation over initiation of pacing for patients presenting in extremis/peri-arrest. Initiate immediate pacing if indicated.
- Bradyarrhythmias associated with ACS – transcutaneous pacing should be the primary management.
- Inotropic support should be considered if pacing fails.



CPG 4.3 TACHYARRHYTHMIAS – BROAD COMPLEX

Broad complex tachycardias are defined as a QRS complex > 0.12 seconds in adults, >0.08 seconds in children and >0.10 seconds for adolescents, and a heart rate > 100 bpm in adults and age dependent in paediatrics. Broad complex tachycardias causing haemodynamic instability requires immediate treatment. VT is the most common broad complex tachycardia.

CLINICAL PRESENTATION

Broad complex tachycardias may present with the following:

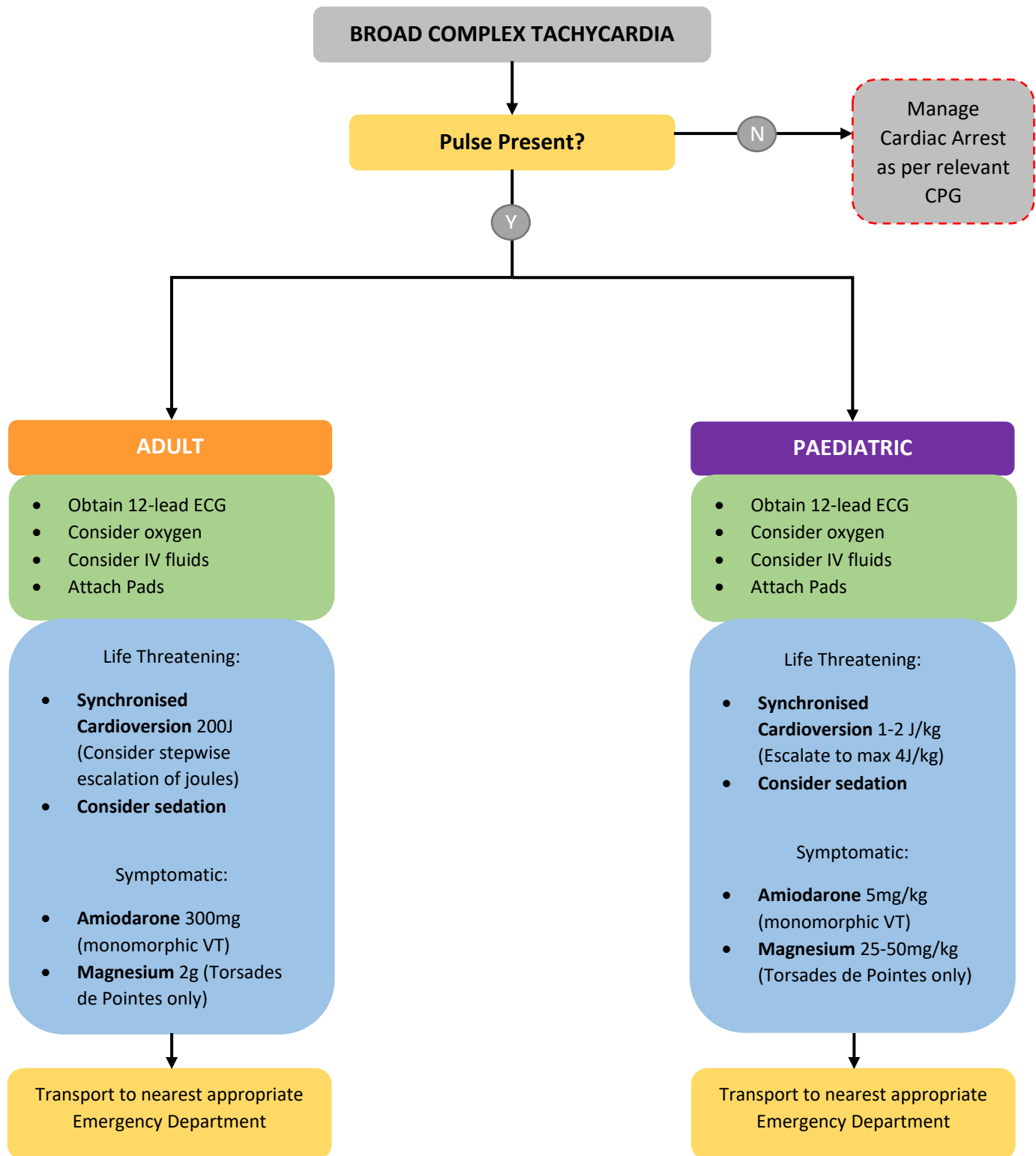
- Palpitations
- Chest pain/discomfort
- Dyspnoea
- Altered LOC
- Syncope
- Haemodynamic instability

ASSESSMENT CONSIDERATIONS

- Patients presenting with haemodynamic compromise associated with broad complex tachycardia require immediate intervention and synchronised cardioversion.
- Escalate energy setting should the first shock fail.
- Consider low dose sedation prior to cardioversion (should time permit).

SPECIAL NOTES

- These patients are at high risk of cardiac arrest. Be prepared to manage the cardiac arrest should the patient become pulseless.
- If synchronisation fails (the monitor is unable to identify an R wave), change the lead and adjust the amplitude. Should this fail, standard (unsynchronised) defibrillation should be performed.
- When performing synchronised cardioversion, opt for the higher energy with the aim of successful cardioversion on the first shock.
- Do not prioritise procedural sedation over providing a sync cardioversion for patients presenting in extremis/peri-arrest. Provide an immediate sync shock if indicated.
- Consider 300mg Amiodarone infusion following three unsuccessful sync shocks in the unstable patient and repeat synchronised shock if indicated.



CPG 4.4 TACHYARRHYTHMIAS – NARROW COMPLEX

Narrow complex tachycardias are defined as a QRS complex < 0.12 seconds in adults, <0.08 seconds in children and <0.10 seconds for adolescents, and a heart rate > 100 bpm in adults and age dependent in paediatrics. Narrow complex tachycardias causing haemodynamic instability requires immediate treatment. Narrow complex tachycardias are either cardiac (usually supraventricular or atrial) or non-cardiac (sinus) in origin. Differentiating between the two will guide treatment.

CLINICAL PRESENTATION

Narrow complex tachycardias may present with the following:

- Palpitations
- Chest pain/discomfort
- Dyspnoea
- Altered LOC
- Syncope
- Haemodynamic instability

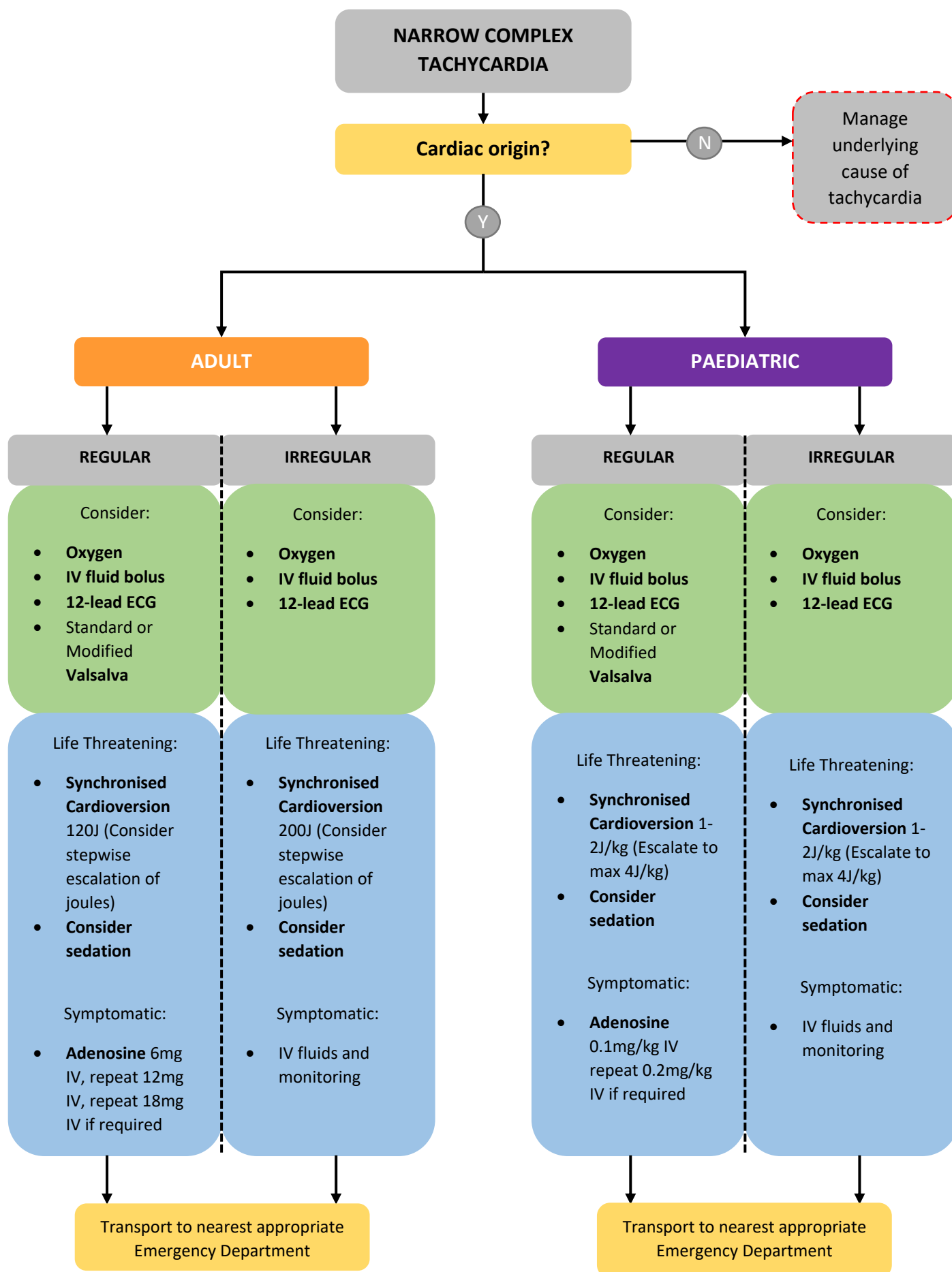
ASSESSMENT CONSIDERATIONS

- The infant rate of SVT is generally >220 bpm and >180 bpm for children.
- Patients presenting with haemodynamic compromise associated with narrow complex tachycardia may require synchronised cardioversion.
- Escalate energy setting should the first shock fail.
- Consider low dose sedation prior to cardioversion (should time permit).
- Atrial Fibrillation >24 hours old increases risk of a thrombotic event. Synchronised cardioversion should only be considered if life-threatening haemodynamic compromise is present.

SPECIAL NOTES

- The modified valsalva manoeuvre should only be considered for patients with a regular narrow complex tachycardia.
- It is important to identify and treat any non-cardiac causes of narrow complex tachycardia, such as pain/anxiety, hyperthermia/fever, anaemia, and drugs.
- If 6mg and 12mg of Adenosine fail to convert the SVT, CCPs may consider 18mg IV. Consider the tolerability/side effects for each patient.
- Following successful conversion of SVT with either the modified valsalva manoeuvre or Adenosine, the CCP may sign off the patient to the AP crew for transport and ongoing monitoring provided the patient is alert, haemodynamically stable and the post conversion 12 lead ECG is a normal sinus rhythm. The CCP must ensure medications are appropriately recorded on the ePCR with the relevant stand down notes.
- When performing synchronised cardioversion, opt for the higher energy with the aim of successful cardioversion on the first shock.
- Do not prioritise procedural sedation over providing a sync cardioversion for patients presenting in extremis/peri-arrest. Provide an immediate sync shock if indicated.

- Consider 300mg Amiodarone infusion following three unsuccessful sync shocks in the unstable patient and repeat synchronised shock if indicated.



CPG 4.5 ACUTE PULMONARY OEDEMA

Acute pulmonary oedema is the rapid build-up of fluid in the alveoli and lung interstitium. APO is generally categorised as cardiogenic or non-cardiogenic. Majority of the APO patients seen by HMCAS are cardiogenic in origin. However, management is similar for the two.

CLINICAL PRESENTATION

Acute Pulmonary Oedema may present with the following:

- Dyspnoea & Tachypnoea
- Diaphoresis
- Anxiety
- Crackles on auscultation – usually heard at the bases, progressing to apices as APO worsens
- Cough
- Tachycardia
- Hypertension
- Hypotension (cardiogenic shock)

ASSESSMENT CONSIDERATIONS

- Blood pressure is a key indicator as to the drug therapy regime to be used when managing these patients.
- Assess the ankles, abdomen, and neck for signs of systemic fluid overload.
- Detailed history taking and assessment are important in order to identify the possible cause of APO. Differentiating between cardiogenic (e.g. ACS, renal failure) and non-cardiogenic (e.g. sepsis, toxins) will guide treatment. Identify and treat potential underlying causes.

SPECIAL NOTES

- CPAP application may be considered in patients with a reduced level of consciousness and adequate ventilation.
- Sedating APO patients may increase mortality. Sedation should be avoided in the APO patient. If a patient is unable to settle on CPAP, oxygenate via NRM at high flow and maximise GTN therapy.
- APO patients in cardiogenic shock may have a fluid deficit. A small fluid bolus (250-500ml) may be administered cautiously. Inotropic/vasopressor support may be required.
- Increase PEEP and Psupport SLOWLY to a maximum PEEP of 15mbar and a combined PEEP and Psupport of 20mbar.
- Do not fix the CPAP mask to the patient until the patient is comfortable.
- Nebulising an APO patient with Salbutamol has been shown to be harmful. Patients presenting with respiratory distress, including wheezing, hypertension, and NO history of asthma/COPD, indicating probable APO, should NOT be nebulised.
- For those patients who are significantly hypertensive, consider starting the GTN infusion at a higher dose.
- Movement of critically hypoxic patients may result in a worsening of their condition. Do not move the critically hypoxic patient until the patient has been sufficiently stabilised.

ACUTE PULMONARY OEDEMA

- **Position** the patient upright
- **High flow NRM Oxygen** to maintain $SpO_2 \geq 94\%$
- **Request CCP assistance**
- **12-lead ECG**
- **IV** – limit fluids
- **GTN** 2 sprays (0.8mg) if SBP > 150mmHg

Low BP (SBP <100mmHg)

Manage cardiogenic shock as per **CPG 4.6**

Normal to High BP

- **GTN S/L** – 1-3 sprays (0.4-1.2mg) followed by 1 spray every 5 minutes until CPAP applied
- **CPAP** – start with PEEP of 5. Increase PEEP and add Psupport as required
- **GTN Infusion** (if no response to S/L GTN) – 100mcg/min

High BP (SBP > 160mmHg)

- **GTN S/L** – 1-3 sprays (0.4-1.2mg) followed by 1 spray every 5 minutes until CPAP applied
- **CPAP** – start with PEEP of 5. Increase PEEP and add Psupport as required
- **GTN Infusion** – 100 - 300mcg/min. Adjust infusion based on the patients response
- **SBP > 200mmHg** – consider starting infusion at a higher dose of 400-500mcg/min for ≈ 2 minutes and titrate down

Transport to nearest appropriate Emergency Department

CPG 4.6 CARDIOGENIC SHOCK

Cardiogenic shock is caused by left or right ventricular failure. Hypotension is associated with poor peripheral perfusion and pulmonary congestion. Cardiogenic shock may result from AMI, certain drugs, electrolyte imbalances and structural abnormalities.

CLINICAL PRESENTATION

Patients in cardiogenic shock may present with the following:

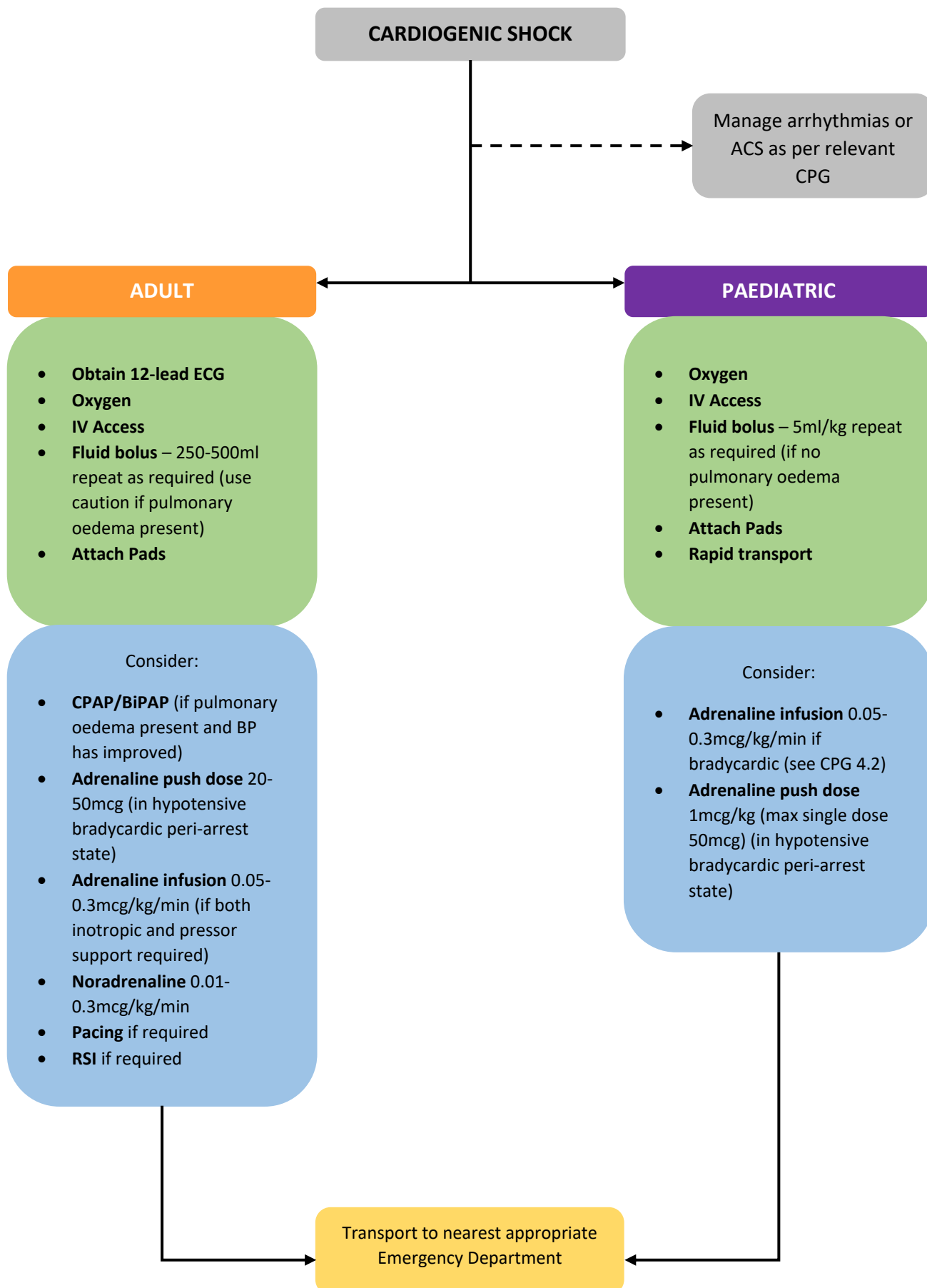
- Acute myocardial infarction
- Chest pain
- Diaphoresis
- Pallor, cold, mottled or cyanotic peripheries
- Reduced level of consciousness
- Tachycardia
- Bradycardia
- Hypotension
- Respiratory distress
 - Pulmonary oedema
 - Tachypnoea
 - Hypoxia
 - Crackles or wheezes

ASSESSMENT CONSIDERATIONS

- The risk of developing cardiogenic shock increases in patients currently experiencing an AMI or following an AMI, particularly amongst those patients who are older, diabetic, hypotensive and have previous history of myocardial infarctions.
- These patients are at increased risk of cardiac arrest. Be prepared to manage a cardiac arrest.
- Identify and treat any reversible causes (e.g. arrhythmias).
- Treatment should focus on maintaining adequate oxygenation, ventilation and perfusion.

SPECIAL NOTES

- Use systolic BP > 90mmHg and MAP of 70-80mmHg for adults as a guide for adequate perfusion.
- Use age dependent SBP for paediatrics as a guide for adequate perfusion (age x 2) + 70.



CPG 4.7 ACUTE AORTIC DISSECTION

Acute aortic dissection is a rare life-threatening emergency that is challenging to diagnose in the prehospital setting. The most common cause of death in the early phase of illness is due to rupture of the aorta into the pericardium, pleural or peritoneal cavities. The role of ambulance service clinicians is to identify patients at high risk, provide supportive care, activate the acute aortic dissection pathway and initiate rapid transport to an appropriate facility.

CLINICAL PRESENTATION

Patients with a suspected acute aortic dissection may present with the following:

- Sudden onset acute chest/back/abdominal pain that is maximal at onset
- Pain may radiate to the neck/jaw, lumbar/abdominal or intrascapular region depending on the location of the dissection
- The pain is often described sharp/ripping/tearing in quality and excruciating
- Difference in SBP of >20mmHg between both arms
- Differences between radial pulses
- Pallor
- Vomiting
- Diaphoresis
- Paraplegia
- Altered sensation in any limbs (numbness, tingling, weakness)
- Syncope
- Neurological signs similar to stroke with associated chest pain

ASSESSMENT CONSIDERATIONS

- Assess for the following risk factors for the development of acute aortic dissection:
 - Hypertension
 - >50 years old
 - Aortic aneurysm
 - Family history of aortic disease
 - Marfan syndrome
 - Ehlers-Danlos syndrome
 - Cardiac/vascular surgery
 - Cardiac catheterisation
 - Bicuspid aortic valve
 - Turner syndrome
 - Cocaine abuse
 - Giant cell arteritis
 - Coarctation of the aorta
 - Pregnancy

- Acute aortic dissection may present similarly to other conditions. Consider alternate differential diagnosis to either rule in or out AAD:
 - Acute coronary syndrome – ST segment depression may occur in AAD, however ST elevation is rare
 - Pulmonary embolism
 - Spontaneous pneumothorax
 - Stroke
 - Acute abdomen
 - Pericarditis
 - Musculoskeletal pain
- Conduct a thorough physical exam and history taking to rule in or out the risk of acute aortic dissection
- Treatment should focus on adequate pain management and supportive care and rapid transport to hospital
 - Patient movement should be kept to a minimum. Handle and move the patient gently. Do not allow the patient to walk



SPECIAL NOTES

- Signs and symptoms of acute aortic dissection are often non-specific and unreliable. Maintain a high index of suspicion based on risk factors, physical examination and presenting signs and symptoms.
- Patients with sudden onset of chest/back/abdominal pain that is described as ripping/tearing in nature, combined with evidence of perfusion deficits or neurological signs should be considered as high risk for acute aortic dissection and treated as such.
- Patient suspected to have acute aortic dissection should be rapidly transported (P1) to HGH Emergency Department with prenotification of arrival.
- HMC emergency departments that identify acute aortic dissection will arrange an immediate transfer to Heart Hospital for further care. These patients should be transported P1 to the confirmed receiving facility.
- Use caution if administering vasopressors and fluid boluses – permissive hypotension is acceptable in these patients. Spikes in BP should be avoided. Be prepared for cardiac arrest.
- Analgesia is an important intervention (particularly opiate analgesia) to keep the patient comfortable and reduce the sympathetic response to pain which can cause progression of the dissection.
- Aspirin, GTN and Clopidogrel should NOT be administered to patients with a suspected or confirmed acute aortic dissection.
- Caution is advised for patients presenting with hypertension and sudden onset severe chest/back/abdominal pain that is sharp/ripping/tearing in nature where there are no ECG changes indicative of ACS. Rule out the potential presence of AAD prior to administering ACS medications.
- Connective tissue disorders such as Marfan syndrome and Ehlers-Danlos syndrome may be present in patients younger than 50 years old. This should be taken into consideration when assessing for risk factors.

SUSPECTED ACUTE AORTIC DISSECTION

- Obtain 12-lead ECG
- Oxygen to maintain SpO₂ ≥94%
- IV Access
- Analgesia (Penthrox/Paracetamol)
- Attach Pads
- Request CCP if unstable or unable to control pain
- Rapid transport to Hospital
- Update Clinical Desk

Hypotensive/Shock

- Fluid bolus – 250-500ml repeat once
- Permissive hypotension is acceptable

Consider:

- Fentanyl
- Ondansetron IV/IM Adult 4mg
- Noradrenaline 0.01-0.3mcg/kg/min. Use lowest possible dose to maintain perfusion

Normo/Hypertensive

Consider:

- Fentanyl
- Ondansetron IV/IM Adult 4mg

Transport to agreed pathway receiving hospital



CPG 5.1-5.6

RESPIRATORY



CPG 5.1 ASTHMA

Asthma is characterised by chronic airway inflammation, hyperresponsive airways and intermittent airway narrowing. Exacerbations may result in varying respiratory and cardiovascular symptoms from mild to severe requiring urgent treatment.

CLINICAL PRESENTATION

Patients presenting with an asthma exacerbation may present with the following:

- Wheeze (possible silent chest in severe cases)
- Dyspnoea
- Tachypnoea
- Chest tightness
- Cough (particularly in paediatrics)
- Accessory muscle use
- Cyanosis

ASSESSMENT CONSIDERATIONS

- Recognition of asthma is based on respiratory symptoms, physical assessment and past history of asthma.
- Identify potential other causes of bronchoconstriction (e.g. anaphylaxis, COPD)
- Attempt to identify the possible trigger/exposure of the asthma exacerbation.
- Wheezes may be less apparent in patients with worsening airway obstruction. A silent chest is a sign of severe obstruction and impending collapse requiring immediate intervention.
- Patients presenting with severe asthma are at risk of sudden deterioration without warning.

ASTHMA SEVERITY TABLE

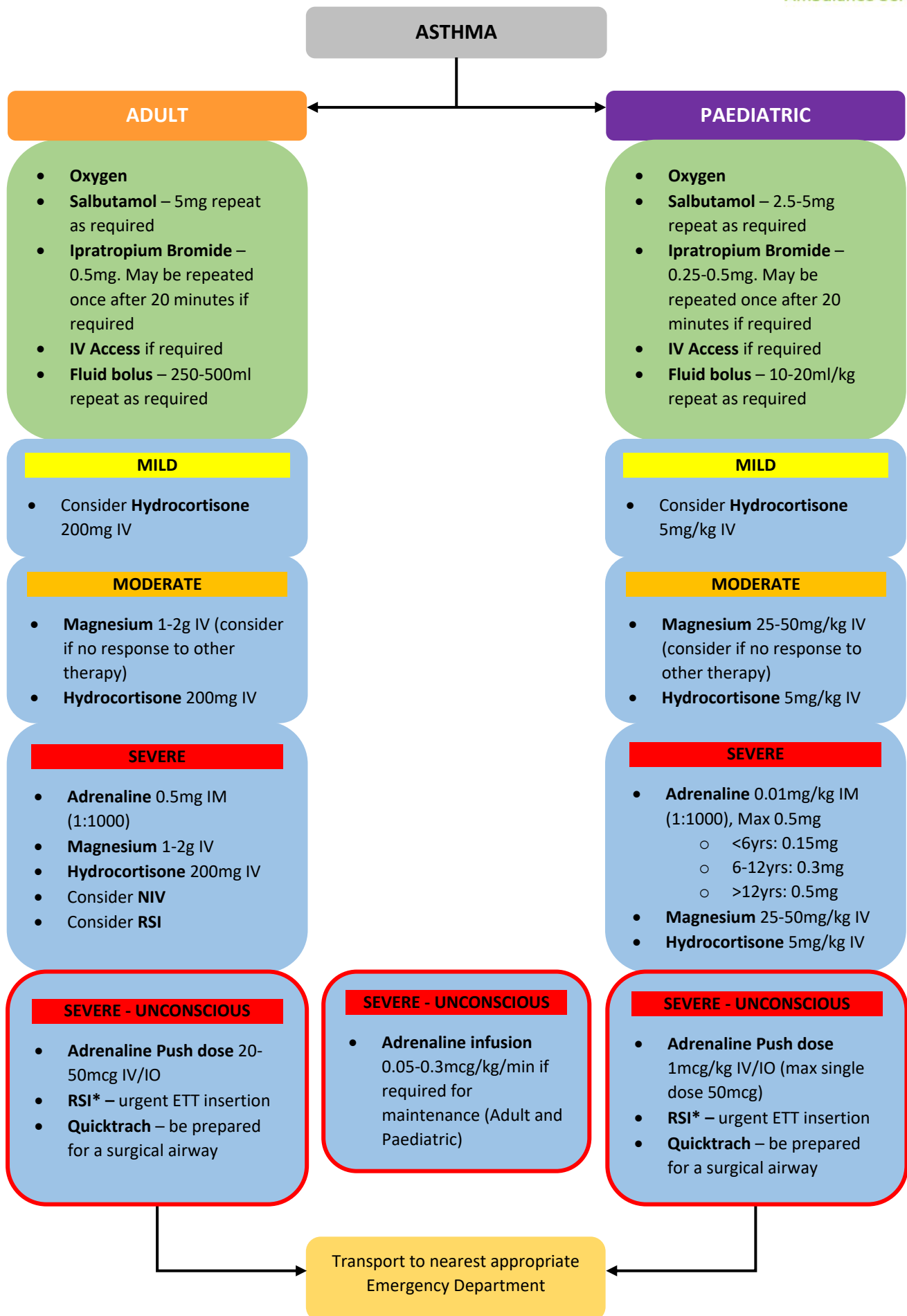
SEVERITY	ADULT	PAEDIATRIC
MILD	• Alert • Expiratory wheeze • Talks in full sentences • RR < 24 bpm	• Alert • Talks/vocalises normally • Wheeze or cough • RR < 30 bpm
MODERATE	• Alert or drowsy • Wheezing in all lung fields • Talks in phrases • RR 24-29 bpm	• Alert but agitated • Wheeze or cough • HR – 120-140 bpm • RR > 30 bpm
SEVERE	• Reduced/decreasing LOC • Severe wheezing or silent chest • Talks in words or unable to talk • RR > 29 bpm	• Agitated and unable to vocalise • Confused • Silent chest • Low SpO₂ • HR > 140 bpm (2-5 years) or >125 bpm (>5 years) • RR > 40 bpm (2-5 years) or >30 bpm (>5 years)

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SPECIAL NOTES

- Ipratropium Bromide must always be administered together with Salbutamol and never on its own. A maximum of 2 doses of Ipratropium Bromide may be given 20 minutes apart
- Movement of critically hypoxic patients may result in a worsening of their condition. Do not move the critically hypoxic patient until the patient has been sufficiently stabilised.
- Use of BVM assisted ventilations in the spontaneously breathing patient should be avoided.
- Consider inline nebulisation in patients that are unresponsive and have poor respiratory effort. Be cautious when providing IPPV to avoid the risk of barotrauma. Allow adequate time for exhalation.
- CCPs may administer hydrocortisone and handover to the AP crew to transport the patient without a CCP escort for stable mild asthma adult patients that have responded to treatment.
- CCPs may consider the use of NIV (CPAP) for adult patients with poor respiratory effort requiring ventilatory support. The use of NIV for these patients involves low to no PEEP and the addition of Psupport.
- The asthmatic patient has the potential to be fluid depleted. Fluid administration should be considered in these patients. A maximum of 1L normal saline for adults and a maximum of two fluid boluses of 10-20ml/kg normal saline for paediatrics. Ensure no signs of fluid overload are present. CCPs may administer additional fluids if required.
- Push dose IV Adrenaline should be considered if unresponsive to 3 x IM Adrenaline doses.
- CCP should consider RSI if patients are in extremis and unresponsive to other therapy.
- The unconscious patient or patient in extremis/peri-arrest require immediate and urgent intervention. CCPs should consider the following:
 - Push dose IV Adrenaline – 1mcg/kg (max single dose of 50mcg). You do not need to administer IM Adrenaline first in such cases.
 - * Urgent establishment of an ETT (age ≥ 8 years). RSI using Rocuronium and low dose Ketamine. No Fentanyl.
 - Be prepared to insert the Quicktrach.
 - A definitive airway should be established within a maximum of 4 minutes from patient contact.
 - An Adrenaline infusion may be required for ongoing maintenance.



CPG 5.2 CHRONIC OBSTRUCTIVE PULMONARY DISEASE (COPD)

COPD is characterised by airflow limitations that progressively worsens and is not fully reversible as well as an abnormal inflammatory response. COPD patients are often at risk of or have significant other comorbidities such as cardiovascular disease, diabetes and cancer. COPD patients are often taking multiple medications (polypharmacy) with limited effect, are generally older and malnutrition and obesity are both common in this patient group.

CLINICAL PRESENTATION

Patients with COPD may present with the following:

- Dyspnoea on exertion
- Productive cough
- Chest tightness
- Wheezes

An acute COPD exacerbation may present with the following:

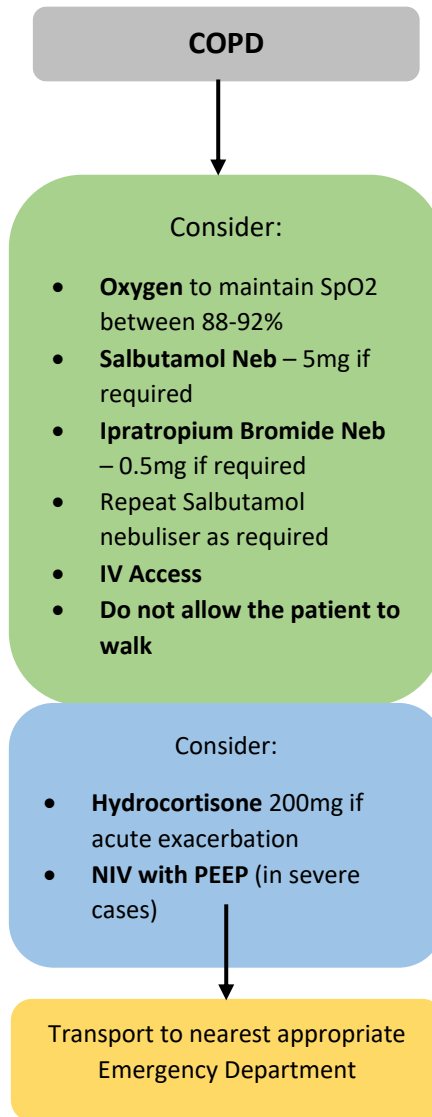
- Change in baseline dyspnoea, cough and sputum production (often associated with a respiratory infection)
- Difficulty speaking
- Anxiety
- Tachypnoea
- Tachycardia
- Cyanosis
- Accessory muscle use, tracheal tugging, intercostal recession and paradoxical abdominal wall motion

ASSESSMENT CONSIDERATIONS

- It is common for COPD patients to have a concomitant illness. Consider additional differential diagnoses such as heart failure and APO, AMI, pulmonary embolism, pneumothorax (spontaneous), pneumonia and anaphylaxis.
- Patients experiencing a COPD exacerbation are at increased risk of hypercapnic respiratory failure secondary to excessive oxygen administration. COPD patients receiving excessive oxygen may result in a worsening VQ mismatch and inhibition of respiratory drive. Hypercapnia and the resultant acidosis may lead to CNS and CVS depression.
- COPD patients often have a lower-than-normal SpO₂ as their baseline and are on home oxygen. Use the lowest dose of oxygen to maintain SpO₂ between 88-92%.
- Nebulised medication should only be administered if bronchoconstriction is present.
- Corticosteroids may be administered if the patient is experiencing an acute exacerbation regardless of whether the patient is taking regular corticosteroids.

SPECIAL NOTES

- COPD patients are at increased risk of adverse events associated with sedative and opiate drug administration. Be cautious when administering these drugs.
- COPD patients have little physical reserve and can deteriorate rapidly with minimal exertion. Use patient movement devices (stair chair, stretcher) when moving these patients.



CPG 5.3 CROUP

Croup is a common viral infection affecting the subglottic airway. The symptoms of croup generally progress over days rather than hours. The severity of croup is determined using the Modified Westley Croup Score.

CLINICAL PRESENTATION

Patients with croup may present with the following:

- Barking cough
- Hoarse voice
- Inspiratory stridor
- Respiratory distress
- Fever (generally mild)

ASSESSMENT CONSIDERATIONS

- Croup generally affects patients between the ages of 6 months – 3 years, however croup can occur in children >3 years.
- The duration of croup symptoms is usually 2-5 days, often worsening at night.
- Rule out other potential differential diagnosis in patients presenting with acute stridor (e.g. epiglottitis, FBAO).
- The Modified Westley Croup Score should be documented on the ePCR pre and post management.

MODIFIED WESTLEY CROUP SCORE

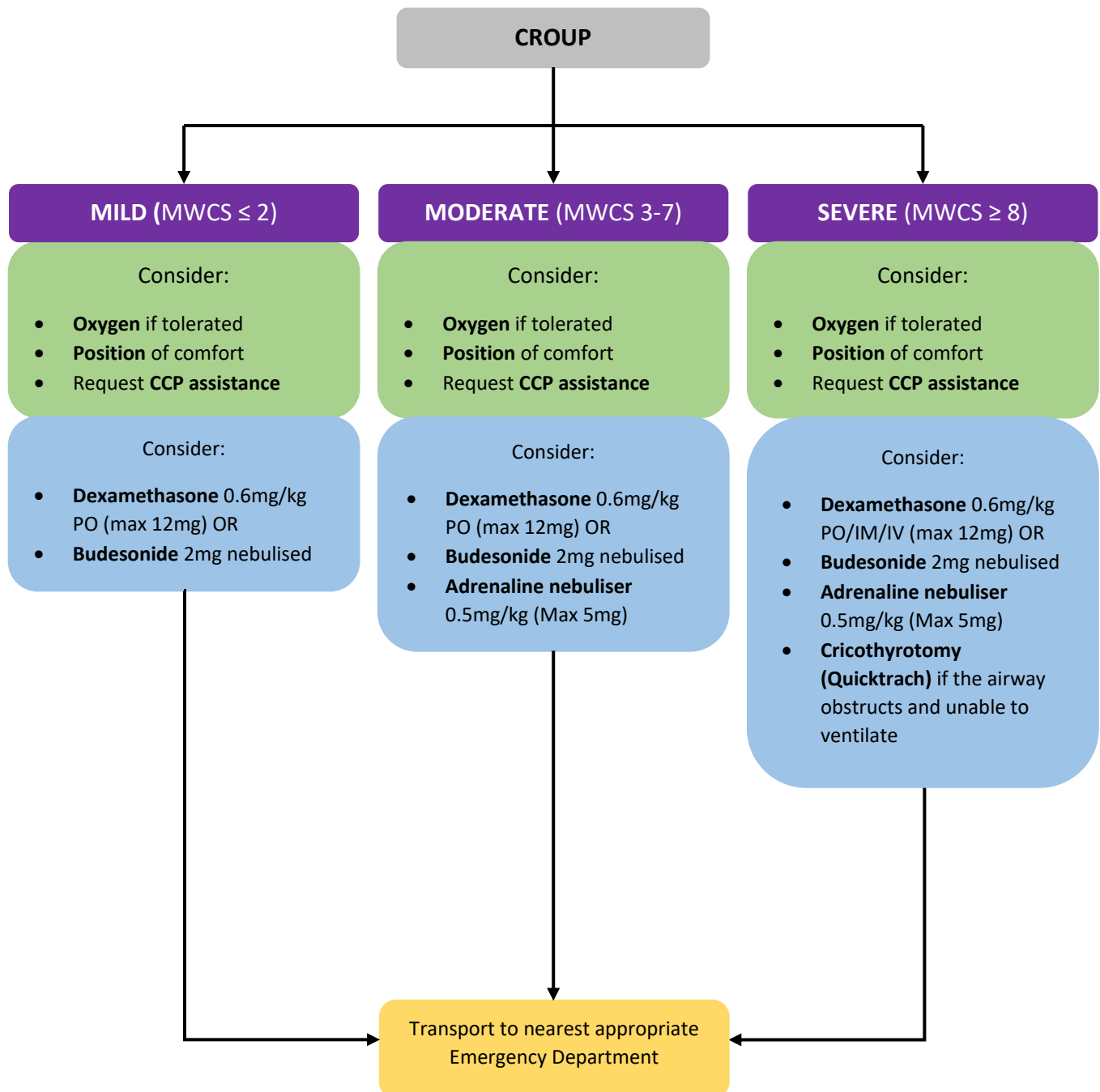
	0 POINTS	1 POINT	2 POINTS	3 POINTS	4 POINTS	5 POINTS
Inspiratory stridor	None	When agitated	At rest			
Intercostal recession	None	Mild	Moderate	Severe		
Air entry	Normal	Mildly decreased	Severely decreased			
Cyanosis	None				With agitation/ activity	At rest
LOC	Awake					Altered

Total possible score = 0-17. **Mild croup** = ≤ 2 ; **Moderate croup** = 3-7; **Severe croup** = ≥ 8



SPECIAL NOTES

- Avoid unnecessary intervention and procedures that may cause further distress or worsening symptoms. This includes vital signs assessment. Transport these patients in a position of comfort, ideally in a parent's arms. P1 transport only if airway compromise develops.
- An IV cannula should only be inserted if absolutely required (e.g. the child requires resuscitation).
- Dexamethasone orally is the first line medication for croup. Budesonide should only be considered if unable to administer Dexamethasone. Do not administer together.
- Mix the Dexamethasone with a small amount of sweet liquid such as fruit juice or oral Dextrose to administer orally via a syringe or clean spoon.
- IM or IV Dexamethasone may be considered in severe/life-threatening cases.
- Oxygen therapy is not routinely required for croup and should only be considered if the patient is desaturating $SpO_2 < 94\%$.
- Adrenaline nebuliser is indicated if stridor or signs of airway obstruction are present.
- CCPs may sign-off mild Croup patients ONLY following either Dexamethasone or Budesonide administration provided the patient is stable and is not at risk of deteriorating.



CPG 5.4 EPIGLOTTITIS

Epiglottitis is an inflammatory disease of the airway. Epiglottitis is a medical emergency. The symptoms of epiglottitis generally progress over hours rather than days.



CLINICAL PRESENTATION

Patients with epiglottitis may present with the following:

- High fever
- Sore throat
- Difficulty swallowing
- Stridor
- Respiratory distress
- Drooling
- Muffled voice



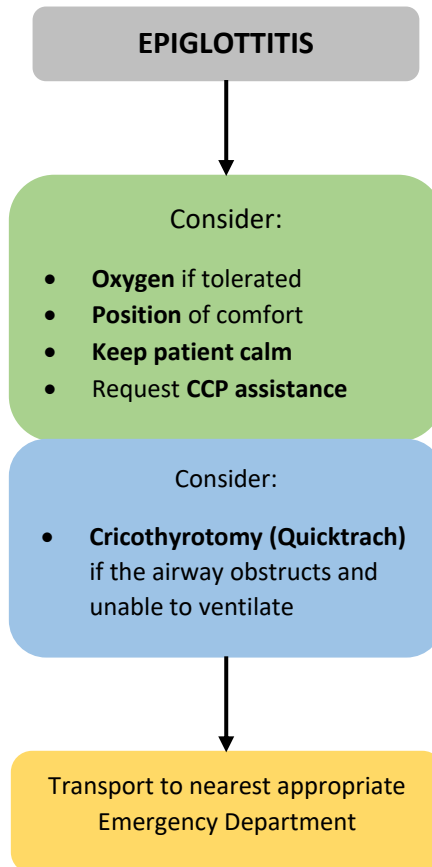
ASSESSMENT CONSIDERATIONS

- Epiglottitis generally affects patients between 2 – 6 years old.
- Do not attempt to examine the patient's throat.
- Do not attempt to insert an IV cannula unless actively resuscitating the patient.
- Any unnecessary assessments or disturbances of the patient may result in complete airway obstruction.
- Keep the patient in a position of comfort, sitting up, preferably in a parent's arms for comfort.
- Only apply an oxygen mask if tolerated by the patient.
- Consider alternative diagnosis (e.g. croup, FBAO)



SPECIAL NOTES

- Do not lie the patient down. This may result in complete airway obstruction.
- Only administer oxygen if tolerated by the patient.
- In severe cases, complete airway obstruction can rapidly develop within 3 – 6 hours.



CPG 5.5 FOREIGN BODY AIRWAY OBSTRUCTION (FBAO)

FBAO is a life-threatening emergency that requires urgent intervention. FBAO is classified as either partial or complete. The severity of obstruction needs to be rapidly identified and managed.



CLINICAL PRESENTATION

Patients with FBAO may present with the following:

- Respiratory distress
- Stridor
- Accessory muscle use, recession and paradoxical breathing
- Restlessness
- Cyanosis
- Unconsciousness
- Bradycardia

History gathering is important in order to identify FBAO. The following signs may indicate FBAO:

- Clutching the neck
- Sudden onset of coughing, gagging, sudden onset of dyspnoea
- History of playing with or eating small items
- Eating



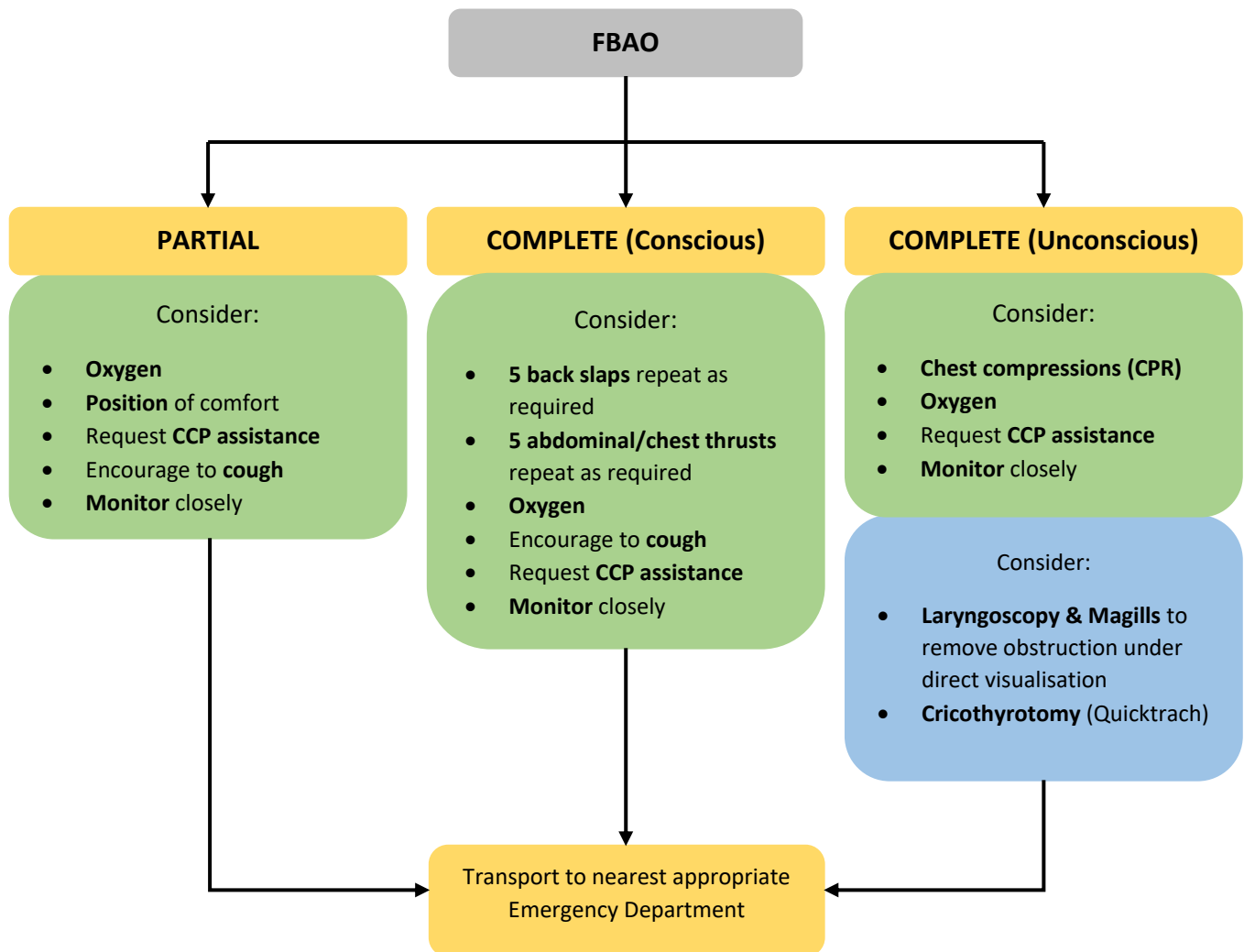
ASSESSMENT CONSIDERATIONS

- FBAO severity is assessed by determining the patient's ability to cough. An effective cough indicates a mild obstruction and the inability to cough indicates a severe obstruction.
- If the patient is still conscious, ask the patient if they are choking.
- Encourage conscious patients to cough if possible.
- Assess the airway to determine if the obstruction is visible after each back slap/chest thrust. The aim is to relieve the obstruction after each back slap/chest thrust rather than giving all five.
- If the patient is unconscious, start CPR. No pulse check is required.



SPECIAL NOTES

- Infants should be placed in the head down position prior to providing back slaps.
- If the patient is in cardiac arrest, use a compression ventilation ratio of 30:2 (manual and mechanical). Regularly assess the airway for the obstruction during compressions. Manage as per relevant cardiac arrest CPG.
- Do not attempt to insert an OPA/SGA until the obstruction has been removed.
- In the presence of a complete airway obstruction, APs should initiate early P1 transport to hospital and meet a CCP en route.



CPG 5.6 HYPERVENTILATION

Hyperventilation, an extreme form of tachypnoea, results in hypocapnia and respiratory alkalosis. It is important to rule out alternative causes of tachypnoea that have the potential to be life-threatening, such as lung pathology (PE, pneumothorax, asthma, pneumonia); Brain pathology (hypoxia, brain stem injury) and systemic illness (heat stroke, anaphylaxis, toxins and metabolic acidosis).

CLINICAL PRESENTATION

Patients with hyperventilation syndrome may present with the following:

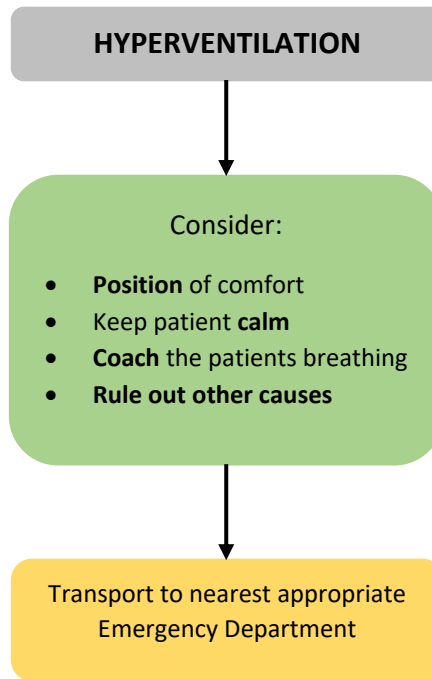
- Tachypnoea (age dependent)
- Paraesthesia around mouth, hands and feet (due to hypocapnia)
- Restlessness
- Pain
- Vertigo
- Carpopedal spasm

ASSESSMENT CONSIDERATIONS

- Hyperventilation due to emotional distress is rare in children. Identify the possible underlying cause of the hyperventilation.
- Search for possible underlying causes of hyperventilation in all patients that are presenting with hyperventilation and manage according to relevant CPG.
- Coach the patient to slow down their breathing and calm down.
- A method to control the patients breathing is to get them to read a passage of text out loud. This distraction forces the patient to regulate their breathing.

SPECIAL NOTES

- Hyperventilation due to emotional distress should not be considered as a diagnosis until all possible other causes of hyperventilation have been excluded.
- Do not use a paper bag to treat hyperventilation.



CPG 5.7 PULMONARY EMBOLISM

Pulmonary embolism (PE) most often originates from DVT, however this is not the only cause. PE may also be caused by a fat, air and amniotic emboli. Patients presenting with signs of a possible PE should also be assessed for signs of DVT.

CLINICAL PRESENTATION

Patients with a pulmonary embolism may present with the following:

- Dyspnoea
- Tachypnoea
- Pleuritic or substernal chest pain
- Syncope
- Cough
- Haemoptysis
- Jugular vein distension
- Hypoxia
- Cyanosis
- Tachycardia
- Hypotension
- Shock

Signs of DVT include (mainly presenting in the lower limbs):

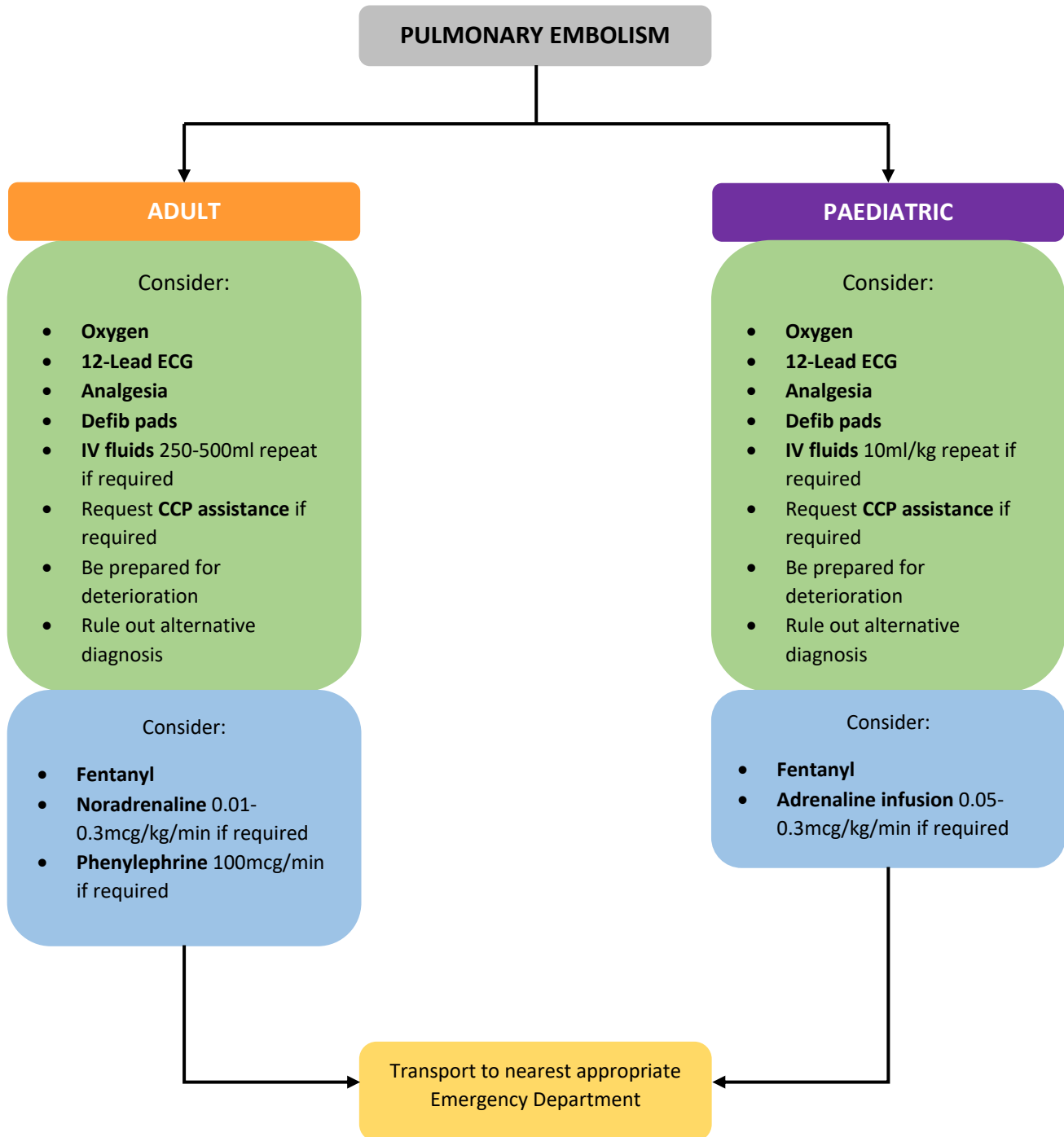
- Unilateral swelling
- Redness, localised warmth to touch
- Tenderness

ASSESSMENT CONSIDERATIONS

- Consider alternative differential diagnosis (e.g. AMI, pneumothorax, pneumonia, heart failure, cardiac tamponade, pericarditis).
- Assess for signs of DVT if PE is suspected.
- Gather detailed history to identify risk for PE:
 - Previous history of DVT or PE
 - Immobile for prolonged period
 - Recent surgery, trauma or hospitalisation
 - Oral contraception use
 - Hormone replacement therapy
 - Cancer
 - Pregnancy
- SpO₂ that fails to improve despite supplemental oxygen is suggestive of a possible PE.
- Right ventricular dysfunction as a result of PE may produce a RBBB.

SPECIAL NOTES

- Administer fluids cautiously, particularly in the presence of right ventricular dysfunction.



CPG 5.8 SEVERE RESPIRATORY INFECTION

Respiratory infections may result from a number of sources. Caution should be taken when managing patients with a potential infectious disease (e.g., COVID-19, MERS etc). Patients with severe respiratory infection will deteriorate if left untreated.

CLINICAL PRESENTATION

Patients with a severe respiratory infection may present with the following:

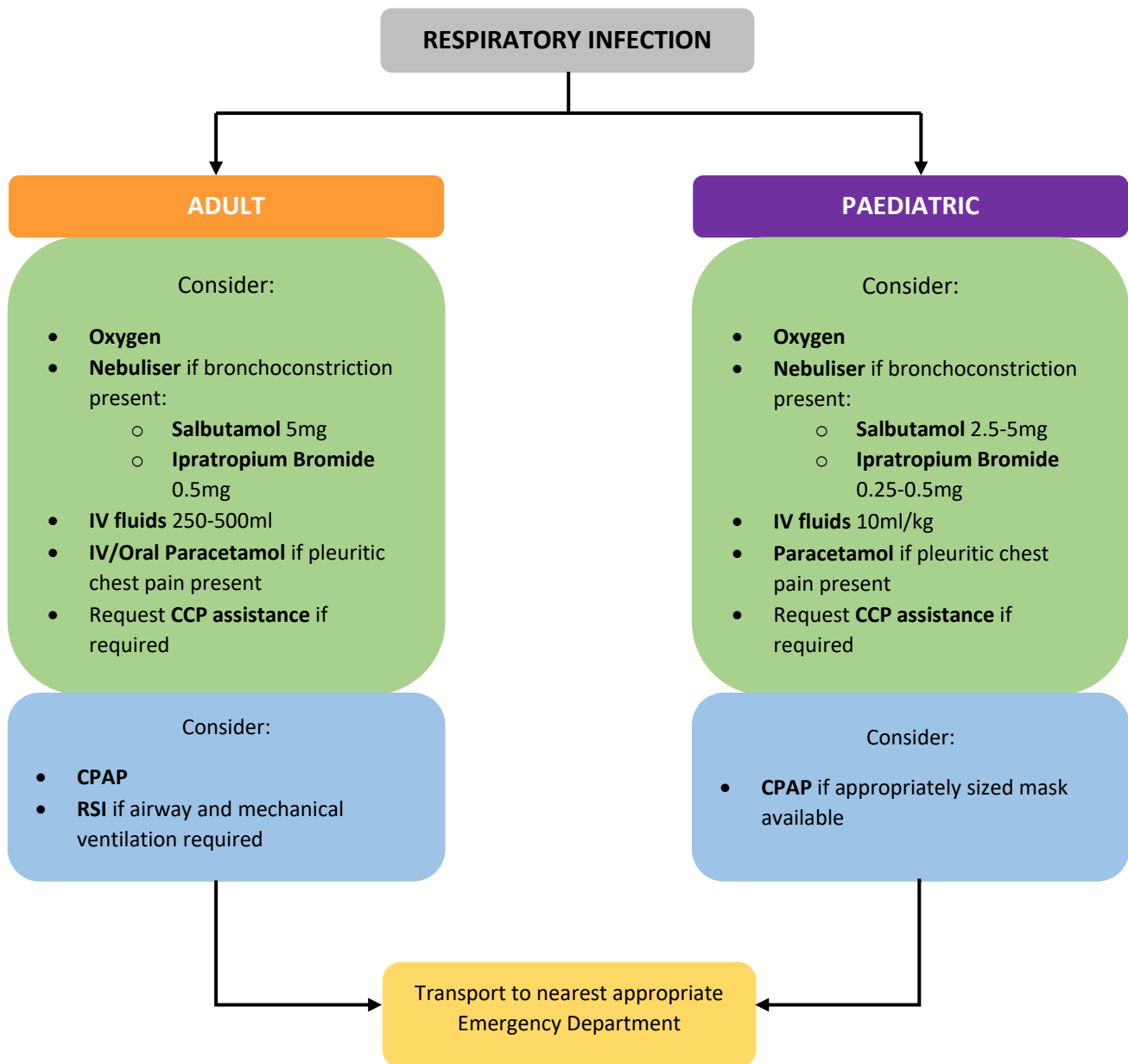
- Cough (often productive)
- Tachypnoea
- Dyspnoea
- Tachycardia
- Hypoxia
- Fever
- Pleuritic chest pain
- Accessory muscle use

ASSESSMENT CONSIDERATIONS

- A detailed assessment and history taking are required to identify potential causes of infection (e.g. recent travel, or exposure to someone with an infectious disease).
- Comorbidities such as asthma and COPD may exacerbate respiratory infections.
- Assess and monitor for signs of sepsis and septic shock. Manage sepsis and septic shock according to CPG 6.6.

SPECIAL NOTES

- Use appropriate PPE when managing patients with a respiratory infection.
- Decontaminate all equipment using appropriate cleaning material after handing over your patient.



CPG 5.9 TRACHEOSTOMY EMERGENCIES

A tracheostomy is a surgical procedure to create a patent airway through the neck. The opening in the neck, known as the stoma, can be either permanent or temporary. Patients that have had a partial laryngectomy may still have a patent upper airway, whereas patients that have had a total laryngectomy will not have a patent upper airway and the trachea has been revised to the front of the neck.

There are various types of tracheostomy tubes that you may encounter, mainly single or double lumen tubes; cuffed or uncuffed tubes; and fenestrated or adjustable tubes. Patients experiencing a life-threatening emergency related to a tracheostomy include either a partial/complete obstruction or a partial/complete dislodgement of the tracheostomy tube. Paramedics should rapidly identify the type of tracheostomy emergency and intervene as required.



CLINICAL PRESENTATION

Patients experiencing a tracheostomy emergency may present with the following:

- Difficulty in breathing
- Tachypnoea
- Low SpO₂
- Grunting, snoring, gurgling or stridor
- Accessory muscle use
- Anxiety, distress, agitation



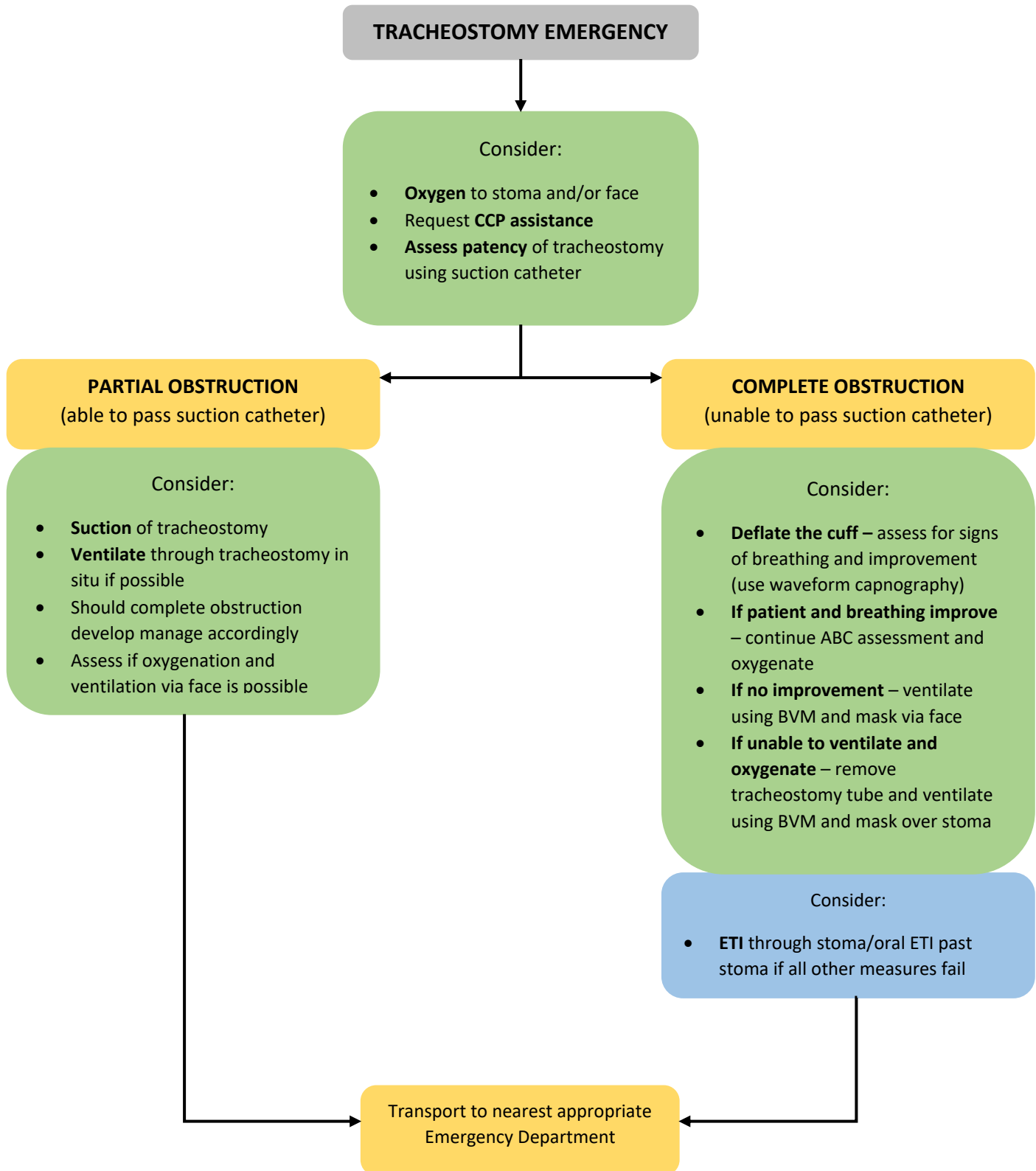
ASSESSMENT CONSIDERATIONS

- The following may contribute to a tracheostomy emergency and should be assessed for:
 - Excessive sputum
 - Coughing
 - Tracheal irritation
 - Dry, hardened secretions
 - Undue movement of the tracheostomy tube
- If required, oxygen may be applied to both the face and the stoma if the upper airway is still patent.
- To assess the patency of the tracheostomy tube or stoma opening, pass a soft tip suction catheter through the opening without applying suction. This is likely to cause the patient to start coughing. Should the catheter not pass through easily, do not attempt to force. Instead, attempt to suction.
- Assess for the need to reinsert the inner cannula (if it has been removed) of a double lumen tracheostomy tube to be able to connect the BVM if required.
- When ventilating through the tracheostomy tube or stoma using a BVM, either attach the BVM directly to the tracheostomy tube if compatible or use a neonatal/paediatric BVM mask placed over the stoma to ventilate. Assess for air leak from the upper airway. Air leak from the upper airway indicates at least partial patency.
- If ventilating through the upper airway only, ensure that the stoma is appropriately sealed with an occlusive dressing.



SPECIAL NOTES

- Care must be taken when passing a suction catheter through a fenestrated tracheostomy tube. There is a risk that the catheter may pass through the fenestrated hole.
- Do not remove the tracheostomy tube unless unable to ventilate through the tube such as in a complete obstruction. If the tube is partially obstructed or dislodged, continue to ventilate through the tube in situ.
- Oral ETI is challenging in these patients. If attempted, pass the ETT beyond the stoma then inflate the cuff.
- If attempting ETI through the stoma, a bougie may be used, however, take note that there is a high risk of creating a false passage, particularly within 1 month of having the tracheostomy created.
- Use a smaller sized ETT if attempting oral or stoma ETI. Preferably one size smaller than the tracheostomy tube. Confirm ETT placement using waveform capnography and auscultation.
- CCPs may only consider ETI for the paediatric patient aged 8 years and older. ET tubes sized for those under 8 years of age are not available within HMCAS.





CPG 6.1-6.6

MEDICAL



CPG 6.1 ABDOMINAL EMERGENCIES (NON-TRAUMATIC)

Non-traumatic abdominal pain is a common condition attended to by the ambulance service. Most presentations are not critical in nature however, it is important to identify and differentiate between those patients who require urgent intervention and transport and those that do not. Time critical abdominal emergencies include ectopic pregnancy, ruptured abdominal aortic aneurism (AAA), peritonitis, testicular/ovarian torsion, uncontrolled GI bleed, acute bowel obstruction and acute pancreatitis.

CLINICAL PRESENTATION

Patients experiencing a high-risk non-traumatic abdominal emergency may present with the following:

- Severe abdominal pain/tenderness
- Altered vital signs
- Nausea and/or vomiting
- Haematemesis and/or melaena
- Pulsating mass/feeling in the abdomen for patients with AAA

Patients that are at higher risk include:

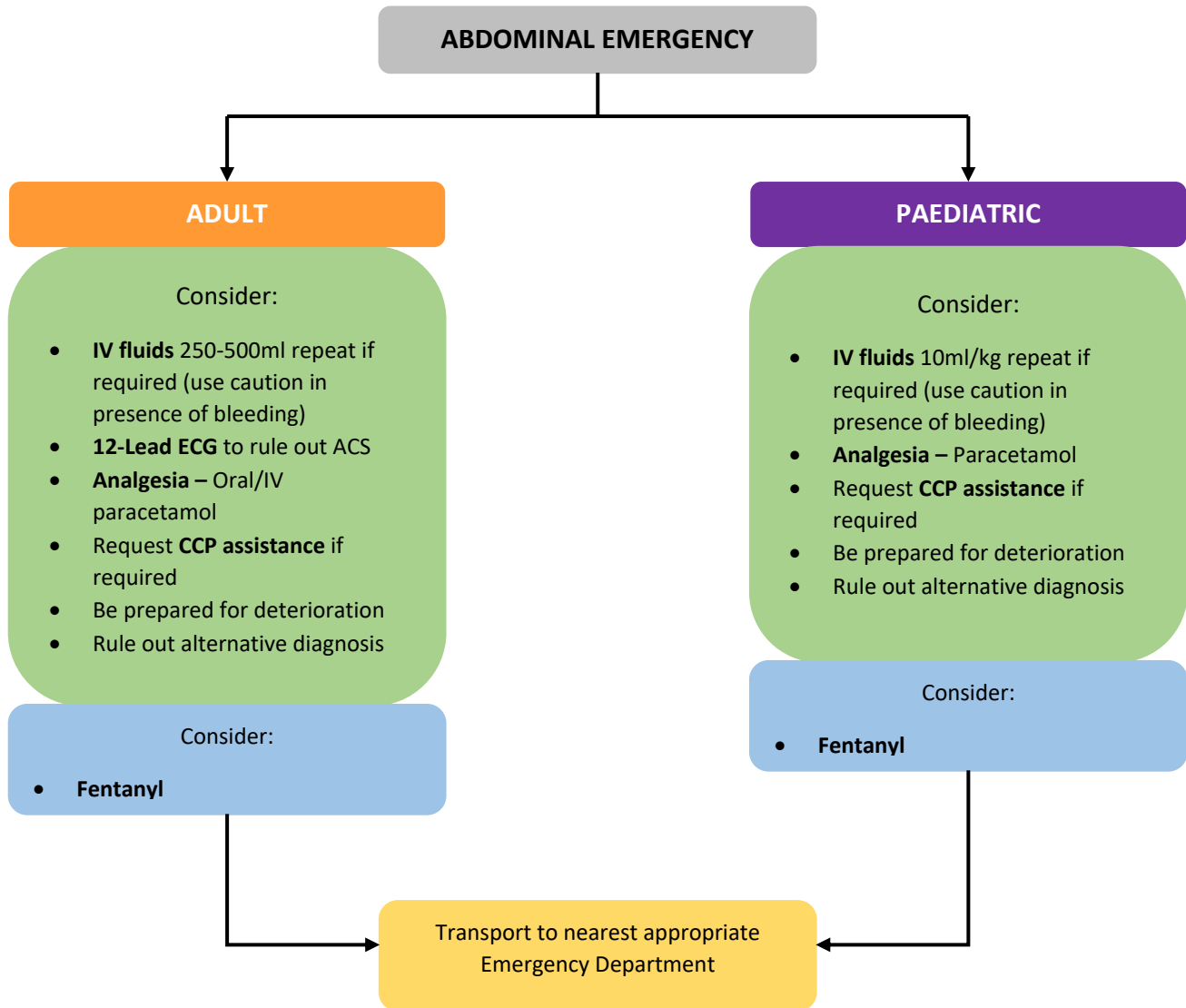
- Females of childbearing age
- Elderly (>60 years old) and paediatrics

ASSESSMENT CONSIDERATIONS

- Cardiac related pain can mimic gastro-oesophageal reflux type abdominal pain.
- Rule out potential cardiac emergencies in patients experiencing abdominal pain. Patients experiencing abdominal pain should have a 12-lead ECG performed if there is a high index of suspicion for cardiac involvement.
- AAAs may present similarly to renal colic type pain, particularly in the elderly patient.
- Assessing BP and pulse strength on both arms may be a useful indication if an aortic dissection is suspected.

SPECIAL NOTES

- TXA is not routinely recommended for patients with a GI bleed. The onset of a GI bleed is difficult to determine. Administering TXA outside of the 3-hour window has no benefit and an increased risk of harm.
- Patients with a AAA often have severe abdominal pain requiring narcotic analgesia. Request CCP if required.
- Use fluids with caution if major bleeding or dissecting aortic aneurysm present.



CPG 6.2 ANAPHYLAXIS & ALLERGIC REACTIONS

Anaphylaxis is a life-threatening emergency requiring immediate treatment. Anaphylaxis is defined as a severe, life-threatening hypersensitivity reaction, characterised by rapidly developing life-threatening airway and/or breathing and/or circulation problems, often associated with skin, mucosal changes, and GI disturbances. An allergic reaction involves a single system (typically skin and mucosal features) versus anaphylaxis which involves two or more systems. The most common triggers for anaphylaxis/allergic reactions are food, drugs/medicine, and venom.

CLINICAL PRESENTATION

Patients experiencing anaphylaxis may present with the following:

- Urticaria, pruritus, flushing, angioedema (localised or general)
- Dyspnoea, wheezes, stridor, rhinitis
- Hypotension, dizziness, bradycardia, or tachycardia, altered LOC
- Nausea, vomiting, diarrhoea, abdominal pain

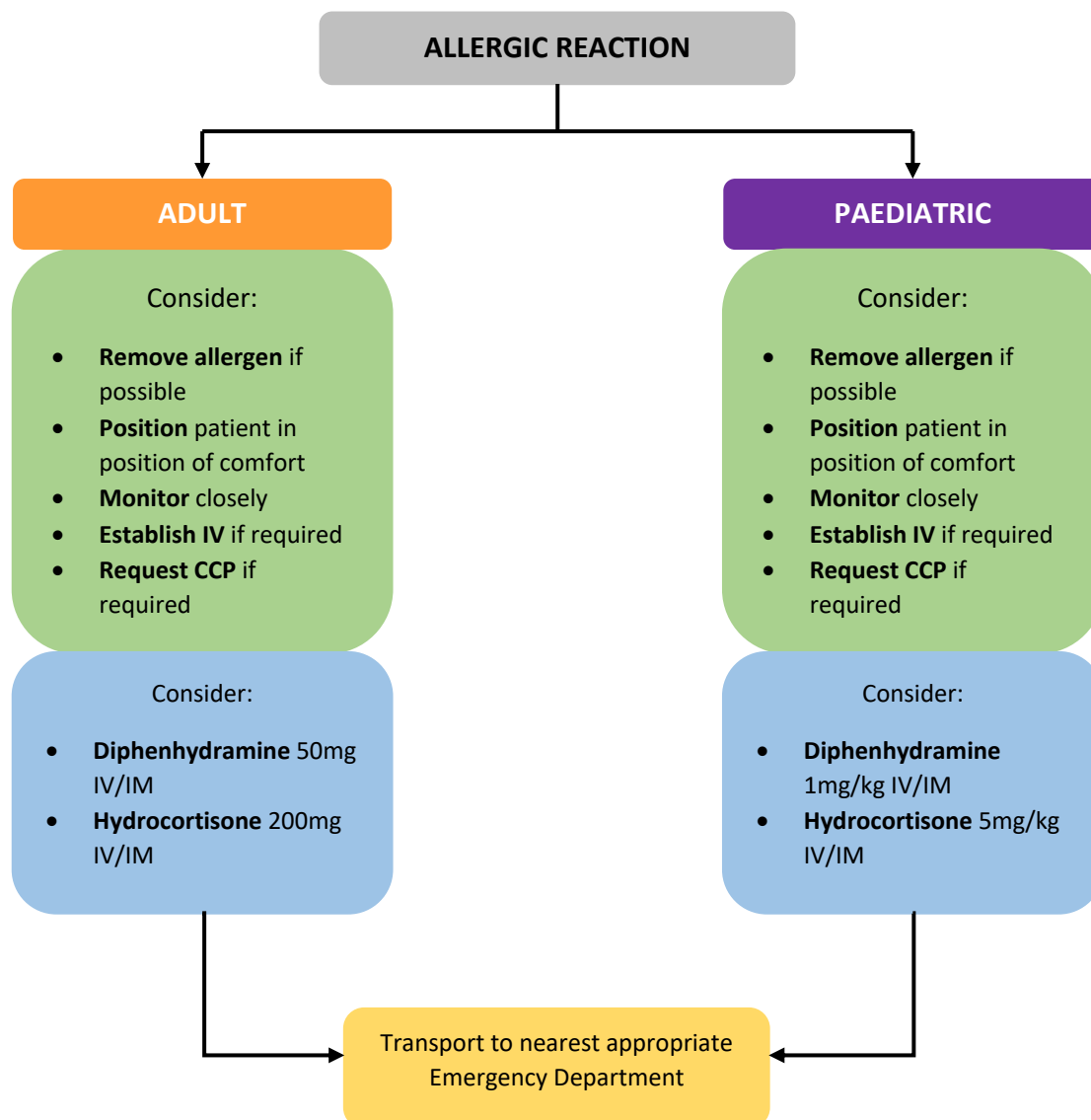
ASSESSMENT CONSIDERATIONS

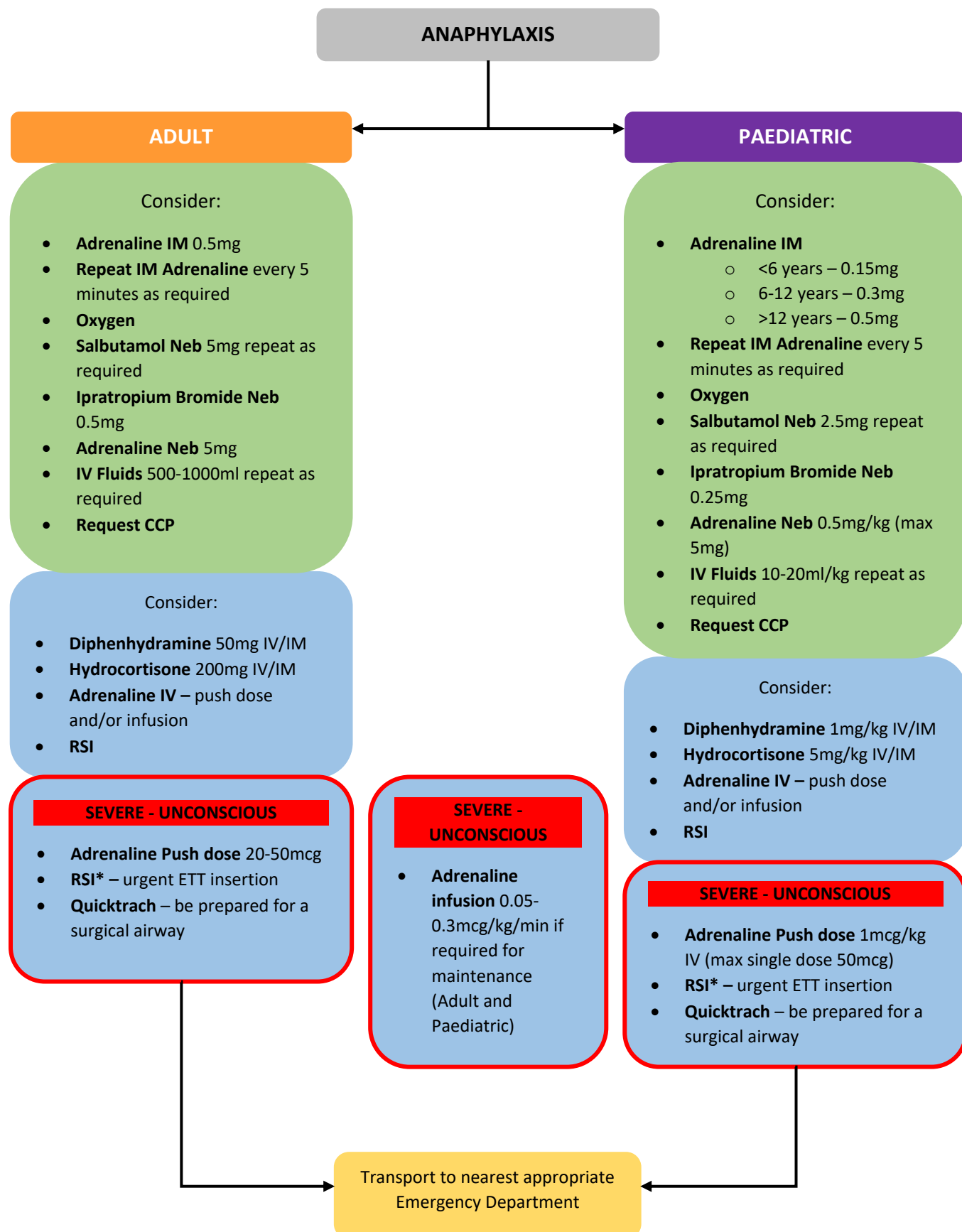
- The presentation of anaphylaxis can be variable and unpredictable. Onset can be rapid or gradual.
- Signs such as rash may not be present if the patient is significantly hypotensive due to reduced cutaneous blood flow.
- Note that repeated doses of IM adrenaline may be required. Continuous monitoring of the patient is required. Patients are at risk of relapse.
- History gathering is important to identify the possible cause of anaphylaxis/allergic reaction as well as to identify previous episodes of anaphylaxis.
- Remove the allergen if still present (e.g., bee sting).
- Position patient in the supine position and elevate their legs if required. Note that patients with respiratory compromise will most likely not tolerate being supine. Position these patients in a position of comfort. Keep pregnant patients in the left lateral position or position of comfort.

SPECIAL NOTES

- Do not be hesitant to administer IM adrenaline, if patients present with signs and symptoms of impending anaphylaxis, even if the diagnostic criteria are not met and there is a high index of suspicion for anaphylaxis, IM adrenaline should be administered.
- Do not allow anaphylaxis patients to walk even if they appear to have recovered.
- Isolated hypotension after exposure to an allergen should be treated as anaphylaxis.
- CCPs may administer Diphenhydramine and/or Hydrocortisone and then handover care to the AP crew for transport for patients with allergic reactions in a stable condition.
- Salbutamol and Ipratropium Bromide neb only if bronchoconstriction is present.
- Nebulised adrenaline should only be considered if upper airway swelling/stridor is present.
- Consider RSI if there is airway involvement and no response to treatment. Predict a difficult airway.
- Push dose IV Adrenaline should be considered if unresponsive to 3 x IM Adrenaline doses.
- CCP should consider RSI if patients are in extremis and unresponsive to other therapy.

- The unconscious patient or patient in extremis/peri-arrest requires immediate and urgent intervention. CCPs should consider the following:
 - Push dose IV Adrenaline – 1mcg/kg (max single dose of 50mcg). You do not need to administer IM Adrenaline first in such cases.
 - * Urgent establishment of an ETT (age $\geq$ 8 years). RSI using Rocuronium and low dose Ketamine. No Fentanyl.
 - Be prepared to insert the Quicktrach.
 - A definitive airway should be established within a maximum of 4 minutes from patient contact.
- An Adrenaline infusion may be required for ongoing maintenance.





CPG 6.3 HYPOGLYCAEMIA & HYPERGLYCAEMIA

The normal blood glucose range is 4.0 – 6.7 mmol/L. This range may vary in the presence of diabetes. Hypoglycaemia is defined as an RBS of <4.0mmol/L in adults and paediatrics and <2.5mmol/L for newborns. Hypoglycaemia can occur in any patient with or without diabetes. Hyperglycaemia associated with diabetes may present as Diabetic Ketoacidosis (DKA) or Hyperosmolar Hyperglycaemic Syndrome (HHS). Hyperglycaemic patients may also be asymptomatic. DKA and HHS generally occur due to acute illness such as ACS, CVA and infection or due to non-compliance with diabetic medication.



CLINICAL PRESENTATION

Patients experiencing Hypoglycaemia may present with the following:

- Diaphoresis and/or pallor
- Hunger
- Tremors
- Tachycardia and/or palpitations
- Altered LOC
- Dizziness
- Anxiety
- Lethargy
- Headache
- Behaviour changes and/or visual disturbances
- Slurred speech
- Seizures (rare)

Patients experiencing Hyperglycaemia (DKA and HHS) may present with the following:

- Lethargy
- Altered LOC
- Seizures
- Dehydration/hypovolaemia
- Hypotension
- Tachycardia
- Pale
- Cool or clammy
- Flushed and hot
- DKA: RBS >10mmol/L
- HHS: RBS >40mmol/L



ASSESSMENT CONSIDERATIONS

Hypoglycaemia:

- Patients with poorly controlled diabetes may still be symptomatically hypoglycaemic despite having an RBS >4 mmol/L. Attempt to determine the patient's normal RBS range. An additional 50ml for adults and 2.5ml/kg for paediatrics of Dextrose 10% may be administered to patient. Avoid hyperglycaemia.
- Hypoglycaemia may produce agitation, aggression and violent behaviour. Use caution when assessing such patients.
- Beta-blocker medication may mask signs of hypoglycaemia.
- All patients with a reduced/altered LOC that is not trauma related should have their RBS assessed.
- Assess for possible cause of hypoglycaemia (e.g. underlying illness, inappropriate use of diabetes medication).

Hyperglycaemia:

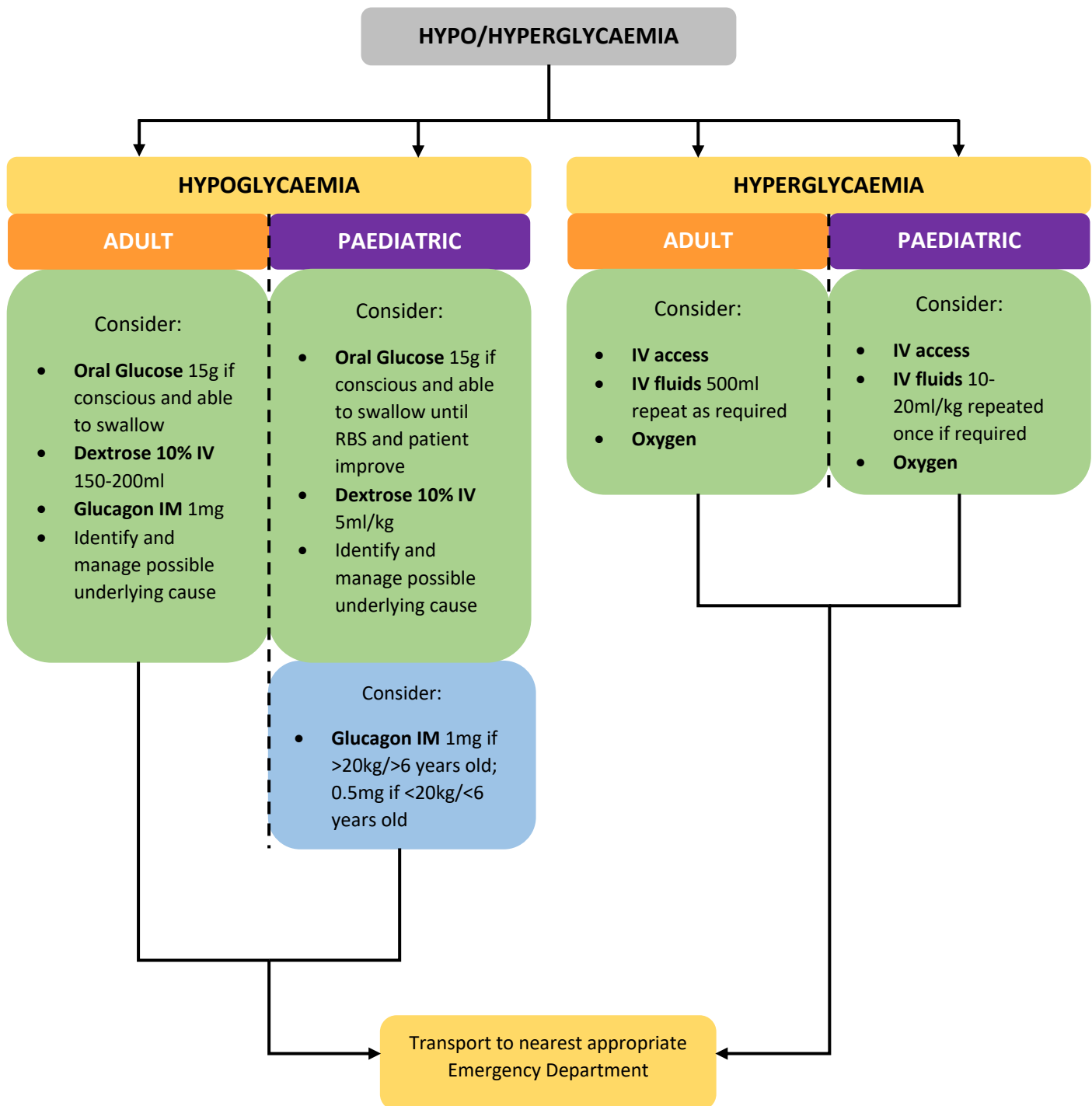
- Hyperglycaemic patients are at increased risk of dehydration and hypovolemia. Assess for such signs and provide fluids as required.
- Patients with DKA and severe metabolic acidosis may present with Kussmaul respirations.

- Assess for potential causes of hyperglycaemia (e.g. infection, CVA, inappropriate use/non-compliance with medication).



SPECIAL NOTES

- Hypoglycaemic patients may present with signs and symptoms that mimic stroke or intoxication.
- Glucagon should be considered in the presence of hypoglycaemia if unable to obtain IV access and the patient is unresponsive. Request for CCP backup if required.
- Consider alternative provisional diagnosis if hypoglycaemic patient does not respond to treatment.



CPG 6.4 NAUSEA AND VOMITING

Nausea and vomiting are nonspecific symptoms commonly associated with a wide range of medical conditions. Identifying the underlying cause is important as the available medication may not be effective.



CLINICAL PRESENTATION

Patients experiencing nausea and vomiting may present with the following:

- Hypersalivation
- Retching
- Clamminess
- Sweating
- Pallor
- Abdominal pain



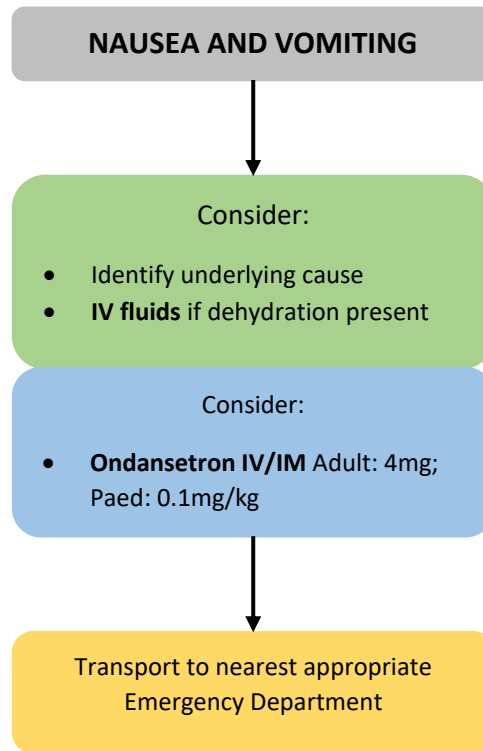
ASSESSMENT CONSIDERATIONS

- Identify the potential cause of nausea and vomiting (e.g. GI disturbances, ACS, medication).
- Antiemetics are not for routine administration. Only administer if the patient is unable to tolerate symptoms.
- Assess for signs of dehydration if the patient has had severe prolonged vomiting.



SPECIAL NOTES

- Prophylactic administration of an antiemetic is prohibited except for long distance transfers where motion sickness may occur and could be detrimental and an appropriate antiemetic is available.
- Ondansetron is not effective against motion sickness.



CPG 6.5 NON-TRAUMATIC SHOCK

Shock is a life-threatening condition requiring rapid recognition and treatment. Non-traumatic shock can be categorised into hypovolaemic, cardiogenic, obstructive, septic and anaphylactic shock. For the management of cardiogenic, septic and anaphylactic shock, see relevant CPG.

Hypovolemic shock is caused by dehydration due to severe vomiting, diarrhoea, exposure to high temperatures and reduced fluid intake or non-traumatic causes of blood loss (e.g. obstetric emergencies).

Obstructive shock is due to severe bronchospasm or airway obstruction with significant air trapping causing an increase in intrathoracic pressure and compression of the heart, reducing venous return and thus cardiac output.



CLINICAL PRESENTATION

Patients experiencing non-traumatic shock may present with the following:

- Hypotension
- Tachycardia
- Bradycardia
- Pallor
- Respiratory distress
- Altered LOC
- As well as signs and symptoms associated with the underlying cause of shock



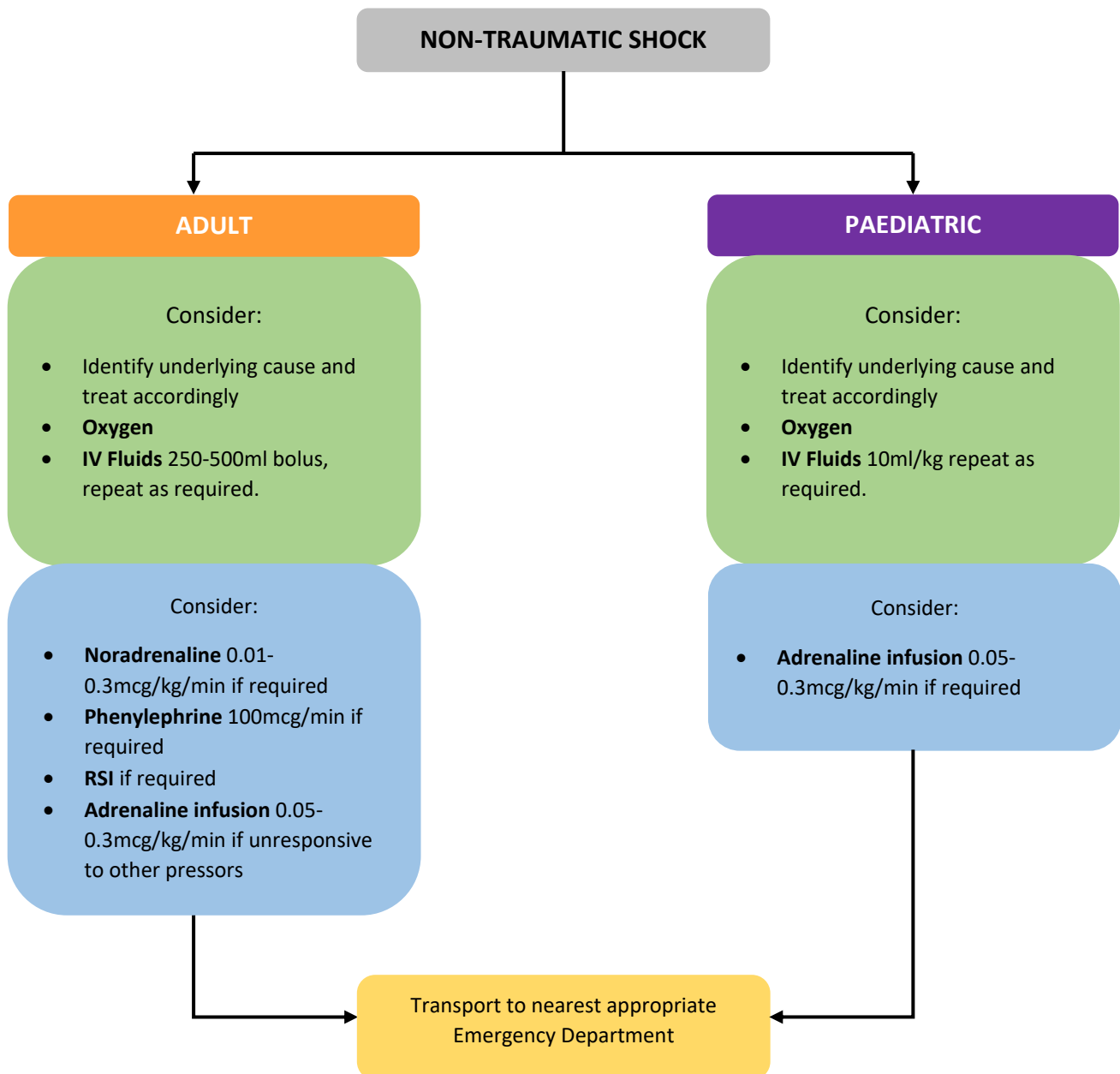
ASSESSMENT CONSIDERATIONS

- Identify, if possible, the underlying cause of shock and treat appropriately.
- Regularly assess perfusion status to determine response to treatment. For adult patients aim for a SBP >90mmHg or MAP of 70-80mmHg. Paediatric patients aim for age specific SBP using (age x 2) + 70. Avoid sudden spikes in BP.
- Assess responsiveness to fluids prior to considering vasopressor support.



SPECIAL NOTES

- Large fluid volumes should be avoided in the elderly and in those with renal failure.
- Patients with heat stroke will require large amounts of fluids. Be aggressive with fluids prior to considering vasopressors.
- If airway protection and ventilation is required, adjust RSI drug dosing according to haemodynamics.
- Should a vasopressor be required, consider a push dose of phenylephrine while preparing a noradrenaline infusion.
- Consider an Adrenaline infusion if there is no response to other pressor infusions



CPG 6.6 SEPSIS

Sepsis is defined as life-threatening organ dysfunction caused by a dysregulated host response to infection. Septic shock is a subset of sepsis with profound circulatory, cellular and metabolic abnormalities. Management of these patients requires prompt recognition, immediate treatment and timely transport to an appropriate facility with prenotification.

CLINICAL PRESENTATION

ADULT

Adult patients experiencing sepsis or septic shock may present with the following (signs and symptoms may vary depending on site/source of infection):

- Respiratory distress/tachypnoea
- Hypotension/poor perfusion
- Tachycardia
- Oxygen required to maintain SpO₂ > 92%
- Non-blanching rash/mottled/ashen/ cyanosis
- Temperature < 35.5°C or > 38.4°C
- Altered LOC
- Anuria last 12-18 hours or significantly reduced urine output

PAEDIATRIC

Paediatric patients experiencing sepsis or septic shock may present with the following (signs and symptoms may vary depending on site/source of infection):

- Tachycardia or bradycardia
- Respiratory distress/tachypnoea/apnoea
- Oxygen required to maintain SpO₂ > 94%
- Non-blanching rash/mottled/ashen/cyanotic
- Altered LOC
- Hypoglycaemia or hyperglycaemia
- Pain and/or restlessness
- Reduced urine output
- Hypothermia or fever

ASSESSMENT CONSIDERATIONS

- Tachypnoea is the most common presenting symptom for sepsis.
- Sepsis should be suspected in the unwell immunocompromised patient.
- Assess temperature. Fever is commonly associated with sepsis however some patients may present with hypothermia which is usually associated with worse outcomes.
- Assess for risk factors such as:
 - Immunocompromised patients
 - Age <3 months or >65 years old
 - Recent trauma, surgery, invasive procedure or wound in the last 6 weeks
 - Indwelling medical devices
 - Patients that are re-presenting to the healthcare services within 48 hours

- Malignancy
- Cardiac/respiratory/neuromuscular disease
- Use qSOFA as a predictor of severity/mortality in the presence of suspected sepsis for adults and the paediatric sepsis screening tool for paediatric patients

Quick Sequential Organ Failure Assessment (qSOFA):

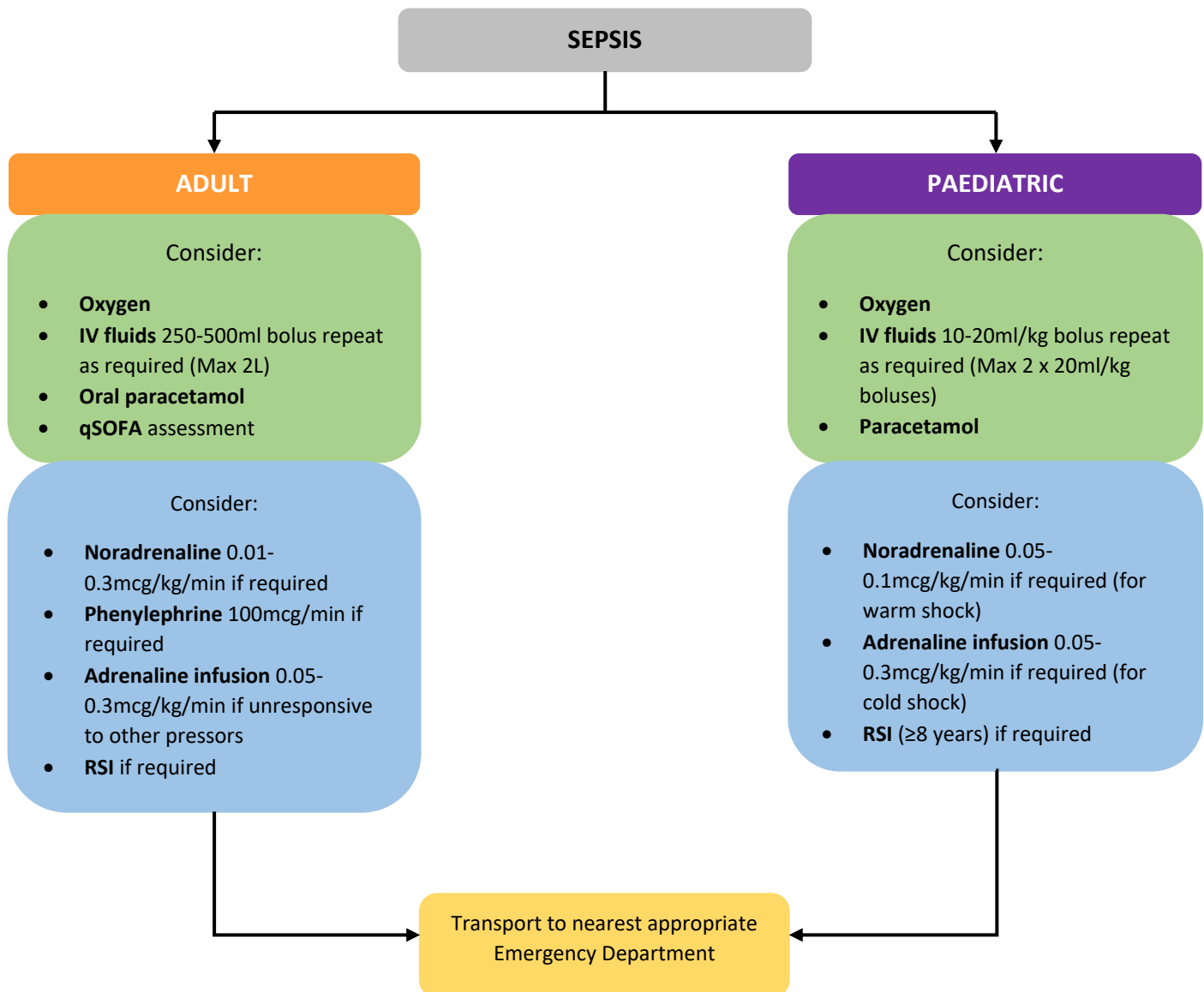
- Altered mental status (GCS < 15)
- SBP ≤ 100mmHg
- RR ≥ 22 breaths/min

PAEDIATRIC SEPSIS SCREENING TOOL								
1	Does this patient have an infection? Is the patient at risk of an infection?		Meningitis Upper respiratory infection Chest infection Urinary tract infection Abdominal pain or distension An infected wound		Cellulitis Septic arthritis Indwelling medical device Chemotherapy <6 weeks ago Immunocompromised Recent organ transplant			
	YES		THINK SEPSIS					
2	HIGH RISK CONDITIONS		Age <3 months; malignancy; immunocompromised, cardiac/respiratory/neuromuscular disease; indwelling medical device; recent surgery/hospitalisation					
	IS THIS PATIENT HIGH RISK?		YES ☐		NO ☐			
3	VITAL SIGNS		INDICATE THE CORRECT AGE GROUP					
	X ONLY IF ABNORMAL	Enter patient vital signs	Age in Months			Age in Years		
			0 – 1mon	1 – 12mon	1 – 2yr	3 – 5yr	6 – 9yr	>10yr
	Heart Rate	bpm	≤100 or ≥180	≤100 or ≥189	≤90 or ≥169	≤80 or ≥139	≤75 or ≥137	≤60 or ≥119
	Resp Rate	rpm	≤30 or ≥60	≤30 or ≥60	≤22 or ≥50	≤20 or ≥45	≤18 or ≥35	≤12 or ≥30
	Systolic BP	mmHg	<60	<70	< (Age x 2) + 70			
	RBS	mmol/L	<2.6	< 4 mmol/L or > 7.7 mmol/L				
	Temp	°C	< 36°C or > 38°C					
	SpO ₂	%	Low SpO ₂ < 94%					
	Mental Status	AVPU	Reduced level of consciousness, confused, drowsiness, floppiness, lethargy, decreased arousability					
	Cap Refill	sec	Prolonged capillary refill > 2 seconds					
	Skin/Rash		Petechiae, any purpura, mottled and cool, flushed					
Total number of X boxes		Patient Identified as High Risk ☐						
4	High risk patients: +1 or more criteria or Standard patients: +2 or more criteria		SUSPECTED SEPSIS				☐	
	Initiate treatment Hospital prenotification Rapid transport		PREHOSPITAL SEPSIS CARE				☐	



SPECIAL NOTES

- Septic patients often require fluids. Provide fluid boluses and monitor fluid response. Use caution if signs of fluid overload are present. A maximum of 2L NS may be administered to the adult patient and a total maximum of two fluid boluses of 20ml/kg for paediatrics or max 2L (dependent on size and age).
- Response to fluid boluses will guide treatment and determine the need for a vasopressor.
- Aim for a target MAP ≥ 65 mmHg/SBP ≥ 90 mmHg for adults and age specific target for paediatrics (age x 2) + 70.
- Should RSI be required, adjust drug dosing according to the patient's haemodynamic status.
- Paediatric patients generally respond to fluid boluses. An vasopressor infusion is seldom required.
- Use EtCO₂ to accurately assess RR and end-tidal CO₂ levels.



CPG 6.7 ADRENAL INSUFFICIENCY

Adrenal insufficiency is an endocrine disorder which results in the reduced production of the adrenal hormone cortisol (a major glucocorticoid). This hormone is essential for the body to be able to respond to stress as well as maintain essential life functions. Unmanaged insufficiency can result in adrenal crisis and potentially be fatal.

CLINICAL PRESENTATION

Patients experiencing adrenal crisis may present with the following:

- Severe weakness
- Confusion or altered level of consciousness
- Syncope
- Postural hypotension
- Abdominal pain
- Nausea and/or vomiting
- Back pain
- Hypoglycaemia

The symptoms described above generally result following exposure to significant physical or psychological stress combined with the risk factors of an acute adrenal crisis. Diagnosis in the prehospital setting can be challenging due to the non-specific signs and symptoms and their gradual progression over time.

ASSESSMENT CONSIDERATIONS

- Patients exposed to significant physical or psychological stress and have any of the following below should be considered as high risk for acute adrenal crisis and should be considered for Hydrocortisone administration:
 - Known history of adrenal insufficiency (Addison's disease, congenital adrenal hyperplasia, panhypopituitarism)
 - Current history of long-term glucocorticoid therapy
 - Recent cessation of long-term glucocorticoid therapy (≥ 4 weeks)
- Examples of significant physical stress triggering the need for Hydrocortisone administration include:
 - Serious illness
 - Infections requiring antibiotics
 - Vomiting and/or diarrhoea
 - Altered vital signs
 - Significant undifferentiated pain
 - Labour
 - Trauma needing medical care
- Recent or current exposure to severe psychological stress may also trigger the need for Hydrocortisone administration
- Detailed history gathering should be performed to determine if the patient has any risks for an acute adrenal crisis

- Long-term systemic use of glucocorticoids, such as for anti-inflammatory or immunosuppressive therapy, is a common cause of adrenal insufficiency. This is due to the mechanism that adrenal glands stop producing endogenous glucocorticoids following long term steroidal therapy (≥ 4 weeks)



SPECIAL NOTES

- Patients diagnosed with adrenal insufficiency are often discharged with Hydrocortisone IM injections and guidance for self-administration. Ambulance Paramedics may administer the patient's own Hydrocortisone to the patient if the patient has not done so already prior to ambulance service arrival if indicated.
- Intravenous administration of Hydrocortisone is preferred. If unable to administer IV, then the IM route is acceptable
- It is preferable that the action plan provided to the patient following discharge from hospital be followed.
- Administration of Hydrocortisone to a pregnant woman is appropriate and should not be delayed if indicated.

ADRENAL CRISIS

ADULT

- **Oxygen** if required
- **IV fluids** 250-500ml bolus repeat as required (Max 2L)
- **Oral Glucose** 15g or **Dextrose 10%** 150-200ml if hypoglycaemic
- **12-lead ECG**
- **Hydrocortisone IV/IM** 100mg if patient has own medication that they have not yet self-administered (Single dose)

- **Hydrocortisone IV/IM** 100mg (Single dose)
- **Glucagon IM** 1mg if Dextrose administration not possible or unavailable

PAEDIATRIC

- **Oxygen** if required
- **IV fluids** 10-20ml/kg repeated once if required
- **Oral Glucose** 15g or **Dextrose 10%** 5ml/kg if hypoglycaemic
- **12-lead ECG**
- **Hydrocortisone IV/IM** if the patient has own medication that they have not yet self-administered (Single dose)
 - 0-4 years – 25mg
 - 5-10 years – 50mg
 - >10 years – 100mg

- **Hydrocortisone IV/IM** (Single dose)
 - 0-4 years – 25mg
 - 5-10 years – 50mg
 - >10 years – 100mg
- **Glucagon IM** if Dextrose administration not possible or unavailable
 - >20kg/6 years – 1mg
 - <20kg/<6 years – 0.5mg

Transport to nearest appropriate
Emergency Department



CPG 7.1

MENTAL HEALTH AND BEHAVIOURAL DISTURBANCES



CPG 7.1 ACUTE BEHAVIOURAL DISTURBANCE

Patients presenting with Acute Behavioural Disturbance (ABD) can pose a significant risk to themselves and others. Paramedics should take into account the special considerations relating to assessment, management and safety when caring for patients with ABD. There are numerous potential causes of ABD, namely, but not limited to:

- **Medical disorders** – hypoglycaemia, sepsis, hypoxia, head injury, dementia, environmental illness, medication interactions or side effects.
- **Psychotic disorders** – schizophrenia, bipolar, psychosis.
- **Mood disorders** – severe depression, social anxiety, PTSD, or specific phobias
- **Substance abuse** – psychostimulants, cocaine, ketamine, LSD, cannabis, alcohol, or withdrawal symptoms.
- **Developmental disorders** – such as autism, learning disabilities etc.
- **Situational** – grief, overwhelming stress.



CLINICAL PRESENTATION

Patients experiencing an ABD may present with the following:

- **Agitation:** The patient may exhibit restlessness, pacing, or an inability to sit still. They may be unable to focus or concentrate on their surroundings.
- **Aggression:** A patient with ABD may become verbally or physically aggressive towards themselves, others, or objects. This aggression can range from shouting and threatening behaviour to physical violence.
- **Confusion:** The patient might appear disoriented, with difficulty understanding time, place, or person. They may have trouble recalling recent events or understanding the current situation.
- **Paranoia:** Patients with ABD may express irrational fears or suspicions about others, believing they are being persecuted or harmed.
- **Hallucinations:** Some individuals may experience auditory or visual hallucinations, perceiving things that are not present. This can contribute to their agitation and distress.
- **Mood Swings:** ABD can lead to rapid and extreme mood changes, with the patient alternating between euphoria, anger, sadness, or fear.
- **Incoherent Speech:** The patient's speech may be disjointed, with rapidly changing topics and difficulty maintaining a coherent conversation.
- **Delusions:** Patients may hold false beliefs that are resistant to reason or contradictory to reality. These delusions can be bizarre or persecutory in nature.
- **Suicidal and/or homicidal:** A patient with ABD may potentially be suicidal or homicidal.
- **Impulsivity:** Individuals with ABD may act impulsively without considering the consequences of their actions, putting themselves or others at risk.
- **Sensory Sensitivity:** Some patients may exhibit heightened sensitivity to sensory stimuli, such as bright lights, loud noises, or tactile sensations, which can lead to increased distress.
- **Physical Restlessness:** Patients may demonstrate constant movement, such as pacing or fidgeting, and be unable to stay still.
- **Lack of Insight:** They may not recognize the severity of their condition or the need for medical or psychiatric intervention.

- **Reduced Inhibition:** ABD can result in lowered inhibitions, leading to socially inappropriate behaviours or a disregard for personal boundaries.
- **Disorganized Thoughts:** The patient's thought process may become disorganized, making it challenging to follow a logical conversation.
- **Altered Perception of Reality:** ABD can cause patients to perceive reality differently, leading to confusion and unpredictable behaviour



ASSESSMENT – RECOGNITION – TREATMENT CONSIDERATIONS

Assessment:

- Assessment of ABD involves a systematic evaluation of an individual's mental and emotional state, behaviour, and circumstances.
- Assess the patient's appearance, behaviour, affect, thought processes, thought content, perception, cognition, insight, and judgment.
- Pay attention to any signs of agitation, aggression, delusions, hallucinations, disorganized thinking, or self-harm/homicidal ideation.
- Assess for any potential underlying cause of the ABD, such as substance use, mental health conditions, trauma, or medical issues. Clues indicating an increased likelihood of an organic aetiology include:
 - >40 years old with first time presentation of psychosis or altered mental state
 - Disorientation/altered level of consciousness
 - Altered vital signs
 - Visual, tactile or olfactory hallucinations
 - Sudden onset
 - Fluctuating conscious state
 - Delirium
- Use the Sedation Assessment Tool (SAT) to determine the potential (not absolute) need for sedation.

Recognition:

- Recognition involves identifying and acknowledging the presence of ABD based on the assessment findings.
- This recognition helps determine the level of risk associated with the individual's behaviour and mental state.
- Recognizing ABD is critical in order to provide appropriate and timely interventions to ensure the safety and well-being of the patient and those around them.

Treatment:

- Treatment of ABD focuses on de-escalation and safety.
- Create a safe environment for the patient and the paramedic.
- Only approach the patient if it is safe to do so.
- Communicate in a gentle, reassuring and non-threatening tone.
- Intervention aims are to reduce agitation, aggression, and emotional distress.
- De-escalation techniques may involve creating a calm and safe environment, communication, and rapport-building, minimizing sensory triggers, and addressing any underlying medical or psychiatric issues.

- In cases where the patient is potentially suicidal or homicidal, immediate intervention and safety measures are essential. This may involve removing harmful objects or escalating to physical and/or chemical restraint.
- Treatment plans may also include symptomatic care and transportation to an appropriate facility.

De-escalation:

- Assess the patient and attempt to **de-escalate** the situation using the following strategies:

Do's

- Safety First: Prioritize your safety and the safety of those around you. Ensure there's a clear exit path if the situation escalates further and be aware of any potential risks.
- Stay Calm and Composed: Your demeanour can set the tone for the interaction. Maintain a calm and composed presence.
- Nominate one person to communicate directly with the patient.
- Maintain a calm and relaxed tone of voice: Speak in a gentle, steady, and soothing tone.
- Use open and non-threatening body language: Keep an open posture by standing or sitting with your arms relaxed and uncrossed.
- Give personal space: Respect the other person's personal space and boundaries.
- Listen actively: Show that you are actively listening to the other person by nodding your head and responding appropriately to their comments.
- Use Distraction Techniques: In some cases, the use of distraction techniques to redirect the patient's attention may be appropriate. Discuss neutral or calming topics to shift focus away from the source of distress.
- Ask Open-Ended Questions: Encourage the patient to share their perspective by asking open-ended questions.
- Simple script: instructions should be one at a time, use as few words as possible. Use repetitive simple statements.
- Match energy levels by responding appropriately and professionally to the situation.
- Empathy and understanding can help defuse a conflict/aggressive situation. Express empathy by acknowledging the other person's feelings and perspective.
- Focus on the issue at hand. Help the patient focus on how to solve their problem.

Don't

- Do not touch the patient.
- Don't make sudden movements: Sudden or aggressive movements can be perceived as a threat. Move slowly and purposefully.
- Do not stand over the patient and avoid pointing fingers, clenching fists, or any other aggressive gestures.
- Avoid raising your voice or speaking in an aggressive or confrontational manner.
- Don't argue or challenge: Refrain from engaging in arguments, debates, or power struggles. It's unproductive and can increase tension.
- Don't make jokes as they may be misunderstood.
- Avoid interrupting: Allow the other person to speak without interruption and then respond calmly and respectfully.
- Do not ignore the patient.
- Avoid getting too close, which may make the other person feel uncomfortable or threatened.

THE USE OF PHYSICAL OR CHEMICAL RESTRAINTS

The use of physical or chemical restraints should only be considered in specific situations when all other alternatives have been exhausted, and there is an imminent risk of harm to the patient or others. The decision to use restraints should follow a thorough assessment and adhere to ethical, legal, and clinical guidelines:

- Imminent Risk of Harm: When a patient poses an immediate and serious threat of harm to themselves or others, and all other de-escalation techniques have failed, physical and chemical restraints may be used to prevent injuries.
- Agitation Puts Medical Staff at Risk: If a patient's agitation and aggression put healthcare professionals at risk and there is no other means to ensure their safety, physical and chemical restraints may be considered.
- Request for CCP, Delta and Police backup.

Physically restrained patients:

- Use the least restrictive method of restraint necessary to ensure safety.
- Obtaining consent from both the patient's family and law enforcement if on scene.
- Physical restraint of patients should preferably be done in conjunction with the police.
- Patients restrained by police must always be accompanied by the police.
- Ensure that the patient is in a comfortable and safe position while restrained. This includes maintaining proper body alignment and providing padding where necessary to prevent injury.
- Position these patients laterally. Do not keep them in the prone position.
- Physical restraints should be removed as soon as safely possible or once chemically restrained.

Chemical restraints:

- Assess the patient using the Sedation Assessment Tool (SAT) to determine the possible need for sedation.
- Request for CCP assistance.

SEDATION ASSESSMENT TOOL

SCORE	RESPONSIVENESS	SPEECH
+3	Combative, violent, out of control	Continual loud outbursts
+2	Very anxious and restless	Loud outbursts
+1	Anxious and restless	Normal
0	Responds easily to name, speaks in normal tone	Normal
-1	Responds only if name called loudly	Slurring or prominent slowing
-2	Responds only to physical stimulation	Few recognizable words
-3	No response	Nil

- Use the patient's responsiveness and speech to determine their SAT score.
- Allocate the appropriate SAT score (+3 to -3) by determining the highest-ranking value (E.g. a patient displaying very anxious and restless behaviour but speaks normally will receive a score of +2).
- A SAT score of +2 or +3 is a good predictor of the need for sedation.

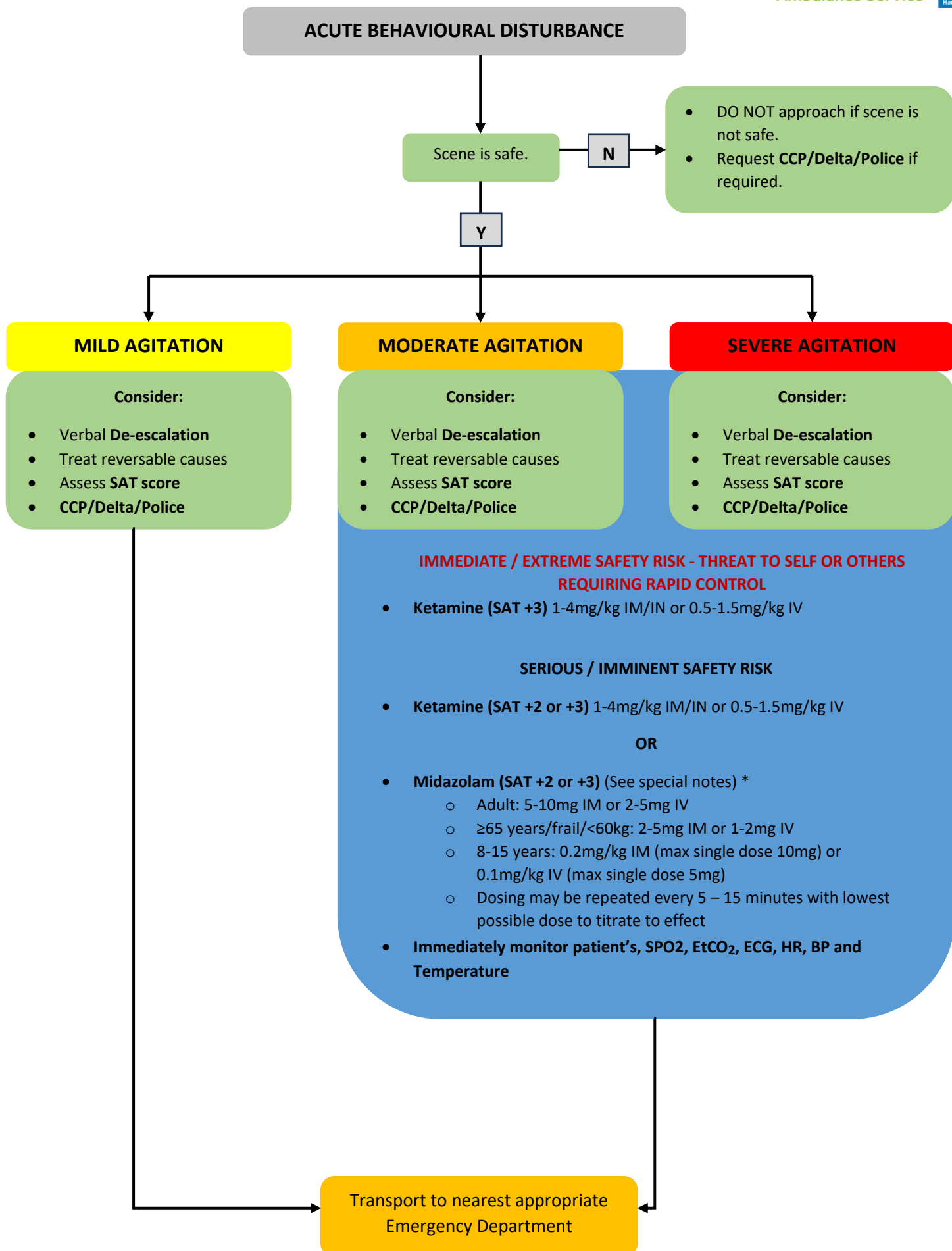
SEVERITY OF AGITATION

Assessing the severity of a patient's agitation will help to determine the degree of intervention required. Agitation assessment should be an ongoing process as the patients' condition is dynamic and the level of risk may escalate or de-escalate during the encounter requiring an appropriate response in care and intervention.

MILD (SAT +1)	MODERATE (SAT +2)	SEVERE (SAT +3)
- Agitated but cooperative - Not aggressive - Anxious, pacing, restless	- Loud outburst - Frequent non-purposeful movements - Not aggressive or violent	- Combative, violent - Danger to patient and/or staff - Non-cooperative - Yelling, loud outbursts, verbally abusive

SPECIAL NOTES

- Do not approach the scene if it not safe to do so. Request CCP/Delta and Police backup.
- A SAT score must be recorded on the ePCR for all ABD patients, including pre and post sedation.
- Verbal de-escalation must be attempted for all ABD patients, even those that are physically restrained.
- Attempt to identify and address any reversible causes.
- Ketamine is the primary drug of choice for all patients with moderate to severe agitation (SAT $\geq$ +2) with ABD unless contraindicated.
- *Midazolam may be used for sedation of the ABD patient in cases where:
 - A Benzodiazepine is better suited (e.g. Cocaine and other sympathomimetic ODs).
 - Ketamine is contraindicated
- Low dose Midazolam may be administered to patients with signs of Ketamine emergence that are unresponsive to non-pharmacological interventions. Prophylactic Midazolam is not to be administered.
- Close monitoring with the full monitoring bundle including temperature is mandatory for all patients following sedation. Monitoring is to be applied immediately after sedation if not already connected.
- Prepare for resuscitation – the patients are at a high-risk for requiring resuscitation. Have all relevant equipment prepared prior to sedation. This includes:
 - BVMR and mask setup, connected to oxygen and ready for use
 - Full monitoring bundle prepared and ready for patient application
 - IV prepared and ready for insertion
- Physical restraint of patients should preferably be done in conjunction with the police.
- IV sedation should only be considered if an IV is already in place prior to the need for sedation. Do not attempt to establish an IV in the agitated patient for the purpose of sedation administration.





CPG 8.1-8.3

TOXICOLOGY AND TOXINOCLOGY



CPG 8.1 BENZODIAZEPINE

Benzodiazepines are a sedative agent that potentiate the effects of the inhibitory neurotransmitter GABA within the CNS. Benzodiazepines are a commonly used medication for self-poisoning and are also frequently abused for recreational purposes.

CLINICAL PRESENTATION

Patients experiencing a benzodiazepine overdose may present with the following:

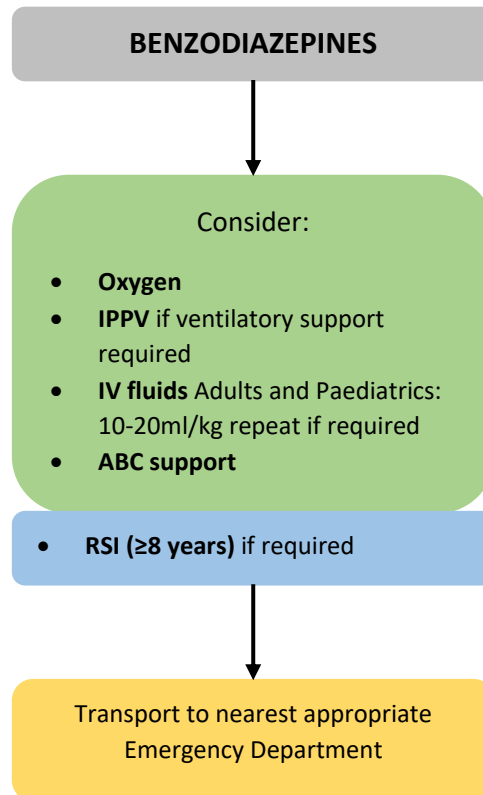
- Ataxia
- Drowsiness
- Slurred speech
- Decreased LOC
- Hypotension
- Bradycardia
- Hypothermia
- Toxicity may be worsened with co-ingestion of CNS depressants (such as alcohol)

ASSESSMENT CONSIDERATIONS

- These patients respond well to supportive care in the presence of isolated benzodiazepine OD.
- The elderly, patients with comorbidities, and patients who have co-ingested other CNS depressants with benzodiazepines are at greater risk of adverse events such as aspiration, airway obstruction and respiratory depression.
- Patients who are profoundly unconscious suggests that there is a high likelihood of co-ingestion with a CNS depressant such as alcohol.
- The unconscious patient may be at risk of aspiration. Continuously assess and monitor the patient. Keep in the lateral position if not intubated to protect the airway.

SPECIAL NOTES

- Any packaging (e.g. boxes of tablets, pill bottles, poison container) of the suspected substance used to overdose/poison, should be taken with the patient to hospital to assist in clearly identifying and managing the patient, only if safe to do so and there is no risk of contamination.
- Cardiac arrest due to overdose/poisoning should have any reversible causes identified and appropriately managed prior to ceasing resuscitation if applicable.
- Intentional poisoning may be associated with an acute behavioural disturbance. If ABD is present, manage appropriately according to ABD CPG 7.1.
- Should RSI be required, consider omitting Fentanyl for induction



CPG 8.2 BETA BLOCKER

Beta blockers are commonly prescribed to patients for cardiac arrhythmias, hypertension and to patients who have suffered a myocardial infarction. Beta blocker toxicity may be accidental or intentional. Significant negative effects such as bradycardia are associated with increased mortality.

CLINICAL PRESENTATION

Patients experiencing a beta blocker overdose may present with the following:

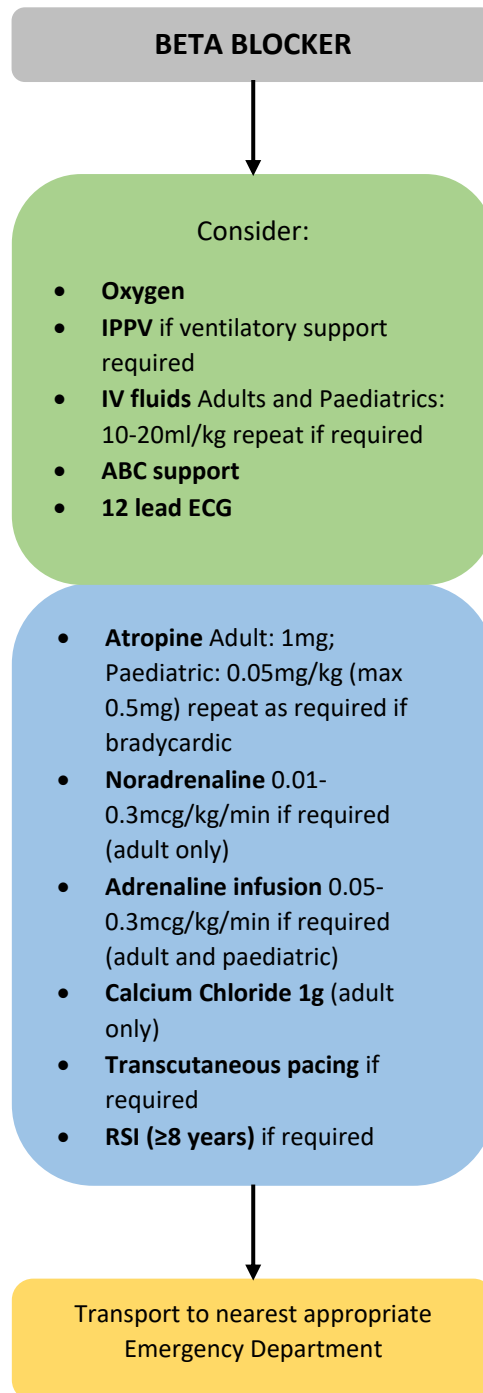
- Bradycardia
- Heart block
- Hypotension
- Cardiogenic shock
- Hypo/hyperglycaemia
- Pulmonary oedema
- Seizures
- Reduced LOC

ASSESSMENT CONSIDERATIONS

- Treatment focuses on supportive care and attempting to accurately identify the type of beta blocker ingested and if any other medicines were co-ingested.
- The elderly, patients with comorbidities, and patients who have co-ingested other cardiac drugs with beta blockers are at greater risk of the effects being compounded.
- Attempt to determine the time of ingestion. Clinical features generally develop within 4 hours after ingestion, depending on the dose.

SPECIAL NOTES

- Any packaging (e.g. boxes of tablets, pill bottles, poison container) of the suspected substance used to overdose/poison, should be taken with the patient to hospital to assist in clearly identifying and managing the patient, only if safe to do so and there is no risk of contamination.
- Cardiac arrest due to overdose/poisoning should have any reversible causes identified and appropriately managed prior to ceasing resuscitation if applicable.
- Intentional poisoning may be associated with an acute behavioural disturbance. If ABD is present, manage appropriately according to ABD CPG 7.1.
- Should RSI be required, consider omitting Fentanyl for induction.
- Propranolol and Sotalol are particularly toxic following an overdose due to their actions on other channels. Be prepared to manage any arrhythmias and cardiac arrest.
- Glucagon is no longer recommended as large doses are required and it only has a short-lived effect.
- Vasopressor and/or inotropic support may be required if unresponsive to other treatment.
- Calcium chloride may transiently improve blood pressure and may be considered if other treatments are ineffective.



CPG 8.3 CALCIUM CHANNEL BLOCKER

Calcium channel blockers are commonly prescribed to patients for cardiac arrhythmias, hypertension and to patients suffering from angina. Calcium channel blocker overdose can result in severe cardiovascular toxicity.

CLINICAL PRESENTATION

Patients experiencing a calcium channel blocker overdose may present with the following:

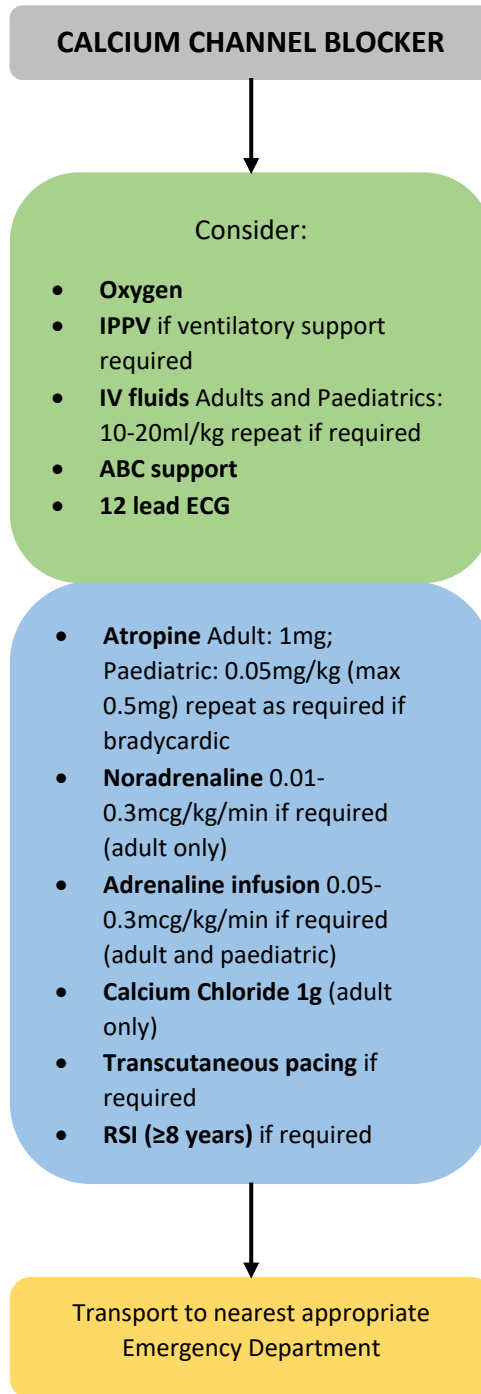
- Bradycardia
- Heart block
- Hypotension
- Cardiogenic shock
- Seizures
- Reduced LOC
- Hyperglycaemia
- Metabolic acidosis

ASSESSMENT CONSIDERATIONS

- The elderly, patients with comorbidities, particularly cardiovascular disease, and patients who have co-ingested other cardiac drugs with calcium channel blockers are at greater risk of the effects being compounded.
- Calcium channel blocker OD can be life-threatening. Verapamil and diltiazem OD are of particular concern due to them being more cardio selective.
- Many forms of calcium channel blockers are in slow-release preparations. Therefore, there may be a significant delay in onset of toxicity following ingestion.

SPECIAL NOTES

- Any packaging (e.g. boxes of tablets, pill bottles, poison container) of the suspected substance used to overdose/poison, should be taken with the patient to hospital to assist in clearly identifying and managing the patient, only if safe to do so and there is no risk of contamination.
- Cardiac arrest due to overdose/poisoning should have any reversible causes identified and appropriately managed prior to ceasing resuscitation if applicable.
- Intentional poisoning may be associated with an acute behavioural disturbance. If ABD is present, manage appropriately according to ABD CPG 7.1.
- Should RSI be required, consider omitting Fentanyl for induction.
- Vasopressor and/or inotropic support may be required if unresponsive to other treatment.
- Calcium chloride may transiently improve blood pressure and may be considered if other treatments are ineffective.



CPG 8.4 CARBON MONOXIDE

Carbon monoxide is a colourless and odourless gas and has 240 times the affinity for haemoglobin compared to oxygen. Common sources of carbon monoxide poisonings include fires, car exhaust fumes and the use of petrol-powered machinery in an enclosed space.

CLINICAL PRESENTATION

Patients experiencing a carbon monoxide poisoning may present with the following:

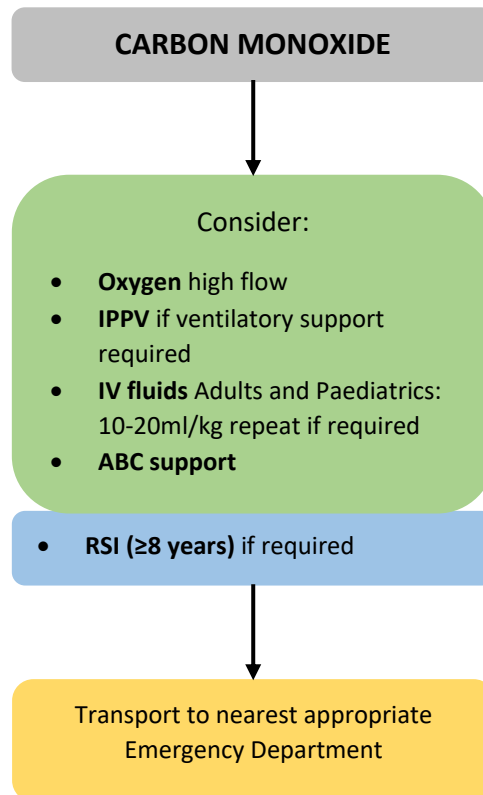
- Headache
- Nausea
- Dizziness
- Tachycardia
- Confusion
- Ataxia
- Syncope
- Seizures
- Altered LOC
- Myocardial ischemia
- Pulmonary oedema

ASSESSMENT CONSIDERATIONS

- The classic cherry red complexion is very rarely seen.
- Pregnant patients and those with cardiovascular disease are at increased risk in the presence of carbon monoxide poisoning.
- SpO2 may display normal saturation % in the presence of carbon monoxide poisoning. If carbon monoxide poisoning is suspected administer high flow oxygen regardless of severity of toxicity.
- When attending a carbon monoxide incident, maintain a high index of suspicion that carbon monoxide is present at the scene. Do not enter the scene until it is safe to do so.
- Cyanide toxicity should also be considered and assessed for when attending to a patient with carbon monoxide toxicity due to a fire.

SPECIAL NOTES

- Intentional poisoning may be associated with an acute behavioural disturbance. If ABD is present, manage appropriately according to ABD CPG 7.1.



CPG 8.5 CYANIDE

Cyanide toxicity can be lethal but is not commonly encountered. Cyanide exposure is most commonly through inhalation of cyanide containing gas or particles, absorption through the skin or deliberate ingestion. The most common exposure is due to smoke inhalation from a fire in an enclosed space.

CLINICAL PRESENTATION

Patients experiencing a cyanide poisoning may present with the following:

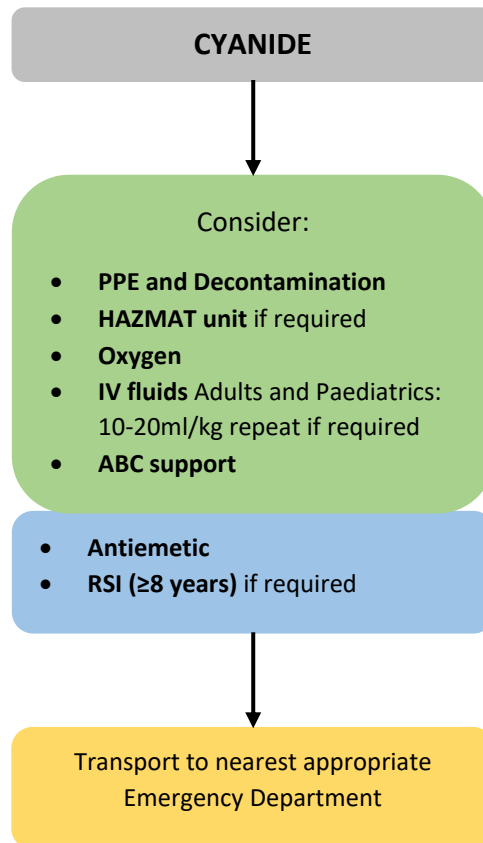
- Anxiety
- Headache
- Nausea and vomiting
- Tachycardia
- Hypertension
- Confusion
- Dyspnoea
- Bradycardia
- Seizures
- Altered LOC
- Pulmonary oedema
- Cardiac arrest

ASSESSMENT CONSIDERATIONS

- The classic cherry red complexion is very rarely seen.
- When attending a cyanide incident, maintain a high index of suspicion that cyanide is present at the scene. Do not enter the scene until it is safe to do so.
- Cyanide toxicity should also be considered and assessed for when attending to a patient with carbon monoxide toxicity due to a fire.
- Onset of clinical features is rapid particularly following inhalation. Symptoms generally appear within 1 hour following ingestion.
- Death will often occur prior to arrival or prior to the opportunity for resuscitation.
- Vomiting following ingestion may emit hydrogen cyanide gas. An antiemetic should be considered if vomiting is present.
- The breath of people exposed to cyanide may have a “bitter almond” odour. However, you must not rely on this sign to rule cyanide toxicity in or out.

SPECIAL NOTES

- Intentional poisoning may be associated with an acute behavioural disturbance. If ABD is present, manage appropriately according to ABD CPG 7.1.
- PPE must be worn to avoid self-exposure.
- Decontamination should occur following dermal exposure. Remove clothing and wash skin with water and soap (if available). Request for HAZMAT unit.
- These patients are at high risk for cardiac arrest.



CPG 8.6 OPIOIDS

Opioids are widely available in the form of analgesic medicines as well as occurring in illicit forms. Both the legal and illegal forms of opioids are abused for their euphoric effects. Tolerance and dependence are commonly seen.

CLINICAL PRESENTATION

Patients experiencing an opioid overdose may present with the following:

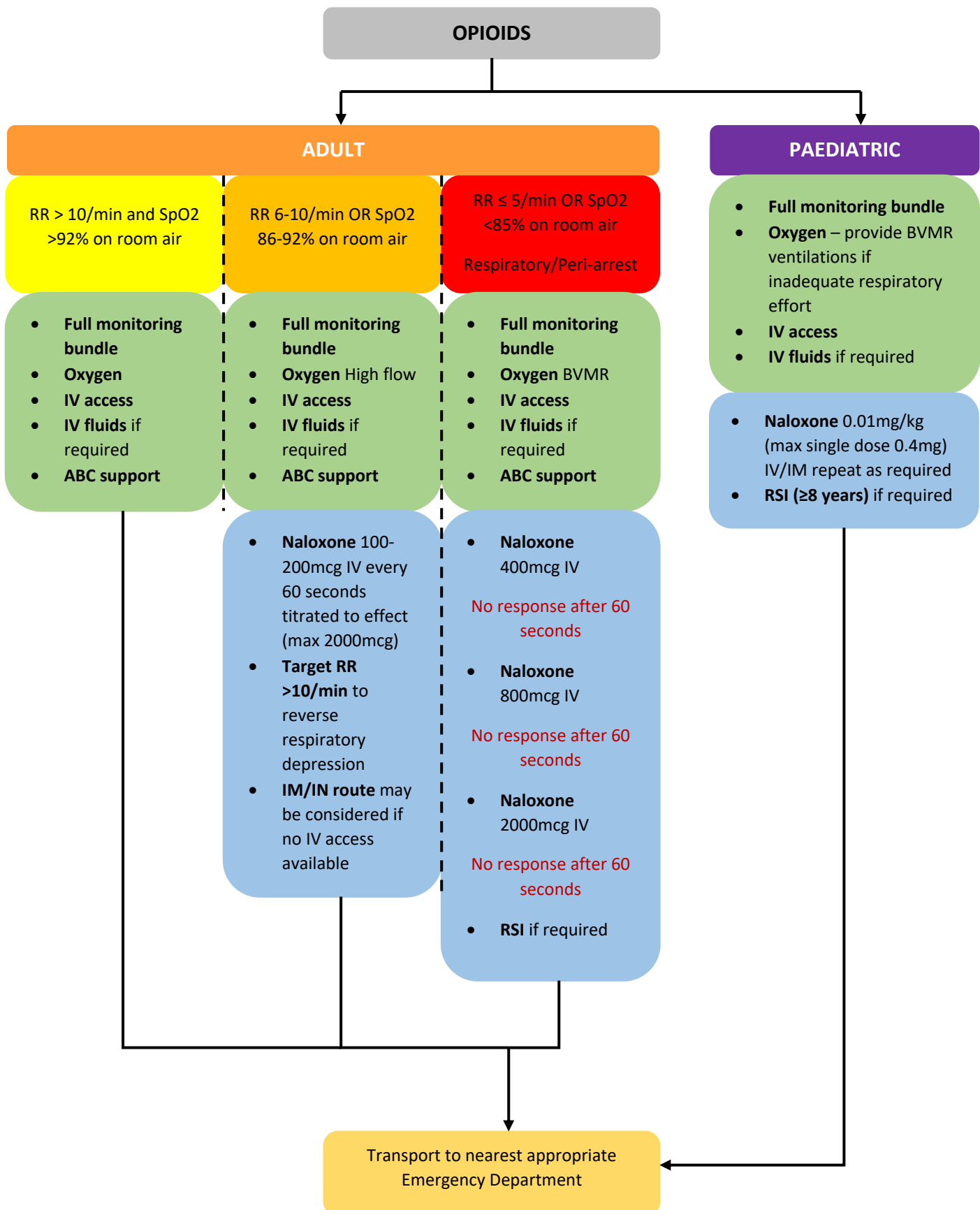
- Miosis
- Drowsiness/sedation/reduced LOC
- Respiratory depression
- Nausea and vomiting
- Hypotension
- Bradycardia
- Convulsions
- Hypothermia due to environmental exposure

ASSESSMENT CONSIDERATIONS

- Assess for respiratory depression and airway obstruction which can be fatal, especially in children and the opiate naïve.
- These patients often only require supportive care, such as ventilation, and naloxone is only required for respiratory depression.
- Naloxone has a relatively short half-life. Monitor the patient for rebound effects of opioid toxicity.

SPECIAL NOTES

- Intentional poisoning may be associated with an acute behavioural disturbance. If ABD is present, manage appropriately according to ABD CPG 7.1.
- Any packaging (e.g. boxes of tablets, pill bottles, poison container) of the suspected substance used to overdose/poison, should be taken with the patient to hospital to assist in clearly identifying and managing the patient, only if safe to do so and there is no risk of contamination.
- The aim of Naloxone administration is to reverse respiratory compromise and return the patients breathing to an acceptable rate, particularly for those that have an opioid dependency. Patients may become combative if the effects of the opioid are fully reversed.
- Suspected opioid toxicity patients require the full monitoring bundle – 4-lead ECG, SpO₂, RR, BP and EtCO₂. Temperature and RBS must also be assessed.
- Alternative diagnosis should be considered if there is no response to Naloxone. Consider RSI if respiratory depression/compromise continues despite Naloxone administration.



CPG 8.7 ORGANOPHOSPHATES

Organophosphates are pesticides that inhibit acetylcholinesterase, resulting in an increased action of acetylcholine, potentially leading to a cholinergic syndrome that can be fatal. Carbamates produce similar effects to organophosphates but are generally less severe.

CLINICAL PRESENTATION

Patients experiencing an organophosphate poisoning may present with the following (DUMBBELS):

- Diarrhoea
- Urination
- Miosis
- Bronchorrhea/bronchospasm
- Bradycardia
- Emesis
- Lacrimation
- Salivation
- Hypotension (this is a late sign and suggests severe toxicity)

Central effects may include:

- Agitation
- Seizures
- Coma

Nicotinic excess effects may include:

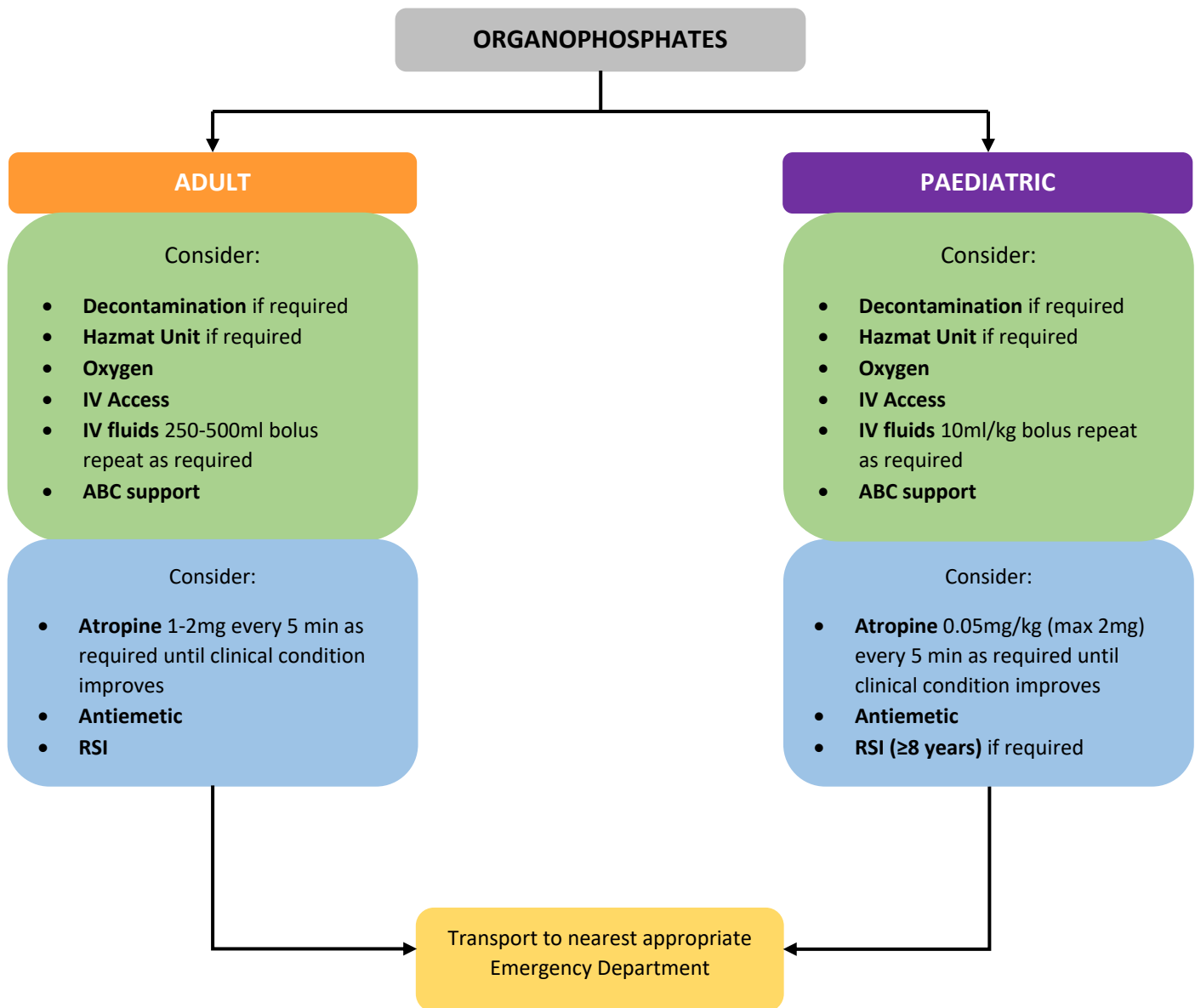
- Fasciculations
- Tremors
- Muscle weakness
- Respiratory muscle paralysis

ASSESSMENT CONSIDERATIONS

- Use appropriate PPE when managing these patients. Standard precautions are usually sufficient.
- Patients with severe/life-threatening toxicity may require large doses of atropine. Atropine is used to block the muscarine effects of acetylcholine.
- Cholinergic syndrome may develop due to nicotinic agents (nicotine patches, tobacco products) and muscarinic agents (mushrooms).
- Note that clinical features may present late – up to 12 hours with some agents.

SPECIAL NOTES

- Intentional poisoning may be associated with an acute behavioural disturbance. If ABD is present, manage appropriately according to ABD CPG 7.1.
- Any packaging (e.g. boxes of tablets, pill bottles, poison container) of the suspected substance used to overdose/poison, should be taken with the patient to hospital to assist in clearly identifying and managing the patient, only if safe to do so and there is no risk of contamination.



CPG 8.8 PSYCHOSTIMULANTS

Psychostimulants act on the sympathetic nervous system. They can be obtained in both the legal and illegal form and can be taken orally, intravenously or inhaled. Common pharmaceutical psychostimulants include caffeine, ephedrine and pseudoephedrine. Illicit psychostimulants include cocaine, methylenedioxymethamphetamine (MDMA), amphetamine and methamphetamine.

CLINICAL PRESENTATION

Patients experiencing a psychostimulant emergency may present with symptoms ranging from mild, moderate to severe:

MILD

- Euphoria
- Restlessness
- Dilated pupils
- Tachycardiac
- Bruxism (teeth grindings seen with MDMA use)

MODERATE

- Paranoia
- Psychomotor agitation
- Diaphoresis
- Hypertension

SEVERE

- Acute behavioural disturbance
- Hyperthermia
- Seizures
- Refractory tachyarrhythmias
- Severe respiratory distress
- Altered LOC
- Severe patients are also at risk of or may present with:
 - Myocardial infarction
 - Intracranial bleed
 - Renal failure
 - Hypertensive crisis

ASSESSMENT CONSIDERATIONS

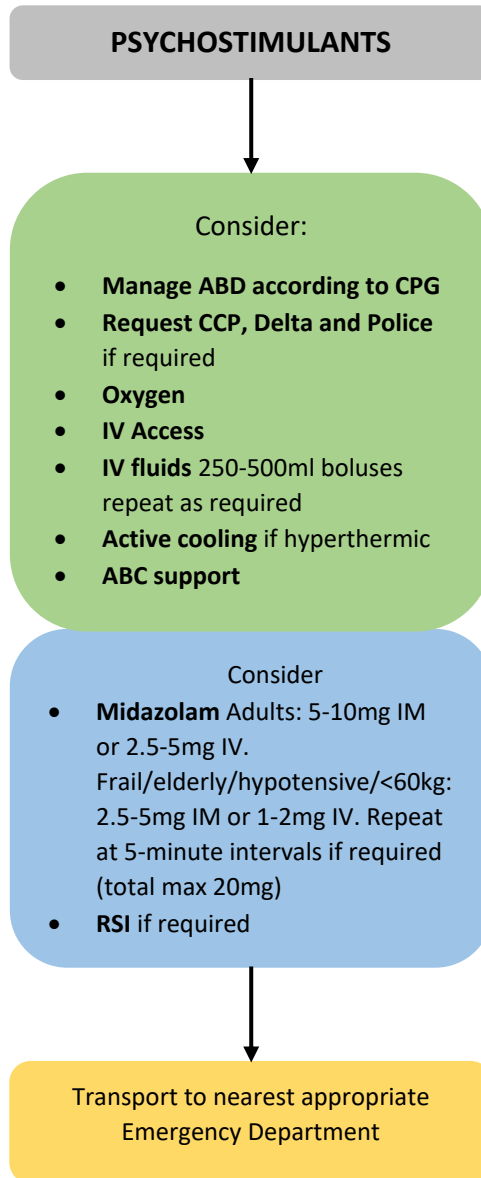
- These patients are at high risk of deterioration and should be monitored closely. Be prepared for resuscitation.
- Management primarily focuses on supportive care.
- Patients may present with an acute behavioural disturbance making assessment challenging. Methamphetamine derivatives are the most common life-threatening psychostimulant emergencies – these patients may present with severe ABD and abnormal vital signs.
- Assess for hyperthermia and perform active cooling if the patient is hyperthermic. These patients may present with extreme hypothermia – active cooling must be aggressive and a priority.

- Consider IV access in all patients presenting with a suspected psychostimulant emergency if/when safe to do so, regardless of severity.
- Reduce external stimulus by creating a calm environment for the patient. Assess for the most suitable place to achieve this, such as in the ambulance or a quiet room.



SPECIAL NOTES

- Psychostimulant emergencies may often present with ABD. Manage ABD as per the ABD CPG.
- These patients are at high risk of deterioration and cardiac arrest post ABD sedation. Be prepared for resuscitation. This includes having the relevant equipment prepared.
- Midazolam is the preferred sedation agent for the patient with a moderate to severe psychostimulant emergency (if able to assess appropriately to identify a psychostimulant emergency). The purpose of Midazolam administration is to reduce physical activity due to agitation, muscle tremors and clonus, which all contribute to a dangerous increase in body temperature as well as managing ABD. Use low dose Midazolam.
- If physically restrained, keep this time to an absolute minimum.
- Use caution when administering fluids if the patient is at risk of fluid overload such as in the elderly or those with cardiac failure.



CPG 8.9 TRICYCLIC ANTIDEPRESSANTS

Tricyclic antidepressants (TCAs) action is mediated by serotonin and norepinephrine reuptake inhibition. TCAs can also inhibit action at the muscarinic, histamine and adrenergic receptors. Overdose can be lethal.

CLINICAL PRESENTATION

Patients experiencing Tricyclic antidepressant toxicity may present with the following:

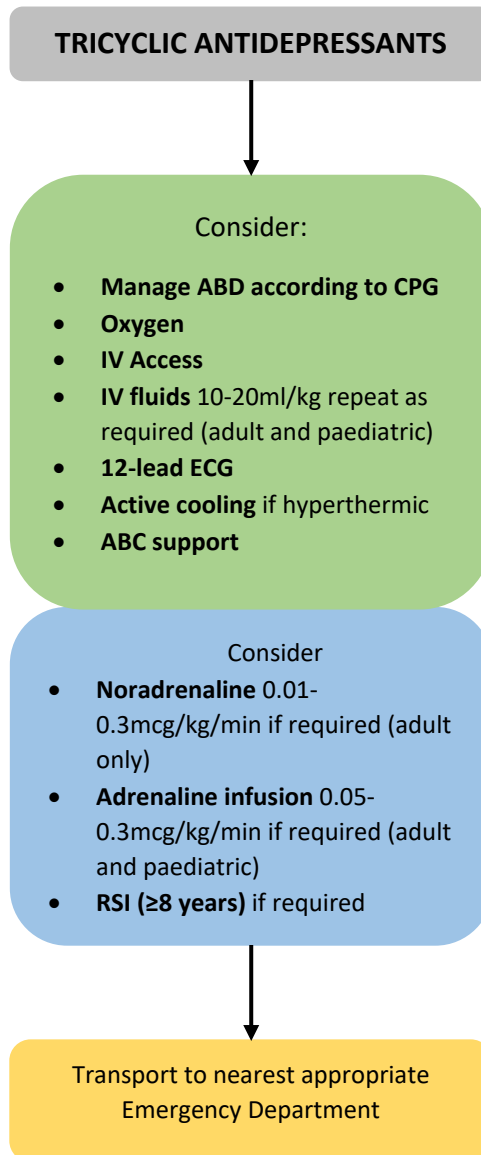
- Agitation/delirium
- Dilated pupils
- Dry, warm, flushed skin
- Hyperthermia
- Tachycardia
- Urinary retention
- Altered LOC/coma
- Seizures
- Hypotension
- ECG changes
 - Prolonged PR and QRS interval
 - Dominant R wave in aVR
 - Ventricular tachycardia

ASSESSMENT CONSIDERATIONS

- Determine the quantity of TCAs ingested to determine the potential or expected severity of toxicity. 10mg/kg in adults and 5mg/kg in children is potentially toxic.
- Rapid deterioration may occur in large overdoses.
- Monitor these patients closely with the full monitoring bundle. Be prepared to manage cardiac arrest.

SPECIAL NOTES

- Any packaging (e.g. boxes of tablets, pill bottles, poison container) of the suspected substance used to overdose/poison, should be taken with the patient to hospital to assist in clearly identifying and managing the patient, only if safe to do so and there is no risk of contamination.
- Manage ABD according to ABD CPG.



CPG 8.10 ALCOHOL INTOXICATION

Alcohol is a CNS depressant with the onset of its effects occurring within minutes. Excessive alcohol consumption can lead to alcohol poisoning. Intoxication may also result in physical injury due to falls as well as acute behavioural disturbances and aggression. Prolonged alcohol abuse may result in chronic medical conditions such as liver disease, cancer and cardiovascular disease, amongst others.

CLINICAL PRESENTATION

MILD ALCOHOL INTOXICATION

Patients with mild alcohol intoxication may experience the following:

- Feelings of wellbeing and relaxation
- Talkative
- Flushing
- Slight reduction in attention, memory and judgement

MODERATE ALCOHOL INTOXICATION

Patients with moderate alcohol intoxication may experience the following:

- Ataxia/loss of balance and/or co-ordination
- Reduction in attention, memory and judgement
- Confusion
- Slurred speech
- Drowsiness or sleepiness
- Dizziness
- Dehydration
- Loss of fine motor skills
- Variations of mood, including risk of aggression and/or ABD

SEVERE ALCOHOL INTOXICATION

Patients with severe alcohol intoxication may experience the following:

- Ataxia
- Severe speech impairment
- Significant reduction of attention, memory and co-ordination
- Poor judgement and decision-making ability
- Vomiting
- Potential for loss of consciousness

ALCOHOL POISONING

Patients with mild alcohol poisoning may experience the following:

- Severe speech impairment
- Severe dizziness
- Loss of consciousness
- Respiratory depression
- Loss of gag reflex
- Bradycardia

- Seizures
- Hypothermia
- Hypoglycaemia



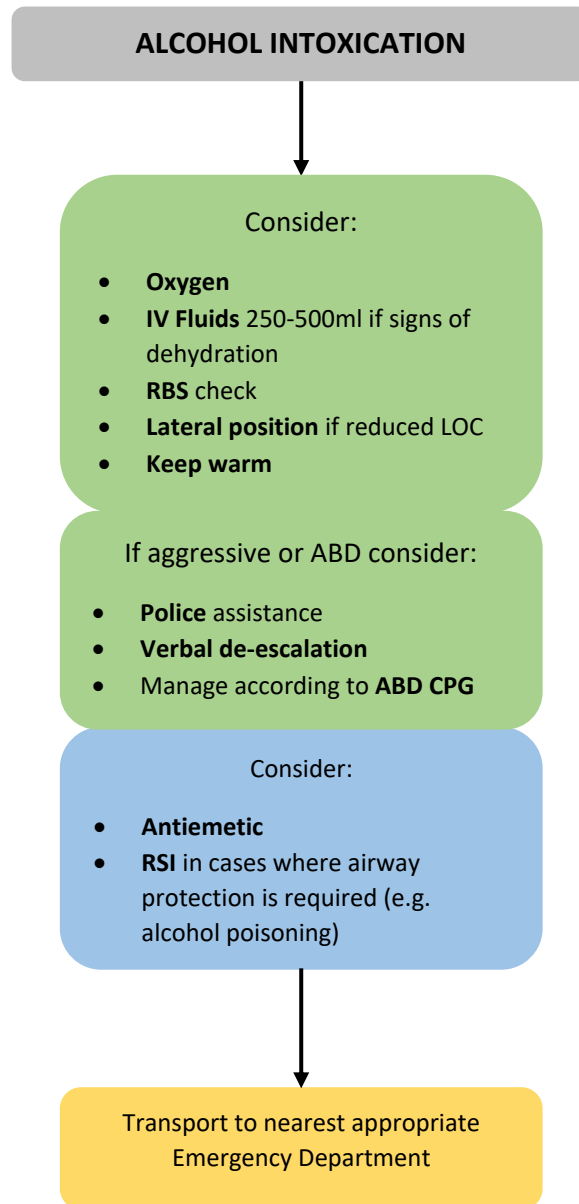
ASSESSMENT CONSIDERATIONS

- Signs of intoxication can mimic other conditions. Obtain a detailed clinical history and events leading up to the call for help in order to rule out other possible provisional diagnosis.
- Attempt to obtain how much alcohol the patient has consumed and if any other substances (medications or illicit substances) were co-ingested with the alcohol.
- Perform a detailed clinical assessment and look for signs of possible trauma.
- Perform an RBS on all suspected intoxicated patients.
- Keep the patient warm as they are at increased risk of hypothermia.



SPECIAL NOTES

- Patients with alcohol dependence will experience alcohol withdrawal if they cease to consume alcohol.
- Alcohol withdrawal syndrome is characterised by insomnia, restlessness, agitation, nausea, sweating, tremors and seizures.
- Alcohol withdrawal syndrome can be life-threatening.



CPG 8.11 ENVENOMATION

Qatar has a number of venomous creatures found both in the sea and on land. These include venomous snakes (horned viper and saw scaled viper), spiders, scorpions and marine animals such as the stonefish and blue bottle/jelly fish found along the coast. Identifying the type of animal that has stung or bitten the patient is important as this will help identify the type of venom (neuro, cyto or haemotoxic venom) and the required treatment.

- **Neurotoxic** – acts on the nervous system, including the brain.
- **Cytotoxic** – localised action at the site of envenomation.
- **Haemotoxic** – acts on the cardiovascular system, including blood clotting abilities.



CLINICAL PRESENTATION

SNAKE BITE

Patients with a snake bite envenomation may present with the following:

- Local pain, swelling and bruising
- Nausea and vomiting
- Abdominal pain
- Headache
- Dizziness
- Collapse
- Neurotoxic
 - Paraesthesia
 - Tremors
 - Seizures
 - Respiratory distress
- Coagulopathy abnormalities
 - Bleeding from bite site, gums, mucus membranes

SPIDER BITE

Patients with a spider bite envenomation may present with the following:

- Pain
- Muscle cramps
- Abdominal pain
- Nausea and vomiting
- Dizziness and syncope
- Respiratory distress

SCORPION STING

Patients with a scorpion sting envenomation may present with the following:

- Pain
- Agitation
- Paraesthesia
- Abdominal pain

- Nausea and vomiting
- Diaphoresis
- Respiratory distress
- Seizures
- Coma
- Cardiac arrhythmias

BLUE BOTTLE/JELLYFISH

Patients with a blue bottle/jellyfish sting envenomation may present with the following:

- Localised pain
- Skin irritation/hives
- Anaphylaxis in rare cases

STONE FISH

Patients with a stone fish envenomation may present with the following:

- Severe pain
- Localised swelling
- Anaphylaxis in rare cases



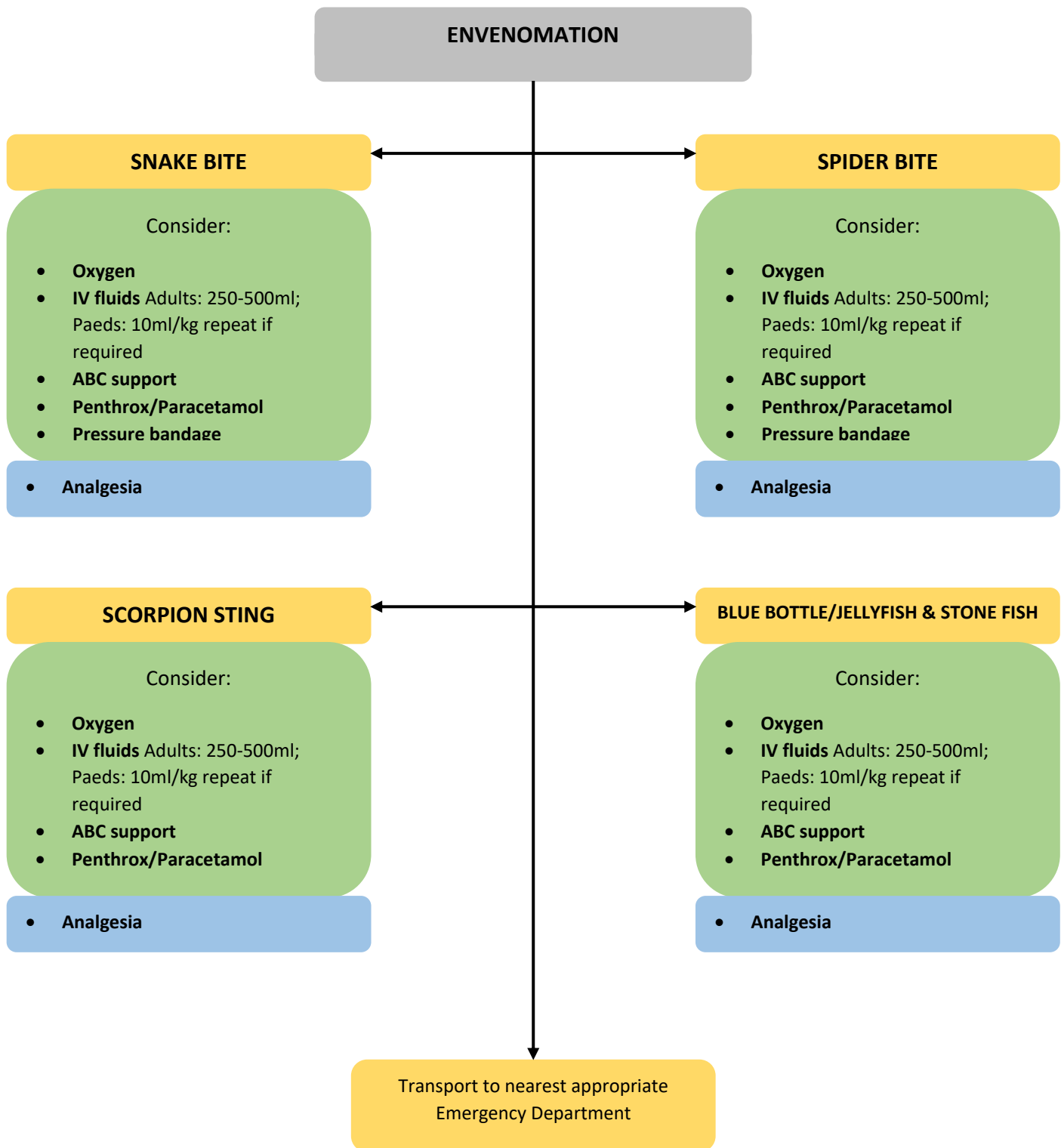
ASSESSMENT CONSIDERATIONS

- Monitor the patient closely for signs of neurotoxicity (respiratory and cardiovascular collapse).
- Limit the patient's movements to prevent the spread of the venom.
- Apply a pressure bandage to the limb from distal to proximal. Observe for excessive swelling.



SPECIAL NOTES

- Attempt to identify the creature that caused the envenomation. DO NOT attempt to catch the creature.
- Provide prenotification to hospital in order for the appropriate antivenom to be prepared.





CPG 9.1-9.2

ENVIRONMENTAL



CPG 9.1 HEAT RELATED DISORDERS

Heat illness results when exposed to high environmental temperatures and the body's thermoregulatory mechanisms are unable to cope. Heat illness can be classified into heat exhaustion or heat stroke. Differentiating between environmental hyperthermia and intrinsic hyperthermia (e.g. infection) is important as management of these patients will differ.

CLINICAL PRESENTATION

HEAT EXHAUSTION

Patients with heat exhaustion may present with the following:

- Temperature of between 37°C and 40°C
- Headache
- Dizziness
- Diaphoresis
- Nausea and vomiting
- Muscle pain/cramps
- Tachycardia, hypotension, tachypnoea

HEAT STROKE

Patients with heat stroke may present with the following:

- Temperature > 40°C
- Abnormal behaviour and/or altered LOC
- Seizures
- Extreme fatigue
- Headache and syncope
- Vomiting and diarrhoea
- Skin hot and dry
- Arrhythmias
- Tachycardia, hypotension, tachypnoea and respiratory distress
- Hypoglycaemia

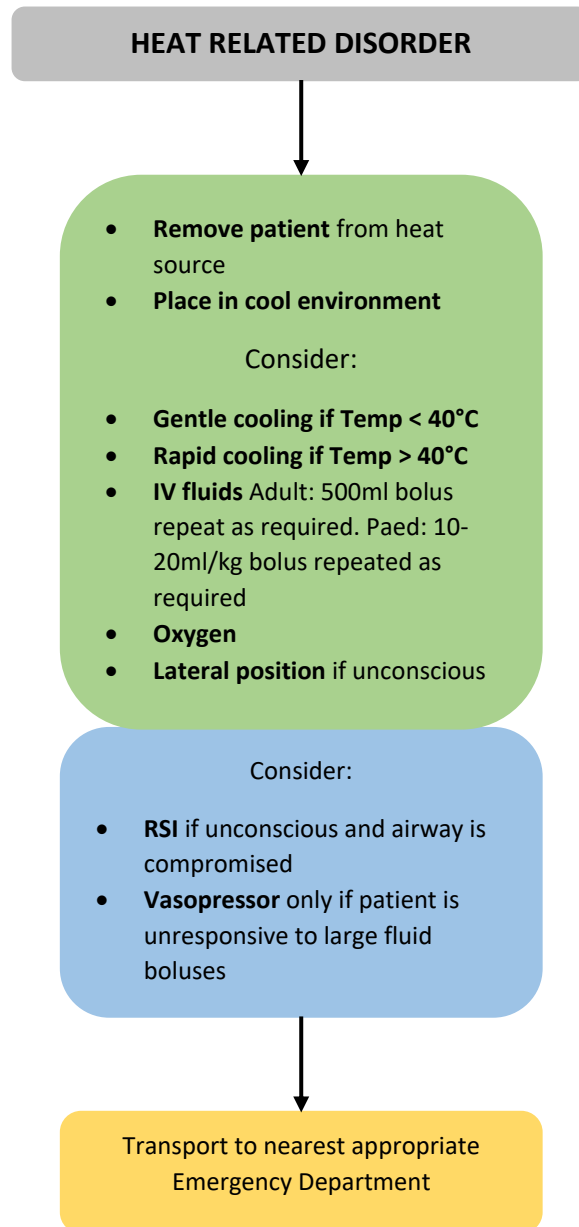
ASSESSMENT CONSIDERATIONS

- Avoid inducing hypothermia and shivering when cooling the patient. If the patient becomes cold, stop the cooling process. Shivering may result in an increase in core body temperature.
- Patients with co-morbidities, the elderly and children are at increased risk of heat illness.
- History taking and detailed assessment are important to identify the cause of hyperthermia as this will guide treatment.
- The "Strip-Spray-Fan-Ice" method is the most effective means of active cooling in the prehospital setting. This method involves removing all clothing, repeatedly spraying the skin with water, fanning the patient or directing A/C onto the patient and placing ice packs at the neck, axilla and groin. When removing the patients clothing, take the patients privacy into consideration.



SPECIAL NOTES

- Patients with heat stroke will require large volumes of IV fluids. Provide a minimum of 2L NaCl for adults and 20ml/kg for paediatrics and assess responsiveness.
- DO NOT administer Paracetamol or Ibuprofen to these patients.
- These patients may present with extreme sinus tachycardias.
- In extremis, vasopressors may be considered with ongoing fluid administration.



CPG 9.2 COLD RELATED DISORDERS

Hypothermia occurs when body heat is lost faster than the body is able to produce heat as the compensatory mechanisms fail. Hypothermia is defined as a core body temperature of $<35^{\circ}\text{C}$. It is important to note that hypothermia can occur at any time of year despite the hot climate in Qatar. All patients should be kept appropriately warm.

CLINICAL PRESENTATION

MILD HYPOTHERMIA

Patients with mild hypothermia may present with the following:

- Temperature of between 32°C and 35°C
- Lethargy
- Ataxia
- Tachycardia
- Tachypnoea
- Normotension

MODERATE HYPOTHERMIA

Patients with moderate hypothermia may present with the following:

- Temperature of between 28°C and 32°C
- Confusion
- Altered LOC
- Bradycardia
- Hypotension
- Muscle rigidity

SEVERE HYPOTHERMIA

Patients with severe hypothermia may present with the following:

- Temperature of $< 28^{\circ}\text{C}$
- Unconscious
- Apnoea
- Dilated pupils
- Absent reflexes
- Arrhythmias

Additional clinical features of the hypothermic patient may include:

- Hyper or hypoglycaemia
- ECG changes
 - VF
 - Bradyarrhythmias
 - Prolonged PR, QRS and QT intervals
 - ST elevation
 - Osborn J wave, T wave inversion



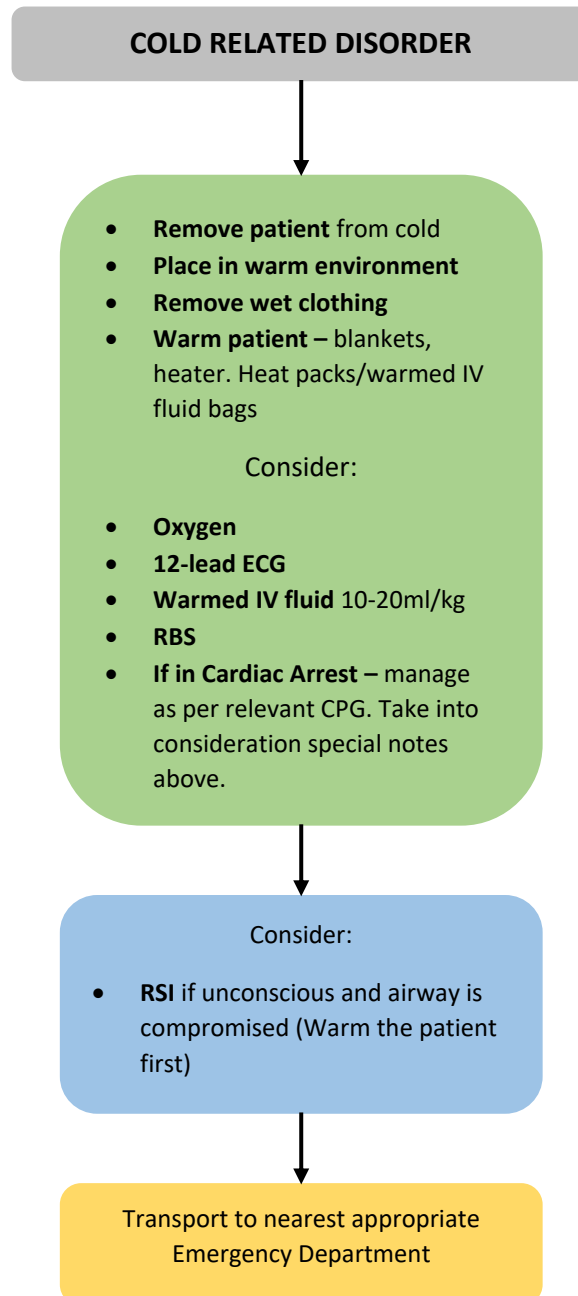
ASSESSMENT CONSIDERATIONS

- The hypothermic patient may have underlying medical conditions that require treatment and should be treated simultaneously (e.g. trauma, seizures, hypoglycaemia).
- A 12-lead ECG should be performed to identify any potential arrhythmias.
- Hypothermic patients should be moved with care as they are at increased risk of developing VF. Move patients into a warm environment.
- Do not submerge the severely hypothermic patient in a warm bath or shower as this may result in haemodynamic collapse.
- Sinus bradycardia as a result of hypothermia should not initially be managed. Warm the patient first and monitor to see if the bradycardia improves.
- Use blankets, heater, heat packs or warmed IV fluid bags (if possible) placed under the axilla, groin and neck to warm the patient. Ideally use warmed IV fluids if administering IV fluids.



SPECIAL NOTES

- Prolonged resuscitation and transport to hospital should be considered for hypothermic patients in cardiac arrest.
- Do not administer any cardiac arrest medications (adrenaline and amiodarone) to the hypothermic cardiac arrest patient until the patient's temperature is $> 30^{\circ}\text{C}$.
- Muscle rigidity due to hypothermia may make intubation difficult. Muscle relaxants do not work effectively if the patient's temperature is $< 30^{\circ}\text{C}$. Warm the patient first.





CPG 10.1-10.10

TRAUMA



CPG 10.1 MAJOR HAEMORRHAGE & HAEMORRHAGIC SHOCK

Rapid recognition and intervention are required when managing a patient with major haemorrhage and haemorrhagic shock. Major blood loss may not always be obvious. Non-compressible bleeds (e.g. intra-abdominal, pelvic bleeds) may be difficult to recognise. Clinicians must be aware of the clinical signs of haemorrhagic shock and take appropriate action.

CLINICAL PRESENTATION

Patients with major haemorrhage and haemorrhagic shock may present with the following, worsening in severity depending on the volume of blood loss:

- Pale, cool extremities
- Clammy
- Tachycardia
- Bradycardia
- Weak/absent peripheral pulses
- Delayed capillary refill time
- Thirst
- Hypotension
- Altered LOC

ASSESSMENT CONSIDERATIONS

- Attempt to identify the source of haemorrhage, whether external or internal and stop bleeding if possible.
- Control any external haemorrhage using compression, elevation and splinting if indicated. Apply a tourniquet to where indicated.
- Apply a pelvic binder to all unstable/suspected fractured pelvises.
- Regularly assess haemodynamic status.
- Assess for indications for TXA administration.
- Aim to maintain a MAP ≥ 65 mmHg for adults (this MAP target includes those patients with an associated TBI). A palpable radial pulse is a good indication that there is adequate perfusion. For paediatric patients maintain an appropriate systolic BP for age using the calculation $(\text{age} \times 2) + 70$. DO NOT attempt to restore normotension
- Assess temperature and maintain normothermia. Do not let the patient get cold.

SPECIAL NOTES

- These patients require rapid treatment and transport to hospital. Do not delay on scene waiting for a CCP. Transport P1 to the TRU and arrange an RV with the CCP if possible.
- Small fluid boluses may be administered if required and BP/MAP response assessed. The target is to administer little to no fluids if possible.
- Vasopressors should only be considered if MAP/BP is unresponsive to fluids. If vasopressors are required, use the smallest possible dose to maintain MAP just above 65mmHg. Avoid causing spikes in the BP.

MAJOR HAEMORRHAGE & HAEMORRHAGIC SHOCK

ADULT

Consider:

- **Control bleeding**
- **Oxygen**
- **IV fluid** 250ml boluses (Max 1L)
- **TXA** 2g
- **Maintain normothermia**
- **Analgesia**
- **Rapid transport to TRU**

Consider:

- **Noradrenaline** 0.01-0.3mcg/kg/min if required
- **Phenylephrine** 100mcg/min if required
- **RSI** if required
- **Adrenaline infusion** 0.05-0.3mcg/kg/min if unresponsive to other pressors
- **Analgesia**

PAEDIATRIC

Consider:

- **Control bleeding**
- **Oxygen**
- **IV fluid** 10-20ml/kg boluses (Max 2 boluses or 1L)
- **Maintain normothermia**
- **Analgesia**
- **Rapid transport to TRU**

Consider:

- **TXA** 15mg/kg (max 1g)
- **Adrenaline infusion** 0.05-0.3mcg/kg/min if required
- **RSI** if required (≥8 years only)
- **Analgesia**

Transport to nearest appropriate
Emergency Department

CPG 10.2 TRAUMATIC BRAIN INJURY

In the prehospital setting, the primary aim when managing the patient with a traumatic brain injury (TBI) is to reduce secondary brain injury as a result of hypoxia, hypercapnia and hypotension. Early recognition, intervention and transport are essential.

CLINICAL PRESENTATION

Patients with a TBI may present with the following:

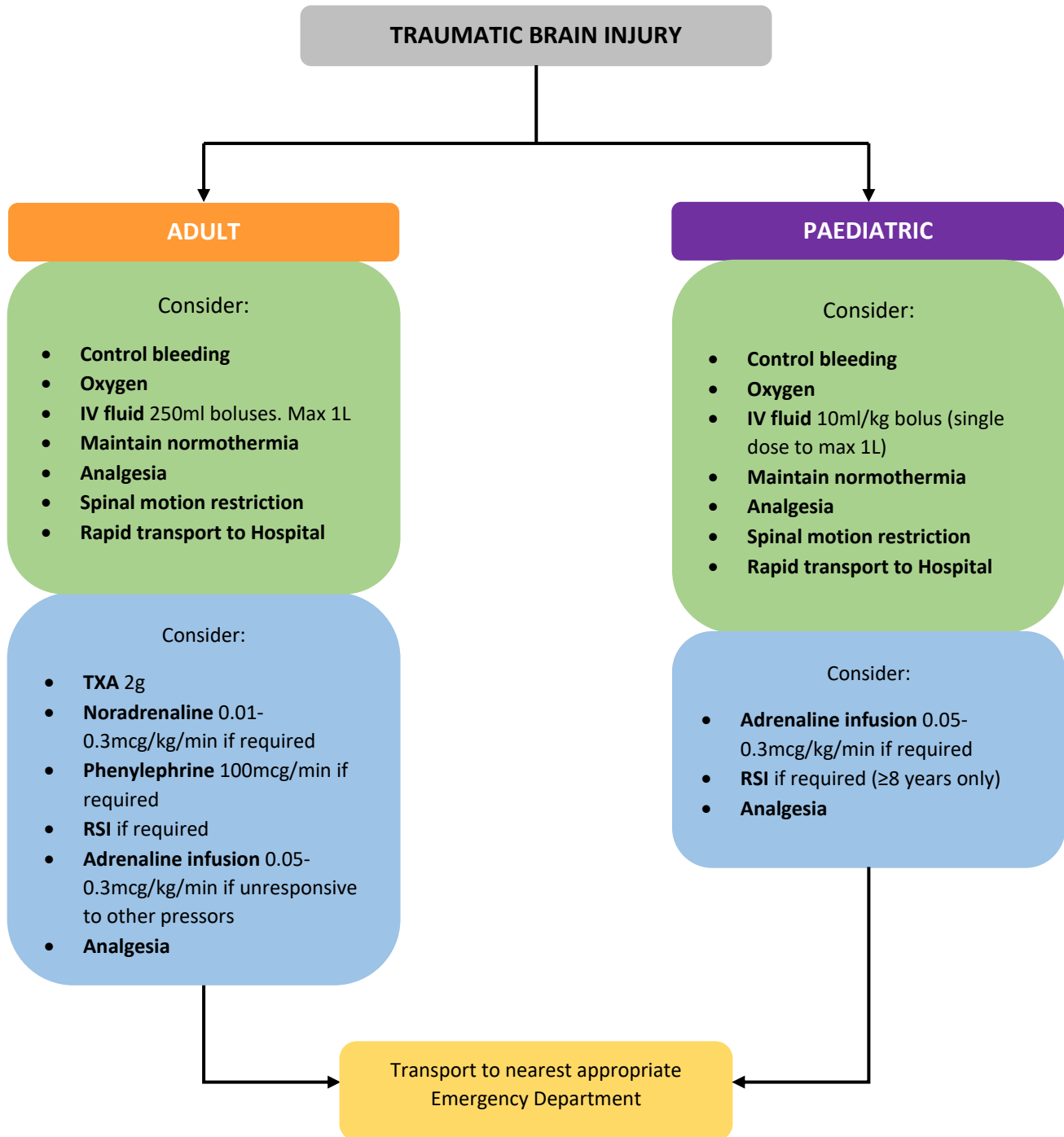
- Trauma to the head (e.g. haematoma, lacerations, abrasions)
- Headache
- Depressed/open skull fracture
- Blood from nose or ears (base of skull fracture)
- Altered LOC/reduced GCS
- Agitation
- Unilateral weakness
- Seizures
- Unequal/unreactive/dilated pupils
- Cushing's reflex (bradycardia, hypertension & irregular breathing)
- Hypotension

ASSESSMENT CONSIDERATIONS

- Assess the patient for other potential injuries.
- RSI should be considered for patients with a TBI, particularly if the patient remains hypoxaemic despite supplemental oxygen, unable to maintain their own airway or an expected worsening clinical course.
- Rule out hypoglycaemia as a potential cause of reduced GCS.
- Avoid hypo or hyperventilation in these patients as abnormal CO₂ levels affects cerebral perfusion.
- Ideally position the patient in the 30-degree head-up position if possible.
- Avoid constrictive devices (tape, collars, ties etc) around the neck. Loosen if present.
- Consider TXA within first 3 hours of injury GCS ≤ 12.
- Manage seizures as required.

SPECIAL NOTES

- Aim for a target MAP of 70-80mmHg for adults and age specific SBP for paediatrics using (age x 2) + 70 presenting with an isolated TBI.
- Adult patients with a TBI and associated polytrauma, aim for a target MAP of ≥65mmHg. For paediatric patients maintain age specific SBP using (age x 2) + 70.
- Avoid causing spikes or drops in BP.
- Use small fluid boluses and vasopressors to achieve target MAP if required.
- RSI should be considered for the patient presenting with a TBI and reduced LOC.



CPG 10.3 SPINAL TRAUMA

Prehospital management of spinal injuries should focus on preventing further injury by applying appropriate spinal motion restriction (SMR), maintaining appropriate haemodynamics and normothermia. The NEXUS criteria should be used to guide SMR decision making. The NEXUS criteria have been validated in both the adult and paediatric populations.

CLINICAL PRESENTATION

Patients with a spinal injury may present with the following:

- Paralysis
- Flaccidity
- Abnormal posturing
- Diaphragmatic respirations
- Local or generalised paraesthesia
- Loss of proprioception
- Hypotension
- Bradycardia
- Priapism
- Thermoregulation

ASSESSMENT CONSIDERATIONS

- All trauma patients with a high mechanism of injury or a high index of suspicion for a spinal injury should be assessed against the NEXUS criteria.
- Use caution when applying the NEXUS criteria to the elderly, those that are intoxicated, and those that are not able to fully understand your questioning. Clinical judgement should be used even if the answer is no to all the NEXUS criteria.
- Patients with spinal injuries are at high risk of other injuries, TBI, haemorrhage and shock.
- Assess and monitor for neurogenic or spinal shock.

NEXUS CRITERIA – SMR DECISION MAKING TREE

1. Neck pain/tenderness or anatomical deformity? **[YES/NO]**
2. Altered mental status GCS <15? **[YES/NO]**
3. Signs of alcohol/drug intoxication? **[YES/NO]**
4. Painful distracting injury present? **[YES/NO]**
5. Focal neurological deficit present? **[YES/NO]**

If the answer is YES to one or more of these criteria, SMR should be applied. Should the answer be NO to ALL of these criteria, SMR may not be required.

If the patient is ambulatory, allow the patient to slowly move themselves onto the stretcher and then apply SMR. The combi-carrier should be placed on the stretcher and the stretcher moved as close to the patient as possible to minimise patient movement.

These criteria cannot be assessed in patients with a language barrier or in children unable to understand. If in doubt, apply SMR.



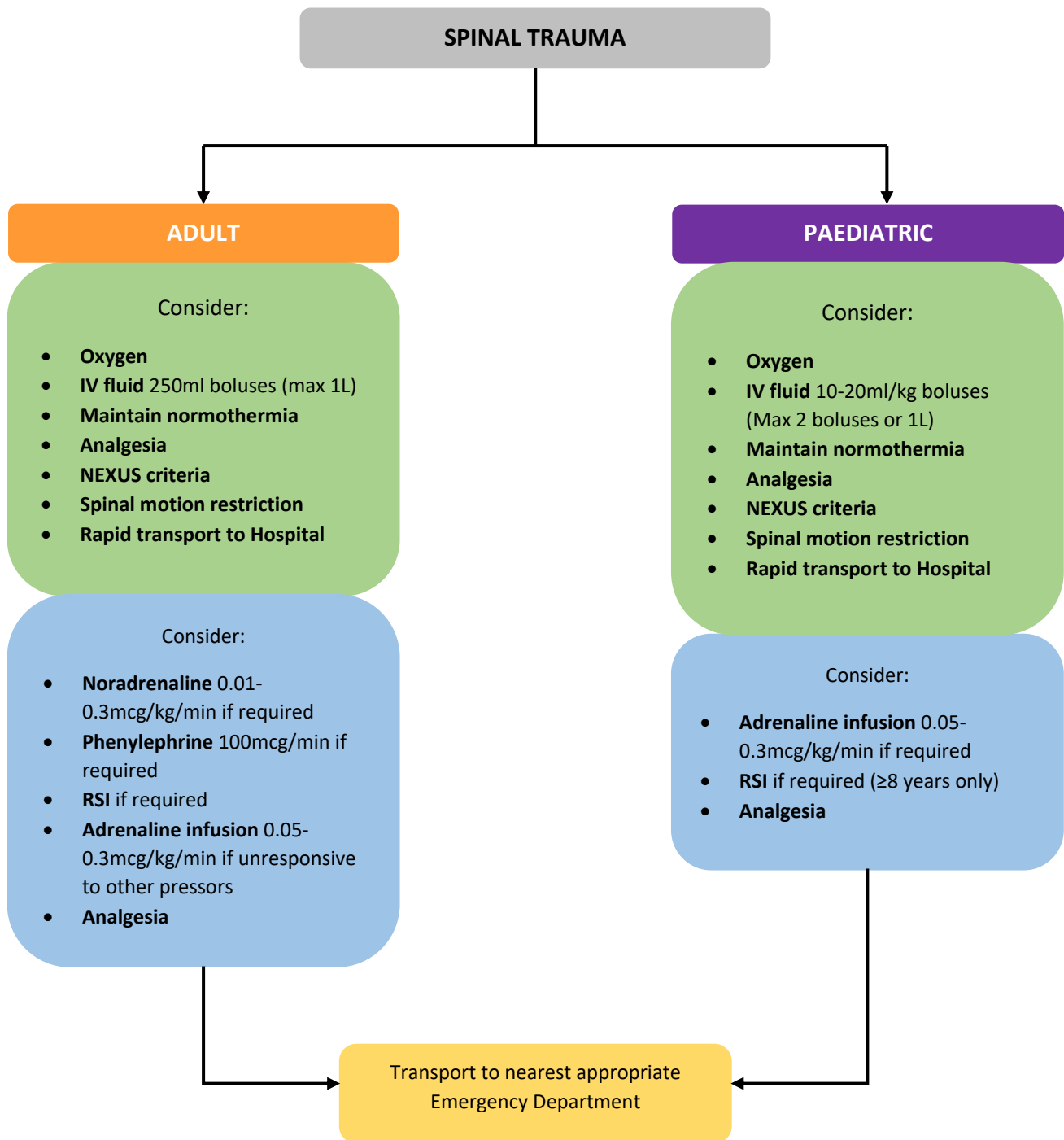
SPECIAL NOTES

- Ensure correctly sized cervical collars are applied to patients. Inappropriately sized collars can cause pain and discomfort and potentially worsen a spinal injury.
- Minimal trauma and at level falls can lead to spinal fractures in the elderly, those with rheumatoid arthritis, severe osteoarthritis, Down's syndrome, cancer or ankylosing spondylitis even though they meet all the SMR clearance criteria.
- Vasopressor support should be considered for patients with neurogenic/spinal shock unresponsive to fluid boluses.



STANDING HEIGHT FALLS IN THE ELDERLY

- Elderly patients are at increased risk of a spinal injury following a fall, including falls from a standing height.
- Elderly patients who fall from a standing height or at a low level should be thoroughly assessed. Clinicians should maintain a high level of suspicion for spinal injury if:
 - Age ≥ 60 years or appear elderly/frail
 - Has pre-existing spinal or vertebral disease (Rheumatoid arthritis, severe osteoarthritis, spinal stenosis, ankylosing spondylitis)
 - History of previous spinal abnormalities (spinal fusion, previous c-spine injury or surgery)
- Apply the principles of spinal motion restriction to protect the c-spine if there is a suspicion or indication of a c-spine injury.
- The use of a backboard/combi-carrier and c-collar may not be appropriate for these patients as it may cause further discomfort and distress. Maintain the patient in a comfortable c-spine neutral position that adheres to the principles of spinal motion restriction. Use padding for support when appropriate to assist with spinal motion restriction to ensure patient comfort.



CPG 10.4 CHEST TRAUMA

Chest injuries are potentially life-threatening yet may not always be immediately apparent. The mechanism of injury should help guide further assessment and provide an index of suspicion of other injuries. These patients require rapid assessment, treatment and transport.

CLINICAL PRESENTATION

Patients with chest injuries may present with the following:

- Penetrating chest/abdominal wounds to the anterior, lateral or posterior aspects
- Bleeding (internal and external)
- Contusions and abrasions
- Haematoma
- Rib and/or clavicle fractures
- Unequal/diminished air entry
- Asymmetrical or paradoxical chest wall movement
- Surgical emphysema
- Sucking chest wound
- Tachypnoea
- Hypoxia
- Hypotension
- Altered LOC
- Jugular venous distension
- Muffled heart sounds

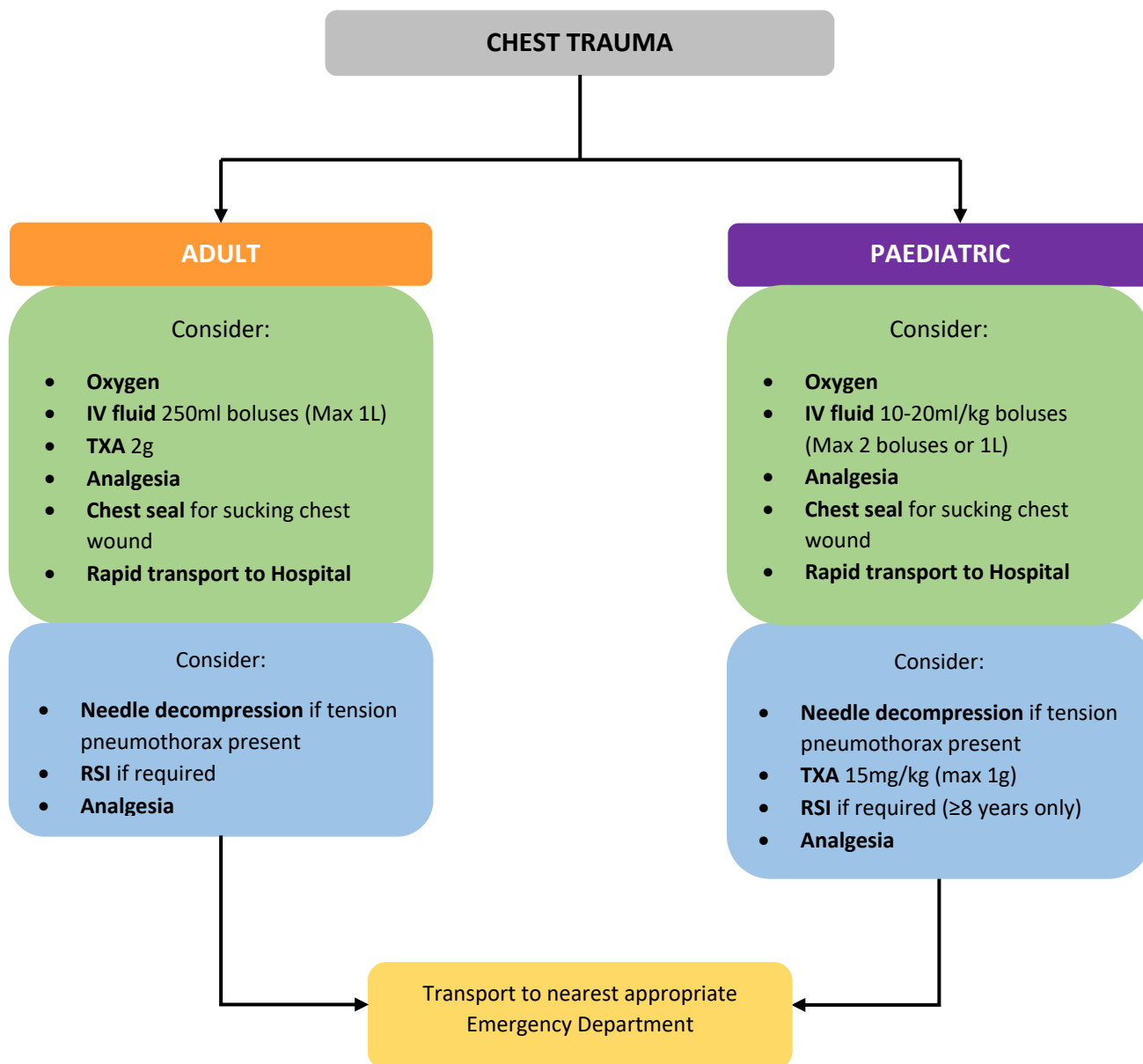
ASSESSMENT CONSIDERATIONS

- Penetrating trauma to the chest may appear minor, however these injuries may be life-threatening and should be treated as such.
- Monitor for signs of tension pneumothorax and decompress the injured side of the chest if indicated.
 - Anterior needle thoracentesis: 2nd intercostal space, above the 3rd rib, midclavicular line. Second needle may be inserted 1cm lateral. May be repeated once.
 - Lateral needle thoracentesis: 4th/5th intercostal space, anterior midaxillary line. May be repeated once. (CCP only skill).
 - Avoid decompressing the chest simultaneously (except in traumatic cardiac arrest). Decompress the injured or most injured side first then reassess.
- Patients with chest trauma may have severe occult bleeding and are at risk of major haemorrhage and shock. Maintain a high index of suspicion for other associated injuries (e.g. abdomen) Consider TXA.
- Effective multimodal analgesia must be administered which may make the work of breathing easier and improve respiratory status.
- Assess for a flail chest and manage with effective analgesia. Do not splint injury with bandages/dressings etc.
- Chest seals should be applied to all sucking chest wounds.



SPECIAL NOTES

- Blunt trauma to the heart (e.g. myocardial contusion) may lead to ECG changes.
- Use caution when using IPPV in the presence of chest trauma. Excessive IPPV may precipitate a tension pneumothorax, especially in patients that have been intubated. Monitor these patients closely.
- If major haemorrhage and haemorrhagic shock present, manage according to CPG 10.1.



CPG 10.5 ABDOMINAL TRAUMA

Abdominal trauma (blunt and penetrating) can result in significant life-threatening injuries. Injuries may appear insignificant as there may be no significant external signs of trauma, however these patients can deteriorate rapidly and unexpectedly.

CLINICAL PRESENTATION

Patients with abdominal injuries may present with the following:

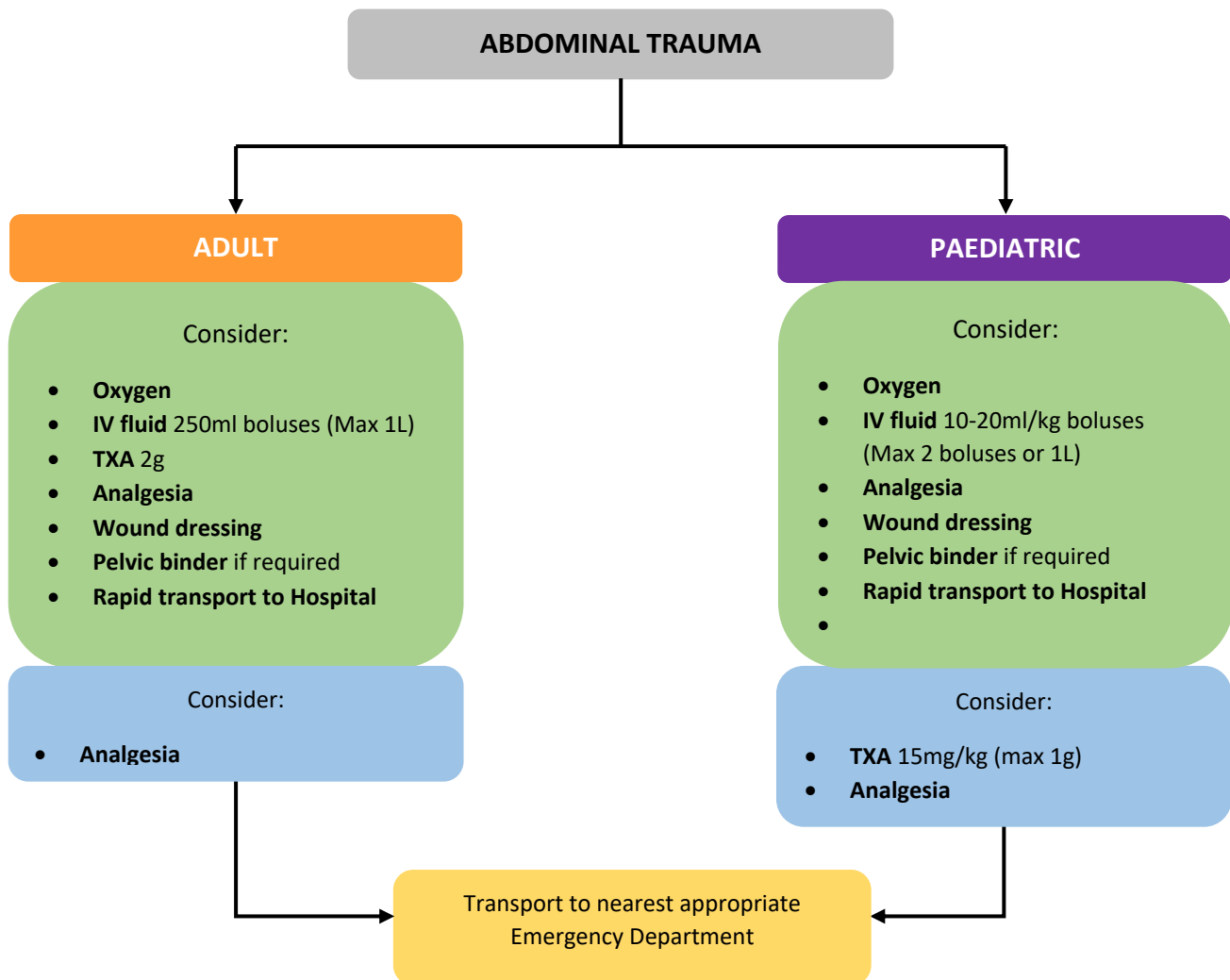
- Abdominal pain/tenderness
- Abdominal guarding
- Abdominal Bruising
- Abdominal distension
- Dyspnoea
- Altered LOC
- Hypotension
- Tachycardia
- Haemorrhage
- Penetrating injuries
- Abdominal evisceration

ASSESSMENT CONSIDERATIONS

- Significant abdominal injuries may present with minor external signs of trauma.
- Some patients with abdominal trauma may only present with unexplained shock in severe cases.
- Identify mechanism of injury to help guide assessment and management.
- Eviscerated bowel should be covered with a moist sterile dressing.
- Patients with abdominal injuries may have other injuries to the pelvis and chest.

SPECIAL NOTES

- Rapid treatment and transport are required for these patients for surgical intervention.
- Limit fluid administration.
- If major haemorrhage and haemorrhagic shock present, manage according to CPG 10.1.



CPG 10.6 PELVIC TRAUMA

Pelvic injuries are potentially life-threatening and require rapid assessment, management and transport. Significant force is required to fracture a pelvis which may result in associated intra-abdominal injuries. Pelvic trauma should be suspected in all patients with a significant mechanism of injury (e.g. RTA, falls from height, crush incidents) and trauma patients with haemodynamic instability.



CLINICAL PRESENTATION

Patients with pelvic injuries may present with the following:

- Pain
- Bruising and swelling – scrotum, vulva, flanks
- Possible bleeding from rectum, urethral meatus, vagina
- Asymmetric pelvis
- Limb shortening
- Reduced/absent sensation or movement in lower limbs
- Hypotension
- Tachycardia
- Shock



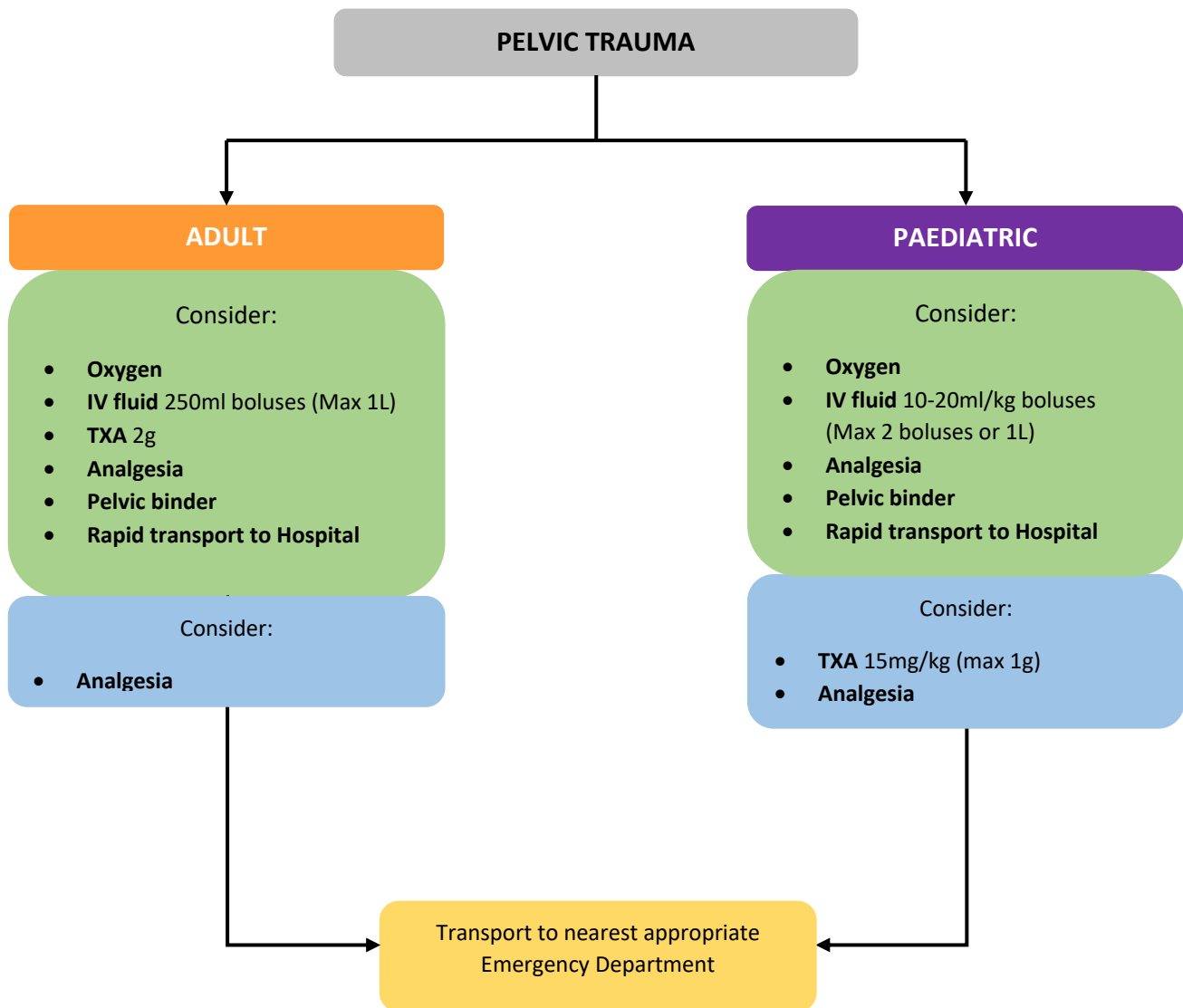
ASSESSMENT CONSIDERATIONS

- Patients with a suspected pelvic fracture should be assessed for associated intra-abdominal and long bone injuries.
- A pelvic binder should be applied to all trauma patients with a mechanism of injury that is suggestive of a pelvic fracture(s) and have any of the following:
 - Hypotension or signs of haemodynamic compromise
 - Decreased LOC
 - Assessment of pelvis indicates likelihood of pelvic fracture
 - Trauma cardiac arrest
- Monitor for major haemorrhage and haemorrhagic shock. Manage the shocked patient according to CPG 10.1.



SPECIAL NOTES

- Pelvic binders should not be applied if an isolated femur neck fracture or isolated dislocation is present.
- Do not palpate or spring the pelvis as this may worsen the potential injury.
- Limit IV fluids.



CPG 10.7 LIMB TRAUMA

Fractured limbs can be extremely painful injuries. Effective analgesia and splinting are required as well as gaining a detailed history of events and the mechanism of injury as there may be associated injuries that the limb fracture is distracting from.

CLINICAL PRESENTATION

Patients with a limb fracture may present with the following:

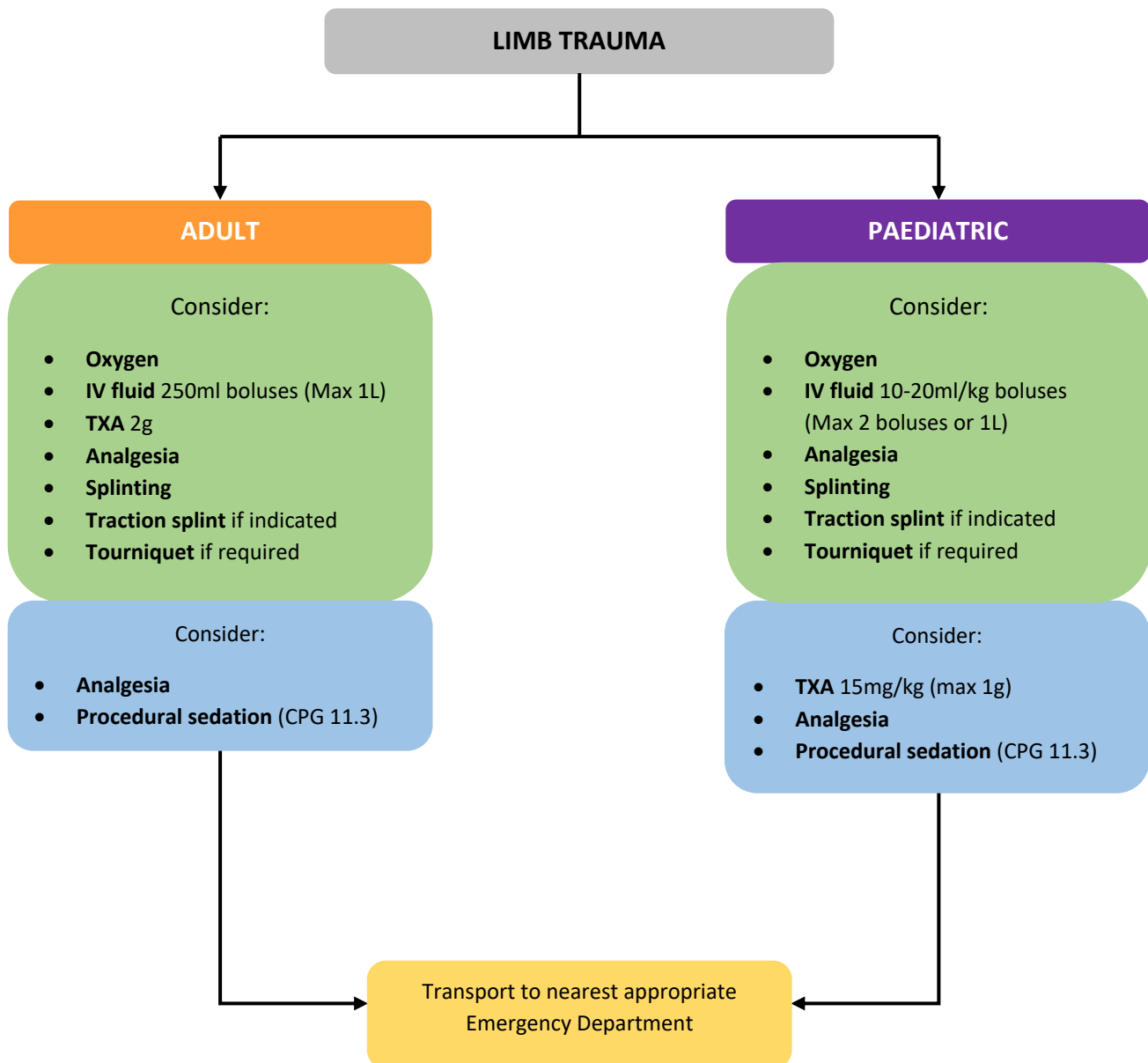
- Pain
- Swelling
- Bruising
- Deformity
- Immobility of limb
- Crepitus
- Tenderness
- Bleeding

ASSESSMENT CONSIDERATIONS

- Considerable force is required to fracture a femur and a detailed assessment should be conducted to identify associated injuries.
- Effective analgesia, ideally multimodal analgesia, should be administered prior to splinting if possible.
- Procedural sedation should be considered for significantly displaced fractures requiring manipulation or for the application of the traction splint.
- Pulse, motor and sensory (PMS) function should be assessed before and after splinting.
- Limbs with compromised vascular supply (absent radial/pedal pulse) should be transported rapidly to the TRU.
- Soft tissue injuries may present with the same signs as above except deformity and crepitus.

SPECIAL NOTES

- The traction splint is the preferred splint for isolated closed mid-shaft femur fractures.
- The vacuum splint is the preferred splint for all other fractures. If vacuum splints are not available, use SAM splints.
- Monitor for major haemorrhage, particularly if significant vascular damage has occurred or femur fractures are present.
- Should procedural sedation be required, manage according to CPG 11.3.



CPG 10.8 TRAUMA IN PREGNANCY

Trauma in pregnancy can result in premature labour, placenta abruption and foeto-maternal haemorrhage. It is important to be aware of the physiological changes that occur during pregnancy and the impact these may have on the pregnant trauma patient.

CLINICAL PRESENTATION

Trauma specific to the pregnant patient may result in:

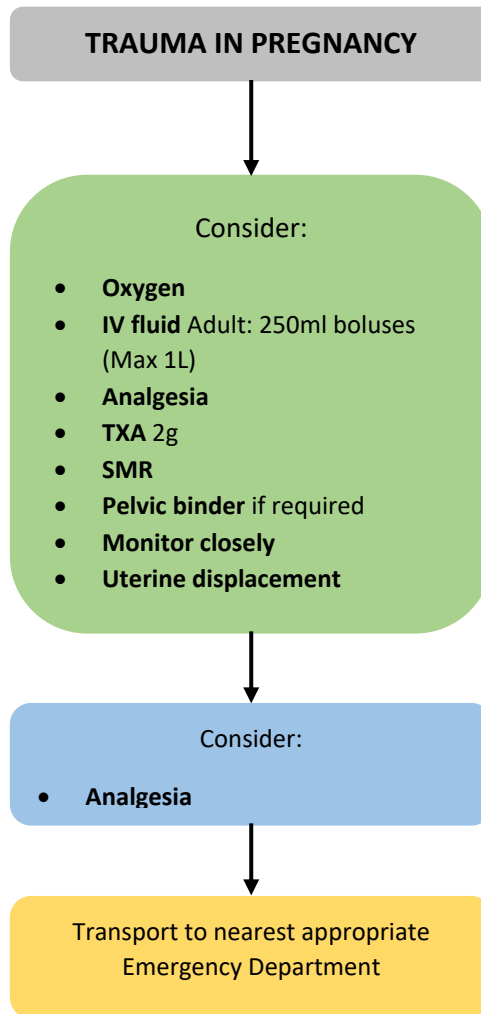
- Premature labour
- Placenta abruption
- Uterine rupture
- Foeto-maternal haemorrhage (major haemorrhage)
- Shock

ASSESSMENT CONSIDERATIONS

- A detailed assessment should be conducted on the pregnant trauma patient, including an assessment of the mechanism of injury.
- An obstetric history should be obtained, including gestational age, the presence of foetal movement and presence of PV bleeding.
- Signs of haemorrhagic shock may have a delayed presentation in the pregnant patient due to the increase in blood volume that occurs during pregnancy. Monitor for major haemorrhage and haemorrhagic shock. Manage the shocked patient according to CPG 10.1.
- Pregnant patients will have a reduction in residual lung volume and reserve as their pregnancy progresses.
- Pregnant patients are at greater risk of aspiration due to the displacement of intra-abdominal organs.

SPECIAL NOTES

- Pregnant patients requiring spinal motion restriction should have their uterus manually displaced to the left to prevent supine hypotensive syndrome. The left lateral tilt may also be performed to relieve pressure on the inferior vena cava.
- Note that the application of the pelvic binder may be challenging or not possible for the pregnant patient in their 3rd trimester.
- Manage specific injuries according to relevant CPG.



CPG 10.9 BURNS

Burns can result in significant injuries and life-threats. Airway and inhalation burns, exposure to toxic smoke and fluid and electrolyte abnormalities due to major burns require rapid assessment and management. Identifying the cause of the burn (e.g. thermal, electrical, chemical etc) and any potential associated trauma (e.g. blast injuries) are important when assessing these patients.

CLINICAL PRESENTATION

Patients with burns may present with the following:

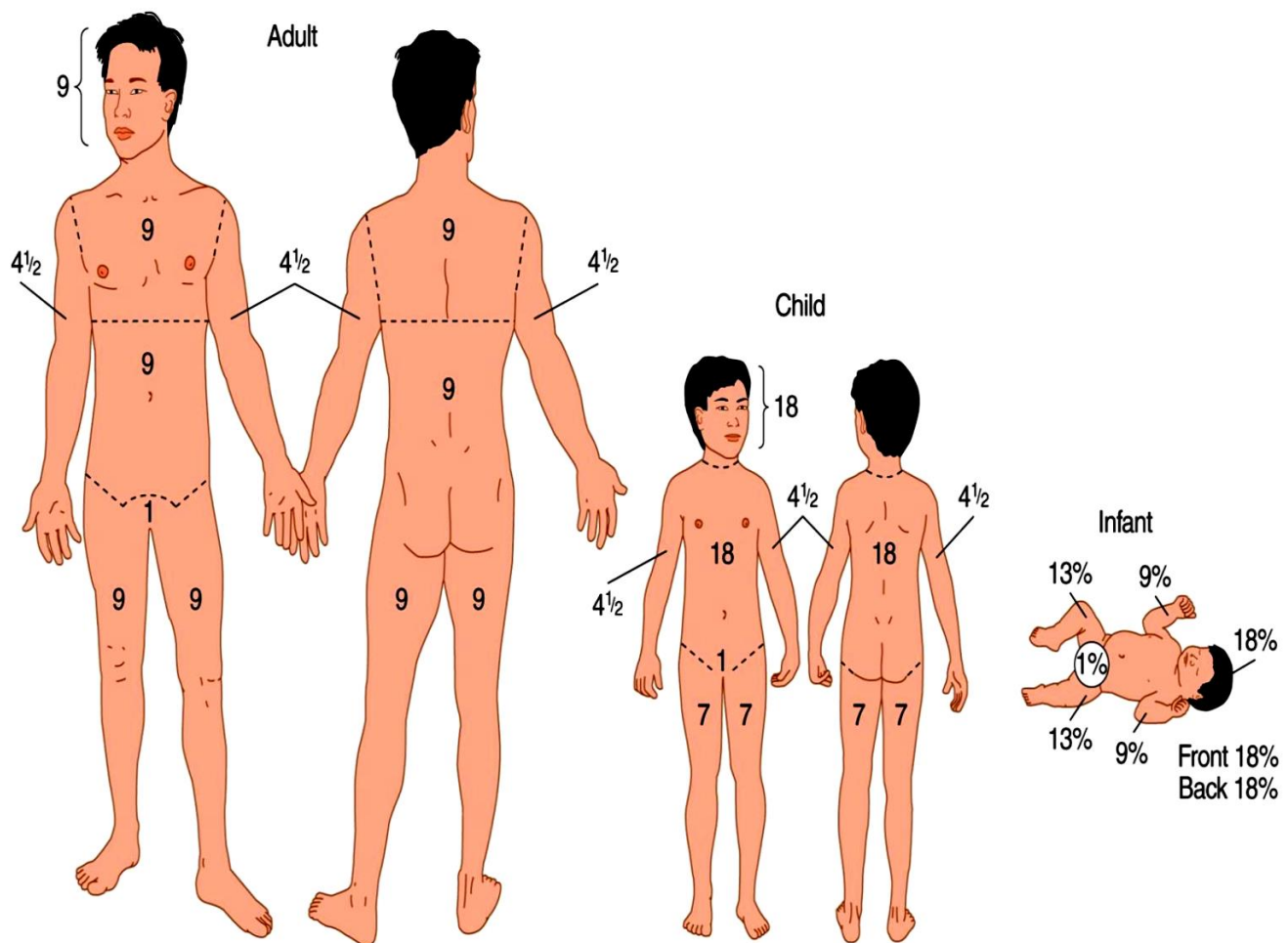
- Pain
- Airway and Inhalation burns:
 - Burns to face/oral burns
 - Singed nasal hair
 - Carbonaceous sputum
 - Tachypnoea, stridor, hoarse voice
- Smoke inhalation
 - Carbon monoxide toxicity
 - Cyanide toxicity
- Circumferential burns resulting in vascular or respiratory compromise
- Associated traumatic injuries
- Hypothermia

ASSESSMENT CONSIDERATIONS

- RSI should be considered early for patients with airway and inhalation burns as respiratory compromise can develop rapidly.
- SpO₂ readings can be inaccurate in the presence of carbon monoxide toxicity. All patients exposed to smoke should receive oxygen.
- Up to 1L of fluid should be administered to adult patients and 20ml/kg in paediatrics within the first hour of injury. Large fluid volumes administered rapidly can increase the risk of tissue oedema and swelling.
- Patients with electrical burns may develop arrhythmias. These patients should have continuous 4-lead ECG monitoring and a 12-lead ECG if required.
- Estimate the total burn area percentage using the diagram below:
- Accurate assessment of burn-depth is difficult in the prehospital setting. A rapid assessment of burn depth should be undertaken. Do not delay transport to perform a detailed burn-depth assessment.

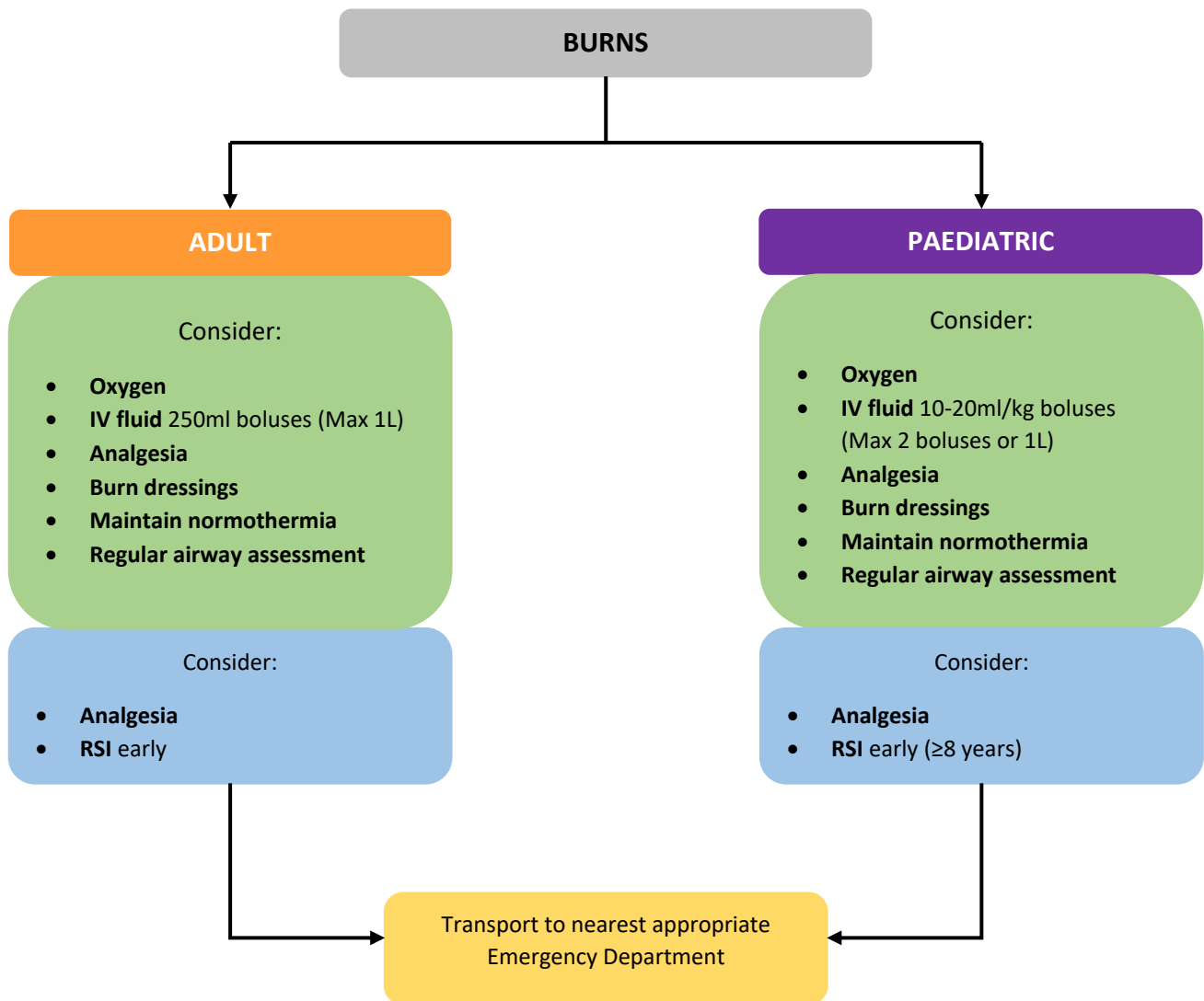
BURN DEPTH ASSESSMENT:

	SUPERFICIAL	PARTIAL THICKNESS	FULL THICKNESS
SKIN APPEARANCE	Erythema	Red or mottled, swelling, blisters	Dark, charred, leathery, white or mottled
SENSATION	Painful	Painful	Painless



SPECIAL NOTES

- Hypovolemia due to burns manifests over hours and will not be seen with in the initial acute phase. Should hypotension be present in the acute stages of the burn injury, consider an associated traumatic injury (e.g. blast injuries).
- Avoid hypothermia. Cool the burns but keep the patient warm.



CPG 10.10 TRAUMA BYPASS CRITERIA

TRAUMA BYPASS CRITERIA

If ONE or MORE of the criteria listed below are met, notify the CCD and transport the patient directly to the HGH Trauma Resuscitation Unit for adults (≥ 14 years) and Sidra Hospital Emergency Department (≤ 13 years) for paediatrics. If none of the listed criteria below are met, transport the patient to the nearest emergency department. Deviation from these criteria must be authorised through the CCD.

MECHANISM OF INJURY

- FALLS:
 - Adults: 6 meters or higher (≥ 20 ft or 2 stories)
 - Paediatrics: 3 meters or higher (≥ 10 ft or 1 story) or twice the patients height
 - Red(T1) or Yellow(T2) code patient that fell from a horse, camel or similar animal
- MOTOR VEHICLE CRASH:
 - Red(T1) or Yellow(T2) code patient from a vehicle with a traumatic death in the same vehicle, except bus or mini-bus type vehicles
 - Red(T1) or Yellow(T2) code patient from a vehicle with significant intrusion of ≥ 30 cm into the occupant compartment or ≥ 45 cm at any site
 - Ejection (partial or complete) from vehicle, including ATV-type vehicles
 - Person trapped in or under a vehicle or extrication ≥ 20 minutes
 - Red(T1) or Yellow(T2) code patient involved in a roll over with roof deformity ≥ 40 cm
- Red(T1) or Yellow(T2) code patient from a motorcycle accident with speed ≥ 30 km/hr
- Red(T1) or Yellow(T2) code pedestrian/paediatric/bicyclist struck by a vehicle (thrown/ run over/ impact speed of ≥ 20 km/hr)
- Burns with evidence of trauma OR $\geq 20\%$ BSA in adults/ $\geq 10\%$ BSA in paediatrics, excluding isolated superficial extremity burns OR any significant neck/facial burns with potential airway compromise
- Hanging or attempted hanging with evidence of trauma (e.g. hanging with a fall greater than height of patient or obvious spinal deformity)
- Drowning or near drowning with evidence of trauma (e.g. diving into shallow water)

VITAL SIGNS AND LEVEL OF CONSCIOUSNESS

- GCS < 15 (consider effects of alcohol/drugs)
- Sustained Systolic BP:
 - < 90 mmHg in adults
 - $<$ age-specific normal value in paediatrics
 - < 110 mmHg for patients ≥ 65 years old
- Respiratory rate:
 - < 10 or > 25 breaths/minute for adults
 - Infants to 1 year old < 20 breaths/minute

Any patient requiring respiratory assistance

ANATOMIC CRITERIA (PATTERN OF INJURY)

- Chest wall instability or deformity, evidence of respiratory distress, subcutaneous emphysema or suspicion of multiple rib fractures
- Traumatic amputation, excluding digits and toes
- Penetrating trauma to head, neck, torso or extremities above elbow or knee

- Open and/or depressed skull fracture
- Suspected pelvic fracture/ unstable pelvis
- Multiple long bone fractures (femur and/or humerus)
- Isolated femur fracture
 - excluding isolated pathologic femur fracture in the elderly (>65) patient
- Crushed, degloved, mangled or pulseless extremity (threatened limb)
- Neurological deficit in patients with spinal trauma or suspected spinal injury
- Trauma Arrest if decision made to continue resuscitation to hospital

Sustained ROSC following trauma arrest

SPECIAL PATIENT OR SYSTEM CONSIDERATIONS

- Patients who have sustained significant trauma but do not fit any of the criteria above but are:
 - ≥65 years old
 - Pregnant ≥20 weeks gestation
- If there is a reasonable concern despite clearing all of the above criteria and following Clinical Desk notification

CPG 10.11 CRUSH INJURY

Crush injuries may occur due to multiple reasons, such as isolated limb trauma, polytrauma, envenomation, drug/toxin exposure, burns, infections and heat stroke. Crush injuries are categorised into three types:

Crush injury – localised tissue injury as a result of prolonged compressive force applied to an area of the body, most commonly limbs.

Compartment syndrome – increased pressure within a muscle, due to the non-elastic nature of the fascia, leading to reduced circulation distal to the injury. Tissue ischemia and necrosis can occur if left untreated.

Crush syndrome – the systemic manifestation of crush injury, as a result of tissue destruction. This results in the destruction of myocytes and release of myoglobin, potassium and uric acid upon reperfusion. Further deterioration into rhabdomyolysis can occur, leading to hypovolaemia, hyperkalaemia, metabolic acidosis and acute kidney failure, leading to death.



CLINICAL PRESENTATION

Patients with a crush injury may present with the following:

- Localised tissue injury due to compressive force
- Localised pain
- Bleeding
- Loss of pulse/motor/sensory function to affected area

Patients with a compartment syndrome may present with the following in addition to the signs and symptoms of a crush injury:

- Palpable tension or swelling of an anatomical compartment
- Pain disproportionate to the injury
- Paraesthesia of skin and paresis of muscle in affected area
- Pallor of skin over affected area

Patients with a crush syndrome present with the following:

- Compartment syndrome
- Haemodynamic instability
- Shock
- Hypothermia
- Hyperkalaemia
- Dysrhythmias
- Renal failure



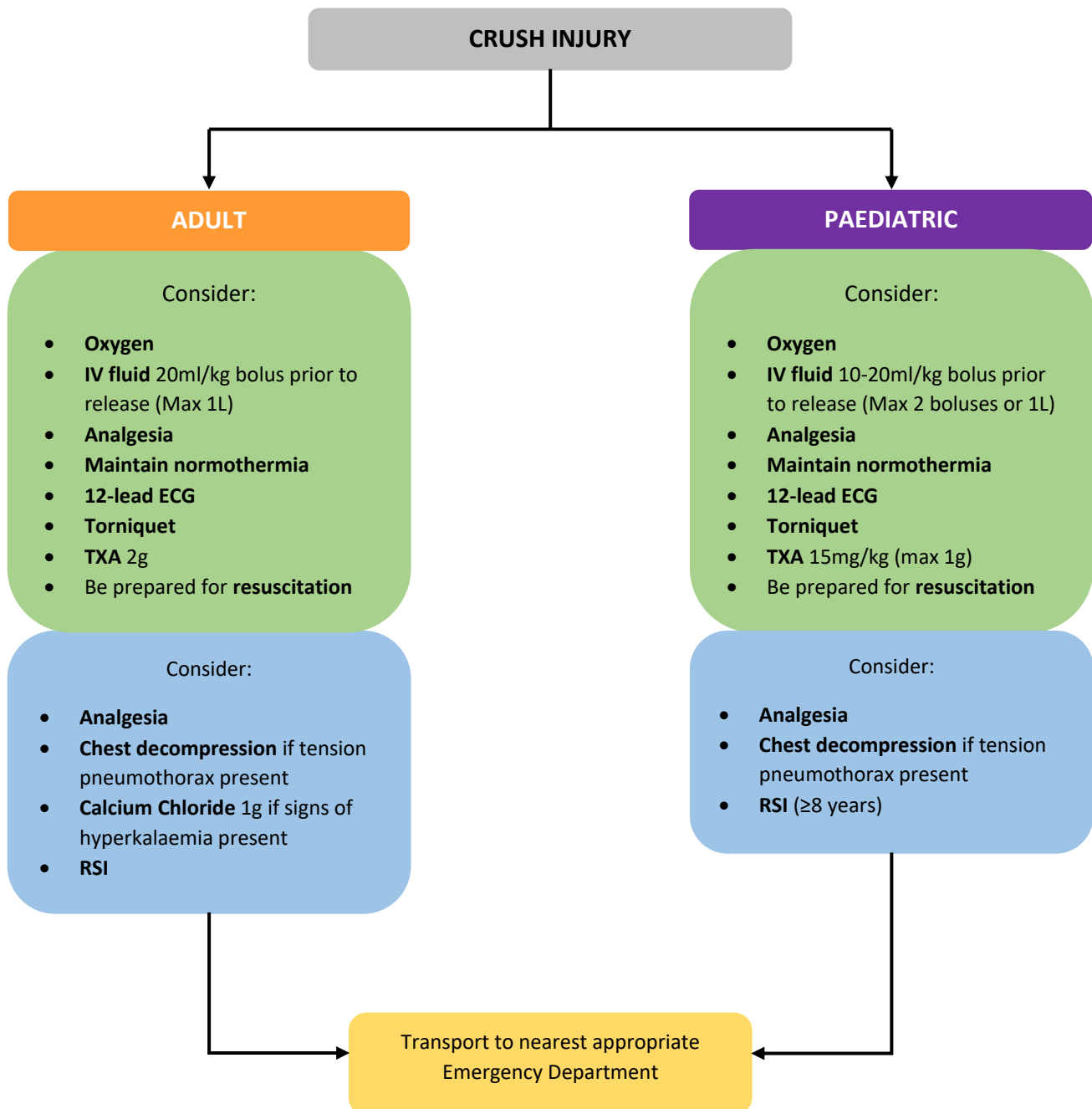
ASSESSMENT CONSIDERATIONS

- A detailed assessment of potential crush injuries should be undertaken
- Maintain a high index of suspicion in any patient that has had a prolonged or significant compressive force applied to them

- Remove the compressive force as soon as possible and be prepared for the development of crush syndrome
- Hypovolaemia following removal of the compressive force should be anticipated and a treatment plan prepared
- A crush injury to the chest should have the compressive force removed immediately
- Crush injury patients are at risk of hypothermia. These patients should be kept warm
- Certain co-morbidities are associated with an increased risk of developing crush syndrome following a crush injury. These should be assessed for:
 - Diabetes
 - Hypothyroidism
 - Bleeding disorders/anticoagulation

SPECIAL NOTES

- Be prepared for the patient to deteriorate following release of the compressive force. Have the resuscitation equipment prepared and ready for use
- Fluid bolus should be administered prior to release of the patient from the compressive force or immediately after if unable to initiate a fluid bolus prior to release
- Adequate fluid resuscitation should be prioritised over permissive hypotension in the setting of a significant crush injury in order to minimise the risk of reperfusion injury
- The application of a tourniquet to a crushed limb should be considered prior to release of the patient if a high risk of significant bleeding is present or immediately after release if significant bleeding is identified. Tourniquets should not be applied as a routine
- Crush injury patients should be closely monitored – ECG, BP, SpO₂, EtCO₂, RBS and temperature. A 12-lead ECG should be performed when practically possible to identify potential evidence of hyperkalaemia
- Whenever possible, release of the compressive force should be controlled
- Patients presenting with traumatic asphyxia should be continuously assessed for signs of a tension pneumothorax – if present, chest decompression should be performed.



CPG 10.12 ONGOING TRAUMA CARE CHECKLIST

Ongoing Trauma Care Checklist

Ver. 4

X

External bleeds

- ☐ External bleeding remains controlled



A

Airway

- ☐ Additional airway support provided if needed



B

Breathing

- ☐ Oxygen & Ventilation strategies optimized
- ☐ Chest injuries treated



C

Circulation

- ☐ Pelvic binder applied
- ☐ IV canula + 3 way stop cock + fluids secured
- ☐ TXA administered if indicated
- ☐ Vasopressors used if required



D

Disability

- ☐ Spinal motion restriction applied
- ☐ Analgesia provided
- ☐ Neurological exam performed (GCS, Pupils, RBS, PMS)



E

Exposure

- ☐ Detailed Physical Examination completed
- ☐ Hypothermia prevented



F

Frequently Reassess

- ☐ Vital monitors attached (BP, SpO₂, EtCO₂, PADS)
- ☐ Bleeding sites reassessed & Vitals trends noted



G

Get positioned

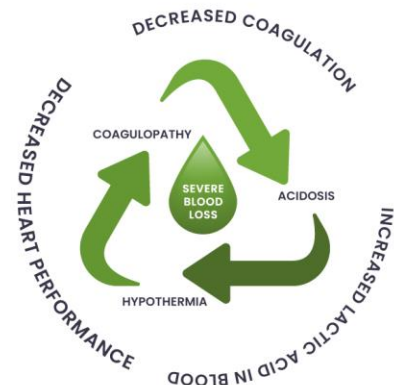
- ☐ Fractures splinted
- ☐ Wounds dressed
- ☐ Patient optimally positioned



H

Hasten Transport

- ☐ Rapid transportation and prenotification to TRU initiated



TBI Neuroprotection

- SpO₂ > 94%
- MAP 70-80 mmHg / MAP 65 mmHg (poly trauma)
- RBS: 4-6.7mmol
- EtCO₂: 35-40 mmHg
- Avoid gag: Attention with OPA
- Position: 30° head up
- C-collar: Properly sized & fitted
- Proper sedation / Analgesia
- Manage seizures



خدمة الإسعاف
Ambulance Service





CPG 11.1-11.5

RAPID SEQUENCE INTUBATION MECHANICAL VENTILATION AND SAFE SEDATION



CPG 11.1 RAPID SEQUENCE INDUCTION

Rapid Sequence Induction of Anaesthesia is used to establish a definitive airway, protect the airway and/or manage oxygenation and ventilation if unable to do so by other means, based on an ad-hoc risk-benefit assessment for the individual patient. **HMCAS CCPs may only perform an RSI on patients ≥ 8 years old in both the medical and trauma setting.**

✓ INDICATIONS FOR RSI

- Loss of airway protection/patency
- Ventilation/oxygenation failure
- Neuroprotection of the TBI and CVA patient
- Anticipated clinical course requiring anaesthesia
- Humanitarian reasons (e.g., major polytrauma and significant BSA burns patients with a relatively high level of consciousness)

✗ CONTRAINDICATIONS FOR RSI

- Hypersensitivity to any of the RSI medications
- Patient in whom intubation and cricothyrotomy would be impossible

! COMPLICATIONS FROM RSI

- Unrecognised oesophageal intubation
- Right main bronchus intubation
- Aspiration
- Hypoxia
- Hypotension
- Laryngospasm
- Oropharyngeal trauma
- Malposition
- Vagal stimulation
- Equipment failure

👂 ASSESSMENT CONSIDERATIONS

- Assess the patient for a potentially difficult airway using the mnemonic **LEON**:
 - LOOK EXTERNALLY
 - EVALUATE 3-3-2
 - OBSTRUCTIONS
 - NECK MOBILITY
- The following monitoring requirements are mandatory:
 - ECG 4-lead
 - Defib pads
 - NIBP
 - SpO₂
 - EtCO₂

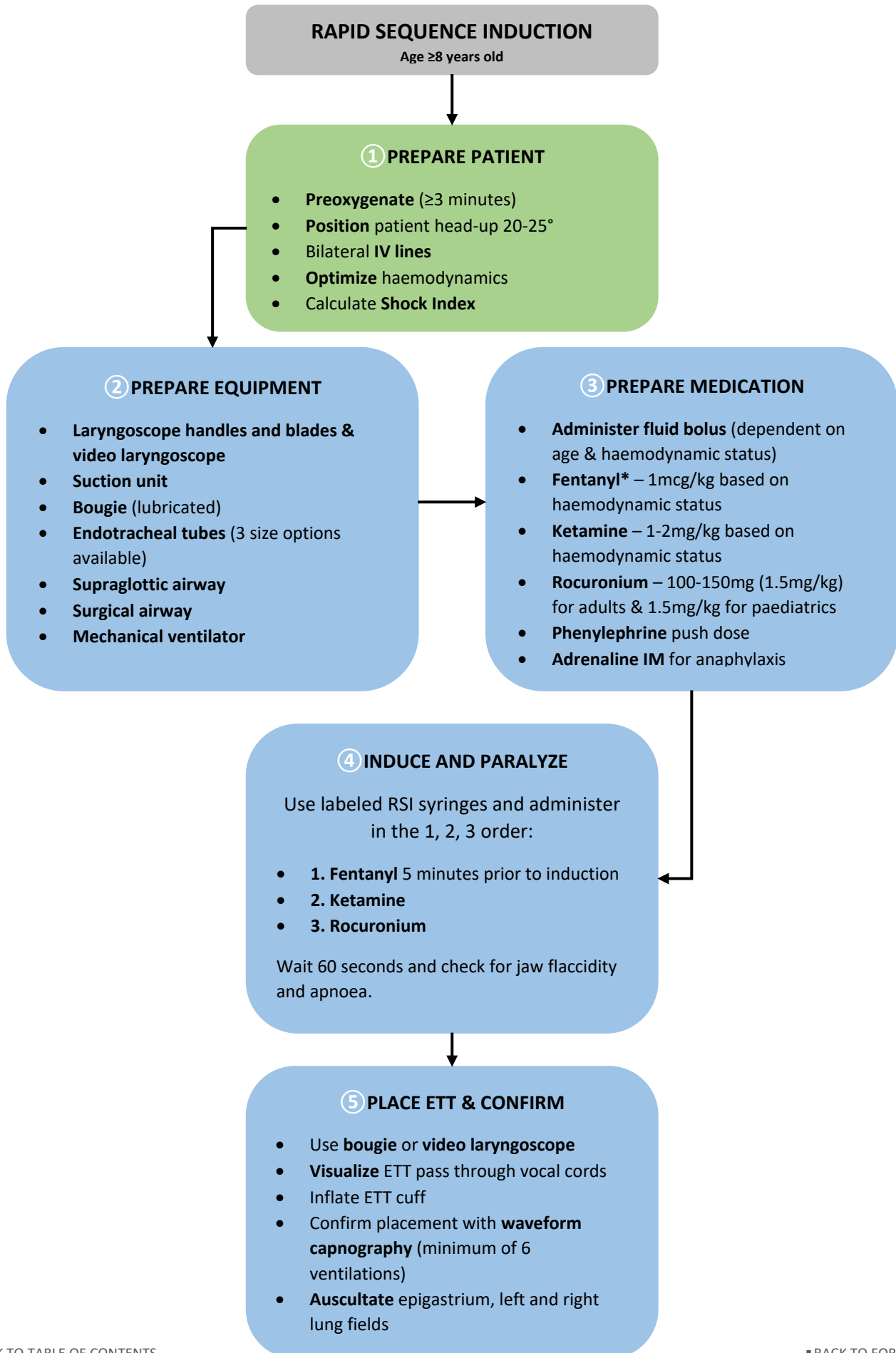
- Post RSI deterioration – perform the quick check using the **DOPE** mnemonic.
 - Displacement
 - Obstructed ETT or ventilator circuit
 - Pneumothorax
 - Equipment failure

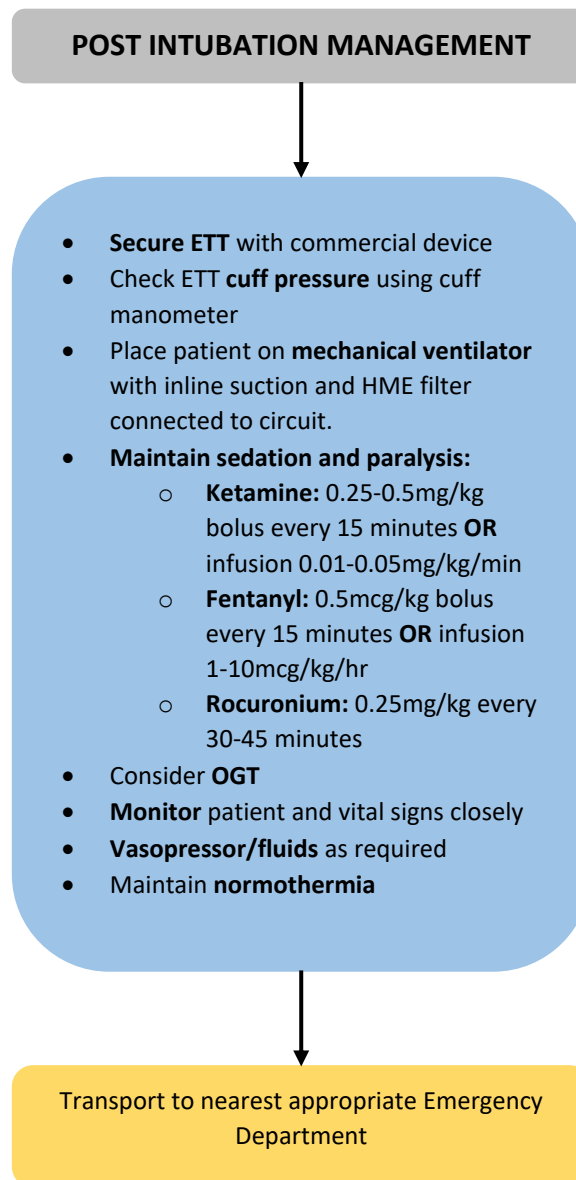
SPECIAL NOTES

- Prehospital RSI should preferably be performed in the back of an ambulance where all the required equipment, oxygen, lighting and shelter from the environment is available
- A bougie (Frova) should be used for all intubations except when using the GlideScope, in which case the GlideRite stylet should be used
- The CCP should conduct a team briefing with all paramedics involved in the RSI process, clearly identifying roles and tasks prior to induction
- Adhere to the ABCDEs of CRM
- The failed intubation drill should be commenced if there is an absence of or inappropriate EtCO₂ reading/waveform or following two failed intubation attempts.
- Consider delaying the RSI for patients that are very high risk (e.g., significant haemodynamic instability despite optimisation)
- Paediatrics desaturate at a much faster rate than adults. Preoxygenate adequately prior to induction.
- Provide at least 3 minutes (preferably longer) of pre-oxygenation using the nasal EtCO₂ and NRB mask. Avoid routinely using the BVM
- Always prepare as if the airway is going to be difficult.
- Shock index is calculated using the formula = HR ÷ SBP. A shock index of: Adults >0.7 | 13-18 years: >0.9 | 8-12 years: >1.0 indicates that the patient is at risk of haemodynamic collapse and post-RSI hypotension. These patients should be resuscitated prior to RSI
- Consider withholding Fentanyl for induction in the severely haemodynamically unstable patient unresponsive to resuscitation, such as in patients with a shock index of ≥1
- The Rocuronium induction dose should always be rounded up to the higher dose to avoid under paralysis (patients estimated to be close to 100kg should rather be administered 150mg Rocuronium)
- The RSI checklist should be used for all RSI cases. At the commencement of the RSI checklist – no one should be moving in or out the back of the ambulance or RSI site. Silence should be maintained throughout to ensure focused attention on the procedure and the patient.
- Consider the insertion of an OGT following RSI provided this will not prolong on scene time

FENTANYL SPECIAL NOTES

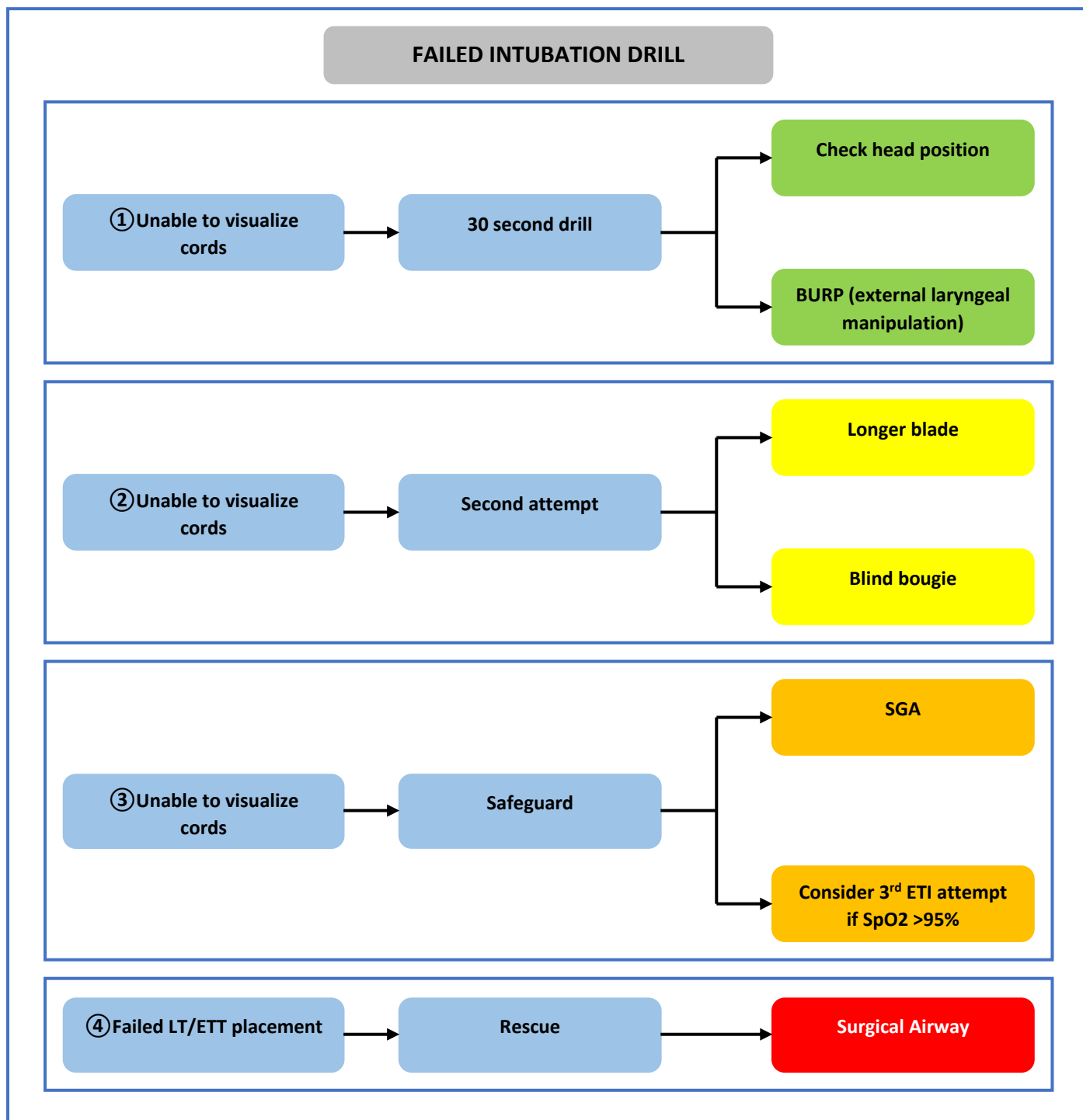
- *Fentanyl for induction dose – 1mcg/kg
- *Fentanyl for induction should be diluted to 10ml
- *Administer the Fentanyl 5 minutes prior to induction and paralysis. Administer Fentanyl slowly over 3 minutes, which allows Fentanyl to be administered 2 minutes prior to Ketamine and Rocuronium. Fentanyl for induction should preferably be administered via a syringe driver whenever possible.





CPG 11.2 FAILED INTUBATION DRILL

The failed intubation drill is a continuation of the RSI drill and mandates alternative techniques after two intubation attempts and progression to the next step if required irrespective of the patient's oxygen saturation. A surgical airway is indicated when ETT and LT placement have failed to establish an airway.



CPG 11.3 PROCEDURAL SEDATION

Procedural sedation involves the administration of sedative and/or analgesic medications to reduce pain and anxiety associated with invasive or painful procedure, enhancing patient comfort and procedural success. This CPG is not applicable to patients presenting with ABD (See CPG 7.1).

SEDATION LEVEL	DESCRIPTION
Light sedation	For anxiolysis. Does not impact the patient's level of consciousness.
Moderate sedation	Reduces level of consciousness yet patient still has purposeful response to voice or light touch.
Deep sedation	Reduced level of consciousness and the patient only has a purposeful response to painful stimuli. May impact on the patient's airway, respiratory and cardiovascular systems.
General anaesthesia	Renders the patient unconscious with no purposeful response to any stimulation. Potential for profound depression of the patient's airway, respiratory and cardiovascular systems.

A risk assessment should be performed prior to performing procedural sedation. The patient must be closely monitored. Light to moderate sedation is the optimal level of sedation to aim for in the prehospital setting. Deep sedation should be avoided. General anaesthesia should only be performed when undertaking an RSI (see CPG 11.1).

	INDICATIONS
- Painful procedures (Transcutaneous pacing or cardioversion) - Trauma patients requiring fracture splinting, extrication or who are distressed or agitated by pain despite appropriate analgesia - Patients posing a risk/danger to themselves or those around them (see Acute behavioural disturbance CPG 7.1) - Maintenance of an advanced airway (see post intubation management section of CPG 11.1)	
	CONTRAINDICATIONS
- Patients with airway compromise where securing the airway will be difficult - Hypersensitivity to any of the medications used for sedation	
	COMPLICATIONS
- Unintentional loss of consciousness - Depressed airway reflexes - Respiratory depression - Cardiovascular system depression - Unpredictable response to administered drug	



ASSESSMENT CONSIDERATIONS

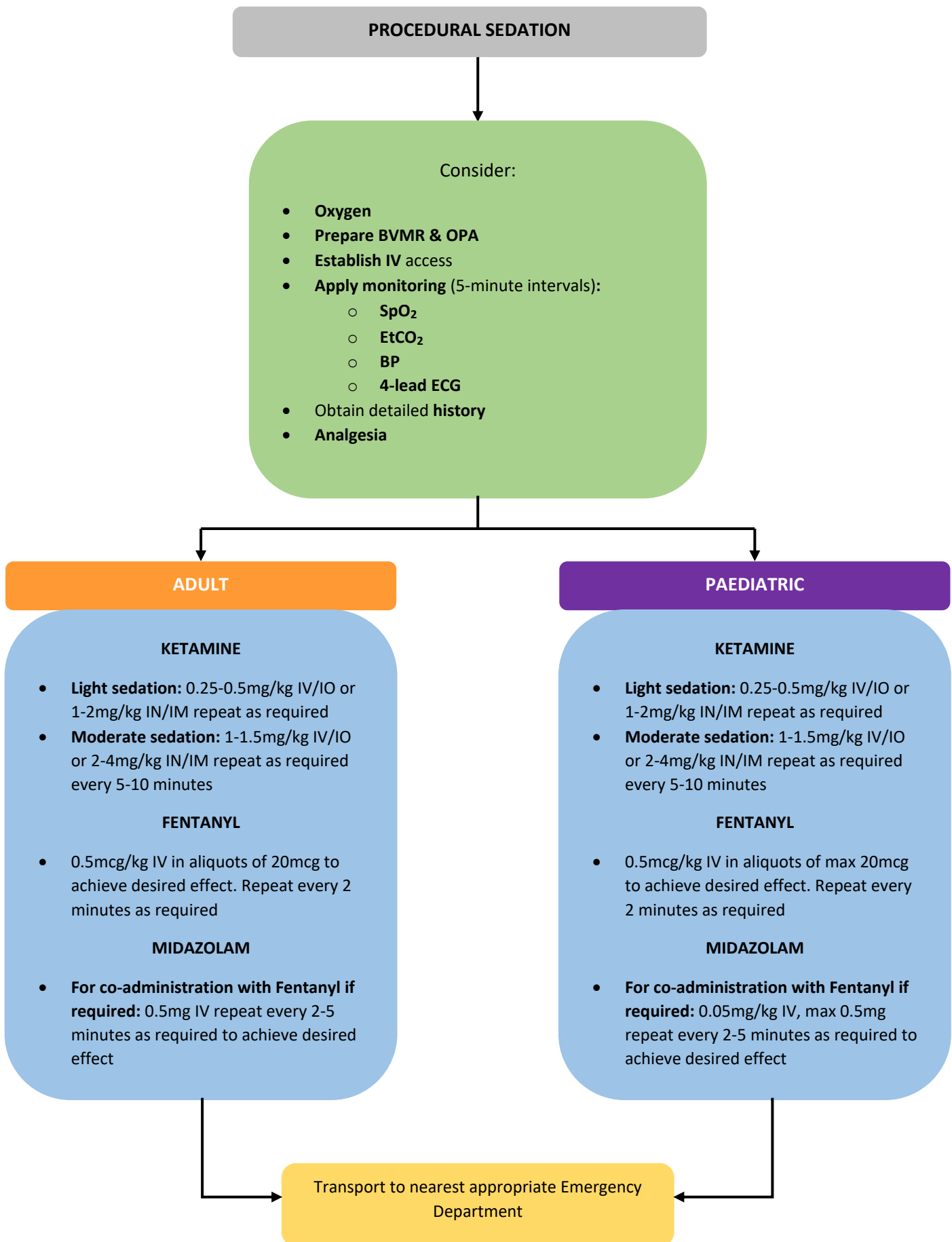
- Gather a detailed history, including co-morbidities, previous anaesthesia and potential adverse reactions.
- Rule out and appropriately manage other factors that may lead to a patient being unnecessarily sedated, such as managing hypoglycaemia and providing adequate pain relief.
- Sedated patients require mandatory continuous monitoring of the following vital signs prior to being sedated or as soon as practical:
 - SpO₂
 - EtCO₂
 - BP (every 5 minutes)
 - ECG (4-lead)
- Oxygen should be administered to patients via NRM while undergoing procedural sedation. The NRM can be replaced with a nasal cannula or removed once the procedure is complete and SpO₂ remains in normal ranges.
- Basic airway equipment must be prepared and ready for use prior to undertaking procedural sedation. This equipment includes the BVMR and an appropriately sized OPA. Additionally, an NRM should be on hand ready for use if required.
- Keep the sedated patient in the lateral position/45-degree head up or most appropriate position based on required procedure or condition (e.g., trauma patient requiring SMR).
- Patients' response to administered medications vary and dosing may need to be adjusted



SPECIAL NOTES

- The patient's airway and ventilatory status should be constantly assessed. The population of Qatar are narcotic naïve and are at risk of developing complications such as apnoea.
- In the prehospital setting, a sedation level of light to moderate should be targeted. Deep sedation is generally not required prehospitally.
- The selection of medications should be determined based on the patient's clinical presentation and the procedure being performed.
- Ketamine disinhibition occurs in a small number of patients. Management of such cases includes providing reassurance and calming words until the appropriate sedation level has been achieved. Low dose midazolam should only be administered if these measures fail.
- Ketamine emergence occurs in a small portion of patients that receive ketamine, with majority of these patients experiencing mild symptoms. Should a patient be experiencing ketamine emergence symptoms, keep them in a quiet calm environment with reduced lighting. Monitor them closely. Should symptoms be more severe, administer low dose midazolam. Prophylactic midazolam is not to be administered.
- If Ketamine is contraindicated, Fentanyl and Midazolam may be used as an alternative.
- Fentanyl is the preferred medication for procedural sedation for patients presenting with myocardial ischemia and requiring transcutaneous pacing or sync cardioversion. Should there be no signs of myocardial ischemia, Ketamine is a suitable option.
- Fentanyl is often co-administered with Midazolam to produce the desired analgesic, amnestic and anxiolytic properties. Low dose Midazolam may be considered in this setting if Fentanyl is not achieving the desired effect.
- Ketamine is the preferred medication for procedural sedation in the absence of contraindications and myocardial ischemia.

- Extra caution should be taken when co-administering Fentanyl and Midazolam. The medication should be diluted and administered slowly in small aliquots to achieve the desired effect while being closely monitored.
- Due care must be exercised when determining drug dosing for elderly patients. The delayed onset times, drug interactions and comorbidities may affect the patient's response to these medications. It is recommended that drug doses be reduced for this patient group and medications are given slower and repeat doses are administered at less frequent intervals



CPG 11.4 MECHANICAL VENTILATION

Prehospital mechanical ventilation should be tailored to the individual patient's needs based on their presenting clinical condition. Mechanical ventilation has the advantage of providing consistent ventilations according to the selected settings, reduces the risk of hyperventilation and frees up manpower to perform additional tasks.

✓ INDICATIONS FOR MECHANICAL VENTILATION

- All patients that have been RSI'ed and have an ETT or LT in place
- Continuation of mechanical ventilation for interfacility transfers
- Non-invasive ventilation (NIV), for patients requiring PEEP and/or pressure support (CPAP and BiPAP) to prevent atelectasis, maximise oxygenation unresponsive to other modes of oxygenation such as in APO and COPD



MECHANICAL VENTILATION MODES

The following modes may be used for mechanical ventilation and adjusted according to the patient's needs (Oxylog 3000⁺):

- Intermittent positive pressure ventilation (IPPV) or control mandatory ventilation (CMV) – are to be used for intubated patients that have been paralysed. Activating the trigger function changes the mode to IPPV-assist, which synchronises ventilation to any spontaneous breaths that the patient may take.
- Synchronised intermittent mandatory ventilation (SIMV) – should be used for intubated patients that are spontaneously breathing. SIMV should be used for interfacility transfer patients or in patients that should not be paralysed.
- Pressure support ventilation (PSV) – provides additional inspiratory pressure during pre-set ventilation modes (SIMV or CPAP/BiPAP) in the spontaneously breathing intubated patient. This mode may be used for interfacility transfers.
- Non-invasive ventilation (NIV) – consists of CPAP or BiPAP and is used for the non-intubated, spontaneously breathing patient.



ASSESSMENT CONSIDERATIONS

- Tidal volume should be set based on the patient's ideal body weight and at a range of 6-8ml/kg. Tidal volume together with respiratory rate should be appropriately adjusted based on the patient's clinical condition and ventilation strategy. Tidal volume can be increased up to 10ml/kg in some settings if required, provided there is no lung injury.
- Higher than normal EtCO₂ levels are to be expected in polytrauma, asthma and post cardiac arrest patients.
- Hyperventilation in the prehospital setting should be avoided. Appropriate tidal volumes and respiratory rates should be set as close to the normal for the patient's age and according to the patient's condition.
- When using PEEP, the following should be considered and assessed:

- Start off with a low PEEP and increase as tolerated by the patient. Dial PEEP according to the comfort of the patient and desired oxygenation.
- Constantly check your plateau pressures to prevent barotrauma. As a general rule of thumb, aim to keep the plateau pressure below 30 cmH₂O.
- Monitor MAP closely when adjusting PEEP.

RECOMMENDED ADULT VENTILATOR SETTINGS FOR COMMON CONDITIONS

SETTINGS	NORMAL LUNGS	ASTHMA/COPD	APO/ARDS
Vent Mode	IPPV	IPPV	IPPV or NIV
Respiratory Rate	12-14 breaths/min*	5-10 breaths/min*	12-16 breaths/min*
Tidal Volume	6-8 ml/kg**	4-6 ml/kg	6-8 ml/kg
Pmax	35-40 mbar	40-45 mbar	40-45 mbar
I:E Ratio	1:2	1:4-1:5	1:1-1:2
PEEP	3-5 mbar	0-5 mbar	Up to 10 mbar
Pressure Support	N/A	N/A	CPAP: Up to 10 mbar BiPAP: Up to 20 mbar
Trigger	3 L/min	3 L/min	3 L/min
FiO ₂	1.0**	1.0**	1.0**

*Respiratory rate should be adjusted according to age for paediatric patients.

** Tidal volume can be increased up to 10 ml/kg if required.

***FiO₂ generally does not need to be adjusted from 1.0 in the acute phase, but may be applicable in some cases. During interfacility transfers, FiO₂ should be matched to the existing ventilator settings and adjusted as required.

SPECIAL NOTES

- For the asthmatic patient that has been intubated it is preferable to ventilate the patient using the BVMR due to air trapping. Once the chest has been sufficiently deflated, the mechanical ventilator may be connected.
- Routine use of the mechanical ventilator during cardiac arrest is not recommended due to the difficulties of synchronising ventilations in time with the indicator from the mechanical chest compression device.
- All patients that are intubated (except if in cardiac arrest) or that are administered a paralysing agent should be placed on a mechanical ventilator.

MECHANICAL VENTILATION

NIV – CPAP

- Select appropriate ventilation circuit
- Connect to oxygen and ensure sufficient supply
- Select appropriate size mask to fit patient
- Select CPAP settings
 - Ensure NIV is 'ON'
 - Start PEEP at 5 mbar. Increase as required to maximum of 10 mbar depending on patient's tolerance
 - Add pressure support as required to a combined maximum PEEP and P_{supp} of 20 mbar
- Coach the patient. Keep them calm. Secure the mask only once the patient is comfortable

CMV or SIMV

- Select appropriate ventilation circuit
- Connect to oxygen and ensure sufficient supply
- Conduct ventilator test cycle
- Input settings according to the ventilator settings table and patient's condition
- Adjust settings based on the patient's clinical response

NIV – BiPAP

- Select appropriate ventilation circuit
- Connect to oxygen and ensure sufficient supply
- Select appropriate size mask to fit patient
- Select BiPAP settings
 - Ensure NIV is 'ON'
 - Start IPAP (P_{insP}) at 10 mbar. Increase as required to maximum of 25 mbar depending on patient's tolerance
 - Start EPAP (PEEP) at 5 mbar. Increase as required to maximum of 10 mbar depending on patient's tolerance
 - Adjust FiO₂ and I:E as required
 - Add pressure support (P_{supp}) if patient requires further support – start at 3-5 mbar, increase by 2 mbar to max of 20 mbar P_{supp}
- Coach the patient. Keep them calm. Secure the mask only once the patient is comfortable

CPG 11.5 RSI QUICK REFERENCE GUIDE & CHECKLIST

ADULT QUICK REFERENCE GUIDE FOR PRIMAR PARALYSIS AND INDUCTION						
S = ± 50 -60 kg; M = ± 70 -90 kg; L = ≥ 100 kg	HAEMODYNAMICALLY UNSTABLE			HAEMODYNAMICALLY STABLE		
	S	M	L	S	M	L
FENTANYL (1 mcg/kg)	0	0	0	50	70	100
KETAMINE (1-2 mg/kg)	50	75	100	100	150	200
ROCURONIUM (1.5 mg/kg)	100	100	150	100	100	150
PAEDIATRIC QUICK REFERENCE GUIDE FOR PRIMAR PARALYSIS AND INDUCTION (8 – 13 YEARS OLD)						
S = ± 30 kg; M = ± 40 kg; L = ± 50 kg	HAEMODYNAMICALLY UNSTABLE			HAEMODYNAMICALLY STABLE		
	S	M	L	S	M	L
FENTANYL (1 mcg/kg)	0	0	0	30	40	50
KETAMINE (1-2 mg/kg)	30	40	50	60	80	100
ROCURONIUM (1.5 mg/kg)	45	60	75	45	60	75

ADULT QUICK REFERENCE GUIDE FOR ONGOING SEDATION AND PARALYSIS			
	S	M	L
FENTANYL (0.5 mcg/kg)	25	35	50
KETAMINE (0.5 mg/kg)	25	35	50
ROCURONIUM (0.25 mg/kg) 30 minutes following 1 st dose or as required	12.5	17.5	25
PAEDIATRIC QUICK REFERENCE GUIDE FOR ONGOING SEDATION AND PARALYSIS			
	S	M	L
FENTANYL (0.5 mcg/kg)	15	20	25
KETAMINE (0.5 mg/kg)	15	20	25
ROCURONIUM (0.25 mg/kg) 30 minutes following 1 st dose or as required	10	10	15

Calculate ideal body weight (IBW) in kg = height (cm) – 100

Round off drug doses to the nearest practical whole number

RSI CHECKLISTS

CHECKLIST 1 : Pre RSI - SOAPPPME				
SUCTIONING v				
Tubing attached	1		3	Yankaeur placed under (R) shoulder
System tested	2		4	Consider multiple suction units
OXYGENATION v				
Nasal Oxygen	1		4	BVM & OPA
NRM	2		5	Inline EtCO ₂
CPAP/BiPAP	3		6	O ₂ cylinders open
AIRWAY v				
Standard Laryngoscope & Blades	1		6	Gastric tube
McGrath & Blades	2		7	Front Of Neck Airway Kit
Gliderite & Bougie	3		8	Syringes attached
Primary & Alternate ETT sizes	4		9	Tube holder
Supraglottic Airway sized	5		10	Laryngoscopy plans A, B & C
POSITION v				
Preoxygenation position optimized	1		2	Intubation position optimized
PHYSIOLOGY v				
Preoxygenation optimized	1		3	Saturation end point
Hemodynamics optimized	2		4	MAP range
PHARMACY v				
Bilateral IVs	1		5	Ketamine dose
IVs patent	2		6	Rocuronium dose
IV fluid preload	3		7	Phenylephrine dose
Fentanyl dose	4		8	Adrenaline dose IV & IM
MONITORING v				
ETCO ₂	1		4	Manual BP cuff
SpO ₂	2		5	4/12 Leads
NIBP	3		6	Pads
EMERGENCY OTHER v				
Ventilator system tested	1		2	Ventilation parameters optimized
CHECKLIST 2 : Per RSI - 60 SECONDS COUNTDOWN				
Nasal Cannula @ 15l/min	1		5	Commence progressive laryngoscopy & intubation
Ensure pharyngeal patency	2		6	Call out O ₂ Sats, HR, Pulse Character, MAP, ECG
Acknowledge ETCO ₂ change & Apnea Alarm	3		7	Cycle NIBP after 60 secs, call out SBP, DP & MAP
Connect Inline ETCO ₂ once apnea is confirmed	4		8	ETT placement: Direct visual, ETCO ₂ waves X 5, A/E
CHECKLIST 3 : POST INTUBATION				
20-30 degree head up stretcher position	1		6	Inline nebulizer attached if applicable
Continuous waveform capnography	2		7	Inline Suction catheter attached
ETT depth, secured well but not kinked	3		8	Naso/Orogastric tube inserted
ETT cuff pressure optimal	4		9	Plan for High & Low-pressure ventilator alarms
Post RSI sedation/Paralysis	5		10	Plan for ventilator failure

Medication Dosing: Adjust doses according to hemodynamic stability, Shock Index, Age & Co-morbidities				
SHOCK INDEX (SI)	MEDICATION	PRIMARY DOSE REGIMEN Intravenous Push (IVP) Intra-venous/osseous Infusion (IV/IO INF)	SECONDARY DOSE REGIMEN Intravenous Push (IVP) Intra-venous/osseous Infusion (IV/IO INF)	
HR/SBP = SI	Fentanyl	INF: 1 mcg/kg Ideal Body Weight	IVP: 0.5mcg/kg	IV/IO INF: 1-10 mcg/kg/hr
Normal SI = < 0.7	Ketamine	IVP: 1 to 2 mg/kg Ideal Body Weight	IVP: 0.5 mg/kg	IV/O INF: 0.01-0.05mg/kg/min
SI > 0.8 Predicts Post RSI Crash Optimize Resuscitation	Rocuronium	IVP: 1.5 mg/kg Total Body Weight	IVP: 0.25 mg/kg	NOT for continuous infusion

Version 4.0



CPG 12.1-12.12

OBSTETRICS



CPG 12.1 ECTOPIC PREGNANCY

Ectopic pregnancy is as a result of the developing embryo implanting outside of the uterus, most commonly in the fallopian tubes. An ectopic pregnancy rupture is a life-threatening emergency, usually occurring between 6-10 weeks of gestation and requires rapid treatment and transport to hospital.



CLINICAL PRESENTATION

Patients with an unruptured ectopic pregnancy may present with the following:

- Amenorrhoea
- Vaginal bleeding
- Abdominal and/or pelvic pain
- Nausea
- Presyncopal symptoms

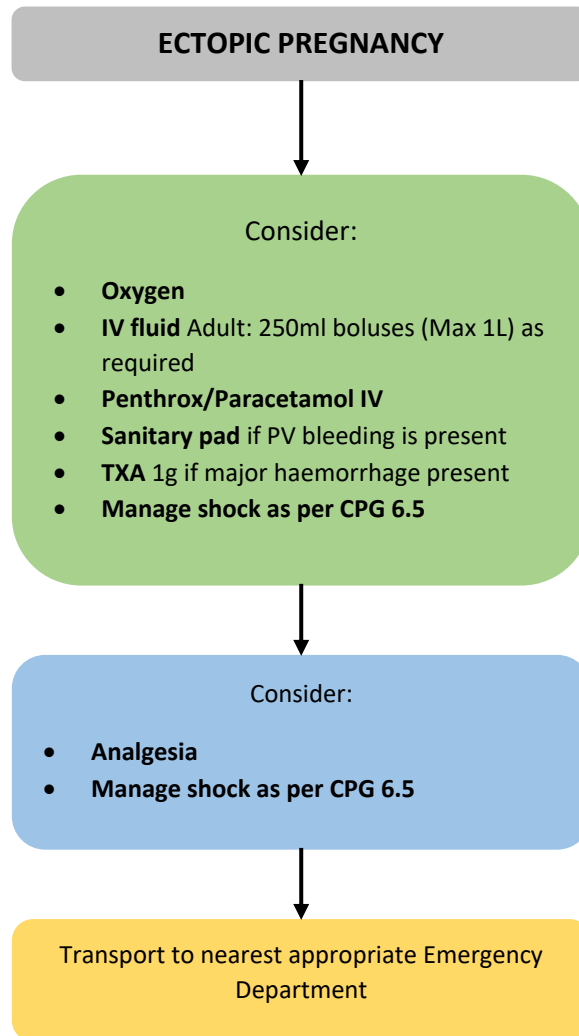
Patients with a ruptured ectopic pregnancy may present with the following:

- Shock
- Syncope
- Acute severe abdominal and/or pelvic pain
- Shoulder pain (Kehr's sign)
- Distended abdomen
- Abdominal guarding and/or rebound tenderness



ASSESSMENT CONSIDERATIONS

- Maintain a high index of suspicion for any female of child-bearing age presenting with the above clinical signs.
- Potential risk factors for ectopic pregnancy include:
 - Pelvic inflammatory disease
 - Smoking
 - Endometriosis
 - In vitro fertilisation/ fertility treatments
 - Sexually transmitted infections
 - Previous history of ectopic pregnancy
 - Tubular surgery
 - Intrauterine contraceptive devices



CPG 12.2 MISCARRIAGE

A miscarriage is defined as the spontaneous loss of the pregnancy before 20 weeks of gestation or a foetal weight of <400 grams. Miscarriage may result in ante-partum haemorrhage and lead to high-risk complications such as haemorrhagic shock and uterine sepsis.

CLINICAL PRESENTATION

Patients with a potential miscarriage may present with the following:

- Lower abdominal pain/discomfort
- PV bleeding
- Hypotension
- Tachycardia
- Pallor
- Dizziness
- Postural symptoms

Signs of a possible intrauterine infection may include the following:

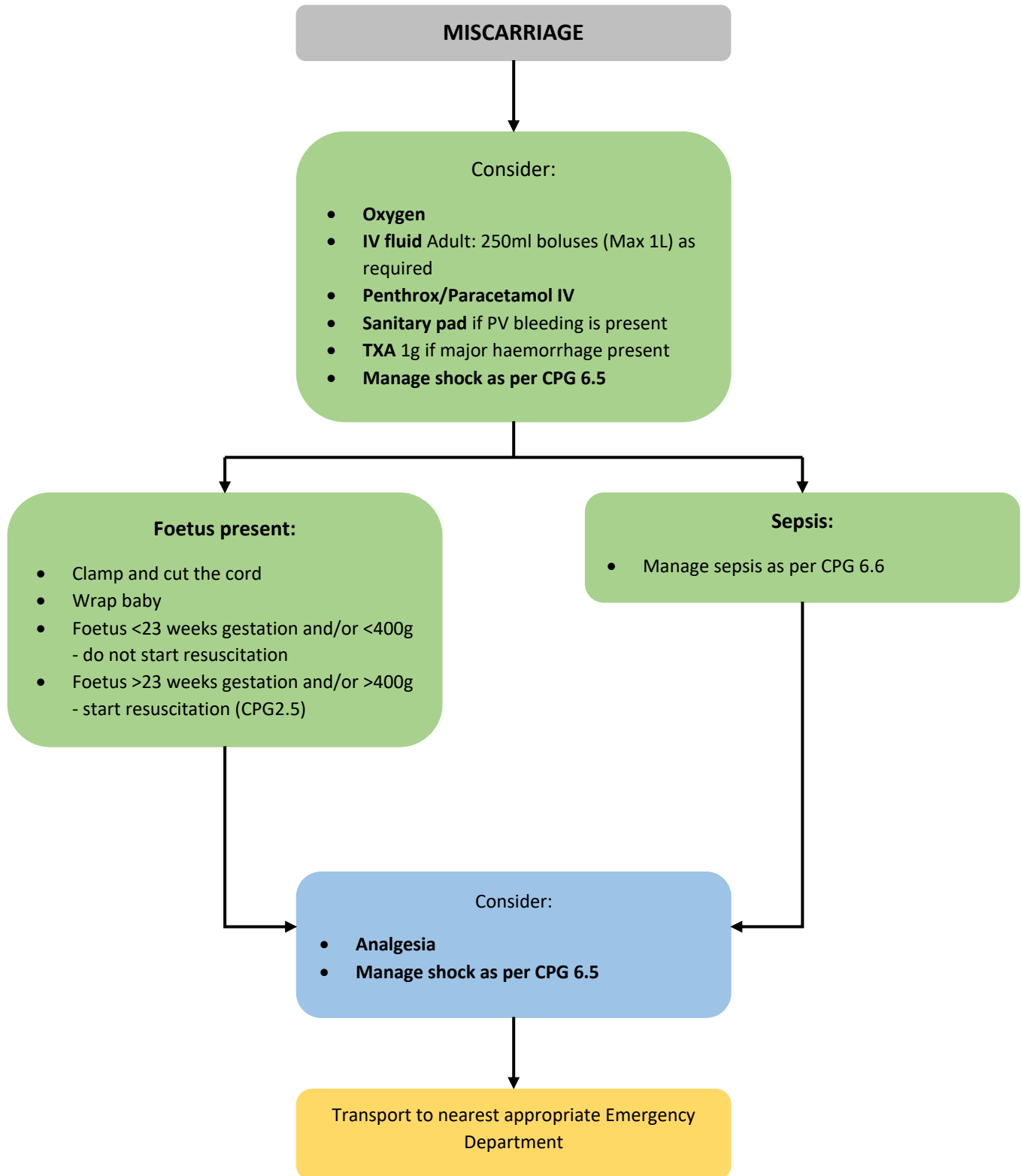
- Fever
- Severe pelvic pain/rigidity/guarding
- Purulent discharge

ASSESSMENT CONSIDERATIONS

- A miscarriage diagnosis can be made only if products of conception (foetus) have been expelled and confirmed.
- If no products of conception have been expelled at the time of assessment, consider alternative differential diagnosis (ectopic pregnancy, normal pregnancy related bleeding, trauma) in addition to miscarriage.
- If possible, collect all expelled tissue and transport with the patient to the receiving facility.
- Risk factors for miscarriage include advanced maternal age, stress, maternal infection, diabetes, smoking, severe hypertension, alcohol consumption, fibroids, renal disease, caffeine, uterine dysfunction, untreated thyroid disease.

SPECIAL NOTES

- A miscarriage that occurs after the first trimester may result in a foetus being expelled from the vagina. Should this occur, clamp and cut the cord, wrap the baby in a blanket.
- Be mindful that this is a psychologically and emotionally distressing event for the mother. Provide appropriate support and counselling to the mother.
- Some pre-term infants less than 20 weeks of gestation may display signs of life, such as gasps or movement. Resuscitation in such cases is futile.
- Resuscitation should not be started on pre-term infants that have a gestational age of less than 23 weeks and/or birthweight of <400 grams.



CPG 12.3 PLACENTA ABRUPTION

Placenta abruption is defined as the premature separation (partially or completely) of the placenta from the uterine wall before delivery of the foetus and should always be considered in the event of vaginal bleeding after 20 weeks gestation.

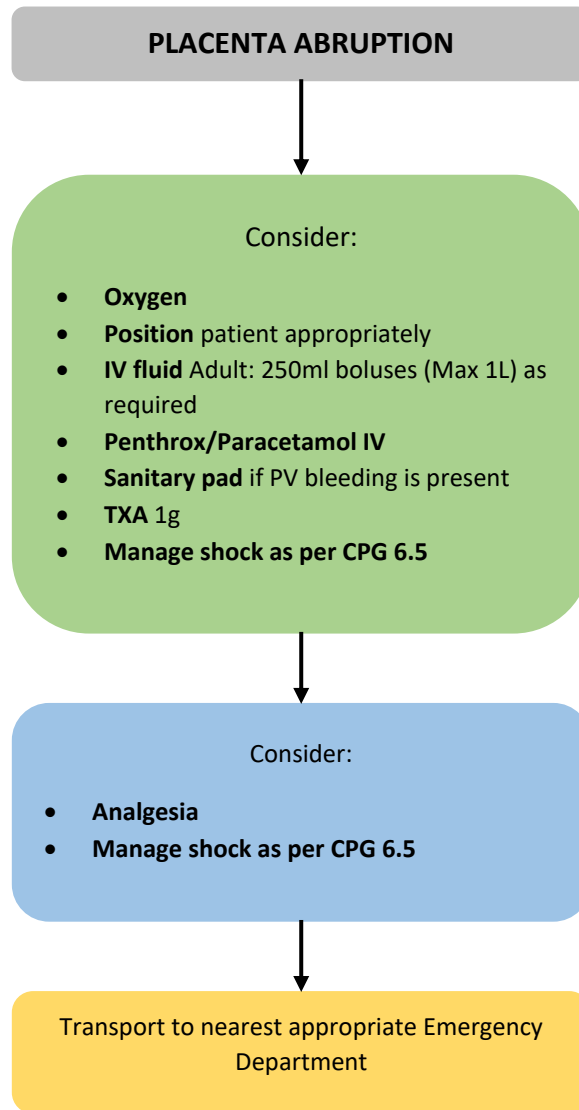
CLINICAL PRESENTATION

Patients with a placenta abruption may present with the following:

- Abdominal/pelvic pain
- PV bleeding (bleeding may be concealed)
- Tetanic uterine contractions (contractions lasting longer than 90 seconds)
- Rigid uterus
- Signs of shock

ASSESSMENT CONSIDERATIONS

- Placenta abruption should be considered as a diagnosis in all pregnant women with abdominal pain, even without clinically obvious signs and PV bleeding.
- Risk factors for placenta abruption include:
 - Gestational hypertension
 - Pre-eclampsia
 - Previous history of abruption or caesarean section
 - Multiparity
 - Advanced maternal age
 - Intrauterine infection
 - Drug use
 - Recent trauma
 - Smoking



CPG 12.4 PLACENTA PRAEVIA

Placenta praevia occurs when the placenta implants either partially or completely in the lower uterine segment. This may result in some degree of placental separation and bleeding. Should the placenta completely cover the internal os, it will physically prevent normal vaginal delivery.

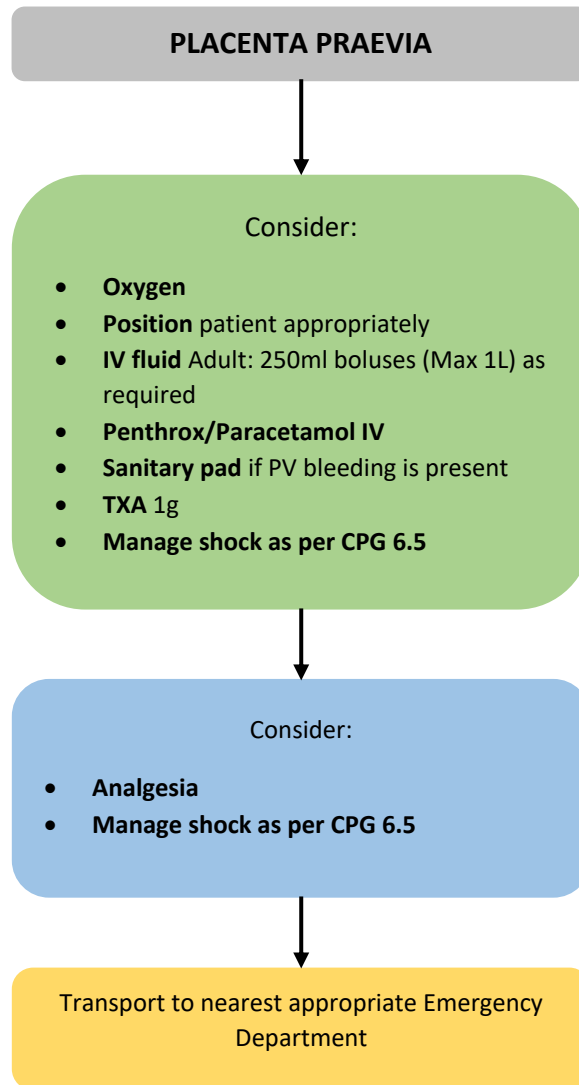
CLINICAL PRESENTATION

Patients with a placenta praevia may present with the following:

- No pain, except the pain associated with contractions
- Bright red blood and several small bleeds
- Soft, non-tender uterus
- Signs of shock (if significant blood loss occurs)

ASSESSMENT CONSIDERATIONS

- Placenta praevia should be considered as a diagnosis in all pregnant women with PV bleeding after 20 weeks gestation.
- Risk factors for placenta praevia include:
 - Short interpregnancy intervals
 - Trauma
 - Previous history of uterine surgery
 - Multiparity
 - Advanced maternal age
 - Fertility treatment
 - Drug use
 - Previous caesarean section



CPG 12.5 PRE-ECLAMPSIA AND ECLAMPSIA

Pre-eclampsia is a multisystem disorder that occurs during pregnancy after 20 weeks gestation and up to 6 weeks post-partum. Pre-eclampsia develops into eclampsia when one or more seizures occurs. Pre-eclampsia and eclampsia can lead to placenta abruption, hepatic and renal failure, and cerebral haemorrhage.

CLINICAL PRESENTATION

Patients with pre-eclampsia are diagnosed by the following:

- Systolic BP ≥ 140 mmHg and/or
- Diastolic BP ≥ 90 mmHg
- Plus, one or more of the following:
 - Neurological
 - Visual disturbances
 - Altered mental status
 - Severe headaches
 - Hyperreflexia
 - Clonus
 - Respiratory
 - Acute pulmonary oedema
 - Cardiovascular
 - Generalised oedema
 - GI disturbances
 - Epigastric pain
 - RUQ tenderness
 - Nausea and/or vomiting
 - Jaundice
- Eclampsia occurs when:
 - The pre-eclamptic patient experiences one or more seizure

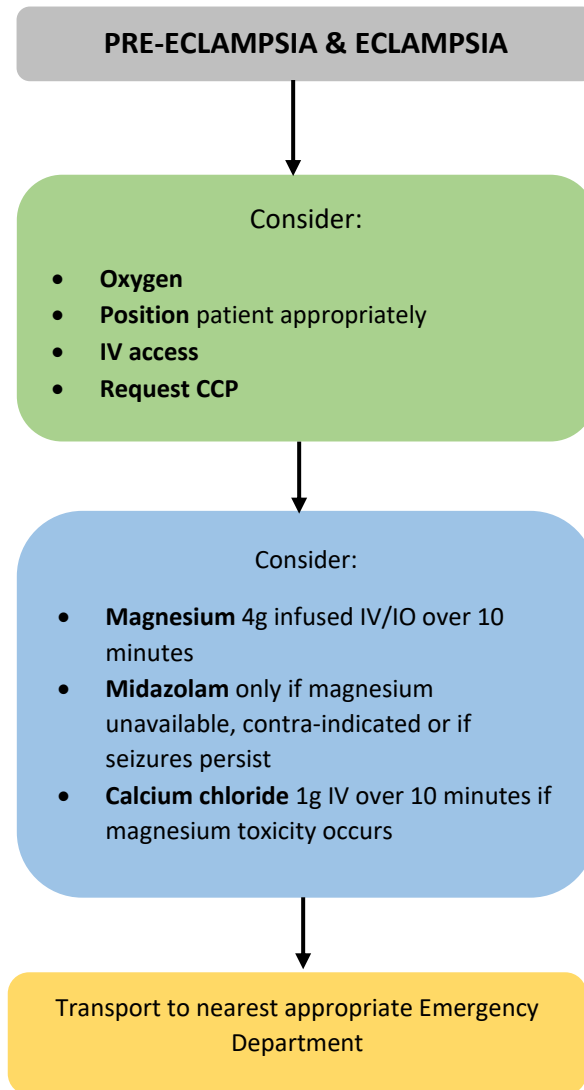
ASSESSMENT CONSIDERATIONS

- Treat all patients with pre-eclampsia as high risk for eclampsia
- Assess the patient for pre-eclampsia risk factors, which include:
 - Primigravida
 - Previous history of pre-eclampsia
 - Gestational hypertension
 - Advanced maternal age
 - Renal disease
 - Diabetes
 - Obesity
 - Multiple pregnancies
- Should pre-eclampsia be suspected, monitor vital signs closely, particularly BP
- Consider comparing auto-BP reading with manual BP reading.

- Keep the patient in a position of comfort, preferably in the lateral position
- Avoid fluid bolus administration due to risk of pulmonary oedema. Fluids should be limited to KVO.
- Note that should these patients require advanced airway management, a difficult airway should be anticipated.
- The pregnant patient that experiences a seizure should be assessed thoroughly and a detailed history taken (in particular a history of epilepsy) to differentiate between eclampsia and epilepsy. Pregnant epileptic patients have a reduced seizure threshold.

SPECIAL NOTES

- The pre-eclamptic/eclamptic patient should preferably be transported P2 to hospital. If P1 transport is required, avoid using the siren as this may precipitate seizures.
- Do not allow these patients to walk.
- Once magnesium has been administered, closely monitor the patient for signs of magnesium toxicity:
 - Nausea, hot flushes, weakness
 - Slurred speech, confusion, blurred vision
 - Loss of deep tendon reflexes/absent patellar reflexes
 - Hypotension
 - Respiratory depression/arrest
 - Cardiac arrhythmias
 - Chest pain
- APs and CCPs should monitor the patient for signs of magnesium toxicity. If magnesium toxicity occurs, stop the magnesium infusion. Consider calcium chloride should symptoms persist.



CPG 12.6 UMBILICAL CORD PROLAPSE

Umbilical cord prolapse can occur after rupture of the membranes and the umbilical cord slips down in front of the presenting part of the foetus, protruding into the vagina. Management of an umbilical cord prolapse requires rapid recognition, intervention and transport.

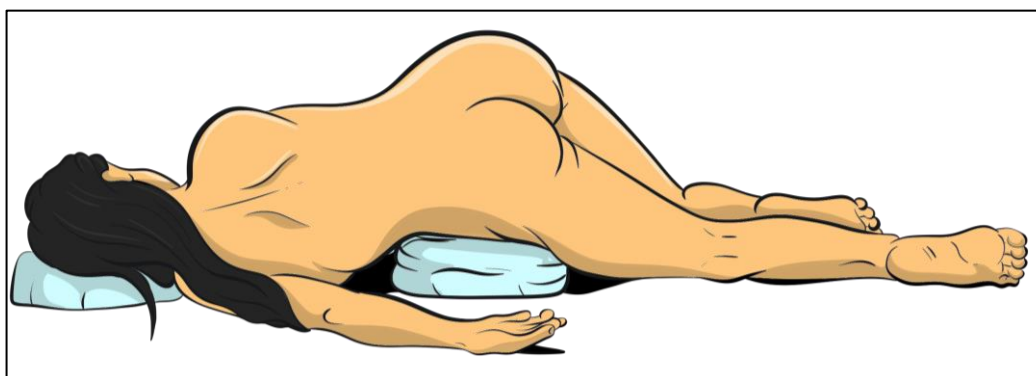
CLINICAL PRESENTATION

Patients with an umbilical cord prolapse present with the following:

- Umbilical cord visible at the vagina on inspection
- Ruptured membranes
- Meconium staining
- Change in foetal movement

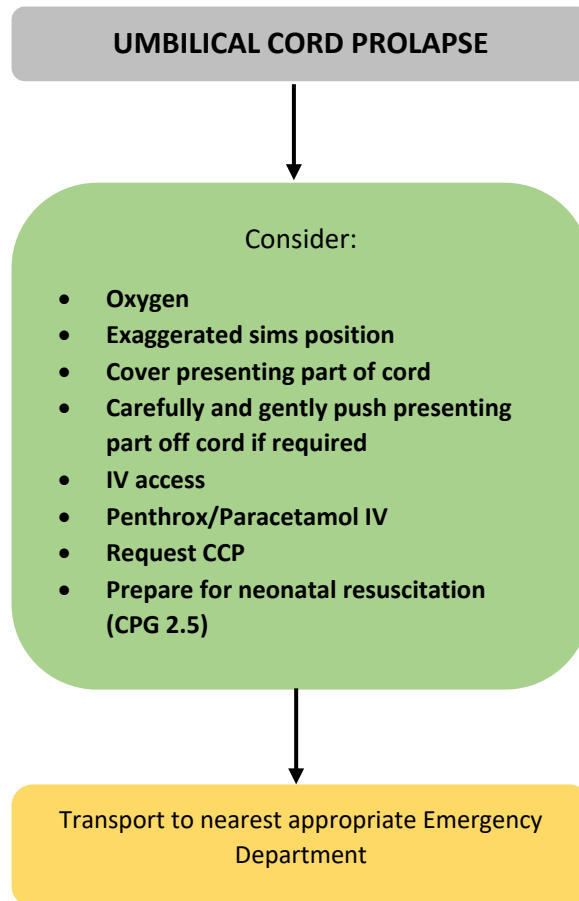
ASSESSMENT CONSIDERATIONS

- Monitor the patient's labour state. Cord prolapse and associated risks worsen as labour progresses.
- Umbilical cord prolapse should be ruled out for all pregnant patients who have ruptured membranes either by visually inspecting or by asking the patient to feel for the cord.
- Constantly monitor the prolapsed umbilical cord for pulsations. Should pulsations stop, alleviate the pressure on the cord by manoeuvring the patient into a different position or by gently pushing the presenting part of the foetus off the cord. Once you feel the cord pulsating, maintain that position until handover at the hospital.
- Cover the cord with a warm moist pad (use sterile water) and keep handling of the cord to a minimum.
- Avoid using the stair chair for these patients if possible. These patients should ideally be moved using the stretcher or walk the patient to the stretcher if necessary.
- Patients with an umbilical cord prolapse should be transported in the *exaggerated sims position*.



SPECIAL NOTES

- Caesarean section is the recommended mode of delivery. Ensure prenotification is provided to the appropriate receiving hospital in order for them to adequately prepare.
- Should delivery be imminent, stop and deliver the baby. Prepare for resuscitation of the neonate.



CPG 12.7 IMMINENT DELIVERY

Recognition of imminent delivery is important to adequately prepare for the delivery of the new-born and manage potential complications. Labour is classified into three stages. **1st stage**: regular contractions of the uterus and full dilation of the cervix. **2nd stage** commences when the cervix is fully dilated and concludes when the new-born is delivered. The **3rd stage** is the delivery of the placenta.

CLINICAL PRESENTATION

Signs of imminent delivery may include the following:

- Increasing frequency and severity of contractions
- Urge to push
- Ruptured membranes
 - In some cases, active rupture of the membrane may be required if the head has been delivered with the membrane still intact
- Bulging perineum
- Crowning – presenting part visible

Signs of placental separation following delivery may include:

- Lengthening of the umbilical cord
- Firming of the uterus
- Gush of blood
- Contractions

ASSESSMENT CONSIDERATIONS

- Gather an appropriate antenatal history prior to delivery of the baby if possible, to pre-empt potential delivery complications.
- Prepare all equipment for delivery in an aseptic technique.
- Keep the mother in a position of comfort for delivery.
- Manage the delivery according to the skill listed in the Clinical Procedure Manual.
- Ensure the time of birth is recorded.
- Once the head has been delivered, finger sweep around the neck to check for the presence of the cord. If the cord is present around the neck, gently slide the cord back over the baby's head.

New-born care considerations should include:

- Thoroughly dry and wrap the new-born in the blanket provided in the maternity prepack. Ensure the head is covered using the beanie provided.
- Assess the APGAR score at 1 and 5 minutes.
- Closely monitor the temperature of the new-born.
- Clamp the cord 30 seconds to 3 minutes post-delivery in the vigorous baby. Do not immediately clamp and cut the cord unless resuscitation is required, in which case wait 60 seconds before clamping and cutting the cord.
- Clamp the cord at 10-15 and 20 cm from the new-born and cut the cord between the clamps at 15 and 20 cm.

- Allow the mother to hold the baby (skin to skin contact) and encourage breast feeding if resuscitation is not required.



SPECIAL NOTES

- Equipment preparation for imminent delivery should include the setup of the neonatal BVM in preparation for possible resuscitation.
- Assist the mother to deliver the placenta naturally. Do not pull on the cord.
- Do not wait on scene to deliver the placenta.
- Transport the placenta (if delivered) together with the mother and baby for inspection by the hospital team.
- Monitor the mother for major haemorrhage and signs of shock.
- Pentrox may be administered to the mother during labour. Avoid administering Pentrox to the mother during the phase of second stage labour where the mother is required to push to deliver the baby.

IMMINENT DELIVERY

- **Position** of comfort for mother
- Assess for **signs of imminent delivery**
- Prepare **equipment** for delivery and neonatal resuscitation
- Consider **oxygen**
- Consider **IV access**
- Consider **Penthrox/Paracetamol IV**
- Prepare for potential **complications**

- Ask mother to pant during delivery of the head – support the baby's head.
- Once head has been delivered – **assess for cord around the neck**. If the cord is around the neck:
 - Attempt to slip cord over head or
 - Slip cord over baby's shoulders and deliver baby through the cord
- After the head has delivered, encourage mother to push during contractions to deliver shoulders and rest of body – support the head and body of baby.

Postnatal care:

- Thoroughly **dry, warm and stimulate** the newborn – record time of delivery
- **Suction** if required
- **Assess breathing** – if breathing effectively continue routine management. If breathing is ineffective refer to resuscitation of the newborn CPG 2.5
- Wrap baby in provided baby blanket and beanie. **Keep the baby warm**
- Assess **APGAR** at 1 and 5 minutes
- Monitor mothers' vital signs and monitor for potential complications (major bleeding)
- **Clamp cord 30 seconds - 3 minutes after birth**. Immediate cord clamping should only occur if resuscitation is required.
- **Clamp the cord** at 10-15 and 20 cm from the newborn then **cut cord** between clamps at 15 and 20 cm

3rd Stage labour management:

- Encourage skin to skin contact between mother and baby while ensuring baby is kept warm.
- Encourage breastfeeding
- Assist mother to deliver placenta naturally – do not apply traction to the cord
- Once placenta has delivered, place in bag and transport to hospital with mother and baby for inspection

Transport to nearest appropriate Emergency Department

CPG 12.8 BREECH PRESENTATION

Breech birth occurs when either the feet or the buttocks of the foetus enter the birth canal first. The most common presenting breech births include complete, frank, footling and knee presentations.

Frank Breech (65%)	Complete Breech (10%)	Incomplete Breech (25%)	
		Footling Breech	Kneeling Breech
			
The baby's hip joints are flexed and knee joints are extended.	The baby's hip and knee joints are flexed.	The baby's hip and knee joints are extended on one or both sides.	The baby's hip joints are extended and knee joints are flexed on one or both sides.

Breech birth poses a high risk to both the foetus and mother. Prepare for resuscitation. Breech presentations should be rapidly recognised and managed as required, with transport initiated to an appropriate facility as soon as practically possible.

CLINICAL PRESENTATION

Signs of breech delivery may include the following:

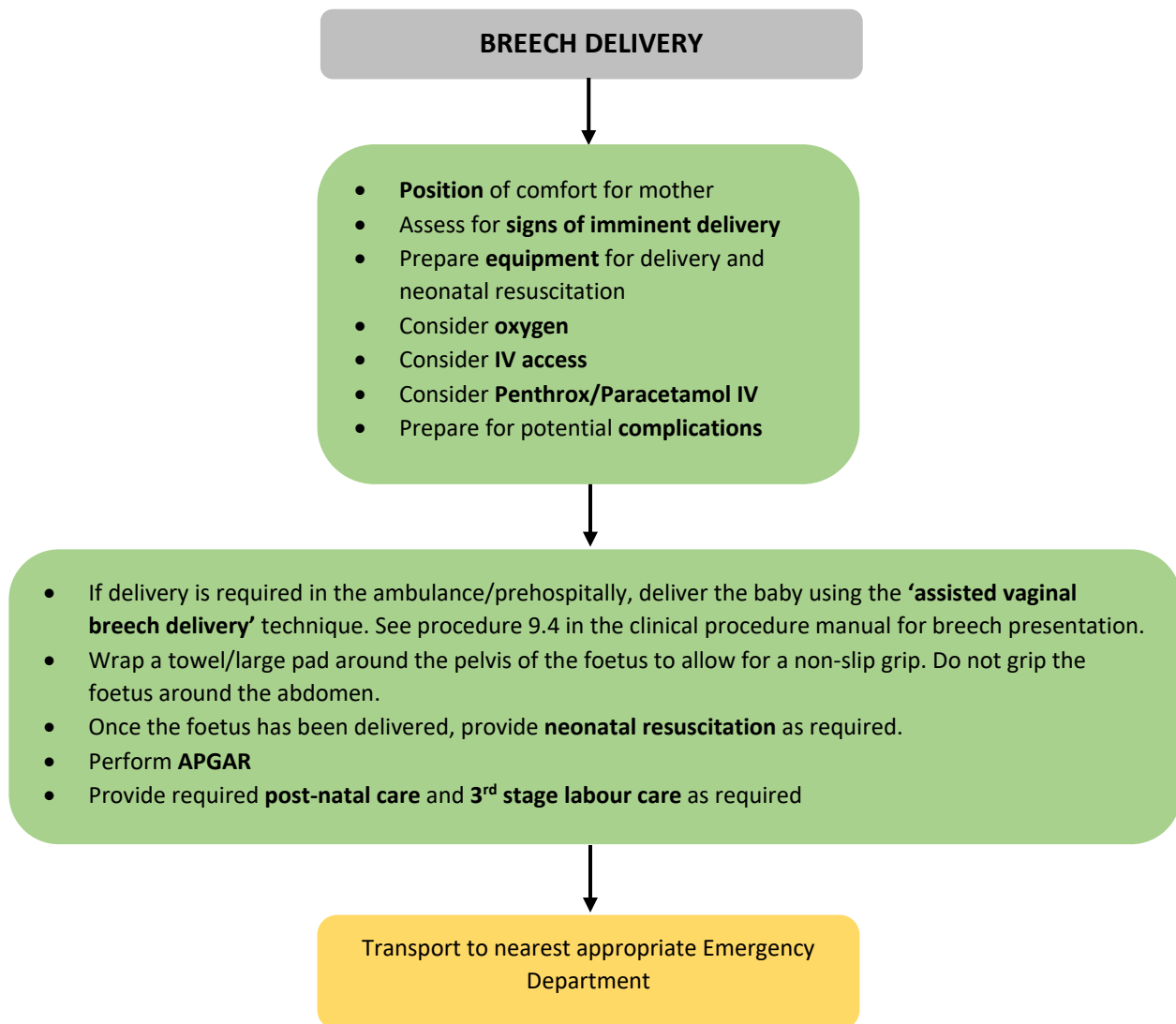
- Appearance of presenting part (depending on type of breech presentation)
- Contractions
- Ruptured membranes
- Signs of imminent delivery

ASSESSMENT CONSIDERATIONS

- Obtain a detailed obstetric history to determine the potential risk of a breech delivery if the presenting part is not showing yet.
- Assess for potential complications of breech delivery
 - Prolapsed cord
 - Meconium staining/aspiration
 - Foetal hypoxia
 - Postpartum haemorrhage
 - Foetal head entrapment
- Do not use rough handling when delivering the baby as this may result in serious injury.

SPECIAL NOTES

- Prepare for neonatal resuscitation if breech presentation is identified.
- Ensure CCD is updated in order for the receiving facility to be prepared for the arrival of a breech delivery.
- Avoid administering Pentrox to the mother during the phase of second stage labour where the mother is required to push to deliver the baby.



CPG 12.9 PRE-TERM LABOUR

The principles of managing pre-term labour are to identify the onset of pre-term labour, provide appropriate pain relief as required and avoid causing any unnecessary distress to the mother.

CLINICAL PRESENTATION

Signs of pre-term labour may include the following:

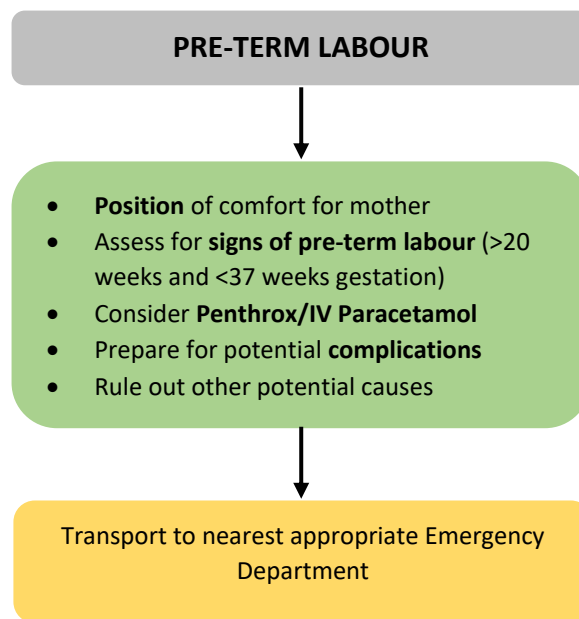
- Onset of uterine contractions after 20 weeks gestation and before 37 weeks gestation
- Uterine contractions that are regular and persistent
- Spontaneous rupture of membranes
- Bleeding or bloody discharge
- Persistent lower back ache

ASSESSMENT CONSIDERATIONS

- Obtain a detailed obstetric history
- Risk factors associated with pre-term labour include:
 - Premature rupture of membranes
 - Multiple pregnancies
 - Assisted conception
 - Infection
 - Trauma
 - Stress
 - Obstetric complications (maternal hypertension, foetal abnormalities and previous miscarriages)
- Monitor mother closely for potential complications such as major haemorrhage

SPECIAL NOTES

- Provide analgesia as required to the mother (Penthrox or IV Paracetamol)
- If delivery is imminent, stop and deliver the baby as per the imminent delivery CPG. Prepare for newborn resuscitation.
- Provide necessary prenotification as required



CPG 12.10 POST DELIVERY (3RD STAGE) COMPLICATIONS

3rd stage labour complications include postpartum haemorrhage (PPH), uterine inversion and retained placenta. These complications can be life-threatening and require rapid recognition, intervention and transport to an appropriate facility. Primary PPH is defined as blood loss of 500ml or more from the genital tract. Secondary PPH is defined as abnormal/excessive bleeding 24 hours to 12 weeks postpartum. Retained placenta is defined as placenta that is not expelled within 30 minutes of the start of the 3rd stage of labour. Uterine inversion occurs when the uterus collapses in on itself and this occurs in varying degrees.

CLINICAL PRESENTATION

Signs of primary postpartum haemorrhage may include:

- PV bleeding of greater than 500ml post-delivery.
- Signs of hypovolaemic shock
- Poor fundal tone/fundal tone above the umbilicus

Signs of secondary postpartum haemorrhage may include:

- Increase in the volume of lochia. Lochia is the normal discharge from the uterus following delivery (e.g. more than 1 soaked pad per hour).
- Bleeding (will change from dark red to bright red)
- Offensive lochia odour
- Fever
- Uterine tenderness

Signs of retained placenta may include:

- Placenta not expelled within 30 minutes following the start of the 3rd stage of labour
- PV bleeding
- Signs of hypovolaemic shock

Signs of uterine inversion may include:

- PV bleeding
- Signs of hypovolaemic shock
- Severe abdominal/pelvic pain
- Visual inspection may reveal mass visible at the vulva (in the case of a prolapsed uterine inversion)

ASSESSMENT CONSIDERATIONS

- Pregnant mothers are often able to tolerate large volumes of blood loss before showing signs of clinical deterioration.
- Assessing the volume of blood loss can be challenging and is often underestimated.
- Use aseptic techniques as these patients are potentially at high risk for infection.
- Uterine inversion is rare and difficult to assess:
 - Incomplete – fundus reaches the cervix
 - Complete – fundus passes through the cervix but does not reach the vaginal opening
 - Prolapsed – the fundus extends through the vaginal opening
- Significant bleeding (>1500ml) often occurs before signs of haemodynamic compromise manifest.



SPECIAL NOTES

- The T's of primary and secondary postpartum haemorrhage management include:
 - Tone – uterine atony
 - Trauma – cervical and genital tract damage post delivery
 - Tissue – retained products
 - Thrombin – coagulation disorder
 - Temperature – this is specific to secondary postpartum haemorrhage. Sepsis should be considered in these patients presenting with fever.
- In the event of a uterine inversion and the placenta and/or uterus is exposed, cover with moist sterile dressing. Do not attempt to replace the inverted uterus.
- Prenotification should be provided to the receiving facility.

3RD STAGE COMPLICATIONS

- **Position** of comfort for mother
- Assess for **signs of major haemorrhage**
- **TXA** 1g if major haemorrhage present
- Consider **IV fluid** 250-500ml Saline boluses (Max 1L)
- Consider **Penthrox/IV Paracetamol**

- **Uterine Inversion:** protect any exposed uterus with moist sterile dressing.
- **Primary and Secondary postpartum haemorrhage:** manage potential causes
 - **Tone** – consider fundal massage
 - **Trauma** – control external bleeding
 - **Tissue and Thrombin** – consider fundal massage

Transport to nearest appropriate Emergency Department

CPG 12.11 APGAR SCORE

The APGAR score is a tool used to determine the condition of a newborn after delivery and determine whether the newborn requires emergency care. All newborns are to have an APGAR score calculated at **1 minute** and **5 minutes** following delivery. Assign a score between 0 – 2 points for each of the five criteria to calculate a total score of between 0 – 10 points. The APGAR score at 1 minute will not be 10, the score should improve over the next 5 minutes.

APGAR SCORE				
	Action	0	1	2
Appearance	<i>Look at skin colour</i>	Blue/pale	Pink (extremities blue)	All pink
Pulse	<i>Count heart rate</i>	Absent	<100	>100
Grimace	<i>Monitor response</i>	No response	Grimace	Vigorous cough/cry
Activity	<i>Look at muscle tone</i>	Limp	Some flexion/extension	Active motion
Respiration	<i>Count and asses</i>	Absent	Slow/irregular	Good cry

ASSESSMENT CONSIDERATIONS

APGAR score of:

- 0 – 3 indicates severe distress
- 4 – 7 indicates moderate distress
- 7 – 10 indicates no distress (normal)

Manage the newborn in distress according to CPG 2.5

CPG 12.12 OBS/GYNE BYPASS CRITERIA

OBSTETRICS & GYNAECOLOGICAL RELATED CONDITIONS/EMERGENCIES

- Patients of any age with an obstetric or gynaecological emergency, with or without complications:
 - Active labour of any gestation
 - Abnormal delivery presentation
 - Prolapsed cord
 - Threatened abortion/ectopic pregnancy
 - Major vaginal haemorrhage (non-traumatic)
 - Pre-eclampsia and eclampsia

NEAREST
OBS/GYN ED

MEDICAL CONDITIONS/EMERGENCIES IN THE PREGNANT PATIENT

All medical conditions or emergencies **NOT** related to Obs/Gyne complications

CLOSEST
APPROPRIATE ED

TRAUMA EMERGENCIES IN THE PREGNANT PATIENT

Any pregnant patient meeting trauma bypass criteria (see trauma bypass criteria CPG)

HIGH TRAUMA
ROOM



CPG 13.1-13.5

OTHER



CPG 13.1 PAIN MANAGEMENT

Pain can be complex to manage. The perception of pain is subjective, varies between patients and is influenced by culture, environment, mood, coping ability and previous pain experiences. Effective pain management is an important part of prehospital patient care requiring appropriate pain assessment, appropriate provision of analgesia determined by a pain assessment and measuring effectiveness of pain management by reassessing the patients' pain after analgesia administration.

CLINICAL PRESENTATION

Signs of pain experienced by a patient may include:

- Self-reported pain
- History of a potentially painful injury or condition
- Distress
- Dilated pupils
- Nausea, vomiting
- Increased heart rate, blood pressure and respiratory rate
- Pallor, muscle tension, guarding and sweating

ASSESSMENT CONSIDERATIONS

- Assessment of pain requires a systematic approach using observation, physical examination, assessment of vital signs and questioning the patient (if possible) regarding their pain.
- All patients should have a pain assessment conducted if indicated and documented in the ePCR.
- Consider the mnemonic OPQRST as part of the pain assessment:
 - **Onset:** "Did your pain start suddenly or gradually get worse?"
 - **Provocation:** "What makes your pain better or worse?"
 - **Quality:** "What does your pain feel like?"
 - **Radiation:** "Point to where the pain is the worst. Where does the pain go from there?"
 - **Severity:** "How bad is your pain?"
 - **Time:** "How long ago did the pain start?"
- HMC Policy CL6078 on Pain screening, assessment and management includes various pain scales of which some are applicable to EMS practice – numeric rating scale and the Wong-Baker FACES pain scale.



- A pain assessment tool is of particular importance if the patient is unable to communicate due to altered LOC, language barrier, cognitive impairment and children.

- All patients should have their pain reassessed and documented in the ePCR following management to gauge the effectiveness of the pain relief provided.
- In certain cases, it may not be possible to completely remove the patient's pain due to the nature of the patient's injuries, movement and transport. Aim to achieve a manageable pain level for the patient and make the patient as comfortable as possible.

SPECIAL NOTES

- The following patients are at risk of under reporting pain are therefore at risk of having pain under treated:
 - Elderly, children and infants
 - Patients with mental health issues
 - Language barrier
 - Altered LOC
- Consider multimodal analgesia to manage pain.
- Certain medications for the management of pain such as Fentanyl and Ketamine can result in sedation of the patient which may result in the loss of protective airway reflexes or ability to independently maintain adequate ventilatory function. These effects are often unpredictable and can be specific to the individual, regardless of the dose administered. Staff should be prepared to appropriately monitor for and manage sedation levels deeper than what was intended. For this reason, the following are required as a minimum standard:
 - Equipment to provide resuscitation must be available at all times.
 - ECG, RR, BP, EtCO₂ and SpO₂ must be continuously monitored any time the patient receives narcotics, benzodiazepines, or dissociative agents by any route.
 - If conscious sedation is being performed, 2L oxygen via nasal cannula should be considered.
 - For inhaled analgesics (Penthrox) continuous SpO₂ and HR monitoring is recommended. EtCO₂ monitoring is not required.
- The use of analgesics carries the risk of adverse effects. Consideration should be given to the following when selecting an appropriate analgesic:
 - Physiological factors – geriatrics, paediatrics and patients with reduced drug clearance (renal or liver impairment).
 - Disease factors – hypovolaemia, respiratory compromise, head injury.
 - Previous history of an adverse response to an analgesic.

PAIN MANAGEMENT

Consider:

- Assess **severity of pain**
- **Minimum monitoring requirements** dependent on medication being administered.
- **IV access**
- **Multimodal analgesia** – combination of medications
- **Supplemental oxygen**
- **BVMR** – have prepared and on standby if administering higher doses of controlled medications
- **Manage underlying cause where possible**

Non-pharmacological treatments:

- **Reassurance** – fear and anxiety worsen the pain experience; explanation and reassurance may aid with the treatment of pain.
- **Splinting** - splinting of fractures may aid relief of pain as well as minimises further trauma and bleeding.
- **Dressings** - burn dressings aid further cooling the burn area and alleviate pain via this mechanism. Care should be taken not to cool burns for more than 20 minutes and prevent the development of hypothermia in large surface area burns.

Pharmacological treatments (see formulary for specific dosing, indications, contra-indications and precautions):

- **Methoxyflurane (inhaled)**
- **Paracetamol (Oral, IV)**
- **Ibuprofen (Oral)**
- **Ketorolac**
- **Diclofenac**

Pharmacological treatments (see formulary for specific dosing, indications, contra-indications and precautions):

- **Fentanyl**
- **Ketamine**
- **Midazolam**

CPG 13.2 MULTIPLE CASUALTY INCIDENTS

The following serves as a quick reference guide for the various triage tools used at a multiple casualty incident (MCI) and the initial reports to provide (WINDSCREEN Report and METHANE Report). For detailed MCI information and action plans refer to the HMCAS Major Incident Response Guide.

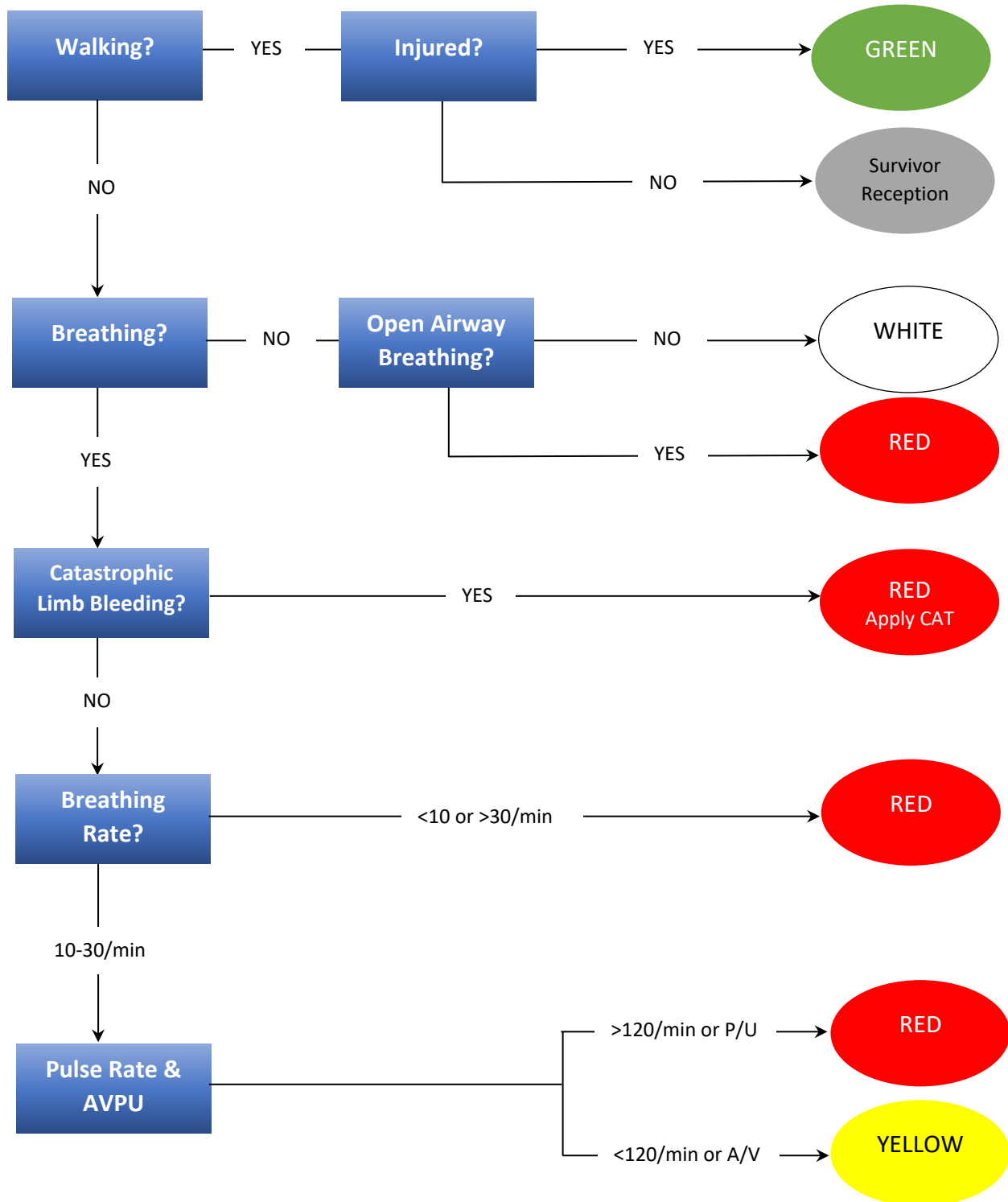
WINDSCREEN REPORT

The WINDSCREEN report is the initial scene size-up performed by the first unit arriving on scene.

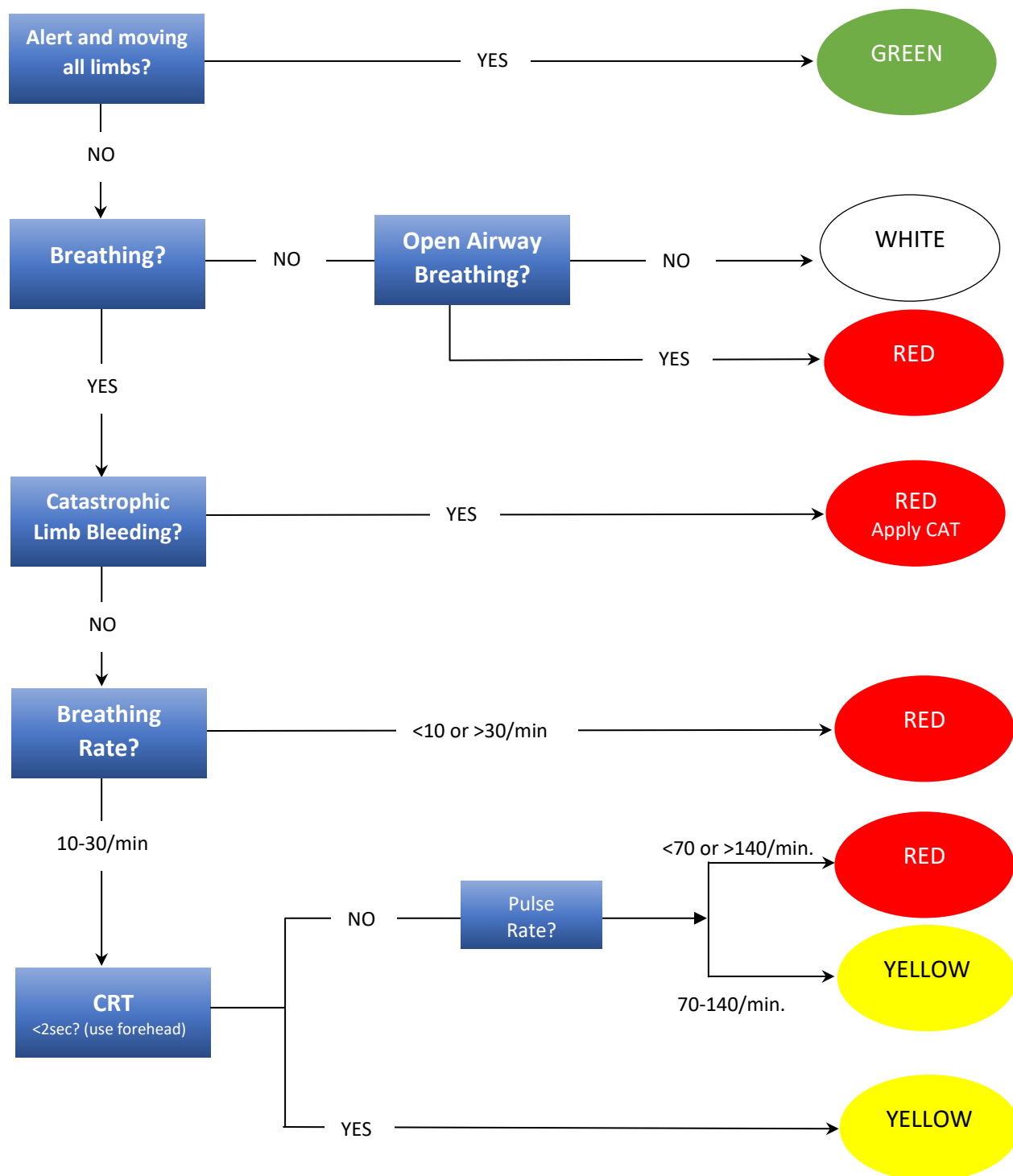
METHANE REPORT

M	Major incident declared with appropriate level (Level 0/1/2/3)
E	Exact location with map reference if possible
T	Type of incident with details of structure and/or vehicle types, size, number
H	Hazards present and potential
A	Access routes and suitable rendezvous point (RVP)
N	Number of patients and severity
E	Emergency services present and required, including equipment

ADULT PRIMARY TRIAGE ALGORITHM (>140cm)



PAEDIATRIC PRIMARY TRIAGE <140cm



SECONDARY TRIAGE

Physiological Variable	Measured Value	Score	Measured Value	Score	Insert Score	
	ADULT		PAEDS			
Respiratory Rate	10-29	4	10-24	4	A	
	>29	3	25-35	3		
	6-9	2	> 35	2		
	1-5	1	< 10	1		
	None	0	None	0		
Systolic Blood Pressure	≥ 90	4	> 90	4	B	
	76-89	3	70-90	3		
	50-75	2	50-69	2		
	1-49	1	< 50	1		
	No BP	0	No BP	0		
Glasgow Coma Scale	13-15	4	14-15	5	C	
	9-12	3	11-13	4		
	6-8	2	8-10	3		
	4-5	1	5-7	2		
	3	0	3-4	1		
Triage Revised Trauma Score					A+B+C	
Revised Trauma Score	PRIORITY		PRIORITY		ADULT	PAEDS
	Priority 1		Immediate		1-10	2-11
	Priority 2		Priority 2		11	12
	Priority 3		Priority 3		12	13
	Priority 0		Priority 0		0	1

CPG 13.3 PATIENT REFUSAL OF TREATMENT OR TRANSPORT

Clinicians may encounter situations where a patient (or their guardian) refuse treatment or transport to hospital by the ambulance service. In such situations, the following guide should be followed.

PATIENT REFUSAL OF TREATMENT OR TRANSPORT

Should a patient refuse treatment or transport from the ambulance service, the following process should be followed:

- Inform the patient of their presenting condition or likely condition and the risks associated with their condition if they refuse treatment or transport to hospital for further care.
- Inform the patient of your clinical assessment findings (if patient allowed assessment).
- Recommend treatment and transport to a healthcare facility.
- Should the patient still refuse treatment or transport from the ambulance service despite providing the patient with appropriate information and advice, complete the ePCR and allow the patient or patients guardian to sign the refusal of treatment or transport section.
- Complete detailed documentation that includes clinical assessment findings, advice and recommendations provided to the patient and the decision of the patient to refuse treatment or transport.
- Should you have any clinical concerns about a patient that is unwell and requires transport to hospital yet still refuses treatment and transport, escalate the matter to your supervisor if there is no CCP on scene.

CPG 13.4 SUBMERSION INCIDENT

Submersion incidents can vary in severity and require rapid identification of potential life threat, treatment and transport to hospital. Drowning is when a person is pronounced dead following submersion or within 24 hours of the submersion incident. Near-drowning is when the person is revived and survives more than 24 hours following the submersion incident.

CLINICAL PRESENTATION

Signs associated with a submersion incident may include:

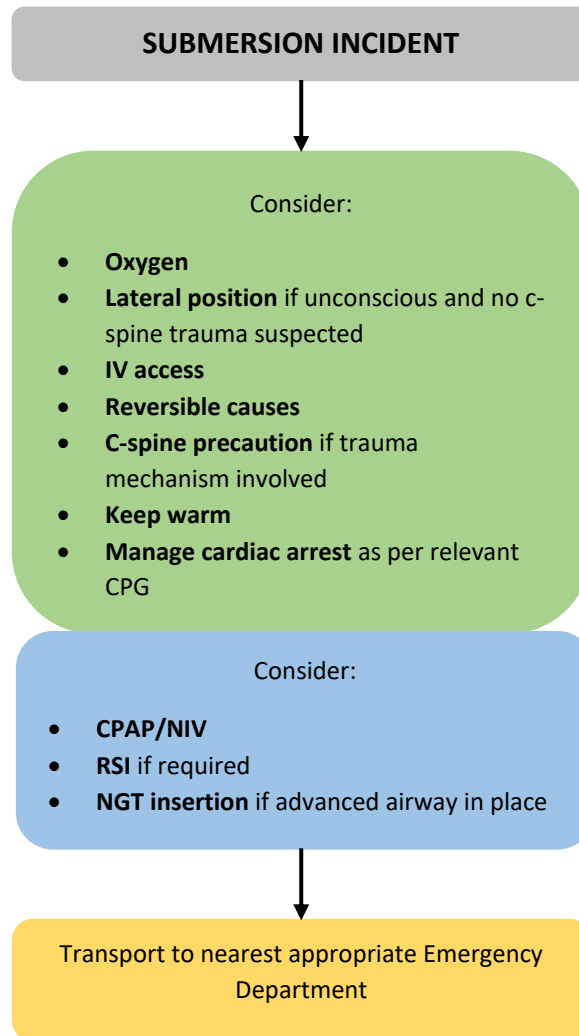
- Altered LOC
- Confusion/agitation
- Dysrhythmias
- Hypotension
- Dyspnoea
- Pulmonary oedema
- Aspiration
- Nausea and vomiting
- Hypothermia
- Trauma (spinal/diving injury)
- Respiratory arrest
- Cardiac arrest

ASSESSMENT CONSIDERATIONS

- Obtain a clear history regarding the circumstances surrounding the submersion incident. This should include the time period of submersion, potential trauma mechanisms, intoxication, seizures, stroke and ACS that may have precipitated the submersion incident.
- Patients that experience a submersion incident may be at risk of secondary complications such as pulmonary oedema and aspiration pneumonia.
- Monitor the patient's temperature as they are at risk of hypothermia. Keep the patient warm.
- Assess for and treat potential underlying causes (e.g., hypoglycaemia, OD, seizures).

SPECIAL NOTES

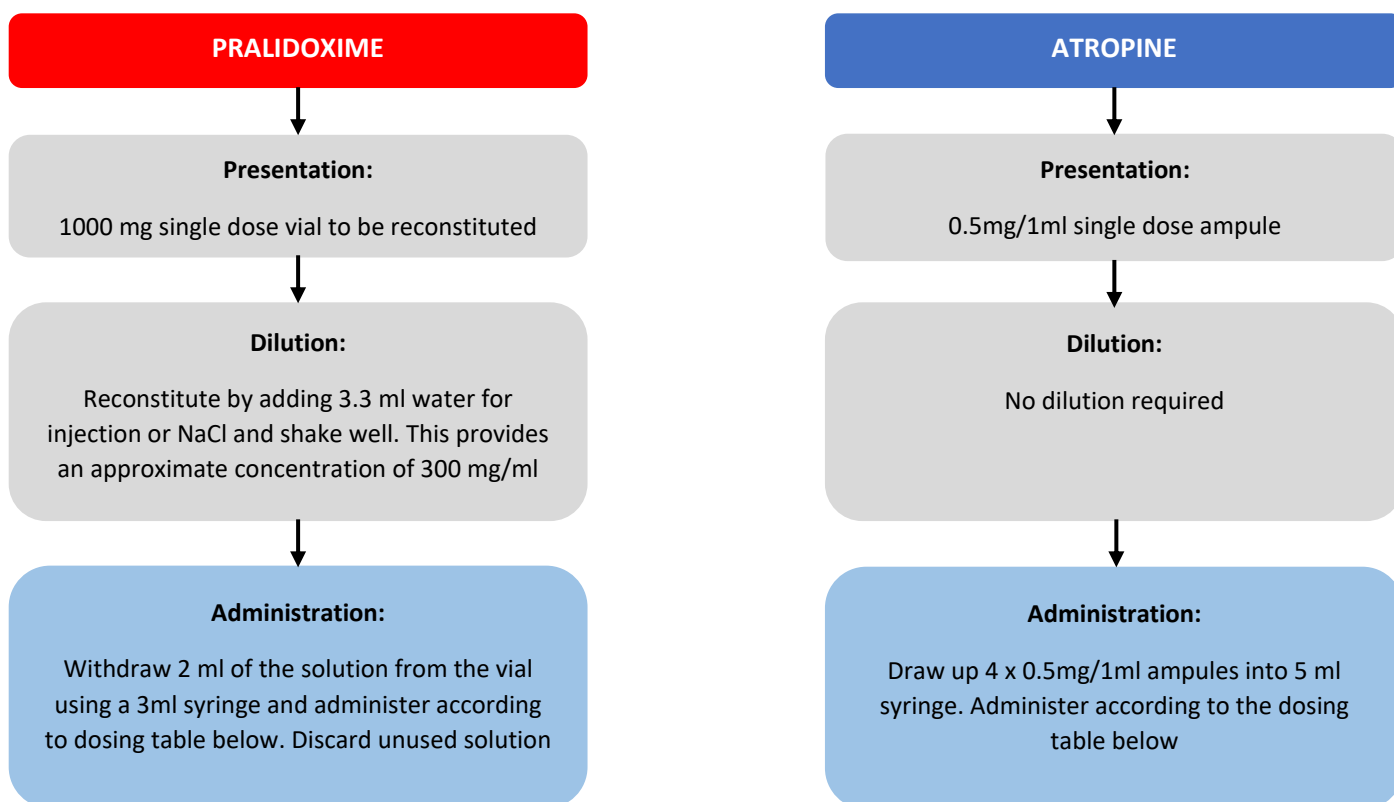
- Consider inserting an NGT for all patients that have an advanced airway inserted to alleviate potential diaphragmatic splinting and gastric distention.
- A c-spine injury should be considered in all dive-related submersion incidents.
- Suctioning of the airway should be avoided as it may exacerbate the hypoxia. Suction the airway only if absolutely necessary.



CPG 13.5 HAZMAT INCIDENT

PRALIDOXIME & ATROPINE DRUG DOSING CHART

In the event of a CBRN incident requiring the administration of Pralidoxime and Atropine, follow the dosing chart below. Pralidoxime and Atropine are only to be administered following Silver Clinical authorisation.



	12 years and older	8 – 11 years	1 – 7 years
1 st Dose	Atropine: 2mg IM	Atropine: 2mg IM	Atropine: 2mg IM
	Pralidoxime: 600mg IM	Pralidoxime: 600mg IM	Pralidoxime: 600mg IM
2 nd Dose	Atropine: 2mg IM	Atropine: 2mg IM	N/A
	Pralidoxime: 600mg IM	Pralidoxime: 600mg IM	
3 rd Dose	Atropine: 2mg IM	N/A	N/A
	Pralidoxime: 600mg IM		
Repeat doses to be administered every 15 minutes. Repeat the dosing according to the above table as per age group. Pralidoxime and Atropine may be administered simultaneously to a patient. Use different injection sites.			
IV Dose	Atropine: 1-2mg every 5 minutes	Atropine: 0.05mg/kg (max 2mg) every 5 min	Atropine: 0.05mg/kg (max 2mg) every 5 min
IV access should only be established once the patient has been decontaminated and in the SAFE ZONE and CBRN PPE has been removed. CCPs may administer IV atropine as required until secretions reduce or clear up. Follow up doses of IV Pralidoxime will be administered in hospital.			

CPG 13.6 IMIST—AMBO HANDOVER

The IMIST-AMBO is a tool designed to enable a quick, to the point and safe patient handover. This handover tool is to be used by paramedics when handing over a patient to colleagues or when handing over a patient at hospital.

I	Identify patient name, age, gender and onset time (this includes time of injury or onset of symptoms)
M	Mechanism of injury or Medical Complaint
I	Injuries/illness identified and/or information related to patient complaint
S	Signs and Symptoms
T	Treatment and Trends
A	Allergies
M	Medications
B	Background History
O	Other Information

HANDOVER CONSIDERATIONS

- Review handover details before arrival at hospital
- Deliver IMIST handover report uninterrupted
- Handover report should be short and detailed, containing all relevant and essential information
- Encourage questions once IMIST handover is complete

CPG 13.7 SEXUAL ASSAULT

The following guideline serves to provide paramedics with guidance on the processes to follow when managing a patient who is a victim of sexual assault or suspected sexual assault. HMC policy CL 7217 defines sexual assault as *“any type of sexual activity that is intentionally committed by one or more persons without the consent of the other. The act is committed either by a physical force, violence, threat, coercion or intimidation; ignoring the objections of another person; causing another’s intoxication or impairment through the use of alcohol or drugs; or taking advantage of another person’s incapacitation, state of intimidation, helplessness, or inability to consent.”*

CONSIDERATIONS

If called to a patient who is a victim of sexual assault or suspected sexual assault, take the following into consideration:

- Conduct a clinical assessment and manage any life threats or trauma/medical conditions as per relevant CPG.
- Request for Delta, CCP and police as required.
- When caring for the victim:
 - Refrain from expressing shock or criticizing the victim or their parents, family, legal guardian, legal representative.
 - Use clear vocabulary to discuss body parts.
 - Avoid any leading statements that can distort information.
 - Provide reassurance and emotional support to the victim:
 - Keep the patient in a private area (such as in the ambulance) where the patient will feel comfortable, safe and free to talk. Limit the number of people with the patient to avoid causing distress/anxiety.
 - Obtain an appropriate history. Questions asked should be medically pertinent and relevant to the current situation.
- As this is a criminal incident, it is important for paramedics to limit any disruption to the crime scene as well as preserve any physical evidence. Therefore, it is important to highlight the following:
 - Patients should be strongly encouraged to avoid showering or washing in order to preserve evidence.
 - Do not destroy or discard any clothing worn by the patient. Should clothing need to be cut off the patient for emergency care, ensure this clothing is collected and handed over either at hospital or to the police on scene.
- Contact the Clinical Desk to provide prenotification to the receiving facility:
 - Adult patients (≥18 years old) should be transported to the nearest emergency department.
 - Patients < 18 years old should be transported to Sidra emergency Department.
 - Female and paediatric patients should be accompanied by a female paramedic to hospital.
- Ensure all clinical details and relevant information disclosed by the patient are documented appropriately in the ePCR and handed over verbally to the receiving medical team at hospital.

CPG 13.8 EPISTAXIS

Epistaxis is defined as bleeding from the nostril, nasal cavity or nasopharynx and is classified as anterior and posterior. Anterior epistaxis is the most common presentation. This CPG focuses on the management of non-traumatic causes of epistaxis.

CLINICAL PRESENTATION

Signs associated with nose bleeds and epistaxis may include:

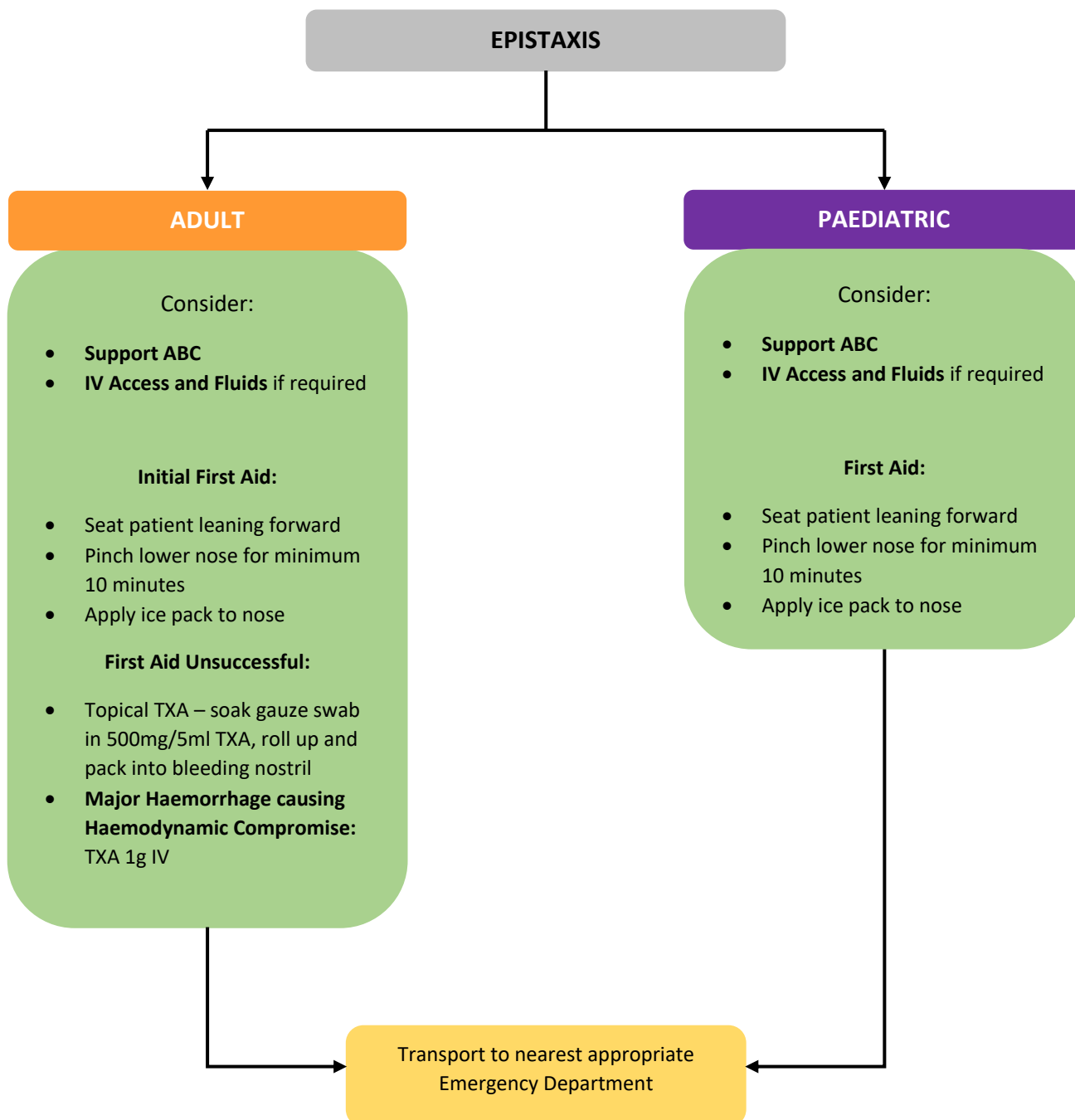
- Bleeding from one or both nares
- Posterior nosebleeds are not as common, and may be associated with haemoptysis or haematemesis
- Significant epistaxis may result in haemodynamic compromise and/or airway compromise

ASSESSMENT CONSIDERATIONS

- A detailed assessment, history gathering and assessment for risk factors:
 - Onset, affected side and amount of blood loss
 - Previous episodes of epistaxis and associated treatment
 - Recent upper respiratory tract infections
 - Recent nasal surgery
 - Anticoagulation medication
 - Co-morbidities that may affect coagulation
 - Alcohol and drug abuse (cocaine in particular)
 - Risk of a foreign body in the nose
- Inspect the nostrils to determine if bleeding site is visible or if a foreign body is present
- Encourage the patient to spit out any blood. Swallowing blood should be avoided
- Assess for haemodynamic compromise and manage accordingly
- PPE should be worn, including face mask and eye protection due to the increased risk of blood droplet exposure

SPECIAL NOTES

- Patients presenting with major haemorrhage not responding to treatment are at risk of haemodynamic deterioration.
- Initial care should focus on basic first aid. Good effective first aid should stop majority of nosebleeds. Care should progress based on response to treatment, unless haemodynamically compromised, in which case rapid treatment and transport are required.
- Topical TXA for adult patients may be considered if First aid is unsuccessful.
- First aid and topical TXA may not be as effective for posterior epistaxis. Early escalation to IV TXA should be considered if bleeding is uncontrolled and significant.
- IV TXA for adults may be considered in severe cases where the patient is haemodynamically compromised (tachycardic, hypotensive, altered level of consciousness)





CPG 14.1-14.3

VULNERABLE PATIENTS



CPG 14.1 ELDERLY PATIENTS (>65 YEARS OLD)

Elderly patients are a growing portion of the population and make up an increasing segment of the Ambulance Service workload. The elderly patient often has multiple comorbidities resulting in the need for both medical care and social care. Assessing and treating the elderly patient is more complex due to the multiple chronic conditions the elderly patient may have. The prehospital assessment should take into account the elderly patients physical and mental state, medications, and social circumstances, all of which may potentially complicate the treatment of what may appear to be a simple ailment. Paramedics should maintain a high index of suspicion when caring for the elderly patient as seemingly uncomplicated signs and symptoms may be the manifestations of a more serious underlying condition.

ASSESSMENT CONSIDERATIONS

It is important for paramedics to understand the aging process and the physiological changes and increased prevalence in chronic conditions in order to conduct a thorough assessment and provide treatment for these patients. The physiologic changes associated with aging can be extensive and most notably affect the:

- **Respiratory system:**
 - Increased risk of disease and infection
 - Reduced response to hypoxia and hypercapnia
 - Reduced respiratory compliance
 - Reduced elastic recoil with alveoli collapse
- **Neurovascular system:**
 - Degeneration of neurons
 - Decrease in brain size and weight
 - Steady decline in sensory perception
 - Decline in hypothalamic function leading to reduced ability to generate and conserve heat
 - Changes in pain perception
 - Increased cognitive impairment
- **Cardiovascular system:**
 - Deterioration in conduction system (increasing prevalence of AF, BBB and sick sinus syndrome)
 - Widespread arteriosclerosis resulting in hypertension
 - Thickening of the myocardium leading to cardiac hypertrophy
 - Decline in cardiovascular reserve
- **Renal system:**
 - Loss of nephrons leading to a reduction in size and weight of kidneys
 - Reduced ability to filter blood
 - Reduced ability to clear medications
 - Marked decline in the subjective feeling of thirst
 - Increased prevalence of infection (particularly UTI's)
- **Miscellaneous:**
 - Loss of muscle tone and strength
 - Increased prevalence of undernutrition and malnutrition
 - Increased prevalence of osteoporosis

- Skin is often fragile, easily torn and elderly patients are at a higher risk of developing pressure sores
- Deterioration of the skin immune system resulting in increased risk of infection

There are numerous factors paramedics should take into account and focus on when assessing the elderly patient.

- **General appearance:** observing the patient can provide you with the patient's general health state. Observing the patient's interactions and movements can provide important information regarding the patient's level of consciousness, mobility and gait, muscle strength, social interactive ability, hygiene and obvious discomfort. Changes in functional status often the first manifestation of a medical problem in the elderly patient.
- **Medications:** elderly patients often require multiple medications due to the numerous chronic diseases they may suffer from. This polypharmacy combined with the risk factors of the physiological aging process, increased number of physicians involved in the patients care, and the concomitant use of medications that cause additive hypotension, sedation or anticholinergic effects, can render older patients susceptible to adverse drug effects.
 - In an emergency (unplanned transport to a health facility) aim to bring all the patient's medication and/or an up to-date list of medication to hospital with the patient.
- **Communication:** remains an essential aspect of all patient care but it is specifically important when treating elderly patients. Whilst subgroups of patients with acute delirium and dementia represent a specific group, communication remains a core skill for all elderly, frail and vulnerable patients. Effective communication is based within holistic and compassionate care. Key elements include:
 - Be aware that elderly patients are less likely to speak
 - Position yourself at eye level so you can make eye contact
 - Speak clearly in short sentences
 - Ensure patients are wearing their hearing aids and glasses
 - Use clear simple language and use repetition if required, but sensitively
 - Avoid patronizing; treat the patient with respect and dignity
 - Use empathy, do not minimize the patient's feelings
 - Try to avoid distractions to help keep their focus
 - Be patient and allow plenty of time for them to respond to questions
 - Provide information in small chunks
 - Be aware that patients may be attached to unusual items such as soft toys
 - Wherever possible arrange for an escort, who is known to the patient, to accompany the patient to hospital
- **Mental status:** assessing mental status is an important part of the elderly patient's assessment. Mental status may be affected by cognitive impairment such as dementia or Alzheimer's, current illness or injury, medications, or relative familiarity with their surroundings. Impaired cognition, even if normal for the patient, may have a significant bearing on the type and direction of medical care. An acute change in mental status in the elderly patient may be the result of delirium. Delirium is characterized by an alteration in consciousness, cognition, or perception that develops over a short period of time (hours to days). Patients with delirium may appear agitated and restless, quiet and withdrawn, or move between these two states. Key risk factors for delirium include:
 - Age >65 years
 - Known cognitive impairment/dementia
 - Severe medical illness and current hip fracture
 - Patients with one or more risk factors should be identified to hospital staff when handing over care. It is also important to note that the lack of adaptive devices such as hearing aids or

eyeglasses can functionally disable patients and may contribute to their altered mental status by limiting their ability to interact with their environment.

- **Atypical clinical presentation:** elderly patients commonly present with different clinical signs when compared to the average adult patient, such as:
 - Silent myocardial infarction
 - Sepsis presenting with hypothermia
 - Infection (especially UTI) induced acute delirium
- **Trauma:** traumatic injuries in the elderly patient are associated with significant morbidity and mortality. The physiological changes associated with aging combined with multiple comorbidities significantly reduces the elderly patient's ability to respond to the physiological stresses of trauma, with the shock state progressing much more rapidly in this patient group compared to younger patients. Falls are a significant risk for this population and appropriate care should be taken to prevent falls when caring for these patients.

CPG 14.2 BARIATRIC PATIENTS

Bariatric patients are increasingly encountered by the Ambulance Service. These patients require special consideration in relation to unique safety, mobilization, transport and clinical care issues. These considerations should be adequately addressed in order to minimize risk to both the patient and the clinician.

ASSESSMENT CONSIDERATIONS

Those who are morbidly obese are at a significantly higher risk for developing chronic diseases such as:

- Cardiovascular disease
- Diabetes
- Hypertension
- Kidney stones
- CVA/TIA
- Pulmonary disease
- Osteoarthritis
- Psychiatric disorders
- Physical disabilities

Prehospital clinicians should be cognisant of the additional needs bariatric patients may require and the challenges associated with caring for these patients, such as:

- Difficulties in face mask ventilation due to increased upper airway resistance, extra adipose weight on the chest and redundant supraglottic tissue
- Large thick tongues, short thick necks and redundant adipose tissue distorting neck anatomy
- Enlarged heart due to excess strain of oxygenating the body's expanse of tissue
- Hypoventilation is a typical finding in bariatric patients caused by impaired chest expansion related to the inability for the diaphragm to fully move down during inhalation. These patients may be relatively hypercapnoeic which may worsen with illness or interventions that reduce respiratory ventilation functioning
- During acute illness rapid oxygen desaturation is a frequent occurrence due to decreased functional residual capacity
- Breath sounds are often distant and difficult to auscultate. Breath sounds should be auscultated on the patients back just medial of each scapula where a lower density of adipose tissue is found
- Cyanosis is best assessed by examining the inside of the patient's eyelids or lips
- Cannulation can be challenging due to thick layers of adipose tissue
- Assess the bariatric patient according to the relevant CPG.

Clinicians may have to adjust or modify certain care techniques when caring for the bariatric patient. Such considerations include:

- Appropriate positioning – these patients should be positioned to best facilitate lung expansion and decrease risk of positional asphyxia. Avoid laying the patients supine, consider “Ramping” the patient
- Blood glucose testing from the finger may not provide enough blood due to thickened fat pads. An alternative site is the earlobe
- SpO2 probes should be attached to the smallest available finger

- Bariatric patients will typically have a 'difficult airway'
 - When managing an emergency airway (non-cardiac arrest) position the patient in a 30° head up position
 - Use a two person BVM technique and oral airways
 - Carefully risk assess the benefits of prehospital Rapid Sequence Induction
- IV access can be difficult to obtain in the bariatric patient. The need for IV access prehospitally should be carefully considered based on the patient's condition and transport time to hospital. Alternatives may be considered such as IO access and the IN and IM medication administration routes.
- Caution should be taken when administering medications with cardiovascular and respiratory depressant effects as they have a limited physiological reserve. Ensure the appropriate monitoring is attached.
- Bariatric ambulances or bariatric capable stretchers may be required to transport these patients. Requesting additional manpower to move the patient is also an important consideration to move the patient safely and to minimize risk of injury to clinicians.
- Transport the bariatric patient in a position of comfort and safety to the nearest most appropriate healthcare facility.



SPECIAL NOTES

- Be sensitive to the fact that patients are often embarrassed about their weight and the associated need for additional assistance with transferring them to hospital
- On identification that the patient requires additional support contact the National Command Center and request additional assistance:
 - Request bariatric unit, including patient movement requirements (wheelchair, bariatric stretcher)
 - Provide estimated weight of patient
 - Request for a supervisor to attend if required

CPG 14.3 PATIENTS RECEIVING RENAL DIALYSIS

Patients who suffer from an acute or chronic kidney injury or illness are at risk of developing renal failure. Treatment of these patients may include dialysis, which can either be provided in-hospital or at home through haemodialysis or peritoneal dialysis. Renal dialysis patients are encountered in both the Emergency and Scheduled Ambulance Services and clinicians should be able to recognize and manage complications associated with renal dialysis.

ASSESSMENT CONSIDERATIONS

- **Haemodialysis:**
 - Haemodialysis involves blood being pumped from the body through special tubing into a dialysis machine from a surgically inserted catheter or arterio-venous fistula. The dialysis machine removes waste products and excess fluid from the blood, acting as a type of artificial kidney. Blood passes through a dialyser (filter), which also assists in balancing fluid, minerals and chemicals in the blood. The machine then returns the filtered and cleaned blood to the body at the same rate as which it is removed.
- **Peritoneal dialysis:**
 - Peritoneal dialysis uses the peritoneal membrane in the body itself as a filter. The peritoneal membrane is rich in vascular supply making it ideal for filtering wastes and excess fluid from the blood. Dialysis solution is instilled into the abdominal cavity via a surgically inserted catheter where over time waste products pass from the bloodstream across the peritoneal membrane and into the dialysis solution. This fluid is then drained and replaced with fresh solution at a later time which is either automated via a machine usually connected to the patient overnight or through a manual process where the patient will instil and drain solutions several times a day.
- On rare occasions clinicians may be called to patients who are still attached to their dialysis machine and will require ambulance intervention and transport. In such cases, follow the emergency evacuation from home dialysis procedure below.

EMERGENCY EVACUATION FROM HOME DIALYSIS

Peritoneal Dialysis:

1. Where possible, use either the patient or the carer that is familiar with the process of connecting and disconnecting from the machine to remove the patient from the dialysis equipment/ device.
2. Where this is not possible, don gloves and ensure an aseptic technique.
3. Turn equipment off at the machine ON/OFF switch (if it is an automated system).
4. Turn tubing flow off from the machine or hanging bag via clamp or roller lock and repeat for the drainage bag.
5. If necessary, (where it cannot be identified how to stop solution flow) clamp the catheter tubing at either side of connector port using available clamps already in situ (if available), or umbilical cord clamps (available in obstetric kit).
6. Disconnect patient from device and place a cap on the end of the catheter.

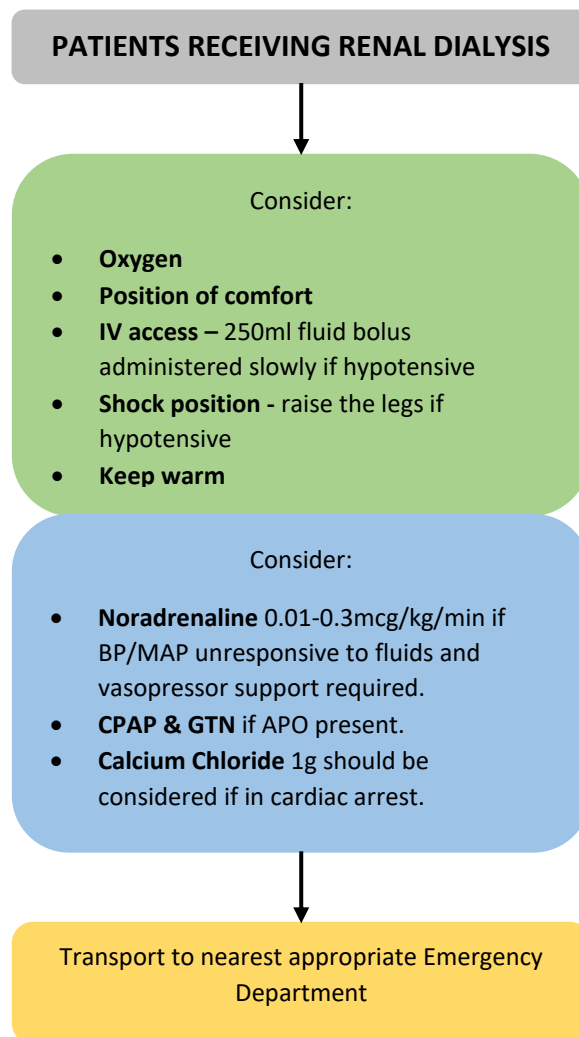
Haemodialysis:

1. Where possible, use either the patient or the carer that is familiar with the process of connecting and disconnecting from the machine to remove the patient from the dialysis equipment/ device.
2. Where this is not possible, don gloves and ensure an aseptic technique.
3. Turn equipment off at the machine ON/OFF switch.
4. Clamp catheter tubing using available clamps already in situ (if available), or umbilical cord clamps (available in obstetrics kit).
5. Disconnect the patient from the device and leave the cannula in situ with or without the extension tubing attached (depending on the type of set).
6. Connect a cap to the cannula or extension tubing where necessary to maintain sterility.
7. Utilize the cannula as required for fluid or medication delivery only as a last resort if unable to establish alternate vascular access (e.g. peripheral IV access).
8. In the event of severe bleeding during the procedure from the therapy site, apply pressure and dressings/ bandages as required and maintain pressure on the site for up to 20 minutes or until bleeding is controlled.



SPECIAL NOTES

- Patients requiring regular renal replacement therapy are often well known to Scheduled Ambulance Paramedics. However, an assessment of vital signs should be completed as per CPG 16.1 due to the impact of dialysis.
- Patients are often tired, fluid depleted and frailer on the post dialysis journey when compared to the initial journey.
 - Beware of increased risk of fall
 - Beware of increased risk of hypotension
- Prior to transport home obtain a handover from nursing staff to include vital signs if the patient is hypotensive (<90mmHg) request dialysis doctor review the patient.
- Use caution when administering IV fluids to patients with renal impairment.
- Renal dialysis patients found in cardiac arrest during, shortly after or those that have missed dialysis may be hyperkalaemic and Calcium Chloride should be considered during the resuscitation.
- Vascular access can be challenging in these patients and fluid administration may not be required. Do not attempt IV access more than twice. Do not establish an IV on the same side as the fistula.





CPG 15.1-15.3

INTER-FACILITY TRANSFER GUIDELINES



CPG 15.1 CONTINUATION OF PREVIOUSLY PRESCRIBED MEDICATION

Critical Care Paramedics (CCP) from both the Emergency Services and Retrieval Services may be required to move patients, between healthcare facilities, with medication infusions not included within their current pharmacopeia and without the support of escorting medical staff. Critical Care Paramedics may comply with this request to meet patient needs where the following is achieved or met:

1. It is in the patient's best interest that the medication is continued during transportation. Stopping the medication, for transportation, is not a viable option.
2. The medication has been prescribed by a DHP licensed doctor or by a doctor entitled to work within Qatar (e.g. US military doctors) and that the prescribing doctor has continued responsibility for the prescribed medication (as is the case when prescribed within a static healthcare facility).

ASSESSMENT CONSIDERATIONS

The primary aim of this CPG is to facilitate safe, effective transportation of patients who are receiving medication not included within the CCP's current pharmacopoeia. This CPG is designed to be flexible to reflect the wide scope of potential requests and the variable experience of CCPs.

There are no drugs that are exempted by this policy on the understanding that the following criteria are met:

- The medication is required for the safe and/ or optimal transportation of the patient.
- The medication is prescribed by the referring doctor AND the CCP has had an opportunity to review the prescription.
 - The referring doctor may prescribe a dosing range if required for the CCP to follow.
 - The referring doctor may prescribe a fixed dose for the CCP to follow.
- That the referring doctor understands that they remain the responsible clinician for any medication that they prescribe that falls outside of the CCP's normal pharmacopeia that the attending clinician's wishes continued during the transfer.
- That the CCP is prepared to transfer the patient and the CCP understands that they are empowered to discontinue the medication should an adverse event occur.

SPECIAL NOTES

Whilst the referring doctor retains responsibility for the prescription of any medication outside of the CCP's standard pharmacopeia the attending CCP is responsible for ensuring optimal patient safety during transportation. Therefore, the following steps are recommended:

- Confirm that the medication is required for the ongoing management of the patient routine infusions such as hydration fluids (excluding burns) should, where feasible, be discontinued during elected transportation.
- If the CCP believes that an alternative medication, within their pharmacopeia, is an option then this should be raised with the referring doctor.
- Review CPG 15.3 Drug chart for frequently used transfer medications as the requested drug may be listed within this pharmacopeia providing advice and support on drug administration and appropriate dosing regimens.

- If conflict exists as to the appropriateness/safety of the medication, then the transfer should be delayed until the referring hospital can supply an appropriate escort or until such time that the medication is no longer required.
- CCPs may escalate to the duty Consultant Effectiveness or Silver Clinical should there be any concerns relating to emergency inter-facility transfers. For scheduled interfacility transfers of stable patients, consult with either the MICU consultant on-call or Silver Clinical.

Actions once continuation of previously prescribed medication during transportation has been agreed:

- The CCP should review the prescription noting the dose and/or dose range prescribed at the host medical facility.
- The medication and the dose range along with the referring doctors name should be entered into the CCP's ePCR.
- The medication should be reviewed to ensure that:
 - Its compatible with syringe driver or infusion pumps carried by the CCP.
 - Where the current medication is not compatible then either a) a new infusion is drawn up using compatible syringes or b) the current infusion pump is continued during transportation. Option B is the least viable option and requires the following:
 - CCP must be familiar with the infusion device.
 - There is adequate battery life for the duration of the transfer plus 60 minutes OR the device is chargeable within the transferring ambulance.
 - That the referring hospital understands that the ambulance service cannot ensure the timescale for the return of the infusion device.
 - That the CCP understands the dosing range and the concentration of the medication.
 - That there is adequate medication provided to cover the transfer at the maximum pre-scribed infusion rate with the addition of at least an additional 60 minutes of infusion time.

Action on adverse reaction to ongoing medication:

- Treat anaphylaxis and/or hypotension as per relevant CPG.
- Discontinue medication.
- Ensure adverse event is documented in an HMCAS OVA and handed over to receiving healthcare facility.

BLOOD TRANSFUSIONS

The following guidance is for patients receiving a blood transfusion and requires transfer:

- For stable patients requiring a scheduled transfer with an ongoing blood transfusion, APs may perform the transfer following screening and consultation with the Clinical Desk.
- For emergency cases requiring immediate transfer with an ongoing blood transfusion that is stable enough for AP transfer - APs may continue the already established blood transfusion for the transfer. APs should request CCP backup if the patient is unstable or has other infusions outside of the AP scope. Cases screened as unstable or that fall within the CCP scope during call taking should have a CCP dispatched as per SOP.
 - APs and CCPs should establish the start time of the blood transfusion and identify if any adverse reactions have occurred prior to transferring the patient. Most severe reactions generally occur within the first 15 minutes of starting the transfusion. Should an adverse

reaction occur, the blood transfusion should be stopped immediately. If still at the referring facility, notify the transferring doctor of the adverse reaction.

- Adverse reactions to blood transfusion include: tachycardiac, hypertension, fever, rigors, headache, myalgia, altered level of conscious, bronchospasm, pulmonary oedema, and worsening coagulopathy.
- Should an anaphylactic reaction occur, manage according to CPG 6.2
- Transfusion rates should be copied from the referring facility.
- Patients must be continuously monitored, including 4-lead ECG, SpO2, BP, RR, ETCO2 and temperature.
- Blood administration, number of units and if any adverse reactions occur must be documented accordingly in the ePCR.
- If applicable, all blood transfusion documentation from the referring facility must be obtained and handed over at the receiving facility.
- Silver Clinical should be contacted if any adverse reactions occur or if there are any concerns or queries related to the transfer.

CPG 15.2 USE OF SPECIALIST MEDICATION DURING TRANSFER

The aim of this CPG is to facilitate level two and three transfers led by CCP's supported by on-call senior clinicians and forms part of the privileges of critical care paramedics. To this end a group of drugs (extended pharmacopeia Transfer and Retrieval Services) are available for use during inter-facility transfers. The utilization of these medications is limited to CCP following direct consultation with silver clinical and/ or MICU consultant.

ASSESSMENT CONSIDERATIONS

The list of medication within the extended pharmacopeia Transfer and Retrieval Services may be utilized during transportation of patients following on-line consultation. The decision on what drug is best suited for the transfer is subject to online consultation between the attending critical care paramedic and silver clinical and/ or MICU consultant.

- Treatment protocol is agreed with the on-call clinician (adhere to SOP – verbal orders).
- Treatment protocol is by the attending CCP within the transfer record.
- See CPG 15.3 Drug chart for frequently used transfer medications for guidance.

CPG 15.3 DRUG CHART FOR FREQUENTLY USED TRANSFER MEDICATIONS

The following drug chart serves as a quick reference guide for frequently used inter-facility transfer medications for adult patients.

Key: SD = Syringe Driver; IP = Infusion Pump; AS = Ambulance Service Guideline; Max = Maximum.

Drug Name	Diluent	Compatible solutions, concentrations & volumes	Bolus/Loading Dose	Infusion/Titration Ranges	Comments
ADRENALINE 1mg/ml	D5W NS	SD: 4mg/50ml IP: 2mg/250ml AS: 2mg/20ml Max: 16mg/50mg	No Bolus	Range: 0.01-0.1mcg/kg/min (1-10mcg/min) ↑↓ 1mcg/min every 5 min AS – 3.3mcg/min – double PRN	
AMIODARONE 150mg/3ml	D5W only	SD: 300mg/50ml	150-300mg IV prior to initiation of infusion	Initial: 1mg/min for 6hrs Maintenance: 0.5 mg/min for 18hrs	
CISATRACURIUM 10mg/5ml	D5W NS	SD: 50mg/50ml IP: 100mg/250ml	Intubation dose: 0.15-0.2mg/kg	Range: 0.5-10mcg/kg/min ↑↓ 1-2mcg/kg/min every hour	
DEXMEDETOMIDE 200mcg/2ml (Precedex™)	NS	SD: 200mcg/50ml	Bolus 1mcg/kg (Not recommended due to risk of hypotension)	Range: 0.2-1.5mcg/kg/hr ↑↓ 0.1-0.2 mcg/kg/hr every 30 minutes	Bradycardia and hypotension at high doses
DOBUTAMINE 250mg/20ml	D5W NS	SD: 250mg/50ml	No Bolus	Range: 2.5-20mcg/kg/min ↑↓ 5mcg/kg/min	
DOPAMINE 200mg/5ml	D5W NS	SD: 400mg/50ml	No Bolus	Range: 2-20mcg/kg/min ↑↓ 1-5mcg/kg/min every 10-30 min till optimum response	
DOXAPRAM 20mg/ml	D5W NS	SD: 100mg/50ml	Initial: 0.5-1mg/kg may repeat at 5 min intervals (only in patients who demonstrate initial response); max total dose: 2mg/kg	Initial: 5mg/min until adequate response or adverse effect seen; decrease to 1-3mg/min; max total dose: 4mg/kg	
ESMOLOL 2500mg/10ml	D5W NS	SD: 500mg/50ml	Loading dose: 0.5mg/kg over 1 minute	Range: 50-300mcg/kg/min ↑↓ 25-50mcg/kg/min every 10 minutes	
FENTANYL	NS	SD: 500mcg/50ml IP: 500mcg/100ml Max: 2500mcg/50ml	25-50mcg	Range: 1-10mcg/kg/hr	
FUROSEMIDE 20mg/2ml	NS	SD: 500mg/50ml	0.1mg/kg as an initial bolus	IV infusion 0.1mg/kg/hour Doubled every 2 hours to max of 0.4mg/kg/hour	
GLYCERL TRINITRATE 50mg/10ml	D5W NS	SD: 50mg/50ml	No Bolus	Range: 20-300mcg/min ↑↓ 5-10mcg/min every 5 min	

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Drug Name	Diluent	Compatible solutions, concentrations & volumes	Bolus/Loading Dose	Infusion/Titration Ranges	Comments
HEPARIN 5000iu/ml	D5W NS	25,000 units in 250ml	5000 units IV one time as a bolus	Start: 1000units/hr Range: 500-1500units/hr	(100units/ml)
HYDRAZALINE 20mg	NS	SD: 100mg/50ml	5-10mg as a bolus =150ml/hr followed by infusion	Range: 50-300mcg/min =1.5-9ml/hr	Bolus 5mg=2.5ml over 1 min. May need over 20mg initially in severe hypertension.
INSULIN (Actrapid) 100iu/10ml	NS	SD: 50units/50ml	No Bolus	Range: 1-20units/hour	Titrate to effect Min: Every 30 min BSL
KETAMINE	NS	SD: 200mg/50ml	PRN	Range: 0.01-0.05mg/kg/min	
LABETALOL 20mg/ml	D5W NS	SD: 50mg/50ml	10-20mg push over 2 min (Max cumulative dose 300mg total)	Start: 2mg/min continuous infusion ↑↓ 1-2mg/min every 10 min	
MAGNESIUM 10% in 20ml (0.8meq/ml=2g vial)	NS	SD: 2g/20ml or SD: 4g/50ml	No Bolus	Bronchospasm: 2g Eclampsia: 4g (Give 2-4g over 10 min)	
MIDAZOLAM	D5W NS	SD: 45mg/50ml	Give 2-5mg with each dose increase	Sedation: 0.3-1.7mcg/kg/min Status Epilepticus: 1-10mcg/kg/min	
NORADRENALINE 4mg/4ml	D5W NS	SD: 4mg/50ml IP: 4mg/250ml Max: 16mg/50ml	No Bolus	Range: 0.01-0.3mcg/kg/min Titration every 5 minutes	
PHENYLEPHRINE 10mg/1ml	D5W NS	SD: 10mg/50ml IP: 10mg/500ml AS: 5mg/50ml Max: 10mg/40ml	No Bolus	Range: 0.5-9mcg/kg/min ↑↓ 10-20mcg/min every 5 min AS: 100mcg/min	
POTASSIUM CHLORIDE 20mmol/10ml	NS	SD: 2-4mmol/50ml IP: 2-4mmol/100ml	No Bolus	Max: 20mmol/hr	Continuous cardiac monitoring. May irritate peripheral veins
PROPOFOL 1% 200mg/20ml	Do not dilute	SD: 400mg/40ml	No Bolus	Range: 0.01-0.25mcg/kg/min	
REMIFENTANIL	D5W NS	SD: 2mg/50ml Max: 10mg/50ml	No Bolus	Range: 0.01-0.25mcg/kg/min	
SALBUTAMOL 0.5mg/1ml	NS	SD: 6mg/50ml AS: 500mcg/50ml	200-300mcg over 2-5min	2mcg/min	
VECURONIUM	D5W NS	SD: 50mg/50ml	0.05mg/kg	0.5mg/kg/hr	Pts with hepatic or renal disease

					may have reduced excretion resulting in prolongation of neuromuscular blockade. Use lower doses in these patients.
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CPG 15.4 AP TRANSFER OF PATIENTS WITH INFUSION DEVICES

Ambulance Paramedics may transfer patients between facilities with infusion devices (infusion pump/syringe driver) without a CCP escort provided that the patient and medication being infused meets inclusion criteria.

INCLUSION CRITERIA

Patients meeting the following criteria are eligible for interfacility transfer with an infusion device by an AP:

- The patient must be stable for transfer by an AP
- The patient is ≥ 18 years old
- The patient is referred from an HMC hospital that is included in the AP infusion transfer procedure
- The medication being infused must be:
 - Any medication within the AP scope
 - Heparin
 - Glycerol Trinitrate (GTN) or Isosorbide Dinitrate (ISDN)
 - Insulin
 - IV Antibiotic
 - Blood/blood products
- The patient has no more than two (2) simultaneous infusions
- The patient only has peripheral intravenous access (patients that have a central or arterial line are not eligible for AP transfer)

TRANSFER CHECKLIST

Handover

1. Identify correct patient for transfer and receive handover from ward staff.
2. Verbally confirm drug name with ward staff and crew mate during handover. If the drug is not part of the box on the right, please contact the Romeo.
3. Verbally confirm drug infusion concentration and infusion device rate with ward staff and crew mate during handover. **Do not change the infusion rate at any time unless instructed to by CC:E**
4. Confirm time remaining for infusion at set rate on infusion device with crew mate. Ensure enough drug volume remains for duration of transfer.
5. Confirm infusion is actively running with crew mate, prior to patient movement.

Patient movement

6. Attach LP15 monitor to patient (4 lead, NIBP, SpO2).
7. Transfer patient to Alpha unit stretcher and secure patient.
8. Secure infusion device to stretcher.

Safety checks (with Romeo, prior to departure)

9. Confirm that the infusion device door is closed and locked.
10. Confirm that there are no kinks in the infusion line.
11. Confirm that there is no air in the infusion line.
12. For infusions running via infusion pump, confirm that the flow controller on the infusion line is open.

13. Confirm that the infusion line is secured to the patient's IV catheter.
14. Confirm that the infusion is still running.

En-route to receiving facility

15. Document infusion device type in ePCR notes (infusion pump or syringe driver)
16. Document drug name in the ePCR notes
17. Document drug infusion concentration and infusion device rate in ePCR notes
18. Monitor vital signs and if patient deteriorates, contact CC:E immediately

On arrival at receiving facility

19. If your patient is received in the ED, update the Romeo on the number of infusion devices used for transfer, and referring facility
20. If your patient is received in a ward other than the ED, collect the infusion devices after handover, hand them to the Romeo and update them on the referring facility

CPG 15.5 RASS – RICHMOND AGITATION-SEDATION SCALE

The Richmond Agitation-Sedation Scale (RASS) is used to measure the agitation or sedation level of an already sedated patient. The use of the RASS can help guide the need and use of sedation for interfacility transfer patients, predominantly those patients that are mechanically ventilated.

RASS		
Description	Term	Score
Overtly combative or violent; immediate danger to staff	Combative	+4
Pulls on or removes tube(s) or catheter(s) or has aggressive behaviour towards staff	Very Agitated	+3
Frequent non-purposeful movement or patient fighting ventilator	Agitated	+2
Anxious or apprehensive but movements not aggressive or vigorous	Restless	+1
Alert and calm	Alert and Calm	0
Not fully alert, but has sustained eye-opening/eye contact to voice for more than 10 seconds	Drowsy	-1
Briefly awakens with eye contact to voice for less than 10 seconds	Light Sedation	-2
Any movement/eye opening to voice but with no eye contact	Moderate Sedation	-3
No response to voice, but movement or eye opening to physical stimulation	Deep Sedation	-4
No response to voice or physical stimulation	Unroutable	-5

CPG 15.6 RETRIEVAL SERVICE PATIENT TRANSFER CHECKLIST

The Retrieval Service Patient Transfer Checklist is to be completed prior to departure from the facility for all CCP or Doctor led transfers. All applicable points in the checklist are to be successfully checked/passed prior to departure from the facility.

START OF SAFETY CHECK CONFIRM ALL PREPARATIONS ARE COMPLETED	
Airway	1. Tracheal tube secure 2. Note distance at lips 3. Recently suctioned 4. Tracheal cuff pressure
Breathing	1. Bilateral air entry 2. SpO₂ satisfactory 3. FiO₂ set appropriately 4. Ventilator set appropriately 5. P_{aw} and P_{plat} satisfactory 6. EtCO₂ including trace satisfactory 7. Flow curve with no air trapping
Circulation	1. Bp, HR and ECG rhythm satisfactory 2. Traces or intervals satisfactory 3. Inotropes running, lines visible and labelled 4. IV sites accessible
Disability	1. RBS satisfactory 2. Sedation ongoing 3. Sedation level satisfactory (RASS)
Exposure	1. Temperature baseline 2. All lines are unkinked 3. Cables tidy, snag hazard removed 4. Patient covered 5. NGT/OGT (mandatory for air transfer)
Safety and Emergencies	1. BVM on trolley 2. Patient secured 3. Drugs expected to be used are drawn up and on hand 4. Defib pads, pacing or LUCAS required? 5. Confirm no ALARMS on any equipment 6. Confirm Hb and K⁺ adequate 7. Confirm clinical condition satisfactory for transfer 8. Check oxygen supply is sufficient 9. Confirm infection control measures are adequate 10. Patient documentation
CHECK COMPLETE	



CPG 15.1-15.3

SCHEDULED AMBULANCE SERVICE GUIDELINES



CPG 16.1

CLINICAL APPROACH TO THE SCHEDULED AMBULANCE SERVICE TRANSFER PATIENT

The Scheduled Ambulance Service (SAS) transports both adult and paediatric patients between facilities (hospital to hospital) and from/to facilities and the patient's home. The primary aim of the SAS is to provide holistic and social care to patients requiring safe transportation. As such, these patients are a diverse group with unique needs that differ from the Emergency Ambulance Service group. Consent should be obtained prior to performing any assessments unless the assessment is part of emergency care where consent is implied.



ASSESSMENT CONSIDERATIONS

- Many SAS crews will develop close professional relationships with their patients as they will interact with them on a daily basis. However, Standard and Environmental Precautions (see SOP 2.1) should be adhered to.
- With the exception of patients listed below, SAS are not required to complete a routine assessment of vital signs during transportation as this reflects the holistic and social needs of the patient as well as patient feedback received during patient engagement sessions. The following patients require two complete sets of vital signs:
 - All SAS patients are continually observed by a paramedic and therefore a complete patient assessment as per relevant CPG should be completed if the patient deteriorates.
 - A single set of observations may be done upon patient or family request.
 - Two sets of vital signs should be recorded during interfacility transfers between acute healthcare facilities:
 - HMC hospital wards/departments.
 - Sidra hospital wards/departments.
 - Paediatric Emergency Centers or HMC urgent care centers/PHCC.
 - Private hospital wards/departments.
 - Patients going to or from renal dialysis.
 - All tracheostomy patients. Should vital signs assessment cause distress or be refused, the paramedic should attempt to monitor SpO₂ as a minimum.
- All vital signs should be recorded within the ePCR and handed over to the receiving facility.
- All SAS patients are continually observed by a paramedic and therefore a complete patient assessment as per relevant CPG (CPG 1.1-1.8) should be completed if the patient deteriorates.
- A number of elderly patients and/or vulnerable patients may become upset/agitated by the recording of vital signs. Due consideration to the patients' comfort is a priority and if vital sign recording results in patient distress, the paramedic should review the clinical relevance of assessing vital signs and record their decision to withhold vital signs within the ePCR.
- The basic set of vital signs is comprised of the following:
 - Pulse rate
 - Respiratory rate
 - Blood pressure
 - SpO₂
 - Temperature (minimum of one recording)
 - AVPU

NORMAL VITAL SIGNS RANGE FOR ADULTS

VITAL SIGN	NORMAL RANGE
Blood pressure	SBP 110 – 140 mmHg/ DBP 60 – 90 mmHg
Pulse	50 – 99 beats/min
Respiratory rate	12 – 20 breaths/min
Temperature	36 – 37.6°C
RBS	4 – 6.7 mmol/L

NORMAL VITAL SIGNS RANGE FOR PAEDIATRICS

Age Group	Weight	RR (breaths/min)	HR (beats/min)	SBP (mmHg)	DBP (mmHg)	Temp (°C)	RBS (mmol/L)
Preterm	2kg	40-70	110-170	55-75	35-45	36.5	2.6-6
Term – 2 months	3 – 4kg	35-55	110-160	65-85	45-55	36.5	2.6-6
3 – 7 months	5 – 7kg	35-45	110-160	70-90	50-65	36.5	4-6
8 – 12 months	8 – 11kg	22-38	90-160	80-100	55-65	36.5	4-6
1 – 3 years	12 – 15kg	22-30	80-150	90-105	55-70	36.5	4-6
4 – 5 years	16 – 21kg	20-24	70-120	95-110	60-75	36.5	4-6
6 – 7 years	22 – 28kg	20-24	70-120	95-110	60-75	36.5	4-6
8 – 9 years	29 – 34kg	16-22	60-110	100-120	60-75	36.5	4-6
10 – 11 years	35 – 40kg	16-22	60-110	100-120	60-75	36.5	4-6
12 – 13 years	41 – 45kg	12-20	60-100	110-135	65-85	36.5	4-6
≥ 14 years	46 – 50kg	12-20	60-100	110-135	65-85	36.5	4-6

SPECIAL NOTES

PATIENT HANDLING

Patients accessing the Scheduled Ambulance Service typically have unique handling requirements which may require them being supported to walk to the ambulance (as part of their self-independence), use of a range of wheelchairs and/or require the ambulance service to use electric stair chairs and/or ambulance stretchers. The movement of patients to/from an ambulance can be a stressful time for patients, therefore:

- Communicate with the patient as to their wishes.
- If the patient is unable to communicate their wishes, then discuss with the family and/or other care givers as to how the patient is normally handled.
- Adhere to SOP 4.8K regarding reducing risk of falls.
- Ensure comfort and appropriate positioning.
- Continuous visual observation.

ONGOING ASSESSMENT AND MONITORING

- As per initial assessment, vital signs and associated use of cardiac monitoring is only required when transporting SAS in-patients described in the section 'initial assessment'. Where patient/family requests that vital signs are completed only a single set of vital signs is required.

- If a patient deteriorates the National Command Center should be notified. Assistance should be requested if required and clinical care instigated in accordance with the relevant CPG.
- However, all patients being transported by the SAS will be continuously observed by a DHP licensed paramedic and if any clinical deterioration is noted, manage the patient according to the relevant CPG.

TRANSPORT CONSIDERATIONS

- Transport to the agreed facility adhering to Qatar road laws as well as providing the optimal 'comfortable' journey for the patient.
- If a patient deteriorates adhere to relevant CPG. Update the HCC and re-route to the closest emergency department or return to the patients ward if still within the hospital. Contact supervisor as required.
- If the patient refuses transport or further care, follow SOP 4.8D for recording refusal for transport and notify the dispatcher. Contact supervisor as required.



CPG 17.1-17.2

COVID-19 CLINICAL GUIDELINES



CPG 17.1

RESPIRATORY DISTRESS/HYPOXAEMIA IN SUSPECTED/ CONFIRMED COVID-19 PATIENTS

The following guideline serves to guide the management of the confirmed or suspected COVID-19 patient presenting with respiratory distress and/or hypoxaemia.



CLINICAL PRESENTATION

Confirmed/suspected COVID-19 patients with hypoxaemia/respiratory distress may present with the following:

- Dyspnoea
- Tachypnoea
- Low SpO₂
- Tachycardia



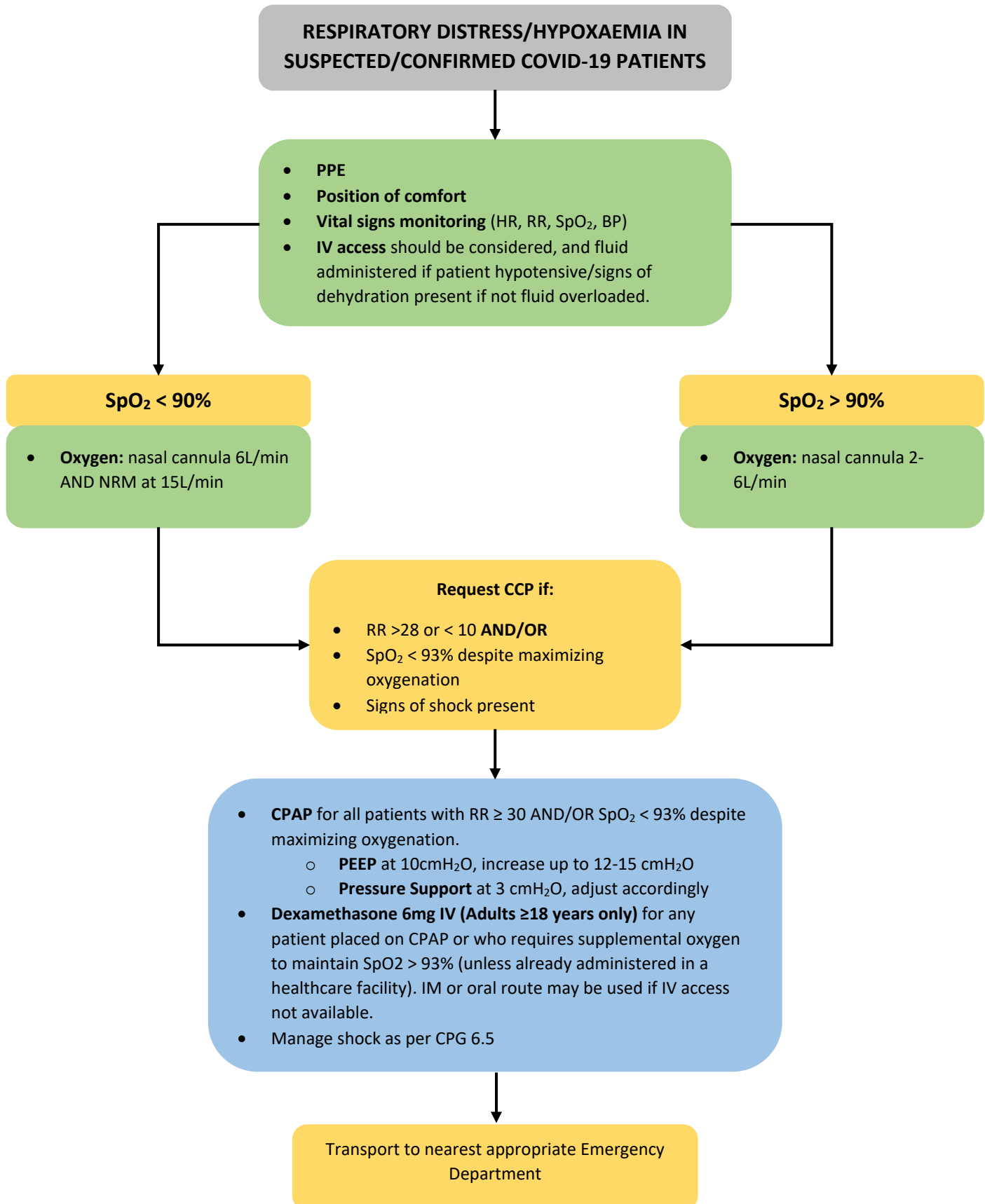
ASSESSMENT CONSIDERATIONS

- It is important to note that despite presenting with critical hypoxaemia, COVID-19 patients may present without proportional signs of respiratory distress (e.g. have critically low SpO₂ but no associated obvious respiratory distress).
- Monitor patients closely. Monitoring requirements include heart rate (4-lead ECG), respiratory rate (use EtCO₂ device), blood pressure, SpO₂ and temperature.
- Consider nebulisation for patients presenting with wheezes that are KNOWN to have asthma/COPD.



SPECIAL NOTES

- Wash and disinfect your hands regularly.
- Wear PPE (N95 mask, face shield, gloves and gown).
- Practice social cleaning of your vehicle and equipment regularly.
- Minimize the number of staff exposed to the suspected/confirmed COVID-19 patient.
- Minimize staff exposed to aerosol generating procedures.
- An HME filter should be attached to the BVM/mechanical ventilator used on all suspected/confirmed COVID-19 patients.
- In the intubated patient, do not squeeze the BVM until the ETT cuff is inflated and the circuit closed.
- CCPs may administer Dexamethasone and handover care to the AP crew for patient who meet the following criteria:
 - SpO₂ >93% on 6L/min or less of oxygen
 - Have a RR <28 breaths/min (once settled in the ambulance and at rest)
 - All other vital signs are stable
 - Patient is not obese or over the age of 59 years.
- Moving and loading patients into the ambulance may increase the exertion placed on these patients. CCPs are to settle these patients in the ambulance and monitor them to ensure their RR and work of breathing reduces after loading. If these patients do not settle, manage them accordingly.
- Avoid RSI unless absolutely necessary. Reduce drug dosing accordingly.



CPG 17.2

CARDIAC ARREST MANAGEMENT OF SUSPECTED/CONFIRMED COVID-19 PATIENTS

The following special considerations should be taken into account when managing cardiac arrest in the suspected/confirmed COVID-19 patient in an attempt to minimise aerosolization of COVID-19 particles. Amendments should be made where applicable.

- **Don PPE** – N95 mask, face shield, gloves, gown.
- **Identify cardiac arrest**
- **Start chest compressions** and manage cardiac arrest as per relevant resuscitation CPG.
- **Do not ventilate via BVMR with a facemask**
 - Focus on high quality chest compressions and defibrillation
 - Continue standard cardiac arrest drug dosing
- Insert **SGA**
- Attach **HME** filter to BVMR and ventilate via the SGA every 6-8 seconds. **Do not** perform compressions during ventilations.



- **ETT** should be inserted as soon as practically possible.
 - Consider via SGA or video laryngoscope
- **Asynchronous ventilations** at 10-12 breaths/minute.
- Continue resuscitation as per relevant resuscitation CPG.

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ADENOSINE

Class

Antiarrhythmic

Action

Slows AV conduction time and is therefore able to interrupt re-entry circuits through the AV node, thus restoring normal sinus rhythm in patients experiencing paroxysmal SVT, including PSVT associated with WPW (Wolff-Parkinson-White Syndrome).

Metabolism

Adenosine is rapidly cleared from the circulation via cellular uptake. Requires no hepatic or renal function to be activated or inactivated.



Indications

- Symptomatic SVT (narrow complex <0.12s).



Contraindications

- Known hypersensitivity to Adenosine.
- 2nd and 3rd degree heart block.
- SSS (Sick Sinus Syndrome)/ Sinus node disease.
- Atrial Fibrillation (AF), including AF with accessory pathways (WPW).



Precautions

- Asthmatic patients (due to mast cell degranulation and histamine release).
- Obstructive lung disease not associated with bronchoconstriction.
- On Digoxin and/or Verapamil.



Side Effects

- Bradycardia
- Hypotension
- Flushing
- Chest tightness
- Shortness of breath

Route of Administration				
Intravenous (IV)				
Intraosseous (IO)				

Special Notes

- The half-life of adenosine is <10s. Thus, adverse effects are generally self-limiting.
- Varying arrhythmias may present at the time of rhythm conversion. These are usually self-limiting and do not require intervention.
- Administer Adenosine via a large vein (e.g. ACF)
- Draw up adenosine in a 5ml syringe and draw up a saline flush in a 20ml syringe. Attach both syringes to a 3-way tap. Administer adenosine rapidly, immediately followed by the 20ml saline flush.

Onset	Duration	Half-life
Rapid	Short	<10 seconds

Dosage

ADULT

Symptomatic SVT					
	III	II	I	IV/IO	6mg – 1 st dose 12mg – 2 nd dose (if required) 18mg – 3 rd dose (if required)

PAEDIATRIC

Symptomatic SVT					
	III	II	I	IV/IO	0.1mg/kg – 1 st dose (max 6mg) 0.2mg/kg – 2 nd dose (if required – max 12mg)

Paediatric Adenosine Dosing Table

Age	Weight	SVT 1st dose		SVT 2nd dose	
		0.1mg/kg IV	Volume	0.2mg/kg IV	Volume
		Max dose: 6mg		Max dose: 12mg	
		Undiluted: 6mg/2ml =0.3mg/0.1ml		Undiluted: 6mg/2ml =0.3mg/0.1ml	
Preterm	2kg	0.3mg	0.1ml	0.6mg	0.2ml
Term-2 months	3-4kg	0.3mg	0.1ml	0.6mg	0.2ml
3-7 months	5-7kg	0.6mg	0.2ml	1.2mg	0.4ml
8-12 months	8-11kg	0.9mg	0.3ml	1.8mg	0.6ml
1-3 years	12-15kg	1.2mg	0.4ml	2.4mg	0.8ml
4-5 years	16-21kg	1.8mg	0.6ml	3.6mg	1.2ml
6-7 years	22-28kg	2.4mg	0.8ml	4.8mg	1.6ml
8-9 years	29-34kg	3mg	1ml	6mg	2ml
10-11 years	35-40kg	3.9mg	1.3ml	7.8mg	2.6ml
12-13 years	41-45kg	4.2mg	1.4ml	8.4mg	2.8ml
≥14 years	46-50kg	4.5mg	1.5ml	9mg	3ml

ADRENALINE (EPINEPHRINE)

Class

Sympathomimetic

Action

A naturally occurring α and β adrenergic stimulant causing an increase in heart rate, increase in the force of myocardial contraction, an increase in the irritability of the ventricles, bronchodilation, peripheral vasoconstriction and stabilisation of the mast cells.

Metabolism

By monoamine oxidase and other enzymes in the blood, liver and nerve endings and is excreted by the kidneys.



Indications

- Cardiac Arrest
- Anaphylaxis
- Severe life-threatening bronchospasm
- Croup
- Non-traumatic shock unresponsive to fluid resuscitation (adrenaline is not 1st line vasopressor)
- Symptomatic bradycardia unresponsive to atropine and/or pacing



Contraindications

- Nil in the emergency setting



Precautions

- Elderly patients
- Cardiac disease, hypertension, acute pulmonary oedema
- Patients on monoamine oxidase inhibitor (MAOI) therapy
- Hypovolemic shock



Side Effects

- Anxiety
- Hypertension
- Palpitations/tachyarrhythmias
- Tremors
- Pupil dilation
- Tissue necrosis in cases of extravasation or repeated IM injections at same site

Route of Administration				
Intravenous (IV)				
Intraosseous (IO)				
Intravenous/intraosseous infusion (IV/IO INF)				
Intramuscular (IM)				
Nebuliser (NEB)				

Special Notes

- Complete monitoring bundle to be used when administering adrenaline.
- Use 1:10,000 PFS for administration during cardiac arrest.
- Dilute 1:1000 concentration to 1:10,000 when administering IV during cardiac arrest.
- When administering IM, use a different site for each injection to avoid local tissue necrosis.
- Cycle BP every 2 minutes when administering via infusion.
- IV/IO push dose may be used as an interim in time critical extremis while preparing infusion.

Onset	Duration	Half-life
30 seconds (IV) 60 seconds (IM)	5 – 10 minutes	2 minutes

Dosage

ADULT

Cardiac Arrest					
				IV	1mg Repeat every 4 minutes
				IO	1mg Repeat every 4 minutes
Upper Airway Swelling					
				NEB	5mg (1:1000) Repeat PRN every 5 minutes

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Anaphylaxis					
			I	IM	0.5mg (1:1000) Repeat PRN every 5 minutes
	III	II	I	IV/IO	1mcg/kg Max single dose of 50mcg Repeat every 1-10 minutes
Severe Bronchoconstriction					
	III	II	I	IM	0.5mg (1:1000) Repeat PRN every 5 minutes
	III	II	I	IV/IO	1mcg/kg Max single dose of 50mcg Repeat every 1-10 minutes
Bradycardia & Inotropic and Vasopressor Support					
	III	II	I	IV/IO INF	0.05 – 0.3mcg/kg/min Mix 1mg (1:1000) in 100ml NS. Draw 50ml into 50ml syringe Titrate to effect
	III	II	I	IV/IO	1mcg/kg Push Dose Max single dose of 50mcg Mix 1mg (1:1000) in 100ml NS (10mcg/ml) Repeat PRN every 1-2 minutes

PAEDIATRIC

Cardiac Arrest					
		II	I	IV	0.01mg/kg Repeat every 4 minutes
	III	II	I	IO	0.01mg/kg Repeat every 4 minutes
Croup (CCP Only) & Upper Airway Swelling					
		II	I	NEB	0.5mg/kg (1:1000) Max 5mg Repeat PRN every 5 minutes
Anaphylaxis					
			I	IM	0.01mg/kg (1:1000) Max 0.5mg Repeat PRN every 5 minutes <6 years: 0.15mg (0.15ml) 6-12 years: 0.3mg (0.3ml) >12 years: 0.5mg (0.5ml)
	III	II	I	IV/IO	1mcg/kg Max single dose of 50mcg Repeat every 1-10 minutes

Severe Bronchoconstriction					
	III	II	I	IM	0.01mg/kg (1:1000) Max 0.5mg Repeat PRN every 5 minutes <6 years: 0.15mg (0.15ml) 6-12 years: 0.3mg (0.3ml) >12 years: 0.5mg (0.5ml)
	III	II	I	IV/IO	1mcg/kg Max single dose of 50mcg Repeat every 1-10 minutes
Bradycardia					
	III	II	I	IV/IO	1mcg/kg (1:10 000) Max 50mcg in a single dose Mix 1mg (1:1000) in 100ml NS (10mcg/ml) Repeat PRN every 2-4 minutes See paediatric dosing table
Inotropic and Vasopressor Support					
	III	II	I	IV/IO INF	0.05 – 0.3mcg/kg/min Mix 1mg (1:1000) in 100ml NS. Draw 50ml into 50ml syringe See paediatric dosing table for starting infusion dose

Paediatric Adrenaline Dosing Table

		Cardiac arrest		Inotrope/Vasopressor	Anaphylaxis/ Severe bronchoconstriction		Croup/ Upper airway swelling	
Age	Weight	0.01mg/kg IV/IO	Volume	0.05-0.3mcg/kg/min IV infusion	0.01mg/kg IM	Volume	0.5mg/kg Neb	Volume
		Bradycardia max dose: 0.5mg			Max dose: 0.5mg		Max dose: 5mg	
		1mg/10ml (1:10 000)		Dilution: 1mg into 100ml NaCl =10mcg/1ml (Withdraw 50mls and administer in a syringe driver using special functions)	Undiluted: 1mg/1ml (1:1000) = 0.1mg/0.1ml		Undiluted: 1mg/1ml (1:1000)	
Preterm	2kg	0.02mg	0.2ml	0.1-0.6mcg/min	<6 years = 0.15mg	0.15ml	1mg	1ml
Term-2 months	3-4kg	0.03mg	0.3ml	0.15-0.9mcg/min			1.5mg	1.5ml
3-7 months	5-7kg	0.06mg	0.6ml	0.3-1.8mcg/min			3mg	3ml
8-12 months	8-11kg	0.1mg	1ml	0.5-3mcg/min			5mg	5ml
1-3 years	12-15kg	0.15mg	1.5ml	0.65-3.9mcg/min				
4-5 years	16-21kg	0.2mg	2ml	0.9-5.4mcg/min				
6-7 years	22-28kg	0.25mg	2.5ml	1.25-7.5mcg/min				
8-9 years	29-34kg	0.3mg	3ml	1.6-9.6mcg/min	6-12 years = 0.3mg	0.3ml		
10-11 years	35-40kg	0.4mg	4ml	1.9-11.4mcg/min				
12-13 years	41-45kg	0.45mg	4.5ml	2.1-12.6mcg/min				
≥14 years	46-50kg	0.5mg	5ml	2.35-14.1mcg/min	>12 years = 0.5mg	0.5ml		

AMIODARONE

Class

Antiarrhythmic

Action

Amiodarone prolongs the duration of the action potential and therefore the refractory period of atrial, nodal and ventricular tissues. It reduces conduction across all cardiac tissue. Amiodarone demonstrates electrophysiological properties across all Vaughan-Williams Class groups, which enables a broad spectrum of activity. Its vasodilatory action decreases myocardial workload and subsequently myocardial oxygen demand.

Metabolism

By the cytochrome P450 enzyme which is present in the liver and GI system. Excreted via the biliary system.



Indications

- Cardiac arrest with refractory VF or Pulseless VT
- Sustained VT with a pulse (haemodynamically stable)



Contraindications

- Known hypersensitivity to Amiodarone or Iodine.
- 2nd and 3rd degree heart block.
- Cardiogenic shock
- Severe sinus node dysfunction



Precautions

- Patients taking β -blockers, Calcium antagonists or Diltiazem.
- Concurrent administration of Fentanyl and Amiodarone may exacerbate bradycardia, hypotension and decreased cardiac output.
- Patients with thyroid dysfunction.



Side Effects

- Bradycardia
- Hypotension
- Nausea and/or vomiting
- Peripheral paraesthesia

Route of Administration				
Intravenous (IV)				
Intraosseous (IO)				
Intravenous/intraosseous infusion (IV/IO INF)				

Special Notes

- Dilute amiodarone with 5% Dextrose
- Treat hypotensive episodes secondary to Amiodarone administration by slowing the rate of infusion and consider concurrent volume replacement.
- Treat bradycardia episodes secondary to Amiodarone administration by slowing the rate of infusion. Consider stopping infusion in severe cases.
- Rapid infusion of Amiodarone can result in liver and renal failure and death.

Onset	Duration	Half-life
5 minutes	30 minutes	14 – 110 days

Dosage

ADULT

Cardiac Arrest					
				IV	300mg after 3 rd shock 150mg after 4 th shock
				IO	300mg after 3 rd shock 150mg after 4 th shock
VT with a pulse					
				IV/IO INF	300mg Dilute 300mg (6ml) with 44ml 5% Dextrose in 50ml syringe Final concentration of 6mg/1ml Infuse at 200ml/hr (20mg/min) over 15 minutes

PAEDIATRIC

Cardiac Arrest					
				IV	5mg/kg after 3 rd shock (Max 300mg) Repeat twice up to 15mg/kg to max 300mg (total 3 doses)
				IO	5mg/kg after 3 rd shock (Max 300mg) Repeat twice up to 15mg/kg to max 300mg (total 3 doses)
VT with a pulse					
				IV/IO INF	5mg/kg Max dose of 300mg Infuse over 20-60 minutes See paediatric dosing chart

Paediatric Amiodarone Dosing Table

Age	Weight	VF/Pulseless VT		SVT/ VT with pulse	
		5mg/kg IV/IO	Volume	5mg/kg IV infusion over 20-60 minutes	Volume
		Max dose: 300mg		Max dose: 300mg	
		Undiluted: 150mg/3ml = 5mg/0.1ml		Undiluted: 150mg/3ml = 5mg/0.1ml	
Preterm	2kg	10mg	0.2ml	10mg	0.2ml
Term-2 months	3-4kg	15mg	0.3ml	15mg	0.3ml
3-7 months	5-7kg	30mg	0.6ml	30mg	0.6ml
8-12 months	8-11kg	50mg	1ml	50mg	1ml
1-3 years	12-15kg	65mg	1.3ml	65mg	1.3ml
4-5 years	16-21kg	90mg	1.8ml	90mg	1.8ml
6-7 years	22-28kg	125mg	2.5ml	125mg	2.5ml
8-9 years	29-34kg	150mg	3ml	150mg	3ml
10-11 years	35-40kg	200mg	4ml	200mg	4ml
12-13 years	41-45kg	225mg	4.5ml	225mg	4.5ml
≥14 years	46-50kg	250mg	5ml	250mg	5ml

ASPIRIN (Acetylsalicylic acid)

Class

Antiplatelet

Action

Aspirin inhibits platelet aggregation by irreversibly inhibiting cyclo-oxygenase, reducing the synthesis of thromboxane A2. This leads to preventing platelet aggregation at the site of vascular injury.

Metabolism

Aspirin is converted to salicylate in the GI mucosa and liver and primarily excreted by the kidneys.



Indications

- Acute Coronary Syndromes



Contraindications

- Known hypersensitivity to Aspirin/salicylates/NSAIDs
- Active GI bleeding and peptic ulcer with active bleeding
- Bleeding or clotting disorders (e.g. haemophilia)
- Suspected dissecting aortic aneurysm
- Patients < 18 years old



Precautions

- History of GI bleeds or peptic ulcers
- Asthma
- Patients already on anti-coagulant therapy (e.g. Warfarin)
- Patients with impaired liver function



Side Effects

- Epigastric pain/discomfort
- GI bleeding
- Nausea and/or vomiting
- NSAID induced bronchospasm
- Dizziness/tinnitus

Route of Administration				
Oral (PO)				

Special Notes

- Aspirin should be administered to suspected ACS even if pain free.
- 300mg of Aspirin may still be administered to patients presenting with ACS even if they are taking regular Aspirin at home.

Onset	Duration	Half-life
≈ 10 minutes	≈ 1 week	2 – 4.5 hours

Dosage

ADULT

Acute Coronary Syndrome					
				PO	300mg

ATROPINE

Class

Anticholinergic and muscarinic antagonist

Action

Atropine blocks acetylcholine receptors at the neuro-effector site, increases the heart rate by increasing the rate of the sinoatrial node and increasing conduction through the atrioventricular node. Atropine acts as an antidote to reverse the effects of cholinesterase inhibitors.

Metabolism

Metabolised by the liver and excreted by the kidneys.



Indications

- Symptomatic bradycardia
- Organophosphate poisoning



Contraindications

- Hypersensitivity to atropine



Precautions

- Atrial fibrillation
- Atrial flutter
- Acute myocardial infarction
- Glaucoma



Side Effects

- Tachycardia
- Palpitations
- Dry mouth/dry skin and reduced bronchial and gastric secretions
- Dilated pupils
- Urinary retention
- Agitation
- Hallucinations
- Heat intolerance or impaired temperature regulation can occur in patients in hot climates

Route of Administration				
Intravenous (IV)				
Intraosseous (IO)				
Intramuscular (IM)				

Special Notes

- Physostigmine is the antidote in the case of atropine overdose.
- Sub-therapeutic doses of atropine may cause paradoxical bradycardia.
- Atropine dosing for organophosphate toxicity can vary greatly between patients.
- Organophosphate toxicity induced tachycardia should not prevent atropine administration in the presence of respiratory distress.

Onset	Duration	Half-life
1 – 2 minutes	Up to 5 hours	3 – 4 hours

Dosage

ADULT

Bradycardia					
	III	II	I	IV/IO	0.5mg Repeat PRN every 3-5 minutes to max of 3mg
Organophosphate Toxicity					
	III	II	I	IV/IO/IM	1mg Repeat PRN every 5 minutes until clinical condition improves

PAEDIATRIC

Bradycardia					
	III	II	I	IV/IO	0.02mg/kg (min single dose 0.1mg; max single dose 0.5mg) Repeat once if required
Organophosphate Toxicity					
	III	II	I	IV/IO/IM	0.05mg/kg (max 2mg) Repeat PRN every 5 minutes until clinical condition improves

Paediatric Atropine Dosing Table

Age	Weight	Bradycardia		Organophosphate poisoning	
		0.02mg/kg IV/IO	Volume	0.05mg/kg IV/IO/IM	Volume
		Min dose: 0.1mg Max dose: 0.5mg		Min dose: 0.1mg Max dose: 2mg	
		Undiluted: 1mg/1ml = 0.1mg/0.1ml		Undiluted: 1mg/1ml = 0.1mg/0.1ml	
Preterm	2kg	0.1mg	0.1ml	0.1mg	0.1ml
Term-2 months	3-4kg			0.15mg	0.15ml
3-7 months	5-7kg			0.3mg	0.3ml
8-12 months	8-11kg	0.2mg	0.2ml	0.5mg	0.5ml
1-3 years	12-15kg	0.3mg	0.3ml	0.75mg	0.75ml
4-5 years	16-21kg	0.4mg	0.4ml	1mg	1ml
6-7 years	22-28kg	0.5mg	0.5ml	1.3mg	1.3ml
8-9 years	29-34kg			1.5mg	1.5ml
10-11 years	35-40kg			2mg	2ml
12-13 years	41-45kg				
≥14 years	46-50kg				

BUDESONIDE

Class

Anti-inflammatory glucocorticoid

Action

Controls the rate of protein synthesis; depresses the migration of polymorphonuclear leukocytes, fibroblasts; reverses capillary permeability and lysosomal stabilization at the cellular level to prevent or control inflammation. Has potent glucocorticoid activity and weak mineralocorticoid activity.

Metabolism

Metabolised in most tissues, but primarily the liver. Rapidly removed from the blood and distributed to muscle, liver, skin, intestines and kidneys. Excreted by the kidneys.



Indications

- Croup: mild, moderate and severe if unable to administer oral Dexamethasone.



Contraindications

- Hypersensitivity to Budesonide



Precautions

- Use with caution in patients with renal failure and hepatic impairment
- Patients with cirrhosis show an exaggerated response to glucocorticoids
- Bronchospasm: paradoxical bronchospasm may occur



Side Effects

- Nausea and vomiting
- Diarrhea
- Hypersensitivity reactions

Route of Administration

Nebuliser (NEB)



Onset	Duration	Half-life
Fairly rapid		2 – 3 hours

Dosage

PAEDIATRIC

Croup					
	III	II	I	NEB	2mg Single dose

CALCIUM CHLORIDE

Class

Electrolyte

Action

Moderates nerve and muscle performance via action potential excitation threshold regulation. Calcium is a positive inotrope. It is required for vasoconstriction to occur in vascular smooth muscles.

Metabolism

Excreted by the kidneys.



Indications

- Suspected hyperkalaemic cardiac arrest
- Severe hyperkalaemia – with haemodynamic compromise and/or significant cardiac rhythm disturbance
- Calcium channel and Beta blocker toxicity
- Magnesium toxicity – with associated hypotension unresponsive to IV fluids



Contraindications

- Hypersensitivity to Calcium chloride
- Not for routine use in cardiac arrest
- Digoxin toxicity



Precautions

- Extravasation
- Rapid administration may cause bradycardia or asystole
- Respiratory acidosis
- Renal impairment



Side Effects

- Local tissue necrosis following extravasation
- Thrombophlebitis
- Bradycardia, arrhythmias, hypotension, syncope, cardiac arrest
- Hot flush
- Dysgeusia
- GI irritation

Route of Administration				
Intravenous (IV)				
Intraosseous (IO)				

Special Notes

- Ensure route of administration (IV/IO) is well sited and free flowing prior to administration. Thoroughly flush line with NaCl after administration.
- Rapid administration may lead to adverse reactions.

Onset	Duration	Half-life
Rapid	30 – 60 minutes	

Dosage

ADULT

All listed indications					
	III	II	I	IV/IO	1g (10ml) given slowly Repeat PRN

CLOPIDOGREL

Class

Antiplatelet

Action

Clopidogrel is a specific and potent platelet aggregation inhibitor. It selectively inhibits the binding of ADP to platelet receptors, inhibiting platelet aggregation.

Metabolism

Metabolised by the liver. Inactive metabolites excreted in urine and faeces.



Indications

- STEMI
- Primary antiplatelet agent for patients with true aspirin allergy



Contraindications

- Hypersensitivity
- Active haemorrhage (excluding menses)
- Prior intracranial haemorrhage
- Patients < 18 years old



Precautions

- Liver impairment
- Patients with increased risk of bleeding



Side Effects

- Haemorrhage

Route of Administration

Oral (PO)				
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Special Notes

- Clopidogrel is only for administration for identified STEMI and not broader acute coronary syndromes.

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Onset	Duration	Half-life
≈ 30 minutes	7 – 10 days	8 hours

Dosage

ADULT

STEMI					
				PO	600mg Single dose

DEXAMETHASONE

Class

Corticosteroid

Action

Long-acting synthetic corticosteroid producing anti-inflammatory and immunosuppressive effects.

Metabolism

Hepatic metabolism and renal excretion.



Indications

- Croup: mild, moderate and severe
- Confirmed or suspected COVID-19 (≥18 years old)
 - Placed on CPAP
 - Requiring supplemental oxygen to maintain SpO₂ >93%



Contraindications

- Hypersensitivity to Dexamethasone
- Systemic fungal infections



Precautions

- None in the setting of croup
- Renal failure
- Patients with cirrhosis show and exaggerated response to glucocorticoids



Side Effects

- Hypersensitivity reactions

Route of Administration

Intravenous (IV)



Intraosseous (IO)				
Intramuscular (IM)				
Oral (PO)				

Special Notes

- Confirmed/suspected COVID-19 patients who have received steroids within the previous 4 hours are to have Dexamethasone withheld.
- The IV solution of Dexamethasone may be administered orally.
- Dexamethasone is the preferred corticosteroid for the treatment of croup

Onset	Duration	Half-life
30 minutes	72 hours	4 – 5 hours

Dosage

ADULT

Confirmed/suspected COVID-19					
	III	II	I	PO	6mg Single dose
	III	II	I	IV/IO/IM	6mg Single dose

PAEDIATRIC

Croup					
	III	II	I	PO	0.6mg/kg Maximum of 12mg Single dose
	III	II	I	IV/IO/IM	0.6mg/kg Maximum of 12mg Single dose

Paediatric Dexamethasone Dosing Table

Age	Weight	Group	
		0.6mg/kg PO/IM/IV/IO	Volume
		Undiluted: 8mg/2ml = 4mg/1ml	
Term-2 months	3-4kg	2mg	0.5ml
3-7 months	5-7kg	4mg	1ml
8-12 months	8-11kg	6mg	1.5ml
1-3 years	12-15kg	8mg	2ml
4-5 years	16-21kg	10mg	2.5ml
6-7 years	22-28kg	12mg	3ml
8-9 years	29-34kg	12mg	3ml
10-11 years	35-40kg		
12-13 years	41-45kg		
≥14 years	46-50kg		

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DEXTROSE (ORAL & IV)

Class

Carbohydrate/hyperglycaemic

Action

Glucose is primary source of energy for the body cells.

Metabolism

Broken down in most tissues and distributed throughout total body water. Stored in the liver and muscles as glycogen.



Indications

- Symptomatic hypoglycaemia
 - RBS < 4mmol/L for paediatrics and adults
 - RBS < 2.6mmol/L for newborns



Contraindications

- Nil



Precautions

- Hyperglycaemia



Side Effects

- Nil

Route of Administration

Oral (PO)				
Intravenous (IV)				
Intraosseous (IO)				

Special Notes

- Patients with an appropriate level of consciousness should be administered oral Dextrose.
- Patients with a reduced level of consciousness should be administered IV/IO Dextrose.

Onset	Duration	Half-life
2 – 4 minutes	20 minutes	

Dosage

ADULT

Hypoglycaemia					
				PO	15g tube Repeat once if patient remains hypoglycaemic
		II	I	IV	15-20g D10W (150-200ml) 50ml D10W in cardiac arrest Repeat if patient remains hypoglycaemic
	III	II	I	IO	15-20g D10W (150-200ml) 50ml D10W in cardiac arrest Repeat if patient remains hypoglycaemic

PAEDIATRIC

Hypoglycaemia					
				PO	15g tube Adjust amount based on patient size (5ml/teaspoon size amounts and repeat as required) Use pea size amount for Newborn Repeat once if patient remains hypoglycaemic
		II	I	IV	5ml/kg D10W 2.5ml/kg D10W in cardiac arrest 2ml/kg D10W in Newborn cardiac arrest Repeat if patient remains hypoglycaemic
	III	II	I	IO	5ml/kg D10W 2.5ml/kg D10W in cardiac arrest 2ml/kg D10W in Newborn cardiac arrest Repeat if patient remains hypoglycaemic

Paediatric Dextrose 10% Dosing Table

Age	Weight	Hypoglycaemia	Hypoglycaemia in cardiac arrest & ROSC
		Newborn: 2ml/kg IV/IO Infant/child: 5ml/kg IV/IO	2.5ml/kg IV/IO (Max single dose of 50ml)
		Dextrose 10%	Dextrose 10%
Preterm	2kg	4ml	4ml
Term-2 months	3-4kg	6ml	6ml
3-7 months	5-7kg	30ml	15ml
8-12 months	8-11kg	50ml	25ml
1-3 years	12-15kg	65ml	30ml
4-5 years	16-21kg	90ml	45ml
6-7 years	22-28kg	125ml	50ml
8-9 years	29-34kg	160ml	
10-11 years	35-40kg	190ml	
12-13 years	41-45kg	200ml	
≥14 years	46-50kg	240ml	

DICLOFENAC

Class

Non-steroidal anti-inflammatory (NSAID)

Action

Inhibits prostaglandin synthesis. Produces anti-inflammatory, analgesic and antipyretic properties.

Metabolism

Metabolised in the liver and excreted in bile and urine.



Indications

- Renal colic
- Mechanical back pain



Contraindications

- Hypersensitivity to NSAIDs and/or Aspirin
- AMI, stroke or bypass surgery in past 3 months
- Active GI ulcers
- Hypovolemia (dehydration/bleeding)
- Bleeding disorders
- ≥6 months pregnant
- Severe hepatic or renal impairment
- ≤18 years of age



Precautions

- Asthma patient
- Hypersensitivity reactions
- Long term use may lead to risk of ACS, CVA and GI bleeding
- Hepatic or renal impairment



Side Effects

- GI disturbances
- Headache, dizziness, blurred vision

Route of Administration

Intramuscular (IM)



Special Notes

- Onset of action 15-30 minutes.
- Administer deep IM in lateral upper gluteus maximus muscle (buttock).
- Do not give IV
- Due to risk of chronic dehydration, administer 250ml of IV fluids or oral fluids following administration of injectable non-steroidal analgesia

Onset	Duration	Half-life
15 - 30 minutes		1.15 hours



Dosage

ADULT

Renal colic/mechanical back pain



II

I

IM

75mg
Single dose

DIPHENHYDRAMINE

Class

Antihistamine

Action

Competes with histamine for H1-receptor sites and effector cells. Also produces anticholinergic and sedative effects.

Metabolism

Hepatic metabolism and renal excretion.



Indications

- Anaphylaxis
- Allergic reaction



Contraindications

- Hypersensitivity to Diphenhydramine
- Patients on Monoamine oxidase inhibitor (MAOI) therapy
- Neonates and premature infants



Precautions

- COPD, Bronchiectasis or obstructive sleep apnoea
- Asthmatics
- Hypotension
- Patients currently taking CNS depressant medications
- Pregnancy
- Cirrhosis or other liver disease



Side Effects

- Hypotension and reflex tachycardia
- Bradycardia
- Drowsiness
- Paradoxical excitation in children
- Dries bronchial secretions
- Blurred vision

- Headache
- Palpitations

Route of Administration

Intravenous (IV)				
Intraosseous (IO)				

Special Notes

- Administer deep IM if using IM route. Must be diluted if administering IV/IO.
- Administer slowly IV/IO.

Onset	Duration	Half-life
Rapid	6 – 8 hours	3.4 – 9.2 hours



Dosage

ADULT

Anaphylaxis/Allergic reaction

	III	II	I	IM/IV/IO	50mg Single dose
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PAEDIATRIC

Anaphylaxis/Allergic reaction

	III	II	I	IM/IV/IO	1mg/kg Max 50mg Single dose
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Paediatric Diphenhydramine Dosing Table

Age	Weight	Allergic reaction/ Anaphylaxis	
		1mg/kg IV/IO/IM	Volume
		Max dose: 50mg	
		Undiluted: 50mg/1ml = 5mg/0.1ml	
Preterm	2kg	5mg	0.1ml
Term-2 months	3-4kg		
3-7 months	5-7kg		
8-12 months	8-11kg	10mg	0.2ml
1-3 years	12-15kg	15mg	0.3ml
4-5 years	16-21kg	20mg	0.4ml
6-7 years	22-28kg	25mg	0.5ml
8-9 years	29-34kg	30mg	0.6ml
10-11 years	35-40kg	35mg	0.7ml
12-13 years	41-45kg	40mg	0.8ml
≥14 years	46-50kg	50mg	1ml

DROPERIDOL

Class

Antipsychotic

Action

Antipsychotic medication and Dopamine antagonist that produces sedative effects.

Metabolism

Hepatic metabolism with biliary/renal excretion.



Indications

- Acute Behavioural Disturbance:
 - SAT $\geq$ +2
 - De-escalation strategies have been unsuccessful
 - Sedation required to prevent harm to patient and others



Contraindications

- Hypersensitivity or adverse drug reaction to Droperidol
- Known Lewy Body Dementia
- Suspected or confirmed Parkinson's disease
- Previous dystonic reaction to Droperidol
- Patients under 8 years old
- Suspected sepsis (this is a relative contraindication)
- Known or suspected QT prolongation, including congenital long QT syndrome



Precautions

- Concurrent use of CNS depressants
- Hypoperfused state



Side Effects

- Hypotension, tachycardia
- QT prollongation (this is rare)
- Extrapyrarnidal effects (e.g. dystonic reactions, although this is rare)

Route of Administration				
Intravenous (IV)		III	II	I
Intramuscular (IM)		III	II	I

Special Notes

- Droperidol is the primary drug of choice for sedation in the context of acute behavioural disturbance.
- Administration of Droperidol or other antipsychotics/sedatives prior to arrival of HMCAS must be taken into consideration and dosing adjusted accordingly.
- Droperidol is suitable for managing ABD in patients with other forms of dementia, excluding Lewy Body Dementia (as listed in the contraindications)
- Droperidol should not be administered if the patient displays obvious signs of resting tremors and cogwheeling rigidity (an abnormal rigor in muscle tissue, characterised by jerky movements when the muscle is passively stretched) which is indicative of Parkinson's disease.
- There is little difference in onset time between IM and IV. IM will likely be the primary route for most cases.

Onset	Duration	Half-life
5 – 15 minutes	4 – 6 hours	N/A

Dosage

ADULT

Acute Behavioural Disturbance - Sedation					
	III	II	I	IM/IV	10mg if 16-<65 years Dose may be repeated once after 15 minutes if required Total maximum dose 20mg 5mg if ≥65 years/frail/<60kg Dose may be repeated once after 15 minutes if required Total maximum dose 10mg

PAEDIATRIC

Acute Behavioural Disturbance - Sedation					
	III	II	I	IM/IV	0.1-0.2mg/kg if 8-15 years Single maximum dose of 10mg Dose may be repeated once after 15 minutes if required Total maximum dose 20mg

FENTANYL

Class

Narcotic analgesic

Action

Acts on the central nervous system by binding with opioid receptors.

Metabolism

Hepatic metabolism and renal excretion.



Indications

- Analgesia for severe pain
- RSI
- Analgesia post intubation



Contraindications

- Hypersensitivity to Fentanyl



Precautions

- Elderly
- Hypotension
- Respiratory depression
- Known narcotic addiction
- Current MOAI therapy
- Hepatic or renal impairment
- Current asthma exacerbation



Side Effects

- Bradycardia
- Drowsiness or altered level of consciousness
- Hypotension
- Respiratory depression and apnoea
- Rigidity of the diaphragm and intercostal muscles
- Nausea and/or vomiting
- Pin point pupils

Route of Administration				
Intravenous (IV)				
Intraosseous (IO)				
Intravenous/intraosseous infusion (IV/IO INF)				
Intramuscular (IM)				
Intranasal (IN)				

Special Notes

- Fentanyl may potentiate respiratory depression and haemodynamic instability, particularly when administered IV in the hypovolaemic or CNS depressed patient.
- In the obese patient, use “ideal body weight”, not actual body weight to calculate dosage.

Onset	Duration	Half-life
< 3 minutes	30 – 60 minutes	2 – 3 hours

Dosage

ADULT

Analgesia					
	III	II	I	IM/IN	1-2mcg/kg
	III	II	I	IV/IO	1mcg/kg at 20mcg aliquots Repeat PRN every 1-2 minutes. Titrate to effect Max total dose of 200mcg
RSI					
	III	II	I	IV/IO	1mcg/kg for induction Consider omitting dose based on haemodynamic instability 0.5mcg/kg for sedation post-RSI
	III	II	I	IV/IO INF	1-10mcg/kg/hr titrated to maintain sedation post RSI Dilute 200mcg (2 x amps) Fentanyl with 16ml NS to a total of 20ml to give a concentration of 10mcg/1ml. Start infusion at desired range and titrate to effect and BP

PAEDIATRIC

Analgesia					
	III	II	I	IM/IN	1-2mcg/kg
	III	II	I	IV/IO	1mcg/kg Repeat PRN every 1-2 minutes. Titrate to effect Max total dose of 2mcg/kg
RSI					
	III	II	I	IV/IO	1mcg/kg for induction Adjust dose based on haemodynamic stability 0.5mcg/kg for sedation post-RSI
	III	II	I	IV/IO INF	1-10mcg/kg/hr titrated to maintain sedation post RSI Dilute 200mcg (2 x amps) Fentanyl with 16ml NS to a total of 20ml to give a concentration of 10mcg/1ml. Start infusion at desired range and titrate to effect and BP

Paediatric Fentanyl Dosing Table

Age	Weight	Analgesia IV/IO		Analgesia IN	
		1mcg/kg IV/IO	Volume	2mcg/kg IN	Volume
		Max dose: 2mcg/kg		Max dose per nostril: 0.3-0.5ml	
		Dilution: 100mcg(2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml		Undiluted: 100mcg/2ml = 5mcg/0.1ml Draw desired dose below and mix with NaCl to allow for use in NAD	
Preterm	2kg	2mcg	0.4ml	5mcg	0.1ml
Term-2 months	3-4kg	3mcg	0.6ml	5mcg	0.1ml
3-7 months	5-7kg	5mcg	1ml	10mcg	0.2ml
8-12 months	8-11kg	10mcg	2ml	20mcg	0.4ml
1-3 years	12-15kg	15mcg	3ml	30mcg	0.6ml
4-5 years	16-21kg	20mcg	4ml	40mcg	0.8ml
6-7 years	22-28kg	25mcg	5ml	50mcg	1ml
8-9 years	29-34kg	30mcg	6ml	60mcg	1.2ml
10-11 years	35-40kg	35mcg	7ml	70mcg	1.5ml
12-13 years	41-45kg	40mcg	8ml	80mcg	1.6ml
≥14 years	46-50kg	45mcg	9ml	90mcg	1.8ml

Age	Weight	RSI		Post RSI	
		1mcg/kg IV/IO	Volume	0.5mcg/kg IV/IO	Volume
		Dilution: 100mcg(2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml		Dilution: 100mcg(2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml	
Preterm	2kg	Not indicated in this age group for intubation		1mcg	0.2ml
Term-2 months	3-4kg			2mcg	0.4ml
3-7 months	5-7kg			3mcg	0.6ml
8-12 months	8-11kg			5mcg	1ml
1-3 years	12-15kg			6mcg	1.2ml
4-5 years	16-21kg			10mcg	2ml
6-7 years	22-28kg			12mcg	2.4ml
8-9 years	29-34kg	30mcg	6ml	15mcg	3ml
10-11 years	35-40kg	35mcg	7ml	20mcg	4ml
12-13 years	41-45kg	40mcg	8ml	20mcg	4ml
≥14 years	46-50kg	50mcg	10ml	25mcg	5ml

FUROSEMIDE

Class

Loop diuretic

Action

Causes venous dilation and reduces venous return. Inhibits absorption of sodium and chloride in the proximal and distal tubules and in the loop of Henle, resulting in diuresis.

Metabolism

Renal excretion.



Indications

- Acute pulmonary oedema with signs of systemic fluid overload



Contraindications

- Hypersensitivity to Furosemide
- Anuria



Precautions

- Elderly – should receive lower initial dose
- Hypotension
- Patients taking Lithium
- Patients taking ACE inhibitors or angiotensin receptor blockers (ARB)



Side Effects

- Hypotension
- Tinnitus and hearing loss
- Hyperglycaemia

Route of Administration				
Intravenous (IV)				

Special Notes

- Furosemide may potentiate the effects of Succinylcholine – a concern in the cardiac failure patient who requires RSI.

Onset	Duration	Half-life
3 – 5 minutes	≈ 2 hours	1.5 hours

Dosage

ADULT

Pulmonary oedema					
	III	II	I	IV/IO	0.5mg/kg Maximum 40mg Single dose

GLUCAGON

Class

Hyperglycaemic

Action

Hyperglycaemic agent that mobilises hepatic glycogen which is released into the blood as glucose. Glucagon has inotropic and chronotropic effects that are not mediated through beta-receptors.

Metabolism

Metabolised by the liver, kidneys and in the plasma.



Indications

- Symptomatic hypoglycaemic - ONLY if unable to obtain IV access or Dextrose unavailable or ineffective



Contraindications

- Hypersensitivity to Glucagon
- Allergy to beef or porcine protein
- Pheochromocytoma
- Neonates



Precautions

- Insulinoma
- Ischemic heart disease



Side Effects

- Nausea and vomiting
- Dizziness, lightheadedness, skin rash, dyspnoea
- Aggravation of myocardial ischemia
- Hypokalaemia

Route of Administration				
Intramuscular (IM)				
Subcutaneous (SC)				

Special Notes

- Ineffective in patients with depleted liver glycogen such as in starvation, adrenal insufficiency, chronic hypoglycaemia, impaired liver function and neonates.
- The effects of Glucagon wear off rapidly. Oral glucose should be provided to the patient once they have regained consciousness. Monitor for secondary hypoglycaemia.

Onset	Duration	Half-life
4 – 7 minutes	Varies	3 – 6 minutes

Dosage

ADULT

Hypoglycaemia					
				IM	1mg Single dose
				SC	1mg Single dose

PAEDIATRIC

Hypoglycaemia					
				IM	0.5mg <20kg or <6 years 1mg >20kg Single dose
				SC	0.5mg <20kg or <6 years 1mg >20kg Single dose

Paediatric Glucagon Dosing Table

Age	Weight	Symptomatic hypoglycaemia (if no IV access or Dextrose ineffective)	
		Reconstituted: 1mg/1ml	Volume
Preterm	2kg	Not for use in neonates (1 month and older use < 20kg dose below)	
Term-2 months	3-4kg		
3-7 months	5-7kg		
8-12 months	8-11kg	<20kg (<6 years) =0.5mg IM	0.5ml
1-3 years	12-15kg		
4-5 years	16-21kg		
6-7 years	22-28kg		
8-9 years	29-34kg	>20kg =1mg IM	1ml
10-11 years	35-40kg		
12-13 years	41-45kg		
≥14 years	46-50kg		

GLYCERAL TRINITRATE – GTN (SL & IV)

Class

Vasodilator

Action

Potent vasodilator that decreases preload by increasing venous capacity, pooling of venous blood in peripheral veins, reduces ventricular filling pressure and decreases arterial blood pressure, reducing afterload. Produces coronary artery vasodilation helping to improve collateral blood flow.

Metabolism

Readily absorbed and metabolised by the liver



Indications

- Acute coronary syndrome
- Acute pulmonary oedema



Contraindications

- Hypersensitivity to GTN
- Heart rate <50
- SBP <90. Use with caution in the elderly
- Ventricular tachycardia
- Acute CVA
- Phosphodiesterase inhibitor medication administration (e.g., Viagra® or Levitra®) in previous 24 hours
- Tadalafil administration (Cialis®) in previous 4 days



Precautions

- Inferior acute myocardial infarction
- Cerebral vascular disease
- Risk of hypotension and/or syncope
- Intoxication
- Heart rate >150 beats per minute



Side Effects

- Dizziness
- Hypotension
- Syncope
- Reflex tachycardia

- Headache
- Flushing

Route of Administration

Sublingual (SL)				
Intravenous/Intraosseous Infusion (IV/IO INF)				

Special Notes

- SL GTN is first line treatment for ACS. **IV GTN may be considered if unresponsive to SL GTN and/or alternative analgesia in the presence of STEMI only.**
- Patients with normal or low left ventricular filling pressures or pulmonary capillary pressures may be hypersensitive to the effects of IV GTN.

Onset	Duration	Half-life
< 2 minutes	20 – 30 minutes	5.5 minutes

Dosage

ADULT

Acute Coronary Syndrome

				SL	0.4mg (1 spray) Repeat PRN every 5 minutes and SBP >100mmHg
				IV/IO INF	5-20mcg/min Dilute to 50mg/500ml = 0.1mg/ml. Infuse at 3-12ml/hr Increase/decrease dose as required by 5mcg/min Titrate to BP

Acute Pulmonary Oedema

				SL	0.4-1.2mg (1-3 sprays) Repeat PRN every 5 minutes and SBP >100mmHg
				IV/IO INF	100mcg/min Dilute to 1mg/1ml. Infuse at 6ml/hr Increase/decrease dose as required by 0.6ml/hr Titrate to BP

HYDROCORTISONE

Class

Corticosteroid

Action

Adrenocortical steroid that produces an anti-inflammatory response. This inhibits the accumulation of inflammatory cells, phagocytosis, lysosomal enzyme release and synthesis and/or release of mediators of inflammation. It also has immunosuppressant properties.

Metabolism

Metabolised by the liver.



Indications

- Asthma (excluding mild)
- COPD – acute exacerbation with respiratory distress
- Anaphylaxis OR moderate/severe allergic reactions
- Adrenal crisis – suspected or high risk of



Contraindications

- Hypersensitivity to Hydrocortisone



Precautions

- Hepatic impairment and/or cirrhosis
- Renal impairment
- Recent myocardial infarction
- Hypertension



Side Effects

- Nausea
- Hypertension
- Cardiac arrhythmias

Route of Administration				
Intramuscular (IM)				
Intravenous (IV)				
Intraosseous (IO)				

Special Notes

- Reconstitute each 100mg vial with 2ml normal saline.
- To be given slowly via IV/IO route (>30s)
- Ambulance Paramedics may administer the patients own Hydrocortisone in line with the dosing regimine if the pateint has not already self-administered

Onset	Duration	Half-life
1 – 2 hours	6 – 12 hours	6 – 8 hours

Dosage

ADULT

Asthma/Anaphylaxis/Allergic reaction/COPD					
				IM/IV/IO	200mg Single dose
Adrenal Crisis					
				IM/IV	100mg Single dose

PAEDIATRIC

Asthma/Anaphylaxis/Allergic reaction					
	III	II	I	IM/IV/IO	5mg/kg (Max 200mg) Single dose
Adrenal Crisis					
		II	I	IM/IV	25mg: 0 – 4 years 50mg: 5 – 10 years 100mg: > 10 years Single dose

Paediatric Hydrocortisone Dosing Table

Age	Weight	Anaphylaxis / Bronchoconstriction	
		5mg/kg IV/IO	Volume
		Max dose: 200mg	
		Reconstituted: 100mg/2ml = 5mg/0.1ml	
Preterm	2kg	10mg	0.2ml
Term-2 months	3-4kg	15mg	0.3ml
3-7 months	5-7kg	30mg	0.6ml
8-12 months	8-11kg	50mg	1ml
1-3 years	12-15kg	65mg	1.3ml
4-5 years	16-21kg	90mg	1.8ml
6-7 years	22-28kg	125mg	2.5ml
8-9 years	29-34kg	150mg	3ml
10-11 years	35-40kg	175mg	3.5ml
12-13 years	41-45kg	200mg	4ml
≥14 years	46-50kg		

IBUPROFEN

Class

Non-steroidal anti-inflammatory drug (NSAID)

Action

A non-selective NSAID which inhibits the synthesis of prostaglandins resulting in analgesic, antipyretic and anti-inflammatory actions.

Metabolism

Metabolised by the liver and excreted by the kidneys.



Indications

- Mild to moderate pain



Contraindications

- Hypersensitivity to NSAIDS
- Current GI bleeding or bleeding peptic ulcer
- Dehydration and/or hypovolaemia
- Renal impairment
- NSAID induced asthma
- Pregnancy
- Heart failure
- Recent coronary artery bypass graft (CABG) surgery
- Chicken pox
- ≤1 year or ≤7 kg



Precautions

- Hepatic impairment
- Asthma
- History of GI bleeding or peptic ulcers



Side Effects

- Nausea
- Dyspepsia

- Tinnitus
- GI bleeding
- Dizziness

Route of Administration

Oral (PO)



Special Notes

- Take into consideration earlier doses of analgesics prior to arrival.
- Consider Ibuprofen administration in conjunction with Paracetamol for additional analgesia.
- Maximum daily dose is 1.2g.
- Administer with food/milk.
- Safe to use while breastfeeding.

Onset	Duration	Half-life
≈ 15 minutes	4 – 6 hours	2 hours

Dosage

ADULT

Analgesia					
				PO	200-400mg Every 4-6 hours

PAEDIATRIC

Analgesia					
				PO	See dosing table or follow dosing instructions on bottle

Paediatric Ibuprofen Dosing Table

Age	Mild to moderate pain relief/ Anti-inflammatory/ Reduce fever	
	Ibuprofen syrup: 100mg/5ml	Volume
Follow dosing on bottle for age groups		
Not for use in <7kg (<1 year old)		
1-2 years	50mg	2.5ml
3-7 years	100mg	5ml
8-14 years	200mg	10ml

IPRATROPIUM BROMIDE

Class

Anticholinergic agent

Action

Antimuscarinic agent that promotes bronchodilation by inhibiting cholinergic broncho-motor tone. Has synergistic effect with β -2 agonists to mediate bronchodilation.

Metabolism

Metabolised by the liver and excreted by the kidneys.



Indications

- Bronchoconstriction



Contraindications

- Hypersensitivity to Ipratropium Bromide or Atropine or any of its derivatives



Precautions

- Glaucoma
- Avoid contact with eyes



Side Effects

- Dry mouth
- Headache
- Palpitations
- Dilated pupils

Route of Administration

Nebuliser (NEB)



Special Notes

- Ipratropium Bromide should not *routinely* be administered on its own.
- Ipratropium Bromide can be administered together with Salbutamol in the same nebuliser.
- Ipratropium Bromide is a single dose. If additional doses are required then request for CCP assistance.

Onset	Duration	Half-life
5 minutes	4 – 6 hours	3 hours

Dosage

ADULT

Bronchoconstriction					
			I	NEB	0.5mg Administer together with salbutamol

PAEDIATRIC

Bronchoconstriction					
			I	NEB	0.25mg <6 years 0.25-0.5mg >6 years Administer together with salbutamol

Paediatric Ipratropium Bromide Dosing Table

Age	Weight	Bronchoconstriction
		0.25-0.5mg Neb
		0.25mg/2ml Neb should be mixed to a volume of 5ml (can be given with Salbutamol)
Preterm	2kg	<6 years = 0.25mg
Term-2 months	3-4kg	
3-7 months	5-7kg	
8-12 months	8-11kg	
1-3 years	12-15kg	
4-5 years	16-21kg	
6-7 years	22-28kg	>6 years = 0.25-0.5mg
8-9 years	29-34kg	
10-11 years	35-40kg	
12-13 years	41-45kg	
≥14 years	46-50kg	

KETAMINE

Class

Anaesthetic & Analgesic (non-barbiturate)

Action

NMDA receptor antagonist. Produces minimal haemodynamic compromise and maintains airway reflexes and respiratory drive (dose dependent). Ketamine produces a dissociative state which may potentially result in disinhibition or emergence phenomenon.

Metabolism

Metabolised by the liver and excreted by the kidneys.



Indications

- Rapid sequence induction
- Analgesia
 - Severe pain
- Sedation
 - Procedural sedation
 - Ongoing sedation post-RSI
 - Acute behavioural disturbance (SAT score ≥ 2)



Contraindications

- Hypersensitivity to Ketamine
- ≤ 3 months old



Precautions

- Elderly patients (consider reducing dose)
- Patients that have already received narcotic analgesia/sedation (consider reducing dose)
- Significant hypovolaemia (consider reducing dose)
- Chronic alcoholism or acute intoxication
- Suspected ACS or acute heart failure where an increase in BP or HR would be detrimental. Reduce drug dosing.
- Uncontrolled hypertension. Reduce drug dosing.



Side Effects

- Emergence phenomenon
- Hypertension
- Tachycardia
- Hypersalivation (particularly paediatrics)
- Laryngospasm
- Respiratory depression (rare – often after rapid administration)

Route of Administration

Intravenous (IV)	IV			
Intraosseous (IO)	IV			
Intravenous/intraosseous infusion (IV/IO INF)	IV			
Intramuscular (IM)	IV			
Intranasal (IN)	IV			

Special Notes

- Avoid rapid IV/IO administration to prevent potential respiratory depression.
- Ensure appropriate monitoring (sedation/analgesia bundle)
- Midazolam for emergence should only be administered if unable to calm patient with reassurance.

Onset	Duration	Half-life
30 seconds	5 – 20 minutes	10 – 15 minutes

Dosage

ADULT

Analgesia

IV	III	II	I	IM/IN	0.5-1mg/kg Repeat PRN
IV	III	II	I	IV/IO	0.25-0.5mg/kg Repeat PRN

Sedation					
	III	II	I	IM/IN	Light sedation: 1-2mg/kg Deeper sedation: 3-4mg/kg
	III	II	I	IV/IO	Light sedation: 0.5mg/kg Deeper sedation: 1-1.5mg/kg
RSI					
	III	II	I	IV/IO	1-2mg/kg for induction Adjust dose based on haemodynamic stability 0.5mg/kg for post-RSI sedation Repeat PRN every 15-20 minutes
	III	II	I	IV/IO INF	0.01-0.05mg/kg/min titrated to maintain sedation post RSI Dilute 200mg (4ml) Ketamine with 16ml NS to a total of 20ml to give a concentration of 10mg/1ml. Start infusion at desired range and titrate to effect

PAEDIATRIC

Analgesia					
	III	II	I	IM/IN	0.5-1mg/kg Repeat PRN
	III	II	I	IV/IO	0.25-0.5mg/kg Repeat PRN
Sedation					
	III	II	I	IM/IN	Light sedation: 1-2mg/kg Deeper sedation: 3-4mg/kg
	III	II	I	IV/IO	Light sedation: 0.5mg/kg Deeper sedation: 1-1.5mg/kg
RSI					
	III	II	I	IV/IO	1-2mg/kg for induction Adjust dose based on haemodynamic stability 0.5mg/kg for post-RSI sedation Repeat PRN every 15-20 minutes
	III	II	I	IV/IO INF	0.01-0.05mg/kg/min titrated to maintain sedation post RSI Dilute 200mg (4ml) Ketamine with 16ml NS to a total of 20ml to give a concentration of 10mg/1ml. Start infusion at desired range and titrate to effect

Paediatric Ketamine Dosing Table

Age	Weight	Analgesia IV/IO				Analgesia IM/IN			
		0.25-0.5mg/kg IV/IO				0.5-1mg/kg IM/IN			
		0.25mg/kg IV/IO	Volume	0.5mg/kg IV/IO	Volume	0.5mg/kg IM/IN	Volume	1mg/kg IM/IN	Volume
		Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml		Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml		Undiluted: 50mg/1ml = 5mg/0.1ml Dilution: Draw undiluted desired dose below and mix with NaCl to allow for use in NAD		Undiluted: 50mg/1ml = 5mg/0.1ml Dilution: Draw undiluted desired dose below and mix with NaCl to allow for use in NAD	
Preterm	2kg	Not for use < 3 months old		Not for use < 3 months old		Not for use < 3 months old		Not for use < 3 months old	
Term-2 months	3-4kg								
3-7 months	5-7kg	1.5mg	0.3ml	3mg	0.6ml	2.5mg	0.05ml	5mg	0.1ml
8-12 months	8-11kg	2.5mg	0.5ml	5mg	1ml	5mg	0.1ml	10mg	0.2ml
1-3 years	12-15kg	3mg	0.6ml	6mg	1.2ml	5mg	0.1ml	10mg	0.2ml
4-5 years	16-21kg	5mg	1ml	10mg	2ml	10mg	0.2ml	20mg	0.4ml
6-7 years	22-28kg	6mg	1.2ml	12mg	2.4ml	10mg	0.2ml	20mg	0.4ml
8-9 years	29-34kg	7.5mg	1.5ml	15mg	3ml	15mg	0.3ml	30mg	0.6ml
10-11 years	35-40kg	10mg	2ml	20mg	4ml	20mg	0.4ml	40mg	0.8ml
12-13 years	41-45kg	10mg	2ml	20mg	4ml	20mg	0.4ml	40mg	0.8ml
≥14 years	46-50kg	12.5mg	2.5ml	25mg	5ml	25mg	0.5ml	50mg	1ml

Age	Weight	Conscious / Procedural sedation (Light sedation)				Conscious / Procedural sedation (Deep sedation)			
		0.5mg/kg IV/IO	Volume	1mg/kg IM/IN	Volume	1mg/kg IV/IO	Volume	2mg/kg IM/IN	Volume
		Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml		Undiluted: 50mg/1ml = 5mg/0.1ml Dilution: Draw undiluted desired dose below and mix with NaCl to allow for use in NAD		Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml		Undiluted: 50mg/1ml = 5mg/0.1ml Dilution: Draw undiluted desired dose below and mix with NaCl to allow for use in NAD	
		Not for use < 3 months old		Not for use < 3 months old		Not for use < 3 months old		Not for use < 3 months old	
Preterm	2kg								
Term-2 months	3-4kg								
3-7 months	5-7kg	3mg	0.6ml	5mg	0.1ml	5mg	1ml	10mg	0.2ml
8-12 months	8-11kg	5mg	1ml	10mg	0.2ml	10mg	2ml	20mg	0.4ml
1-3 years	12-15kg	6mg	1.2ml	10mg	0.2ml	10mg	2ml	20mg	0.4ml
4-5 years	16-21kg	10mg	2ml	20mg	0.4ml	20mg	4ml	40mg	0.8ml
6-7 years	22-28kg	12mg	2.4ml	20mg	0.4ml	25mg	5ml	50mg	1ml
8-9 years	29-34kg	15mg	3ml	30mg	0.6ml	30mg	6ml	60mg	1.2ml
10-11 years	35-40kg	20mg	4ml	40mg	0.8ml	40mg	8ml	80mg	1.6ml
12-13 years	41-45kg	20mg	4ml	40mg	0.8ml	40mg	8ml	80mg	1.6ml
≥14 years	46-50kg	25mg	5ml	50mg	1ml	50mg	10ml	100mg	2ml

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Age	Weight	RSI Haemodynamically UNSTABLE patient		RSI Haemodynamically STABLE patient		Post RSI	
		1mg/kg IV/IO	Volume	2mg/kg IV/IO	Volume	0.5mg/kg IV/IO	Volume
		Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml =5mg/1ml		Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml =5mg/1ml		Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml =5mg/1ml	
Preterm	2kg	Not indicated in this age group for intubation		Not indicated in this age group for intubation		Not for use < 3 months old	
Term-2 months	3-4kg						
3-7 months	5-7kg					3mg	0.6ml
8-12 months	8-11kg					5mg	1ml
1-3 years	12-15kg					6mg	1.2ml
4-5 years	16-21kg					10mg	2ml
6-7 years	22-28kg					12mg	2.4ml
8-9 years	29-34kg	30mg	6ml	75mg	15ml	15mg	3ml
10-11 years	35-40kg	35mg	7ml	75mg	15ml	20mg	4ml
12-13 years	41-45kg	40mg	8ml	100mg	20ml	20mg	4ml
≥14 years	46-50kg	50mg	10ml	100mg	20ml	25mg	5ml

KETOROLAC

Class

Non-steroidal anti-inflammatory (NSAID)

Action

Reversibly inhibits cyclooxygenase 1 & 2 (COX 1 & 2) enzymes resulting in decreased formation of prostaglandin precursors. Produces anti-inflammatory, antipyretic and analgesic properties.

Metabolism

Metabolised by the liver and excreted by the kidneys and in bile.



Indications

- Analgesia
 - Renal colic
 - Mechanical back pain
 - Muscular-skeletal pain
 - Migraine (CCP only)



Contraindications

- Hypersensitivity to NSAIDs
- ≤18-year-old
- Severe heart failure
- Severe renal failure
- Severe hepatic failure
- NSAID administration in previous 6 hours
- Recent myocardial infarction or stroke (< 3 months)
- Active GI bleed or peptic ulcers
- Hypovolaemia (active bleeding or dehydration)
- Taking anti-coagulant medications
- ≥6 months pregnant
- Peripheral arterial disease
- Taking Selective Serotonin Reuptake Inhibitor (SSRI) antidepressants



Precautions

- Asthma
- Elderly
- Long term NSAID therapy
- Chronic hepatic or renal impairment



Side Effects

- GI disturbances
- Headache
- Dizziness
- Vertigo

Route of Administration

Intramuscular (IM)				
Intravenous (IV)				
Intraosseous (IO)				

Special Notes

- Due to risk of chronic dehydration, administer 250ml of IV fluids or oral fluids following administration of injectable non-steroidal analgesia.
- Consider alternative analgesic in the elderly or low body weight patient.
- If administering IV/IO, dilute Ketorolac with 2ml normal saline (10mg/1ml). Discard 2ml (20mg) and reserve remainder (10mg/1ml) for administration.
- If administering IM, draw up all of the 30mg/1ml into 1ml syringe and discard 0.7ml (20mg) and reserve 0.3ml (10mg) for administration.

Onset	Duration	Half-life
< 10 minutes	6 – 8 hours	33 – 63 hours

Dosage

ADULT

Analgesia					
			I	IM	10mg Single dose
		II	I	IV	10mg Single dose
	III	II	I	IO	10mg Single dose

MAGNESIUM SULPHATE

Class

Electrolyte

Action

Produces vasodilation and bronchodilation by inhibiting smooth muscle contraction. Magnesium also possesses anticonvulsant and ant-arrhythmic properties.

Metabolism

Filtered by the kidneys and primarily excreted in urine.



Indications

- Pre-eclampsia and Eclampsia
- Moderate - severe asthma
- Torsades de pointes



Contraindications

- Hypersensitivity to Magnesium
- Atrioventricular (AV) block
- Addison's disease
- Infective hepatitis



Precautions

- Renal impairment
- Myasthenia gravis



Side Effects

- Drowsiness
- Skin flushing and sweating
- Magnesium toxicity may result in respiratory depression, hypotension and loss of tendon reflexes

Route of Administration				
Intravenous infusion (IV INF)				
Intraosseous infusion (IO INF)				

Special Notes

- Ensure IV/IO line is well sited and flowing freely prior to administration. Flush the line thoroughly with normal saline following administration.

Onset	Duration	Half-life
Immediate	30 minutes	Varies

Dosage

ADULT

Severe asthma/Torsade's de pointes					
	III	II	I	IV/IO INF	2g Infused over 10 minutes. May repeat once to a max dose of 4g.
Pre-eclampsia/Eclampsia					
	III	II	I	IV/IO INF	4g Infused over 10 minutes

PAEDIATRIC

Severe asthma/Torsade's de pointes					
	III	II	I	IV/IO INF	25 – 50mg/kg (Max dose of 2g) Infused over 20 minutes

Paediatric Magnesium Sulphate Dosing Table

Age	Weight	Bronchoconstriction/ Torsades de Pointes		Bronchoconstriction/ Torsades de Pointes	
		25mg/kg IV/IO (over 20 minutes)	Volume	50mg/kg IV/IO (over 20 minutes)	Volume
		Max dose: 2g		Max dose: 2g	
		Undiluted: 100mg/1ml		Undiluted: 100mg/1ml	
Preterm	2kg	50mg	0.5ml	100mg	1ml
Term-2 months	3-4kg	75mg	0.75ml	150mg	1.5ml
3-7 months	5-7kg	150mg	1.5ml	300mg	3ml
8-12 months	8-11kg	250mg	2.5ml	500mg	5ml
1-3 years	12-15kg	350mg	3.5ml	700mg	7ml
4-5 years	16-21kg	450mg	4.5ml	900mg	9ml
6-7 years	22-28kg	625mg	6.25ml	1250mg	12.5ml
8-9 years	29-34kg	800mg	8ml	1600mg	16ml
10-11 years	35-40kg	1000mg	10ml	2000mg	20ml
12-13 years	41-45kg	1100mg	11ml		
≥14 years	46-50kg	1200mg	12ml		

METHOXYFLURANE (PENTHROX®)

Class

Analgesic

Action

A volatile, self-administered inhaled analgesic for short-term pain relief. Methoxyflurane is more susceptible to metabolism than other halogenated ethers and has a greater propensity to diffuse into fatty tissue.

Metabolism

Metabolised by the liver and excreted primarily by the lungs.



Indications

- Mild, moderate or severe pain
- Labour and obstetric pain
- Cardiac chest pain with no evidence of cardiovascular instability



Contraindications

- Hypersensitivity to Methoxyflurane
- ≤1 year old
- History of malignant hyperthermia
- Patients who have already received 6ml in previous 24 hours
- Cardiovascular instability (e.g. hypotension, arrhythmias, heart failure)
- Altered level of consciousness due to any cause
- Clinically significant renal impairment
- Clinically evident respiratory depression



Precautions

- Intoxicated or drug affected patients
- Renal and/or hepatic disease



Side Effects

- Altered level of consciousness
- Cough
- Renal/hepatic failure (as a result of repeated administration of high doses)

Route of Administration

Inhaled (INH)



Special Notes

- Do not exceed a dose of 6ml/24 hour period or 3ml/24 hour period for patients with renal and/or hepatic disease.
- Patients < 5 years administered Methoxyflurane may result in deep sedation.
- Methoxyflurane should not be used to deliberately sedate/reduce the LOC of a patient.
- Patients receiving Pentrox need to be able to follow instructions and self-administer the medication.
- APs may only administer a single dose. CCPs may administer a second dose once on scene if required.

Onset	Duration	Half-life
1 – 3 minutes	5 – 10 minutes	



Dosage

ADULT & PAEDIATRIC

Analgesia

				INH	3ml May repeat once after 25 minutes Max of 6ml/24 hours (2 doses)
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METOCLOPRAMIDE

Class

Antiemetic

Action

Stimulates the motility of the upper GI tract, increases the tone and amplitude of GI contractions resulting in accelerated gastric emptying.

Metabolism

Metabolised by the liver and excreted by the kidneys.



Indications

- Nausea and vomiting due to:
 - ACS
 - Migraine
 - Gastroenteritis



Contraindications

- Hypersensitivity to Metoclopramide
- GI bleeding
- Bowel obstruction or perforation
- Pheochromocytoma
- Epilepsy
- GI surgery in previous 4 days
- <18 years old



Precautions

- Abdominal pain of unknown cause
- Hepatic/renal impairment – use reduced dose



Side Effects

- Anxiety/restlessness
- Lethargy
- Dry mouth

- Muscle tremor
- Extrapramidal reactions
- Drowsiness

Route of Administration

Intramuscular (IM)	IV			
Intravenous (IV)	IV			
Intraosseous (IO)	IV			

Special Notes

- Metoclopramide is not effective for motion sickness.
- Metoclopramide is only to be considered if Ondansetron is unavailable. Ondansetron remains the primary choice for antiemetic.

Onset	Duration	Half-life
1 – 3 minutes	1 – 2 hours	5 – 6 hours

Dosage

ADULT

Nausea & Vomiting

IV	III	II	I	IM/IV/IO	10mg Single dose given slowly
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MIDAZOLAM

Class

Benzodiazepine

Action

A short acting CNS depressant that produces amnesia, anaesthesia, hypnosis and sedation. Midazolam enhances the action of gamma-amino butyric acid (GABA).

Metabolism

Metabolised by the liver and excreted by the kidneys.



Indications

- Seizures (generalised and focal)
- Sedation
 - Procedural – Temporary transcutaneous cardiac pacing or cardioversion
 - Ketamine emergence
 - Combative patient
 - Acute behavioral disturbance (SAT score ≥ 2) if ketamine contraindicated or ineffective
- Anxiolysis in analgesia



Contraindications

- Hypersensitivity to Benzodiazepines



Precautions

- Reduced doses for patients with renal impairment, hepatic impairment, cardiac failure, shock and the elderly should be considered
- Myasthenia gravis
- Respiratory depression/compromise
- Pregnancy
- Acute alcohol intoxication



Side Effects

- Hypotension
- Respiratory depression

Route of Administration				
Intravenous (IV)				
Intraosseous (IO)				
Intramuscular (IM)				
Intranasal (IN)				
Oral (PO)				

Special Notes

- Patients that receive a benzodiazepine prior to paramedic arrival should have the amount administered taken into account prior to administering additional Midazolam.
- Midazolam for sedation of any nature should only be considered if Ketamine is contraindicated or unavailable. **Ketamine remains the primary drug of choice for sedation.**
- Oral use of Midazolam is an acceptable route for mild sedation.

Onset	Duration	Half-life
5 – 15 minutes (IM) 1 – 3 minutes (IV)	Varies	2.5 hours

Dosage

ADULT

Seizures/Sedation					
	III	II	I	IN	0.2mg/kg Max of 1ml per nostril/dose
	III	II	I	IM	0.2mg/kg Repeat PRN after 15 minutes Max single dose of 10mg
	III	II	I	IV/IO	0.1mg/kg Repeat PRN after 5 minutes Administer in aliquots of 5mg to maximum total of 20mg 3mg for anxiolysis, TCP or cardioversion
	III	II	I	PO	5 – 15mg Mixed with juice/water

PAEDIATRIC

Seizures/Sedation					
	III	II	I	IN	0.2mg/kg Max of 1ml per nostril/dose
	III	II	I	IM	0.2mg/kg Repeat PRN after 15 minutes Max single dose of 10mg
	III	II	I	IV/IO	0.1mg/kg Repeat PRN after 5 minutes Maximum single dose of 5mg to maximum total of 10mg

Paediatric Midazolam Dosing Table

Age	Weight	Seizure management IV		Seizure management IN/buccal	
		0.1mg/kg IV/IO	Volume	0.2mg/kg IN/buccal	Volume
		Max dose: 10mg			
		Dilution: 15mg (3ml) + 12ml NaCl = 15mg/15ml = 1mg/1ml		Undiluted: 15mg/3ml = 5mg/1ml = 0.5mg/0.1ml Dilution: Draw undiluted desired dose below and mix with NaCl to allow for use in NAD	
Preterm	2kg	0.2mg	0.2ml	0.5mg	0.1ml
Term-2 months	3-4kg	0.3mg	0.3ml	0.5mg	0.1ml
3-7 months	5-7kg	0.6mg	0.6ml	1mg	0.2ml
8-12 months	8-11kg	1mg	1ml	2mg	0.4ml
1-3 years	12-15kg	1.5mg	1.5ml	3mg	0.6ml
4-5 years	16-21kg	2mg	2ml	4mg	0.8ml
6-7 years	22-28kg	2.5mg	2.5ml	5mg	1ml
8-9 years	29-34kg	3mg	3ml	6mg	1.2ml
10-11 years	35-40kg	3.5mg	3.5ml	7mg	1.4ml
12-13 years	41-45kg	4mg	4ml	8mg	1.6ml
≥14 years	46-50kg	5mg	5ml	10mg	2ml

NALOXONE

Class

Opioid antagonist

Action

Competes for the same receptor sites as opioids, preventing or reversing the effects of opioids.

Metabolism

Metabolised by the liver.



Indications

- Opioid overdose resulting in respiratory depression



Contraindications

- Hypersensitivity to Naloxone



Precautions

- Administer with caution to opioid dependent patients as it may produce sudden withdrawal syndrome
- Neonates



Side Effects

- Acute withdrawal syndrome:
 - Nausea and/or vomiting
 - Sweating
 - Tachycardia
 - Hypertension
 - Tremors
 - Seizures
 - Agitation
- Acute pulmonary oedema

Route of Administration				
Intravenous (IV)		III	II	I
Intraosseous (IO)		III	II	I
Intramuscular (IM)		III	II	I
Intranasal (IN)		III	II	I

Special Notes

- Administering Naloxone to opioid dependent patients should be done cautiously and only if respiratory depression is present.
- The duration of effect of the opioid may be longer than that of the Naloxone, potentially resulting in renarcotisation. Repeat doses may be required.
- To minimise the risk of refusal to go to hospital and/or patient aggression (upon waking), aim to restore effective breathing without waking the patient fully (titrate effect using 100mcg aliquots).

Onset	Duration	Half-life
3 – 5 minutes (IM) 1 – 3 minutes (IV)	≈ 60 minutes	60 minutes

Dosage

ADULT

Opioid Overdose					
	III	II	I	IV/IO	0.4 – 2mg (Max total 10mg) Repeat PRN every 2-3 minutes Administer aliquots of 100mcg titrated to effect
	III	II	I	IM	0.4 – 2mg (Max total 10mg) Repeat PRN every 2-3 minutes
	III	II	I	IN	2mg given in 1ml boluses, alternating between nostrils to allow for absorption

PAEDIATRIC

Opioid Overdose					
	III	II	I	IV/IO/IM	0.01mg/kg (Max single dose: 0.4mg; Max total 2mg) Repeat PRN every 2-3 minutes. Use guide below:

Paediatric Naloxone Dosing Table

Age	Weight	Opioid overdose	
		0.01mg/kg IV/IO/IM	Volume
		Max single dose: 0.4mg Max total dose: 2mg	
		Dilution: 0.4mg(1ml) + 3ml NaCl = 0.4mg/4ml = 0.1mg/1ml	
Preterm	2kg	0.02mg	0.2ml
Term-2 months	3-4kg	0.03mg	0.3ml
3-7 months	5-7kg	0.06mg	0.6ml
8-12 months	8-11kg	0.1mg	1ml
1-3 years	12-15kg	0.15mg	1.5ml
4-5 years	16-21kg	0.2mg	2ml
6-7 years	22-28kg	0.25mg	2.5ml
8-9 years	29-34kg	0.35mg	3.5ml
10-11 years	35-40kg	0.4mg	4ml
12-13 years	41-45kg		
≥14 years	46-50kg		

NORADRENALINE

Class

Sympathomimetic

Action

A vasopressor that acts on α receptors primarily, and to a lesser extent β adrenergic receptors, resulting in increased vascular tone.

Metabolism

Primarily metabolised by monoamine oxidase at synaptic level.



Indications

- Adults: Shock of all pathologies unresponsive to adequate fluid resuscitation
- Paediatrics: Fluid refractory septic shock only



Contraindications

- Hypersensitivity to Noradrenaline



Precautions

- Hypertension
- Hypovolaemia
- Concurrent MAOI therapy



Side Effects

- Reflex bradycardia
- Hypertension
- Extravasation necrosis

Route of Administration

Intravenous/intraosseous infusion (IV/IO INF)



III

II

I

Special Notes

- Noradrenaline must be diluted with D5W or NS.
- Must be administered using a syringe driver through a dedicated, well sited, free flowing line.
- In event of extravasation, stop infusion immediately and disconnect. Gently aspirate extravasated solution (DO NOT flush the line). Remove the cannula, elevate the extremity and report the extravasation to emergency department staff at handover.
- Titrate infusion to MAP/BP target.
- Label ALL syringes and solution bags.

Onset	Duration	Half-life
30 seconds	5 – 10 minutes	2 minutes

Dosage

ADULT

Shock					
	III	II	I	IV/IO INF	0.01 – 0.3mcg/kg/min Increase rate by 0.05mcg/kg/min if required Dilution: add 2 x 4mg/4ml amps into 500ml NS, creating 8mg/500ml (16mcg/ml). Draw 50ml of Noradrenaline solution into 50ml syringe and start infusion at above dose.
Interfacility Transfers					
	III	II	I	IV INF	Follow hospital prescribed dilution and dosing.

PAEDIATRIC

Interfacility Transfers					
	III	II	I	IV INF	Follow hospital prescribed dilution and dosing.
Septic Shock					
	III	II	I	IV/IO INF	0.05– 0.1mcg/kg/min Increase rate by 0.05mcg/kg/min if required Dilution: add 2 x 4mg/4ml amps into 500ml NS, creating 8mg/500ml (16mcg/ml). Draw 50ml of Noradrenaline solution into 50ml syringe and start infusion at above dose.

ONDANSETRON

Class

Antiemetic

Action

A serotonin 5-HT₃ receptor antagonist which primarily reduces vagus nerve activity (which activates the vomiting center in the medulla oblongata). Further blocks serotonin receptors in the chemoreceptor trigger zone.

Metabolism

Metabolised by the liver and excreted by the kidneys.



Indications

- Significant nausea and/or vomiting



Contraindications

- Hypersensitivity to Ondansetron
- Congenital long QT syndrome
- Current apomorphine therapy (used for Parkinson's treatment)



Precautions

- Hepatic impairment
- Intestinal obstruction
- Should only be administered to pregnant women in first trimester for extreme uncontrolled hyperemesis



Side Effects

- Headache
- Constipation
- Flushing
- Dysrhythmias

Route of Administration				
Intravenous (IV)		III	II	I
Intraosseous (IO)		III	II	I
Intramuscular (IM)		III	II	I

Special Notes

- Ondansetron is not for routine administration. It should only be administered if significant nausea and/or vomiting is present.
- Ondansetron must be diluted with 10ml normal saline if administering via the IV/IO route.
- Ondansetron is not effective for the management of motion sickness.

Onset	Duration	Half-life
5 minutes	Several hours	3 – 4 hours

Dosage

ADULT

Nausea & Vomiting					
	III	II	I	IV/IO/IM	4mg Single dose. Given slowly.

PAEDIATRIC

Nausea & Vomiting					
	III	II	I	IV/IO/IM	0.1mg/kg Maximum dose of 4mg. Given slowly.

Paediatric Ondansetron Dosing Table

Age	Weight	Nausea and vomiting	
		0.1mg/kg IV/IM	Volume
		Max dose: 4mg	
		Dilute: 4mg (2ml) + 8ml NaCl= 4mg/10ml = 0.4mg/1ml	
Preterm	2kg	0.2mg	0.5ml
Term-2 months	3-4kg	0.4mg	1ml
3-7 months	5-7kg	0.6mg	1.5ml
8-12 months	8-11kg	1mg	2.5ml
1-3 years	12-15kg	1.4mg	3.5ml
4-5 years	16-21kg	1.8mg	4.5ml
6-7 years	22-28kg	2.6mg	6.5ml
8-9 years	29-34kg	3.2mg	8ml
10-11 years	35-40kg	3.6mg	9ml
12-13 years	41-45kg	4mg	10ml
≥14 years	46-50kg		

PARACETAMOL

Class

Analgesic and antipyretic

Action

Paracetamol is a p-aminophenol derivative that produces analgesic and antipyretic activity.

Metabolism

Metabolised by the liver and excreted by the kidneys.



Indications

- Mild to moderate pain (excluding ischemic cardiac pain)
- Fever causing distress (oral use only)



Contraindications

- Hypersensitivity to Paracetamol
- Environmental heat illness



Precautions

- Hepatic impairment
- Renal impairment



Side Effects

- Nausea

Route of Administration

Intravenous (IV)				
Intraosseous (IO)				
Oral (PO)				

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Special Notes

- Take into consideration paracetamol that was previously self-administered or administered by the carer, parent or guardian.
- The dose of Paracetamol should not be repeated.

Onset	Duration	Half-life
10 – 60 minutes	4 hours	≈ 2 hours

Dosage

ADULT

Analgesia					
				PO	500 – 1000mg (tablet)
		II	I	IV	500 – 1000mg
	III	II	I	IO	500 – 1000mg
Fever					
				PO	500 – 1000mg (tablet)

PAEDIATRIC

Analgesia					
				PO	15mg/kg (syrup)
		II	I	IV	Paediatric: 15mg/kg Infant: 7.5mg/kg
	III	II	I	IO	Paediatric: 15mg/kg Infant: 7.5mg/kg
Fever					
				PO	15mg/kg (syrup)

Paediatric Paracetamol Dosing Table

Age	Weight	Pain and/or fever (oral administration)		Moderate pain (IV administration)	
		15mg/kg oral	Volume	Infants: 7.5mg/kg IV Child: 15mg/kg IV	Volume
		Paracetamol syrup: 120mg/5ml		Max dose: 20mg/kg Undiluted: 1000mg/100ml = 10mg/1ml	
Preterm	2kg	Not recommended < 2 months old		15mg	1.5ml
Term-2 months	3-4kg			20mg	2ml
3-7 months	5-7kg	84mg	3.5ml	45mg	4.5ml
8-12 months	8-11kg	132mg	5.5ml	75mg	7.5ml
2-3 years	12-15kg	180mg	7.5ml	200mg	20ml
4-5 years	16-21kg	240mg	10ml	270mg	27ml
6-7 years	22-28kg	360mg	15ml	375mg	37.5ml
8-9 years	29-34kg	420mg	17.5ml	480mg	48ml
10-11 years	35-40kg	480mg	20ml	570mg	57ml
12-13 years	41-45kg	600mg	25ml	630mg	63ml
≥14 years	46-50kg	720mg	30ml	720mg	72ml

PHENYLEPHRINE

Class

Sympathomimetic

Action

Direct action on α -1 receptors causing vasoconstriction of arterioles.

Metabolism

Metabolised in the liver primarily by monoamino-oxidase deamination. Excreted by the kidneys.



Indications

- Hypotension following drug induced vasodilation (RSI, sedation)
- Traumatic brain injury with inadequate MAP
- Shock unresponsive to fluid therapy



Contraindications

- Hypersensitivity to Phenylephrine
- Severe hypertension
- Bradycardia
- Heart block
- Cardiogenic shock
- Hyperthyroidism



Precautions

- Hypertension
- Cardiac dysfunction/disease



Side Effects

- Reflex bradycardia
- Atrioventricular (AV) block
- Myocardial ischemia

Route of Administration				
Intravenous (IV)		III	II	I
Intraosseous (IO)		III	II	I
Intravenous/intraosseous infusion (IV/IO INF)		III	II	I

Special Notes

- In time critical situations or when starting an infusion is not practical, bolus doses of phenylephrine may be administered. If multiple bolus doses are required, an appropriate vasopressor infusion should be administered.
- Side effects encountered are generally short-lived and should resolve after infusion has been stopped.
- Blood pressure must be closely monitored every 2 minutes. Maintain a constant BP and avoid spiking BP.

Onset	Duration	Half-life
Immediate	15 – 20 minutes	2 – 3 hours

Dosage

ADULT

Hypotension/Shock					
	III	II	I	IV/IO	25 – 100mcg (0.25 – 1ml) Administered as a bolus Repeat PRN every 2 – 5 minutes
	III	II	I	IV/IO INF	100mcg/min (infuse at 60ml/hr) Dilution: mix 10mg into 100ml NS. Draw 50ml into 50ml syringe. Infuse at rate above (use special functions feature) Reduce rate by 2.5 – 5ml/hr (4.15 – 8.3mcg/min) as required. Increase rate by 5 – 10ml/hr (8.3 – 16.6mcg/min) as required. Once BP has stabilised, reduce rate to 30ml/hr (50mcg/min).

PROCHLORPERAZINE (STEMETIL)

Class

Antiemetic

Action

Dopamine antagonism has direct action on labyrinthine apparatus and on the chemoreceptor trigger zone.

Metabolism

Metabolised in the liver and excreted by the kidneys and bile.



Indications

- Nausea and/or vomiting due to motion sickness
- Vomiting due to inner ear disorders (vertigo)



Contraindications

- Hypersensitivity to Prochlorperazine
- CNS depression
- Circulatory collapse
- Blood dyscrasia
- Hepatic impairment
- Renal impairment
- Pheochromocytoma
- Paediatrics



Precautions

- Elderly
- Epilepsy
- Hyperglycaemia
- Parkinsons
- Hypothyroidism



Side Effects

- Blurred vision
- Constipation

- Dry mouth
- Urinary retention
- Hyperglycaemia
- Seizures in epileptics
- Drowsiness
- Akathisia
- Parkinsonism

Route of Administration

Intramuscular (IM)



III

II

I

Special Notes

- Prochlorperazine is only for use by Retrieval Services during non-emergency transfers.

Onset	Duration	Half-life
5 – 10 minutes	3 - 4 hours	≈ 24 hours



Dosage

ADULT

Nausea and/or vomiting



III

II

I

IM

12.5mg
Administered deep IM

ROCURONIUM

Class

Non-depolarizing muscle relaxant

Action

Competes with acetylcholine and blocks the receptors at the motor neuron endplate in striated muscle.

Metabolism

Metabolised by the liver with hepato-biliary excretion.



Indications

- Paralysis for RSI
- On-going paralysis post-RSI
- Facilitation of mechanical ventilation



Contraindications

- Hypersensitivity to Rocuronium
- Muscular dystrophies and myotonias
- Personal or family history of malignant hyperthermia



Precautions

- Cardiac or respiratory dysfunction may be potentiated
- Renal and hepatic dysfunction may lead to prolonged neuromuscular blockade
- Elderly may have a slower onset and prolonged duration of action
- Burn victims may develop resistance and require more frequent dosing



Side Effects

- Hypotension
- May induce episode of malignant hyperthermia in susceptible individuals
- Rash
- Pain at injection site

Route of Administration				
Intravenous (IV)		III	II	I
Intraosseous (IO)		III	II	I

Special Notes

- Rocuronium is not expected to modulate the cardiovascular effects of other anaesthetic drugs.
- Ability to adequately ventilate the patient must be established prior to administration of Rocuronium.
- The action of Rocuronium can be antagonised by acetylcholinesterase inhibitors (Neostigmine) or nonselective relaxant binding agents (Sugammadex).
- Age restrictions for RSI standardised to ≥ 8 years for both medical and trauma.

Onset	Duration	Half-life
60 – 90 seconds	$\approx 30 - 45$ minutes	14 – 18 minutes

Dosage

ADULT

Paralysis for RSI					
	III	II	I	IV/IO	1.5mg/kg Use standard dose of 100-150mg for induction
Paralysis post-RSI/Facilitate mechanical ventilation					
	III	II	I	IV/IO	0.25mg/kg Repeat PRN every 30 – 45 minutes

PAEDIATRIC

Paralysis for RSI [Medical and Trauma ≥ 8 years]					
	III	II	I	IV/IO	1.5mg/kg
Paralysis post-RSI/Facilitate mechanical ventilation					
	III	II	I	IV/IO	0.25mg/kg Repeat PRN every 30 – 45 minutes

Paediatric Rocuronium Dosing Table

Age	Weight	PRE intubation paralytic		POST intubation paralytic	
		1.5mg/kg IV	Volume	0.25mg/kg IV	Volume
		Undiluted: 10mg/1ml (1mg/0.1ml)		Undiluted: 10mg/1ml (1mg/0.1ml)	
Preterm	2kg	Not indicated in this age group for PRE intubation		1mg	0.1ml
Term-2 months	3-4kg			1mg	0.1ml
3-7 months	5-7kg			2mg	0.2ml
8-12 months	8-11kg			2mg	0.2ml
2-3 years	12-15kg			4mg	0.4ml
4-5 years	16-21kg			5mg	0.5ml
6-7 years	22-28kg			6mg	0.6ml
8-9 years	29-34kg	50mg	5ml	10mg	1ml
10-11 years	35-40kg	60mg	6ml		
12-13 years	41-45kg	70mg	7ml		
≥14 years	46-50kg	75mg	7.5ml		

SALBUTAMOL

Class

B-adrenergic agonist

Action

Direct acting sympathomimetic mainly affecting β_2 -adrenoceptors. Mainly acts as a bronchodilator but also has inotropic and chronotropic actions.

Metabolism

Metabolised by the liver and excreted by the kidneys.



Indications

- Bronchoconstriction



Contraindications

- Hypersensitivity to Salbutamol



Precautions

- Acute pulmonary oedema
- Ischemic heart disease
- Tachyarrhythmias



Side Effects

- Anxiety
- Tachyarrhythmias
- Tremors
- Hypokalaemia and metabolic acidosis

Route of Administration

Intravenous/intraosseous infusion (IV/IO INF)	IV	III	II	I
Nebuliser (NEB)	IV	III	II	I

Special Notes

- The use of IV versus nebulised Salbutamol should be gauged based on the adequacy of tidal volume.
- IV Salbutamol is generally reserved as an alternative when first line medications are ineffective.
- COPD patients requiring nebulisation should receive the lowest appropriate oxygen flow rate (6L) to produce aerolisation.

Onset	Duration	Half-life
2 – 5 minutes	16 – 60 minutes	1.6 hours

Dosage

ADULT

Bronchoconstriction					
			I	NEB	5mg Repeat 5mg dose PRN
			I	IV/IO INF	250mcg infused over 10 minutes Dilution: 500mcg (1ml amp) mixed with 49ml NS in a 50ml syringe (10mcg/ml). Use syringe driver infusing at 150ml/hr.

PAEDIATRIC

Bronchoconstriction					
			I	NEB	2.5 – 5mg Repeat 2.5 - 5mg dose PRN
			I	IV/IO INF	10mcg/kg infused over 10 minutes Maximum dose of 250mcg Dilution: 500mcg (1ml amp) mixed with 49ml NS in a 50ml syringe (10mcg/ml). Use syringe driver special functions

Paediatric Salbutamol Dosing Table

Age	Weight	Bronchoconstriction		Bronchoconstriction	
		2.5-5mg Neb	10mcg/kg IV over 10 minutes	Volume	
		2.5mg/3ml nebule Neb should be mixed to a volume of 5ml (can be given with Ipratropium Bromide)	Max dose: 250mcg		
			Dilution: 500mcg/1ml + 49ml NaCl = 10mcg/1ml (Use special functions on syringe driver)		
Preterm	2kg	2.5mg	20mcg	2ml	
Term-2 months	3-4kg		30mcg	3ml	
3-7 months	5-7kg		60mcg	6ml	
8-12 months	8-11kg		100mcg	10ml	
2-3 years	12-15kg		150mcg	15ml	
4-5 years	16-21kg		200mcg	20ml	
6-7 years	22-28kg	2.5-5mg	250mcg		
8-9 years	29-34kg				
10-11 years	35-40kg				
12-13 years	41-45kg				
≥14 years	46-50kg				

SUCCINYLCHOLINE (SUXAMETHONIUM)

Class

Depolarising neuromuscular blocking agent

Action

A short acting muscle relaxant that mimics acetylcholine as it binds with the cholinergic receptors at the motor endplate resulting in total skeletal muscle paralysis with fasciculations.

Metabolism

Metabolised by pseudo-cholinesterase in the plasma.



Indications

- Paralysis for RSI



Contraindications

- Hypersensitivity to Succinylcholine
- Personal or family history of malignant hyperthermia
- Skeletal muscle myopathies
- Major burns ≥ 24 hours old
- Upper motor neuron injury
- Extensive denervation of skeletal muscle



Precautions

- Increased intraocular pressure
- Hyperkalaemia and other electrolyte abnormalities
- Significant burns
- Bradycardia (if second dose administered)



Side Effects

- Increased intraocular pressure
- Increased intragastric pressure
- Elevated serum potassium levels
- Malignant hyperthermia

Route of Administration				
Intravenous/intraosseous (IV/IO)		III	II	I

Special Notes

- Succinylcholine is not the primary paralysis agent of choice and should only be used if the patient is allergic to Rocuronium or if no alternative is available.
- Succinylcholine will not be routinely supplied to CCPs.

Onset	Duration	Half-life
30 – 60 seconds	3 – 5 minutes	

Dosage

ADULT

Paralysis for RSI					
	III	II	I	IV/IO	150 – 200mg Use 150mg for small adults (<60kg) and 200mg for adults over 60kg.

TRANEXAMIC ACID (TXA)

Class

Antifibrinolytic

Action

Blocks the formation of plasmin from plasminogen. At high concentrations, TXA non-competitively inhibits plasmin. Inhibits the process of fibrinolysis.

Metabolism

Metabolised by the liver and excreted by the kidneys.



Indications

- Adult (>14 years old) major trauma < 3 hours old with any of the following:
 - Suspected major haemorrhage (AP and CCP)
 - Major haemorrhage with SBP <90 mmHg and/or HR >110 beats/min (AP and CCP)
 - Traumatic brain injury with GCS ≤ 12 (AP and CCP)
- Major haemorrhage due to ante/post-partum haemorrhage < 3 hours old (AP and CCP)
- Paediatric (<14 years old) major trauma < 3 hours old with any of the following:
 - Suspected or confirmed major haemorrhage (CCP Only)
- Epistaxis (Adults only)



Contraindications

- Hypersensitivity to TXA
- Injury > 3 hours old
- Significant blood loss is unlikely



Precautions

- Nil



Side Effects

- Headache
- Nausea and/or vomiting
- Seizures
- Rapid bolus may produce hypotension

Route of Administration				
Intraosseous (IO)				
Intravenous (IV)				
Topical				

Special Notes

- TXA is not for routine use in traumatic cardiac arrest.
- Basic first aid is the primary treatment for epistaxis. Topic TXA may be considered if this fails. IV TXA may be considered if haemodynamic compromise is present.

Onset	Duration	Half-life
5 – 15 minutes	7 – 8 hours	2 hours

Dosage

ADULT

Traumatic Major Haemorrhage					
				IV/IO	2g (4 x 500mg amps) Add 2g to 100ml NS and infuse over 10 minutes
				IV	2g (4 x 500mg amps) Add 2g to 100ml NS and infuse over 10 minutes
Traumatic Brain Injury GCS ≤ 12					
				IV/IO	2g (4 x 500mg amps) Add 2g to 100ml NS and infuse over 10 minutes
				IV	2g (4 x 500mg amps) Add 1g to 100ml NS and infuse over 10 minutes
Ante/Post-Partum Major Haemorrhage					
				IO	1g (2 x 500mg amps) Add 1g to 100ml NS and infuse over 10 minutes
				IV	1g (2 x 500mg amps) Add 1g to 100ml NS and infuse over 10 minutes
Epistaxis					
				Topical	500mg (1 x 500mg amps) Soaked into a gauze swab

				IV	1g (2 x 500mg amps) Add 1g to 100ml NS and infuse over 10 minutes
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PAEDIATRIC

Traumatic Major Haemorrhage					
				IV/IO	15mg/kg infused over 10 minutes Up to a maximum of 1g

VECURONIUM

Class

Non-depolarising neuromuscular blocking agent

Action

Moderate to long-acting skeletal muscle relaxant that competes with acetylcholine at post-synaptic acetylcholine receptors preventing the binding of acetylcholine producing skeletal muscle paralysis.

Metabolism

Metabolised by the liver and excreted by the kidneys and bile.



Indications

- Paralysis for RSI
- On-going paralysis post-RSI
- Facilitation of mechanical ventilation



Contraindications

- Hypersensitivity to Vecuronium



Precautions

- Myasthenia gravis or myasthenic syndrome
- Renal impairment
- Hepatic impairment (reduce dose)
- Cardiovascular disease



Side Effects

- Bradycardia
- Flushing
- Pruritis
- Skin rash

Route of Administration				
Intravenous/intraosseous (IV/IO)		III	II	I

Special Notes

- Ability to adequately ventilate the patient must be established prior to administration of Vecuronium.
- The action of Vecuronium can be antagonised by acetylcholinesterase inhibitors (Neostigmine).

Onset	Duration	Half-life
≈ 1 minute	45 – 60 minutes	

Dosage

ADULT

Paralysis for RSI					
	III	II	I	IV/IO	0.3mg/kg
Paralysis post-RSI/Facilitate mechanical ventilation					
	III	II	I	IV/IO	0.1mg/kg Repeat PRN every 45 minutes Take into consideration paralytic agent used for primary RSI

PAEDIATRIC

Paralysis for RSI [Trauma ≥ 8 years; Medical ≥ 14 years]					
	III	II	I	IV/IO	0.3mg/kg
Paralysis post-RSI/Facilitate mechanical ventilation					
	III	II	I	IV/IO	0.1mg/kg Repeat PRN every 45 minutes Take into consideration paralytic agent used for primary RSI

Paediatric Vecuronium Dosing Table

Age	Weight	PRE intubation paralytic		POST intubation paralytic	
		0.3mg/kg IV/IO	Volume	0.1mg/kg IV/IO	Volume
		Reconstituted: 10mg/10ml =1mg/1ml		Reconstituted: 10mg/10ml =1mg/1ml	
Preterm	2kg	Not indicated in this age group for PRE intubation		0.2mg	0.2ml
Term-2 months	3-4kg			0.3mg	0.3ml
3-7 months	5-7kg			0.6mg	0.6ml
8-12 months	8-11kg			1mg	1ml
2-3 years	12-15kg			1.5mg	1.5ml
4-5 years	16-21kg			2mg	2ml
6-7 years	22-28kg			2.5mg	2.5ml
8-9 years	29-34kg	9mg	9ml	3mg	3ml
10-11 years	35-40kg	11mg	11ml	4mg	4ml
12-13 years	41-45kg	12mg	12ml	4mg	4ml
≥14 years	46-50kg	14mg	14ml	5mg	5ml

RETRIEVAL SERVICES EXTENDED PHARMOCOEPIA

Generic Drug Name	Concentration	Packaging	Administration
Metoprolol	1mg/ml	Vial	See CPG 15.3
Ceftriaxone	1g	Vial	See CPG 15.3
Cisatracurium	20mg/10ml	Vial	See CPG 15.3
Dexamethasone	8mg/2ml	Vial	See CPG 15.3
Dextrose 50%	50%/50ml	Vial	See CPG 15.3
Dobutamine	250mg/20ml	Vial	See CPG 15.3
Doxapram	20mg/5ml	Vial	See CPG 15.3
Enoxaparin (Clexane)	40mg	Vial	See CPG 15.3
Glucagon	1g	Vial	See CPG 15.3
Haloperidol	5mg/ml	Vial	See CPG 15.3
Heparin	5000IU/ml	Vial	See CPG 15.3
Hydralazine	25mg	Vial	See CPG 15.3
Hyoscine	20mg/ml	Vial	See CPG 15.3
Insulin regular	100IU/10ml	Vial	See CPG 15.3
Ketorolac	30mg/ml	Vial	See CPG 15.3
Labetalol	100mg/20ml	Vial	See CPG 15.3
Lignocaine 1%	200mg/20ml	Vial	See CPG 15.3
Phenytoin Sodium	250mg/5ml	Vial	See CPG 15.3
Potassium Chloride	20mmol/10ml	Vial	See CPG 15.3
Propofol 1%	20ml	Vial	See CPG 15.3
Sodium Bicarbonate 8.4%	50ml	PFS	See CPG 15.3

MEDICATION PRIVILEGES

MEDICATION	Level 4 	Level 3 	Level 2 	Level 1 
Adenosine	✓			
Adrenaline NEB	✓	✓		
Adrenaline IM	✓	✓	✓	
Adrenaline IV	✓	✓		
Adrenaline IO	✓			
Adrenaline INF	✓			
Amiodarone IV	✓	✓		
Amiodarone IO	✓			
Amiodarone INF	✓			
Aspirin	✓	✓	✓	✓
Atropine	✓			
Budesonide	✓			
Calcium Chloride	✓			
Clopidogrel	✓	✓	✓	
Dexamethasone	✓			
Dextrose PO	✓	✓	✓	✓
Dextrose IV	✓	✓		
Dextrose IO	✓			
Diclofenac IM	✓	✓		
Diphenhydramine	✓			
Fentanyl	✓			
Furosemide	✓			
Glucagon IM	✓	✓	✓	
Glucagon IV	✓			
Glycerol Trinitrate SL	✓	✓	✓	✓
Glycerol Trinitrate INF	✓			
Hydrocortisone	✓			
Ibuprofen	✓	✓	✓	✓
Ipratropium Bromide	✓	✓	✓	
Ketamine	✓			
Ketorolac IM	✓	✓	✓	
Ketorolac IV	✓	✓		
Magnesium Sulphate	✓			
Methoxyflurane	✓	✓	✓	✓
Metoclopramide	✓			
Midazolam	✓			

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Naloxone	✓			
Noradrenaline	✓			
Ondansetron	✓			
Paracetamol PO	✓	✓	✓	✓
Paracetamol IV	✓	✓		
Paracetamol IO	✓			
Phenylephrine	✓			
Prochlorperazine	✓			
Rocuronium	✓			
Salbutamol NEB	✓	✓	✓	
Salbutamol INF	✓			
Succinylcholine	✓			
Tranexamic Acid	✓	✓		
Vecuronium	✓			

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